01. INTRODUCTION : THE WUWAI

DURATION : 08:15 STUDY SHEET

COURSE SUMMARY

This introductory video to the WUWAI of biodynamic embryology presents the therapist's role as a facilitator rather than an active intervener. The concepts of primary respiration and primary respiratory movement are explored as key elements of the healing process. By focusing on discovering the 'neutral', the practitioner learns to centre themselves to allow the therapeutic potency to emerge, particularly through an encounter with the patient's neutral. Furthermore, the notions of gravitational force and electromagnetic force are discussed to bring each patient back into their body, thereby promoting better self-regulation and access to health rather than illness.

INTRODUCTION TO THE WUWAI OF BIODYNAMIC EMBRYOLOGY

TRUST AND ABSENCE OF INDUCTION

In **biodynamic embryology**, the therapist intervenes very little through **direct induction**. It is more about trusting an inherent force that transcends us.

Not giving induction and not actively intervening is a challenge, as classical training has often taught us the opposite. We are not only in contact with the tissue, but also with the **space** surrounding it.

PRIMARY RESPIRATION AND PRIMARY RESPIRATORY MOVEMENT

A key concept is the distinction between **primary respiration** and **primary respiratory movement**. Primary respiration is omnipresent, not limited to the physical body, but present in the totality of existence.

It can be considered the **"breath of life"**. It is essential to experience this and dare to enter this space.

PRINCIPLE OF HEALTH AND BIODYNAMICS

Biodynamic embryology is part of one of the great principles of **osteopathy**: the body's capacity for **self-regulation** and its homeostatic principle of health.

Our work in biodynamics consists of working with health, rather than against disease. This is a paradigm shift, as we are often inclined to seek out the lesion.

My approach invites you to seek health. This is not always easy, as the attraction to the lesion is strong. We do not disregard it, but we do not dwell on it.

THE SECRET: DISCOVERING THE NEUTRAL

The secret to immersing oneself in this process is to discover what the **neutral** is.

PRACTITIONER'S NEUTRAL

Firstly, there is the **"practitioner's neutral"**. This is not simple, as we are not spontaneously in this state.

I have gradually guided you towards a state of neutrality through **meditation**, the "therapist's neutral".

This state is necessary, but it is not yet sufficient to bring forth therapeutic power, nor to fully express the breath of life. The practitioner must first be centred and discover this state of neutral.

MEETING OF NEUTRALS

Once we have discovered this state of neutral and are in contact with the patient, we must open our space to the different fields:

- **Zone A:** The physical body.
- **Zone B:** The fluidic system, which I have emphasised throughout the seminar.
- **Surrounding spaces:** The space we are in, then the different spaces up to the broadest image you can conceive.

After the "therapist's neutral", the second step, and this is where the difficulty lies, is to bring forth the **"patient's neutral"**.

This is the key. You have your own neutral, but if the patient does not enter their own neutral, that is, a state of disengagement, the therapeutic forces cannot emerge.

EMERGENCE OF THERAPEUTIC FORCES

The emergence of therapeutic forces occurs when these two neutrals meet. It is like the formation of "one" and "two" to reach the "whole", which could be called the **"three"**.

When the therapist's neutral meets the patient's neutral, the therapeutic forces – primary respiration, the breath of life – allow the inherent power of the system to emerge.

This emergence manifests with the force of its originality. It is at this moment that the encounter with the embryonic forces occurs. This is the same energy that the body will seek to regenerate, restructure, and regain its function.

THE FOUR BODIES

We will realise that we need to gradually bring people "back to earth".

1. FORCE OF GRAVITY (CONTACT WITH THE EARTH)

This is what is called the **force of gravity**. Being well anchored in one's force of gravity means being well in contact with the earth. This contact is very important.

Some are too "in the earth", as if already "buried", while others are not "on the earth", they float. This fundamental contact must be restored.

2. ELECTROMAGNETIC FORCE (BRINGING BACK INTO THE BODY)

Next, there is the **electromagnetic force**. This involves bringing people back into their bodies.

Following shocks (such as whiplash or other accidents), it is possible for people to be "out of their bodies". Although not visible, they are as if disconnected. It is about bringing them back into their electromagnetic field.

3. ENERGETIC BODY (THE ATMOS, THE NON-FORM)

The third step is to bring people into their **energetic body**, into the "atmos". The "atmos" represents the non-form.

This energetic body is often perceived as a vibrational state, a frequency, a vibration.

4. LIGHT BODY (FULL FUNCTION)

Fourth point, it is about bringing people into their **light body**. Being completely luminous means being fully in one's function.

It is therefore about bringing people back to their full functionality. The more you are in tune with this, the more you attract the right energy around you, what I call **"right synchronicities"**.

THE "MEETING POINT"

To reach this level, we need what is called the **"meeting point"**: a point of encounter between the therapist and the patient.

If there is no such meeting point, there is no therapeutic emergence. Sometimes, patients can be treated without a true "meeting point" occurring; in this case, nothing essential will happen.

Some patients arrive and we already know what we are going to do, but this is not a meeting point. The important thing is to discover and bring forth this neutral in your patient.

This requires a particular space and attention to the possibility of this emergence, through touch, presence, and levelling.

When I work with a child, I always get down to their level to seek this meeting point. If this point is created, the child very quickly enters the emergence of their neutral.

It is then that the therapeutic force can begin to express itself.

LAWS OF NATURE AND CORRECTIONS

We do not impose, we seek the neutral, the patient's neutral. We surrender to what I call the **laws of nature**.

It is very important to have this trust in the process. Corrections will emerge through your presence, movements will occur.

These corrections are similar to what **Sutherland** described, and they are close to the biodynamic movements and the motility of embryonic development. These are the same phenomena that will manifest.

LONGITUDINAL FLUCTUATION

The first very important thing is to restart this **longitudinal fluctuation**. This is one of the primary powers the body uses.

If you study Sutherland, he talks about it, particularly when working at the base, to revitalise this longitudinal fluctuation.

THE WUWAI: NON-DOING

I repeat again: what is your neutral? I ask you this question: what is neutral for you? Because it is from this neutral that we will work.

This is where the Chinese have a fantastic word: the **"wu bei"** (or **"wu wei"**). This does not mean "doing nothing", but rather "not doing" (the **"non-doing"**). This is a very important distinction.

Some therapists only use their knowledge, others rely more on their imagination. I ask you to use both.

We need this knowledge, your background, and we need this imagination. We must combine them and work with them progressively.

02. INTRODUCTION THE MESODERM

DURATION : 05:50 STUDY SHEET

COURSE SUMMARY

This video introduces the concept of the mesoderm and its central role in embryonic development and human health. The instructor explores the growth and longitudinal power that emerge from this tissue, highlighting the importance of synchronisation between the inside and outside of the body. By integrating the notion of bodily fluids, he demonstrates how this fluidic approach can transform our understanding of health and therapy, while emphasising the body's capacity for self-regulation. Finally, he discusses key concepts such as the notochordal axis, the development of the viscerocranium, and the importance of water in body structure.

INTRODUCTION TO THE MESODERM

What is happening here demonstrates the unity and **synchronicities** of development. The importance of this coiling around the **cardio-pleuroperitoneal cavity** is paramount. We can say that the **mesoderm** is an articulation between the **ectoderm** and the **endoderm**.

The therapeutic path and the path of practice are a **mesodermal** path. These processes, these therapeutic lines, develop in a process of **growth**. It is a single movement, not a succession of distinct movements. We can connect to this great movement.

THE NOTION OF GROWTH AND LONGITUDINAL DEVELOPMENTAL POWER

It is essential to observe this notion of **growth** and this **longitudinal developmental power**. This developmental power is an encounter between the inside and the outside, and vice versa.

The **therapeutic force** comes from the outside. The body has a capacity for **self-regulation** to maintain its health, and we work with that.

In the stomach, there are **villi**. These villi are made up of cells, a layer of **epithelial** cells. This layer contains cells that produce **mucus** and cells that produce **acids**.

There is always, and this is very common throughout the body, a layer of **stem cells**. In the stomach tissue, these are **regenerative** cells that use this power of self-regulation.

Remember that your body is continuously regenerating. The **gastric mucosa**, the cells, the skin... all of this is constantly regenerating and uses this embryonic power.

Through the encounter of neutrals, this principle that I propose allows for the sublimation of this process. In this principle of **structure-function**, I speak of **form**.

Am I in good form today? This maintains this notion of health. The notion of **time**, the notion of **tissue**, and the notion of **fluids** are interconnected.

THE IMPORTANCE OF FLUIDS IN THE BODY

When we began studying **embryology**, I mentioned that your mental image was built around **bone**. In reality, our body is built around **fluids**.

This means that we are moving more and more towards the perception of this fluid body. In your body, you have **volumes** that can express a mental state. In fact, you function according to these volumes: it is the **void** and the **fullness**.

SUMMARY AND COURSE ORIENTATION

1. **First session:** Study of the **notochordal axis** and the development of the **neural tube**.

2. **Second session:** Study of the development of the **viscerocranium** and the importance of the **base of the skull** in this process.

3. **Now:** We will essentially study everything that is in green, which we will call the **internal man**. That is, the entire **support system**, and especially the **fluid system**.

I remind you that you are made up of a very large percentage of **water**. But how does this water get into the body?

This fluid body will be dependent on an **electrical body**. We could say: I first need something undifferentiated, then an **electrical field**, and finally a **structural field**.

We will first go into this void, to feel the electrical field, and to help the fluids restructure. Then, these fluids will **densify**.

We start from a level of **light** towards a level of **density**, to one day rediscover our light.

THE FORM OF WATER AND ITS THERAPEUTIC IMPLICATIONS

70% of body mass is **equine**. But water can express itself as **vapour**, as **gas**. There is water that can **crystallise**. This is also important: it is the **form of water**.

You have already heard of this Japanese study which, depending on thought, structured the quality of the form of water. This means that our **thoughts** are important.

It must be realised that water is **structured**, it is bound. This is what maintains **cohesion**. These phenomena give rise to **dried**, desiccated or hydrated fields.

When we work at the level of the **brain**, of the **cranium** on the fluid body, we will gradually learn to feel this energy level. Is the body dried out or not?

Structural techniques can momentarily dehydrate the body if they are not done with awareness. On the other hand, **biodynamic** techniques tend to rehydrate the body.

This notion of **rebalancing** in terms of my **ions**, in terms of my **macromolecules**, in terms of how water will structure itself, forms shells that provide cohesion.

If you have **dehydration** in your body, you lose your capacities. Remember that it takes **six months** to rehydrate. Rehydrating means drinking spring water, quality water. Biodynamics helps to restructure the body's water quality.

THE HOMUNCULUS INTERNI AND MESODERMAL TRAJECTORIES

The **Homunculus Interni**, the mesodermal man, is linked to **trajectories**.

- On the **vascular** level.
- On the level of **urogenital** development trajectories.
- On the development of **locomotor** trajectories.

I like the idea of trajectory because it gives you therapeutic help and support for understanding.

03. MESODERM AND NOTOCHORD FORMATION

DURATION : 03:43 STUDY SHEET

COURSE SUMMARY

This video guides you through the fascinating process of mesoderm formation and the crucial role of the notochord in embryonic development. You will learn how these structures interact at a cellular level, particularly through the primitive streak and bottle cells, leading to the formation of primitive mesenchymal tissue. Furthermore, you will explore practical aspects related to the coccyx, discovering how impulses and electromagnetic fields influence tissue health. The session includes a practical approach to notochordal movements while highlighting the interconnection between the sacrum, coccyx, and even the heart, thus deepening your understanding of biodynamic embryology and osteopathy.

ESTABLISHMENT OF THE MESODERM AND NOTOCHORD

THE ESTABLISHMENT OF THE MESODERM

The first crucial step is the **establishment of the mesoderm**. We observe the appearance of the **primitive streak** and the **notochordal axis**.

There is an aspiration field between the **ectodermal** tissue and the **endodermal** tissue, creating a movement where a space will be filled.

Mesodermal tissue originates from **epiblastic** tissue, not ectodermal tissue. The **bottle cells** are drawn towards the centre.

This is how the space between the epiblastic and hypoblastic tissue becomes the **primitive mesenchymal tissue**, the future mesoderm.

ROLE OF THE NOTOCHORD

The **notochord** influences this process. It is an electrical field that acts.

In the previous lesson, we worked at the level of the **sacrum**, focusing on the notochordal axis.

We will explore the process in more depth. The notochord, positioned here, is linked to **Hensen's node**.

This generates a wave-like movement, a **cranio-caudal** movement, where the primitive streak progressively regresses backwards.

This primitive streak is found at the **sacrococcygeal** base, constituting an energetic vestige.

Thus, the establishment of the mesoderm is observed.

THE PRACTICE: THE COCCYX AND IMPULSES

We will practice by focusing on the **coccyx**, as this area concentrates energy.

The first impulse occurs on the **hypoblastic** tissue. It is at this moment that the epiblast becomes **ectoderm** and the hypoblast becomes **endoderm**.

This means that the epiblast informs all my tissues, and it is itself completely integrated by my notochord. It is an **electromagnetic** field that informs the body.

When we consciously touch this level, we are connected to this force that the body has established.

DEVELOPMENT OF MESODERMAL TISSUE

We arrive at the second impulse, the second wave, which will form the **mesodermal tissue**.

I thus obtain an initial notochordal information which, with my primitive streak, generates this aspiration field.

The latter creates the space of my primitive streak, in contact with the establishment of my **embryonic pedicle** and my fields of **permeation** and **infusion** that are organising themselves.

It is in this movement that the integration of my mesoderm takes place. This occurs between the **fourteenth** and **sixteenth day**.

When you practice, you first perceive this kind of **"Tai Chi"**, this notochordal wave.

Then, underneath, at the same time as this movement, there is an invasion underneath. It is a **double movement**.

One can position oneself on the sacrum or on the coccyx. Subsequently, we will address other very specific balance points at the level of the **heart**.

The presumptive aspect of the heart is determined by this field of action.

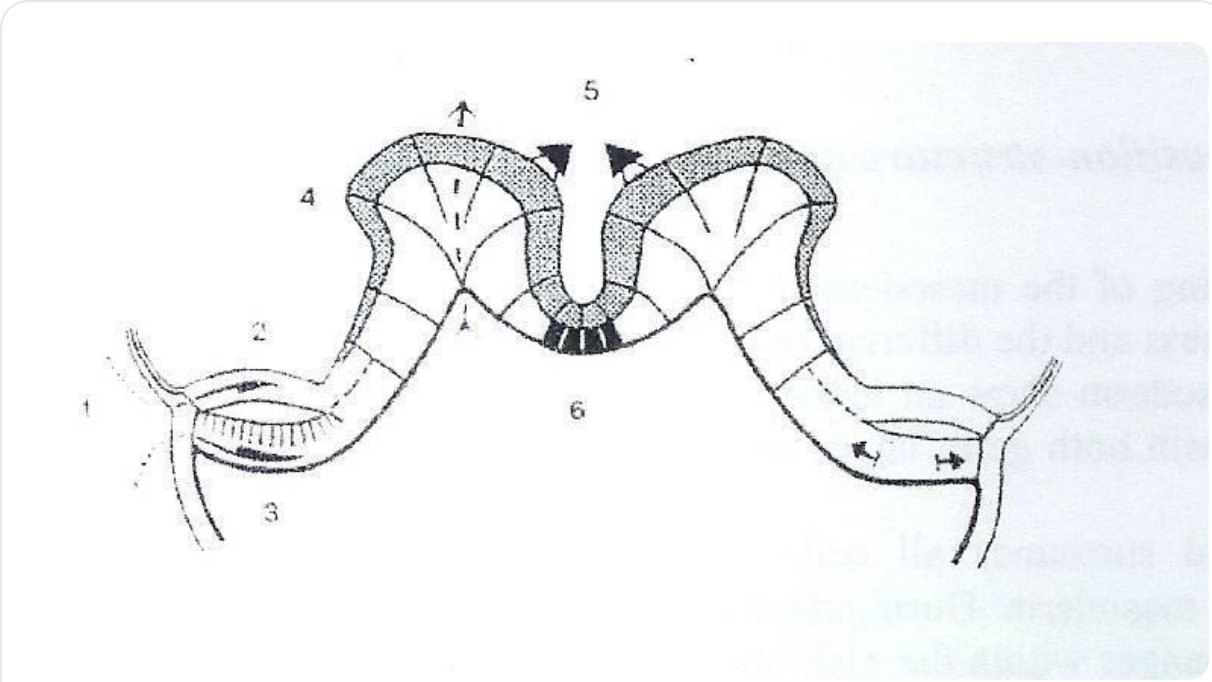


FIG. — Beginning of neurulation

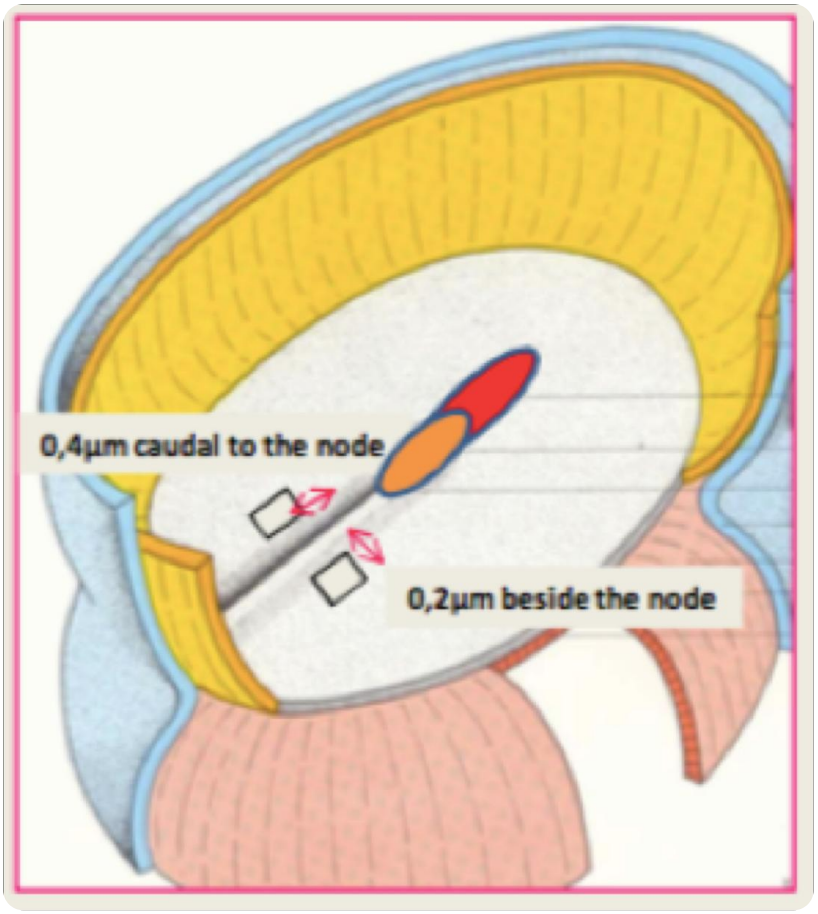


FIG. — Early embryonic development

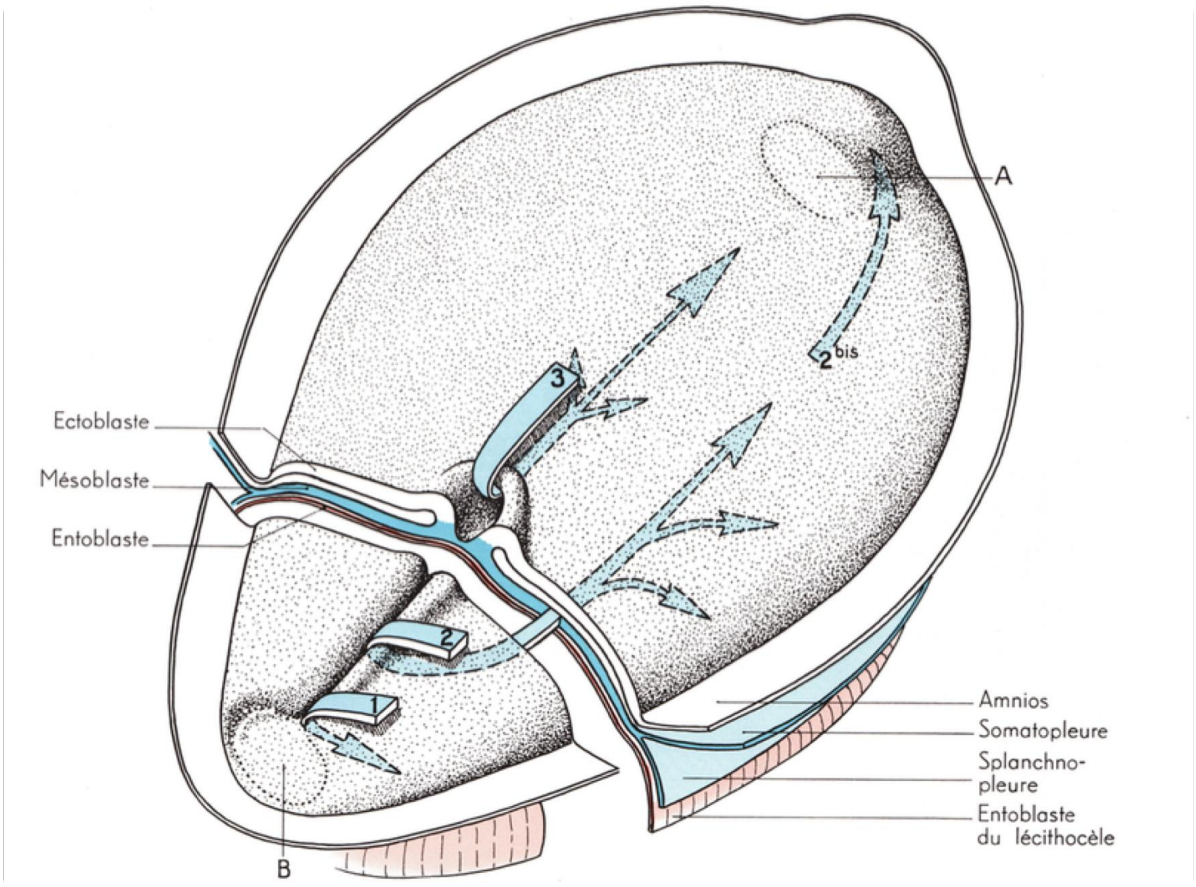
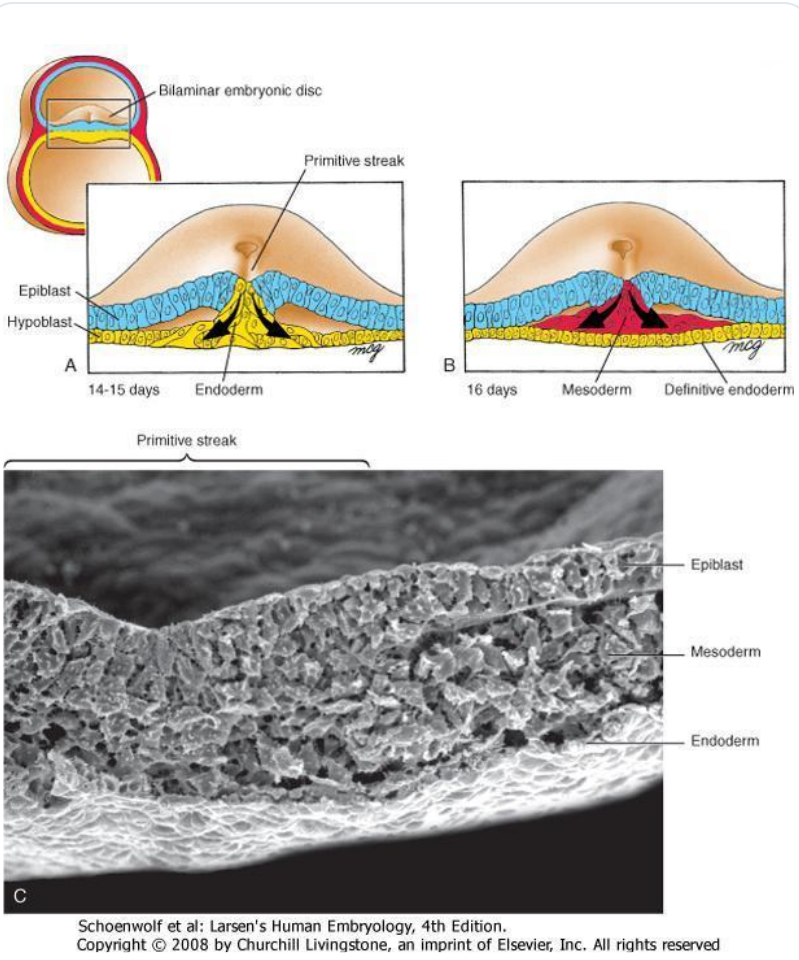


FIG. — Human gastrulation 3 weeks



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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FIG. — Human gastrulation

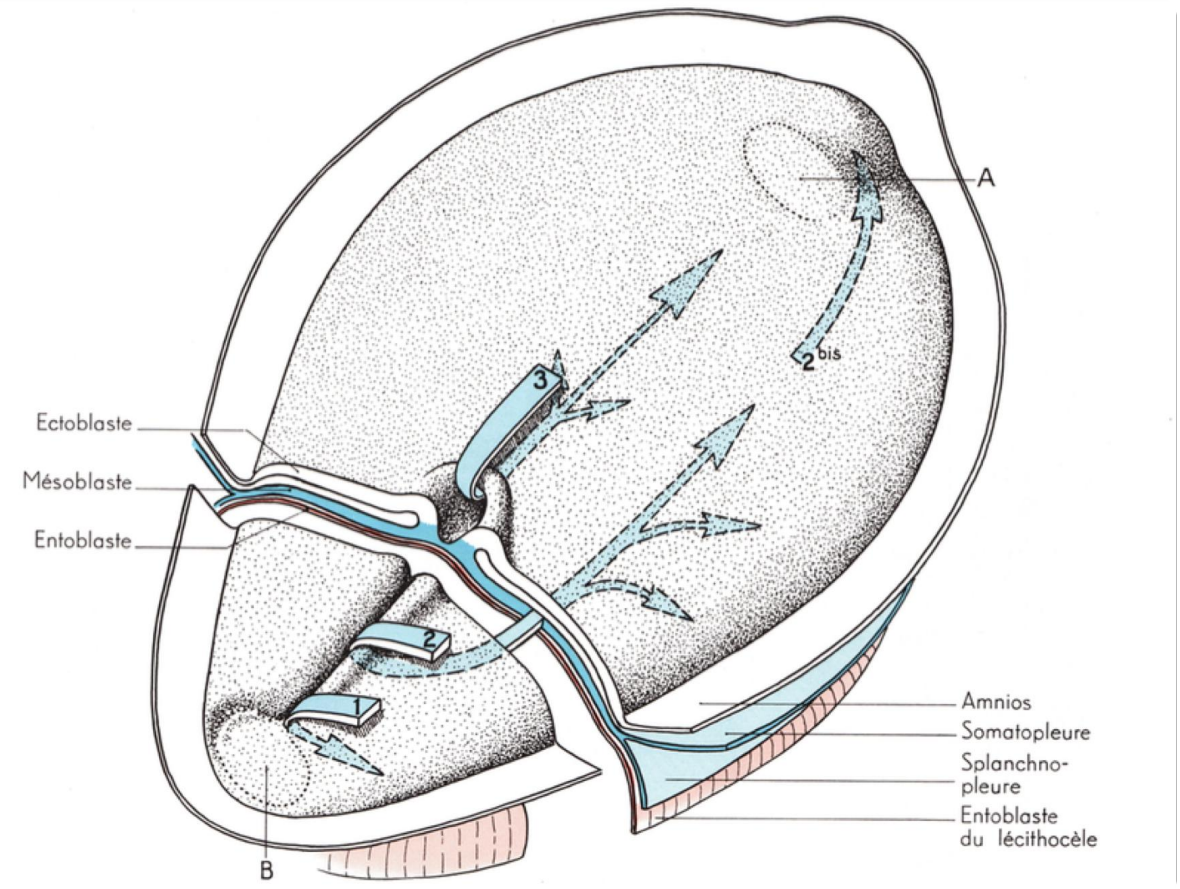


FIG. — Human embryonic gastrulation

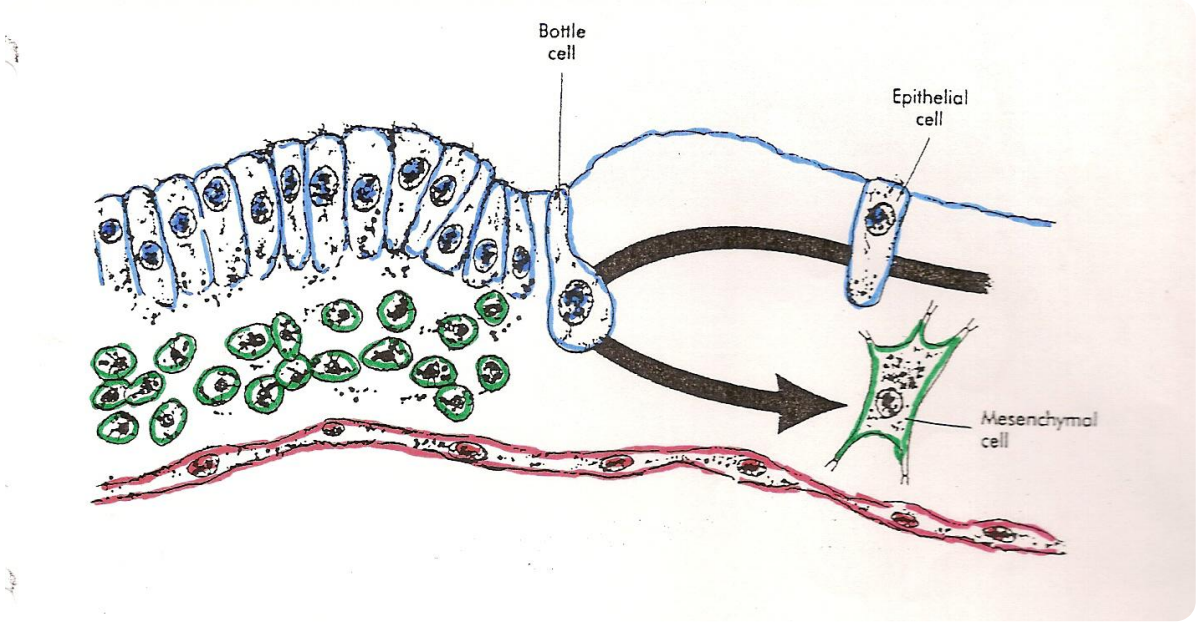


FIG. — Mesenchymal migration

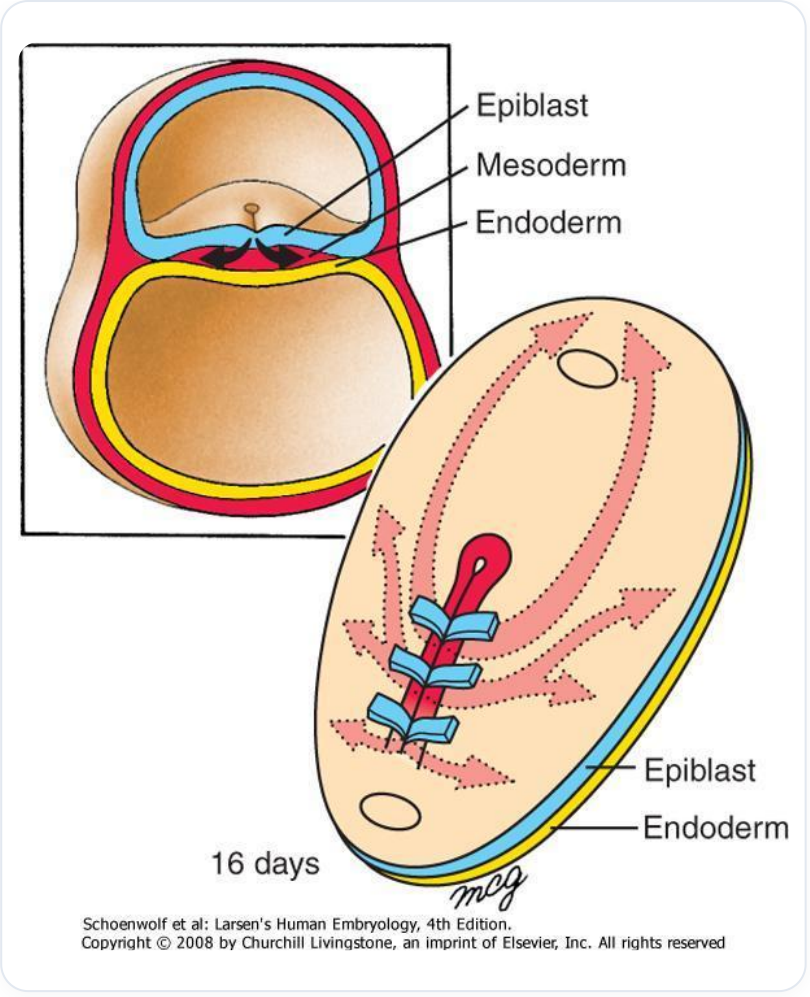


FIG. — Gastrulation 16 days

04. THE CIRCULATORY SYSTEM UPDATE

DURATION : 03:59 STUDY SHEET

COURSE SUMMARY

This video offers a profound update to the traditional model of the circulatory system, demonstrating that there are in fact three distinct venous systems. You will learn about the complex interactions between blood and the heart, as well as the different pathways blood can take, illustrating the intrinsic intelligence of circulation. The theories of pioneers such as M. Herveillat and M. Batson allow us to rethink the roles of veins and understand the cardinal, vitelline, and umbilical systems in detail. Practical implications for common pathologies, such as lower back pain and local acidity, will also be addressed, providing you with concrete tools to enrich your osteopathic practice.

THE CIRCULATORY SYSTEM: AN UPDATE

THE DEVELOPMENT OF VASCULAR PATHWAYS

The traditional image of the **circulatory system** is that of a circuit starting from the heart via the **arteries** and returning to it via the **veins**. While this is true, there are actually three different venous systems:

- **A vertebral system:** exclusively in relation to the vertebral system.
- **A portal system:** draining the entire **ectodermal**, **mesodermal**, and **endodermal** systems.
- **A third type of venous system.**

Blood does indeed return to the heart, but it can also turn back or take other pathways depending on the pressures and depressions at the heart. For example, if the portal system is overloaded and cannot ensure its return, the blood will take another route. It will use a **caval return system** via diaphragmatic exchange areas.

This means that pressures can adjust. Blood does not follow a simple closed circuit but can take various routes. There is an **intrinsic intelligence** to this system, always aiming to return to the heart.

A RENEWED VISION OF THE VENOUS SYSTEM

Blood does indeed leave the heart and return to it. However, our understanding of the venous system, and by extension the circulatory system, dates back to the 17th century. This conception has now been **renewed**.

The pioneers of this new understanding are **M. Herveillat** and **M. Batson**. A question arises: "But I was always told that veins contain valves to prevent blood from flowing backwards." In reality, these valves are only present in the **large veins**, and not throughout the entire network. There are very few small valves of this type.

THE DIFFERENT VENOUS SYSTEMS TO STUDY

We will study:

- **The cardinal veins:** which will become subcardinal and supracardinal.
- **The vitelline system:** in relation to the liver.
- **The umbilical system:** which will be remapped into the vertebral-caval-portal system and reintegrated into osteopathic practice.

The cardinal system will change so much that we will see in detail how it forms tissues and structures that will integrate into both **ectodermal** and **mesodermal** structures.

A method has been developed to put this into practice. It is crucial to have understood its meaning.

PRACTICAL AND CLINICAL IMPLICATIONS

Stasis and changes in **abdominal pressure** can have significant consequences. For example, **low back pain** can be the result of an overload of the portal system.

It is then necessary to "unload" the vertebral system due to this portal overload. This is where the entire mechanism of **discopathies** comes into play.

Any form of stasis will lead to **acidity**, even local, which will need to be resolved. The same pattern is always used. For example, a **sacral tumour** is almost always accompanied by **metastases** to the left kidney, left lung, and brain. A constant trajectory is observed.

In **prostate problems**, the same pattern can be found. Furthermore, there is **vascular continuity** between the vertebral sinuses and the sacral sinuses of the lesser pelvis.

A blood test taken from one arm does not reflect the totality of the blood that could be taken from a venous, arterial, or other system.

FIG. — Cardiovascular system

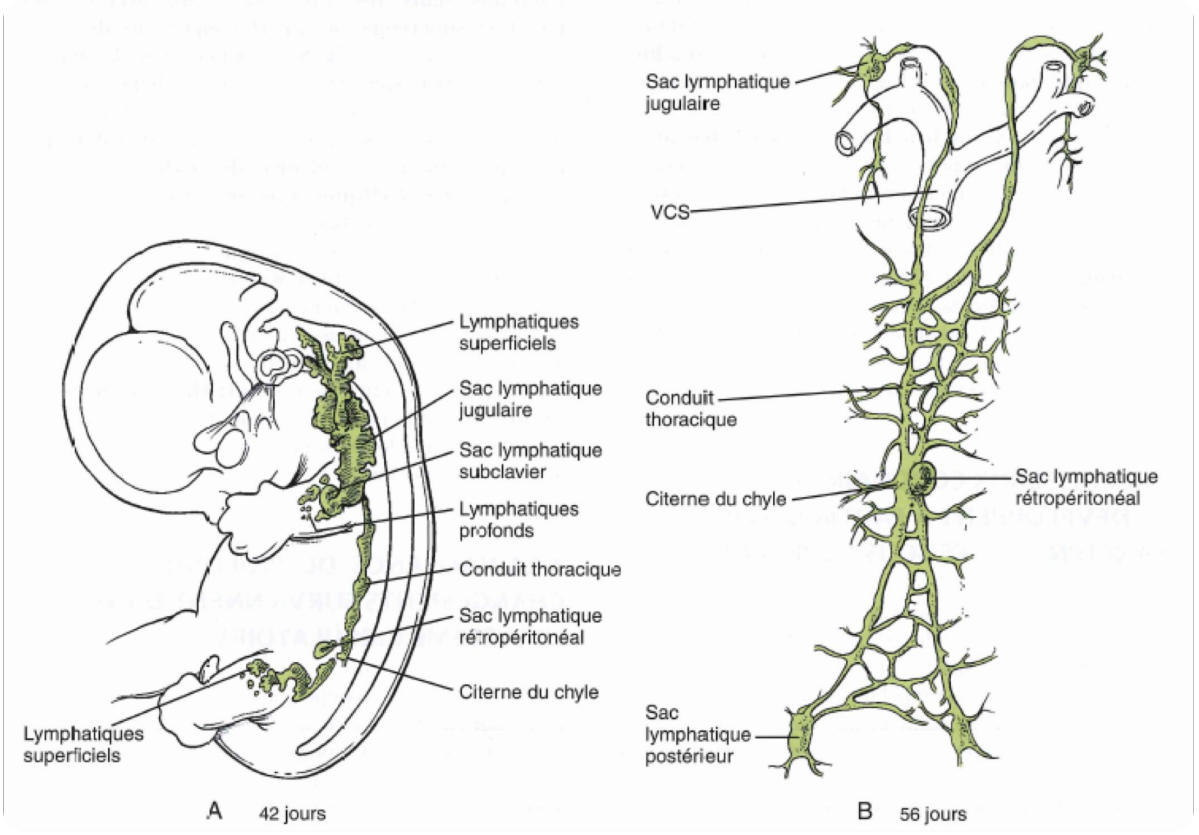


FIG. — Lymphatic system development

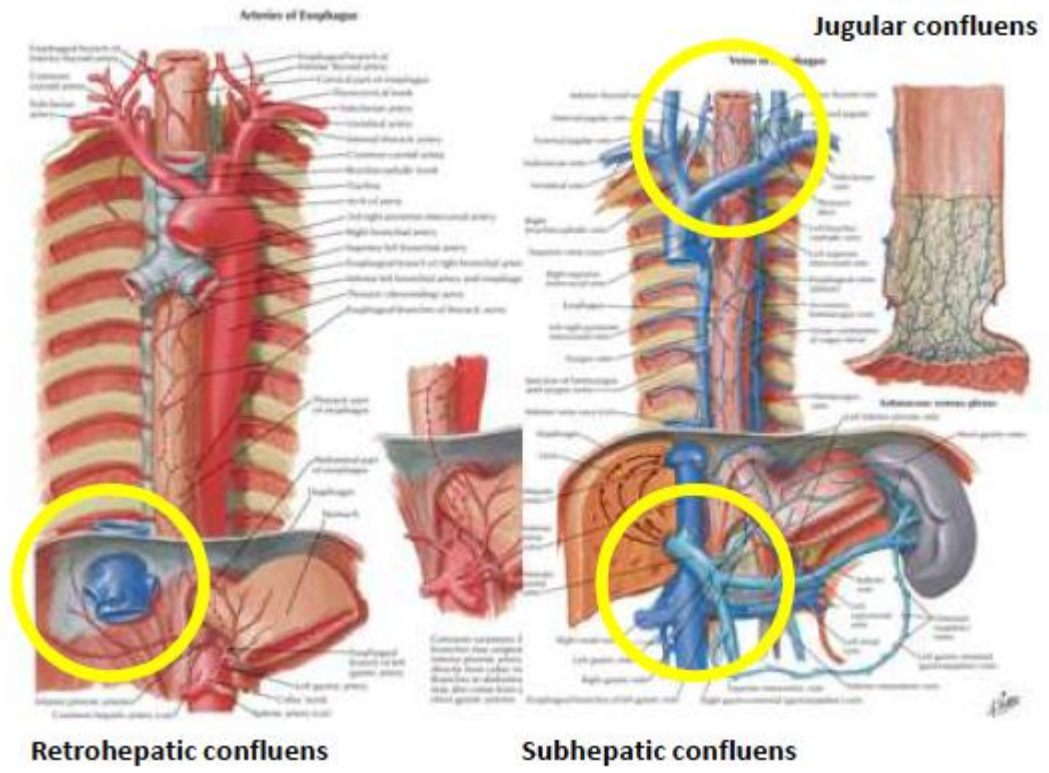


FIG. — Esophageal collateral veins

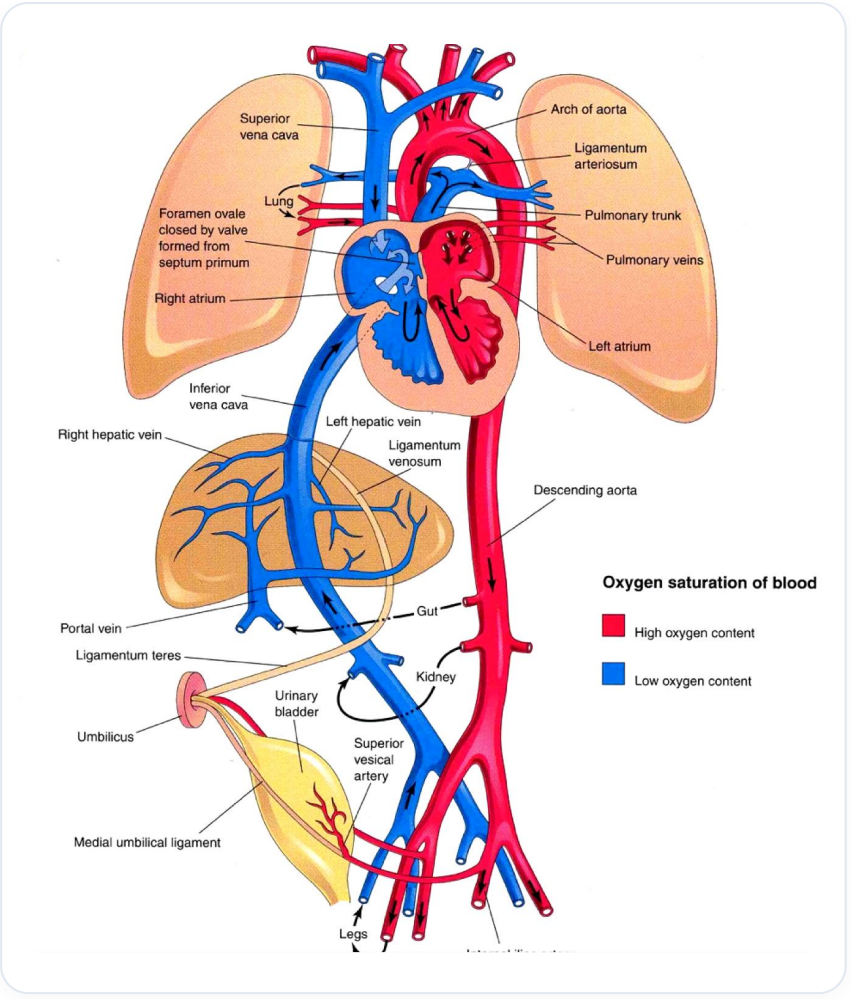


FIG. — Human fetal circulation

05. CONSTRUCTION OF THE PRIMITIVE VESSELS

DURATION : 13:32 STUDY SHEET

COURSE SUMMARY

This video explores the fascinating construction of primitive vessels within the embryo, highlighting the movement of fluids and the embryonic forces acting from the outside to the centre. Key concepts include the vital role of the amniotic cavity, the importance of an intercellular substance, and the four components necessary for vascular formation. The teaching also addresses the development of primitive arterial and venous systems, polarity, and the phases of assimilation and dissimilation, while emphasising how these interconnected processes form the basis of embryonic vascular networks. This knowledge is essential for any practitioner wishing to understand the anatomical and physiological foundations of human development.

CONSTRUCTION OF PRIMITIVE VESSELS

1. FLUID MOVEMENT AND EMBRYONIC FORCES

At the moment the embryo penetrates the **uterine mucosa**, all fluidic force comes from the outside and centres within the embryo. The **embryonic force** develops from the periphery towards the centre, whether fluidic, venous, or arterial. Everything converges towards the centre.

2. ROLE OF THE AMNIOTIC CAVITY AND YOLK SAC

The integration of fluids is enabled by the **amniotic cavity**. The anlage of the vascular system appears in the **splanchnopleuric mesoblast**, at the level of the wall of the **yolk sac**, the **chorion**, and the **intra-embryonic mesenchyme**. This manifests as small vascular islets in the yolk sac. The amniotic cavity allows the integration of this vitelline cavity towards the centre, marking the vitelline, placental, and neonatal stages.

3. THE FOUR ESSENTIAL COMPONENTS FOR VASCULAR FORMATION

For **kinetics** and vessel formation to occur, four components are always necessary:

1. The presence of an **intercellular substance**.

2. A **local chemical concentration**.

3. The presence of **vacuolisation**.

4. The presence of **trajectories**.

Blechmit compares this to a **rainbow**, which requires water, rain, sun, and a certain angulation. If a single element is removed, the rainbow immediately disappears. The same applies to vessels. If a trajectory or a concentration is removed, the process stops. The vascular system is in constant adaptation throughout life.

4. DEVELOPMENT OF THE VASCULAR NETWORK

A large network develops from the yolk sac, but also from the chorion. The embryo initially develops in a **"racket"** shape, with an **embryonic pedicle**. The **notochord**, located below the primitive streak, influences the neural plate. The cells of the neural plate do not grow at the same rate as the surrounding cells, leading to the development of a **groove**.

5. RESTRUCTURING OF FLOW AND VESSEL FORMATION

This groove space leads to a restructuring of the **trophic metabolic flow** within the mesoderm, in the form of bilateral fluidic trajectories. In mesodermal development, small **vacuoles** appear. This is a growth process where the fluidic mesodermal tissue is aspirated. These vacuoles grow, organising a trajectory with significant concentration and intercellular substances.

All these components allow for the formation of a vessel. These structures continue to grow to form what is called a **corrosion field**. When these fields meet, the tissue nourishing the membrane yields, allowing the progressive development of a vessel.

6. POLARITY AND PHASES OF ASSIMILATION AND DISASSIMILATION

A field of aspiration, given by this wave and methodical process, is essential. Two movements are at work: an **autochordal** movement and an underlying aspiration movement. The notochord reorganises the ectoderm into a groove, organising bilateral trajectories around the neural tube. The vacuoles meet, forming the first vessel, called the **assimilation vessel**. Cells feed on it.

A **polarity** is observed: food arrives from the ectoderm, creating an aspiration. Initially, congestion is low, but it becomes more significant in a later phase.

7. APPEARANCE OF PRIMITIVE ARTERIAL AND VENOUS SYSTEMS

This congestive phase recreates an elimination phase. An **arterial** type vessel appears through the assimilation phase (aspiration field). Then, through the return phase and the elimination of **catabolites**, a second level develops: a **descending network**.

There is thus a system of first assimilation of information, then a second plane of disassimilation. The first develops the **primitive arterial system**, the second the **secondary venous network**. These two returning fluidic trajectories are the anlage of the **cardinal vein** and the two **primitive aortas**. The venous system is more lateral and external than the primitive aortas.

8. PRIMITIVE EMBRYONIC NETWORKS

The first vascular network to develop is that of the **primitive aortas**, under the dependence of developmental movement. The uplift between the ectoderm and the endoderm aspirates from the primitive streak, which becomes the mesoderm.

The mesoderm, through the development of the neural plate, reorganises and creates trajectories with vacuolisations. These vacuolisations, through the aspiration field, create the necessary components to form a **double aorta** that will join towards the midline. Laterally and anteriorly, a return occurs in the form of trajectories of the cardinal veins.

Thus, the first embryonic network is a primitive aortic network of nutritional assimilation, and the return is the disassimilation phase. This first vascular network is therefore composed of:

1. The **primitive aortas**.

2. The network of **cardinal veins**.

The whole is integrated by the **embryonic pedicle** and the **vitelline system**. The vitelline system is crucial for the visceral system.

9. THE ROLE OF THE SPLEEN AND LIVER IN VOLUMETRIC AND VASCULAR BALANCE

The four components for all vessels are essential. Clinical cases show that volumetric changes can recreate vascular fields. For example, in an anaemic patient due to an iron problem, the **spleen**, a volume regulator, can increase blood volume by bringing back red blood cells.

The spleen can "reopen" and re-establish a concentration, reactivating old vascular networks that had atrophied. Any malformation of the spleen is linked to a malformation of the vascular system. The spleen is fundamental for rebalancing the volume of the system, by adapting its volumetry to metabolic needs.

10. SYNTHESIS OF PRIMITIVE VASCULAR COMPONENTS

The assimilation phase, organised by the notochord and neural tube, reorganises a groove and a trajectory. It includes the necessary components:

- The **trajectory**.
- The **vacuole**.
- The **metabolic concentration**.
- The presence of **intercellular substances**.

These elements give rise to a primitive vascular system of assimilation (primitive arterial system) and, through the return, to a disassimilation system (venous system). The first primitive aortas and the two cardinal veins (primitive cardinal veins) constitute this first vascular network.

The ectoderm reorganises the vascular plane, but the first aspiration field is linked to a metabolic field (permeation and infusion movements). Fluids must always be brought back to the centre.

The spleen is a great motor, but for a functional spleen, the **liver** is essential. The liver, through its growth and orientation, is the motor of organisation. It is a global (sub-diaphragmatic) and left-right organisational centre, structuring the gastro-hepato-pancreatico-splenic space. The splenic system depends on the liver.

FIG. — Lateral embryonic folding

06. THE HEART : INTRODUCTION AND ORIGINS

DURATION : 11:43 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth introduction to the central role of the heart in embryonic development. The heart, which begins to beat as early as day 21 of intrauterine life, plays a crucial role in information distribution, influencing the neural tube and the digestive system via autocrine and paracrine mechanisms. The interdependence between the child's heart and the mother's health is also discussed, highlighting that children are energetically linked to their mother until the age of 7, a significant moment in their development. The primitive origins of the heart, its asymmetrical position, and the polarities established from fertilisation are also explored, illustrating the complex interaction between biology and embryology, and how this influences the individual's future health.

THE HEART: INTRODUCTION AND ORIGINS

THE HEART, ENGINE OF DEVELOPMENT

Your **heart** begins to beat on the **21st day** of intra-embryonic life. After three weeks, it is the first system to function and will maintain this function throughout its life: the **distribution of information**.

As soon as the heart beats, the process of **growth** accelerates. The first to receive this great information from the heart is the **neural tube**, the entire brain.

It is the heart that is the engine, the one that will draw in. There is a very important and functional relationship between the heart and the brain. It is the heart that will give strength to the brain.

Studies have shown that in the **deep ventricular system**, there is a direct pulsatile vibration from the heart that informs the brain. There is a direct link, at very subtle levels.

THE HEART AND THE DISTRIBUTION OF INFORMATION

The heart distributes information. But what information? It distributes **"nourishment"**, in close relation with the **digestive system** and the link with the mother.

You bring trophic information organised by your heart and redistributed by your system. You are therefore dependent on your **endoderm**, your digestive system, in your development.

Autocrine, **paracrine**, digestive, and only then does the system visibly appear.

THE CHILD'S HEART AND THE MATERNAL LINK

When we work on this link with a child, it is often necessary to treat the **mother**. If the mother is not free in her function, the child cannot free themselves either, because they will be energetically dependent, at least until **7 years old**.

At the cellular energetic level, all the cells in your body are renewed at 7 years old. This is called **"the new body"**. It is their first birth, the moment when they can have **"reason"**.

It is the moment when their **frontal sinus** opens and their **thyroid** becomes completely electrical, because it has re-informed their heart. This is why one can feel the heart at a distance.

It is informed by the **neural crest** and has transmitted its information. At 7 years old, the child can defend themselves in their environment, they find their first independence.

PRIMITIVE ORIGIN OF THE HEART

The heart anlage appears in the **splanchnopleural mesoblast**, on a primitive plane. Again, it will be dependent on the entire **vitelline system**.

Here we find a presumptive force of the heart before **gastrulation** and before the vascular plane, along the **cardiogenic** cells of the epiblast. It already has its full field of possibilities.

When I take a **sacrum** or a **coccyx** and I put myself into the system, I already have the history of the heart presenting itself. So, I must begin to treat the heart by first releasing the **primitive streak**, under this first wave that develops.

SYMMETRY BREAKING AND HEART POSITION

The breaking of **symmetry** occurs as soon as the **notochordal** process is established. There is a nodal flow with small cilia that bring cerebral vesicles from right to left, via the sonetic **HH**, the **FGH8**, and finally a change in **retinoic acid**.

This induces a change at the **calcium** level which gives a different integration in the genetic information. Ultimately, there is a breaking of symmetry from right to left, where important information is given at this level.

This breaking of symmetry will be one of the factors that will position the heart to the left in the future. This is done by **morphogenetic molecules**, with an increase in calcium levels over an entire lateral plate, and thus all the information of genetic integration will be different.

This asymmetry occurs between the base of the skull and the sacrum, with the primitive streak. The deviation is present: it is a flow that comes from right to left.

POLARITY AND SPERM IMPACT

This is found at another level by the **follicular nurse cells** which release enzymes. They already create **polarity** within the oocyte, in the cytoskeleton.

A first polarisation occurs on a **cranio-caudal** and **dorsoventral** plane, even if the oocyte is not yet informed of its external environment.

We observe that the site of **sperm** penetration will provide re-information in the longitudinal axis of the embryo, where this presumptive force will be created.

This integration is already information from the outside by the force of the sperm that restructures the force of the cytoskeleton on an almost **electromagnetic** plane. Some describe that this is found in the apical part of the embryo.

This means that a presumptive field of information is captured. And it is in this presumptive field that the future **cardiogenetic** phase will be found.

The sperm already represents, in a way, the cardiac force, the pulsation that is preparing there. But there is no form here yet, it is a pre-form.

EMBRYO ORGANISATION AND HEART FORMATION

We will very quickly see that an entire organisation of the embryo will develop with the famous **"ola"**, with a fulcrum here, and a fulcrum there, with the primitive streak, and with the reintegration of this system.

The importance of these points is crucial. The second point will be the moment when I have the wave, that is, the passage of the establishment of my **third embryonic chamber**, the **coelom** or the **external lecithal cell**.

This will reorganise the embryo and create an **S-shaped** structure. It is at this moment that I will have the integration in relation to the attachment point, and this is what gives my primitive streak. In an embryo, the heart is there.

THE HEART, PERICARDIUM AND DIAPHRAGM

The heart has found its place, it has integrated the **pericardium** and the **septum transversum**, with these **diaphragmatic pillars** which will become a fulcrum for the cardiac system.

Treating a heart means releasing the diaphragm, since its orientation does this. The diaphragm truly becomes our therapeutic tool, since the pericardial cells are the same cells that will be found on the plane of the septum transversum, i.e. the diaphragm.

Where do they originate? I have my heart, and there I have small **mesenchymal cells** that will form my diaphragm. The ectoderm grows fully, and it balances, and will form this whole system.

THE IMPACT OF RECOGNITION AND FERTILITY

The sperm arrives, and the **"key"** is there for the encounter. The pill has killed the concept, the possibility of conceptualising. The hormonal system has been cut off for years.

More and more women are having difficulty conceiving. A medical level intervened saying: "Now we can help you." They do what is called **"IMSI"**, where a needle is taken, pricked and passed through the membrane.

We hope to be in the right place in relation to polarity. Except that we do not have the first recognition, which is a membrane, coronary, tissue, molecular recognition.

Probably, these children will have more and more fertility problems. So, they will no longer know how to reproduce or recognise each other. The first recognition has been removed.

These children who come for consultation, they must be recognised. Recognised at what level of the body? Their heart. They must be contacted by looking into their heart.

If you do, the child knows it. It's the only thing you can give. "I can recognise you, make you reborn." Otherwise, these people will be searching their whole lives.

METABOLIC POWER AND INITIAL DEVELOPMENT

This already happens in your **primitive vascular system**, on the fifth or sixth day. The **zygote** must have metabolic power in a specific pole.

The **cellular trophoblast** must transform into **syncytiotrophoblast**. This syncytiotrophoblast has lytic activity to be able to penetrate the mucosa. If the embryo does not have this power, it cannot enter the mucosa.

Secondly, the mother must be welcoming. But she is not the one doing the work for the embryo. Has it received this power?

Its metabolic power is determined first by a phase of **cellular dedifferentiation**. During the first three or four days, the zygote does not grow.

However, it increases its **cell division**, at cleavage, at high speed. There is an overall increase in the membrane, and a small loss of water to the outside.

This is where it will create this first cavity called **"germ of the sky"**, the **blastocoel**, which is a specific connection to your **"outer sky"**.

Then, at the cellular level, there will be an increase in the membrane. If the membrane increases, exchanges increase. Membranes are spaces for exchanges, always polarised.

If I have a cell, I have a nucleus, and I see that my nucleus is concentrated. This is where I have my greatest exchange force. Instead of having 50, I will have 3000.

Cells in this division phase become **200 times smaller** than the two original blastomeres. Relatively, my membrane increases.

If I take my membrane in relation to my first two blastomeres, let's say it's 100 metres. If I take all these small membranes, I increase enormously.

I increase my metabolic power. If I have sufficient metabolic power, I have sufficient strength to enter.

METABOLIC POWER FOR LIFE

I will always need this metabolic power, all my life. It's a bit like the **"battery"** you will receive. And you must trust that.

It regenerates up to a certain point through your **sleep**, your **diet**, your **breathing**, your right thoughts, etc. You degenerate it if you do not respect this and if you are intoxicated.

You lose your fundamental substrates of this power. So, you need this. The heart originates practically in the contact of the sperm, in this encounter, symbolically.

Afterwards, it is found at the level of the primitive streak. Then, information already from the cardiac fields that are formed here in relation to the presence of a healthy oedema, it is in the growth wave of the development of the cardiac fields.

And then we will see something that is very important, it is all this developmental movement.

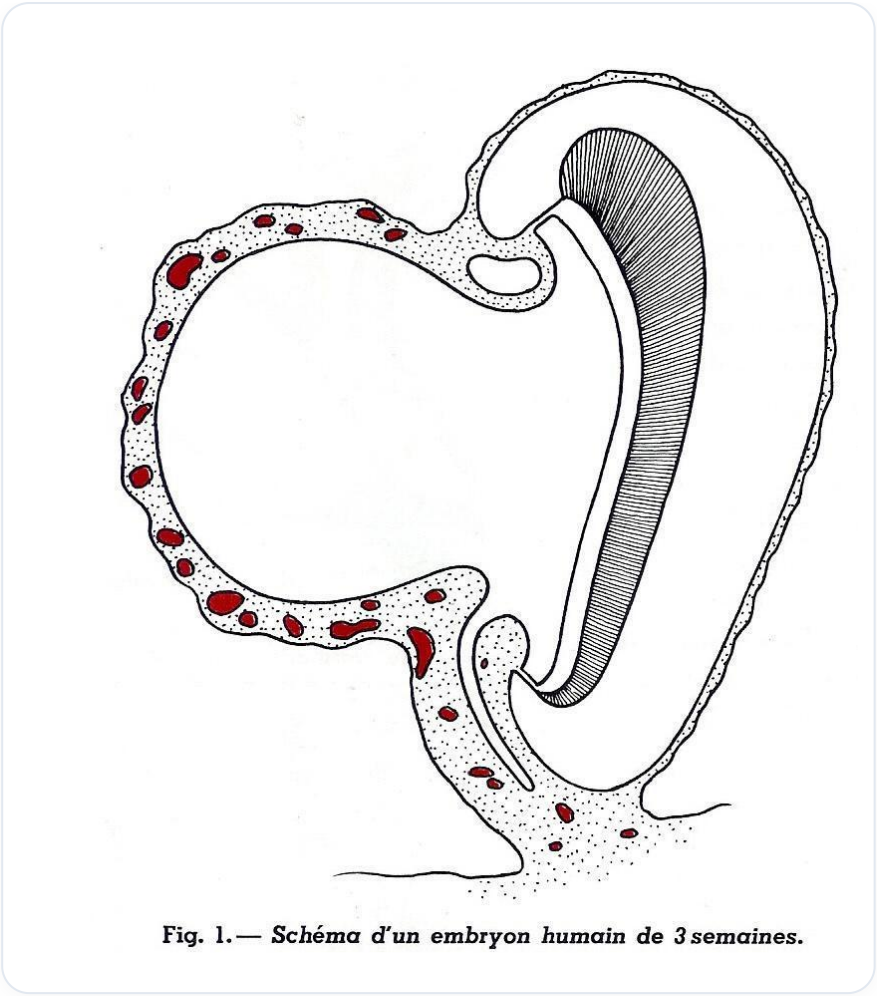


Fig. 1.— *Schéma d'un embryon humain de 3 semaines.*

FIG. — Human embryo 3 weeks

07. THE HEART : SETTING UP THE DEVELOPMENTAL MOVEMENT

DURATION : 10:08 STUDY SHEET

COURSE SUMMARY

This video explores the initial formation of the heart through a process of congestion and reorganisation of the mesoderm. The course details how a concentration of primitive angiogenic cells gives rise to a cardiac tube, with embryonic flexion aligning the aortas towards the centre. The interactions between the neural tube and the amniotic cavity are also highlighted, particularly the importance of convergence on the midline. Furthermore, the video addresses the concept of cardialisation, illustrating how the morphogenetic movements of the brain and surrounding cavities influence cardiac development and other key embryonic structures, positioning the understanding of embryologists and osteopaths in relation to biodynamics.

THE HEART: ESTABLISHING DEVELOPMENTAL MOVEMENT

THE INITIAL FORMATION OF THE HEART

Initially, a phase of **congestion** is linked to the reorganisation of the **mesoderm** in relation to the **neural tube**. This upward trajectory leads to a concentration of primitive **angiogenic cells** upwards.

This process leads to the formation of a **cardiac tube**, initially in the form of primitive left and right aortic tubes. The differential growth rate between the internal and external tissues causes the embryo to **flex**.

This flexing movement brings the aortas closer to the centre, where they meet to form the heart. This phenomenon begins from the **precardiac zone**, located at the level of the coccyx, before cerebral development.

Initially, two aortas develop under the influence of the neural tube, which acts as a motor, and the enormous surrounding **amniotic cavity**. The differential growth of the aortas brings them closer to the midline, in front of the **notochord**.

These two tubes gradually meet at the midline. If this convergence does not occur correctly, the specific internal nutritive tissue, essential for formation, can lead to the appearance of a **corrosion field**.

This first meeting point will structure the beginning of the heart. It is a convergence on the midline of these two levels. It is crucial to adopt an "outside-in" perspective to understand this process.

Upstream of the heart, a layer of cells, precursors of the **pericardium** and the **septum transversum**, is already stretching. These **mesenchymal** cells establish an essential relationship between the heart and the diaphragm.

CEREBRAL COILING AND CARDIALISATION

The developing **brain** coils around the heart, while the amniotic cavity exerts a circumferential thrust. In the centre, this brings back the vessels that meet in front of the notochord.

The growth rate between the cardiac tube (pure mesoderm, both motor and organiser of growth) becomes a field of restriction in relation to the pushing amniotic cavity.

This thrust is informed by the neural tube, which receives the greatest trophic information, grows the fastest, and induces the flexing movement of the embryo. This is the system that coils around the heart.

The heart remains in place; it is the surrounding system that coils. This process involves a slowing of the growth rate and a drawing together of fluids, from the periphery to the centre.

We observe here that the movement of **cephalisation** leads to the space of **cardialisation**, which will progressively lead to **diaphragmatisation**, then to **hepatisation**.

This hepatisation will then lead to **adrenalisation**, then to **gonadisation**. This is a very interesting developmental movement, linked to the development of the amniotic cavity which will completely envelop the "B zone".

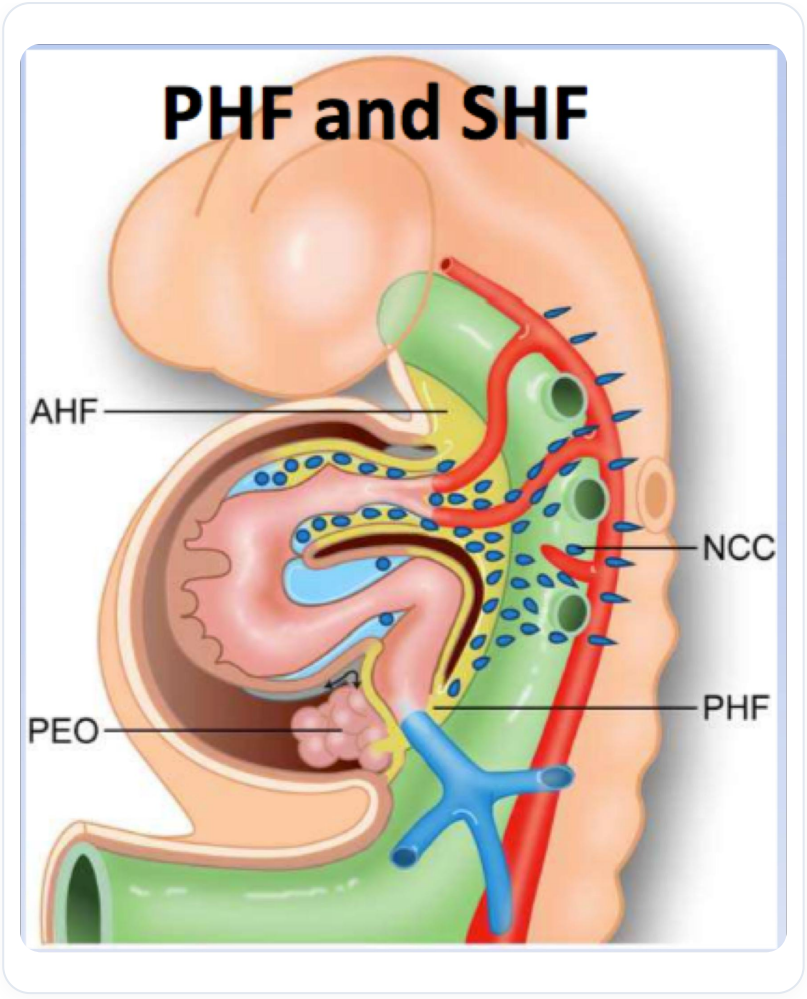


FIG. — Embryonic heart development

Cephalisation leads to cardialisation, which leads to diaphragmatisation, then to the entire posterior part of the peritoneal cavity.

This includes, for example, the movement of the midline in women that will form the **uterus** and **fallopian tubes**, and in men, a movement that, without going all the way, will form the **prostate** on the median plane.

Then, the **perineal** system will develop, with a return to a rebalancing of the **liver**. The liver will perform its tilting and lateral movement, becoming a major source of organisation for overall development and the closure of the anterior midline.

MIDLINE APPROXIMATION AND THE ROLE OF THE AMNIOTIC CAVITY

It is essential to remember the **approximation** towards the midline and the initial development of the heart towards this anterior midline. Simultaneously, a significant process of cephalisation is underway.

The amniotic cavity, the developing cerebral **ectoderm**, the cardiac zone, and the diaphragmatic zone integrate. The **vitelline vesicle** progressively meets at the embryonic pedicle to form the **umbilical cord**.

All of this constitutes a developmental movement organised by the amniotic cavity. The process is a balance between the aortic system (the vascular axis) and the amniotic cavity.

The brain coils around the heart. There is a movement of integration, then a movement to form the diaphragm, the posterior parietal peritoneum, and a return around.

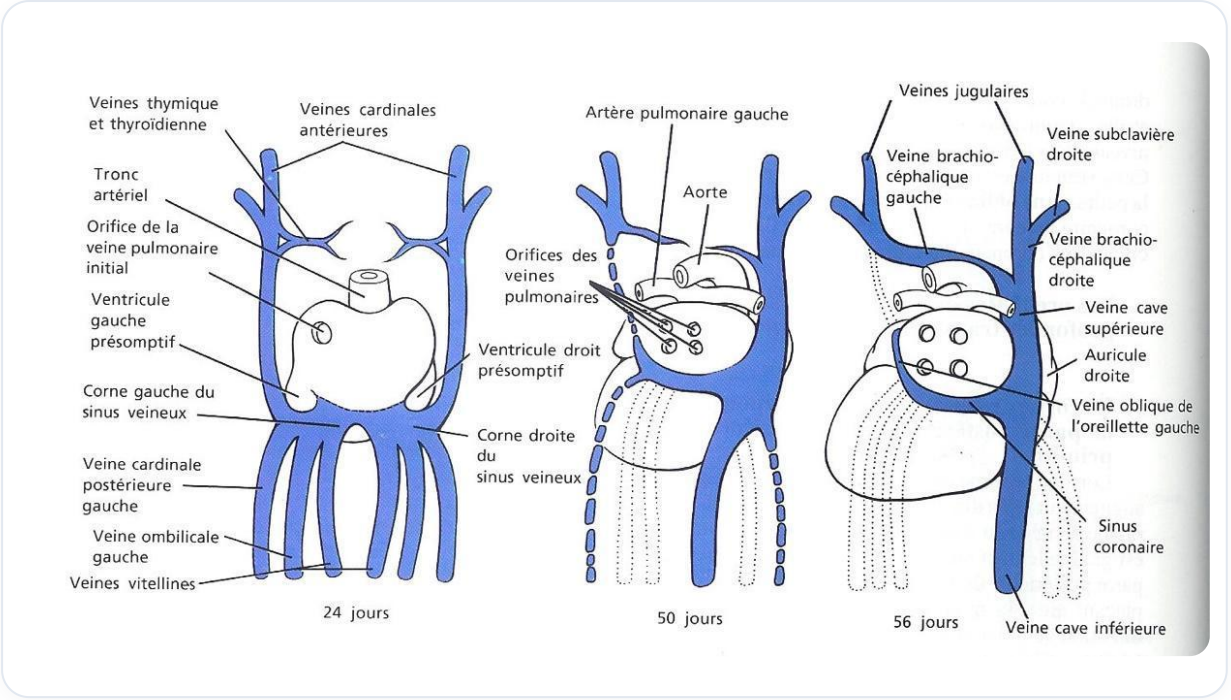


FIG. — Cardiac venous development

In the embryonic position, the coccyx, brain, and heart constitute points of support. The embryonic anlage depends on the **primitive streak**, which can be deviated at birth, at conception, or even pre-conceptionally.

The brain and heart are intimately linked. The heart was already present and remains in place. It is the amniotic cavity that forms around it in this movement.

INTEGRATION OF FORCES AND THE AMNIOTIC CAVITY

Work will be done on zone B, also integrating **lateral forces**, i.e., the force of transverse thoraco-abdominal delimitation in the integration of the circulatory system, from the periphery to the centre.

Zone B, in this integration, is the amniotic cavity. A baby floats in its amniotic cavity. This membranous envelope will disappear, but its energetic aspect remains.

This "ball" around oneself is essential. The amniotic cavity forms and integrates the neural tube. In case of dysfunction (whiplash, craniosacral problem, or other), it is crucial to return to the origin.

Treat the amniotic cavity. Re-establish the spaces between the sacrum and the heart. Redo the entire movement and identify the blockages. Then, you can rework and let the system operate in the recognition of neutrals or intervene yourself.

FROM THE AMNIOTIC CAVITY TO THE CENTRAL NERVOUS SYSTEM

A cross-section reveals the amniotic cavity above the **neural groove**. The fluid present is **primitive amniotic fluid**, which will form the neural tube. This fluid will be integrated into the tube.

The closure of the **anterior neuropore** and the **posterior neuropore** occurs around the 28th day, when the neural tube is completely closed. One must focus on zone B, because it is this that integrates the **CSF** (cerebrospinal fluid), and not the other way around.

CSF appears when the embryo implants in the mucosa. At the time of this implantation, an **exudate** appears, and it is this amniotic cavity that will create the neural tube, the neural plate, and the digestive tube.

Functionally, one cannot separate the rest of this cavity; it is always integrated. The **external coelom**, through the developmental movement of the embryo, will largely integrate into the **internal coelom**.

The rest will leave as an exudate to the outside, pushed back by the amniotic cavity. At a certain point in growth, the external coelom no longer develops.

The **neuroenteric canal** is information between the vitelline vesicle and the amniotic cavity, unrelated to the external coelom. The external coelom will be integrated into the internal coelom to form the peritoneal cavity.

However, at some point, there will be no further growth of this cavity. It seems to be pushed back, but it is actually an exudate, because the amniotic cavity grows so fast that it has integrated the internal coelom.

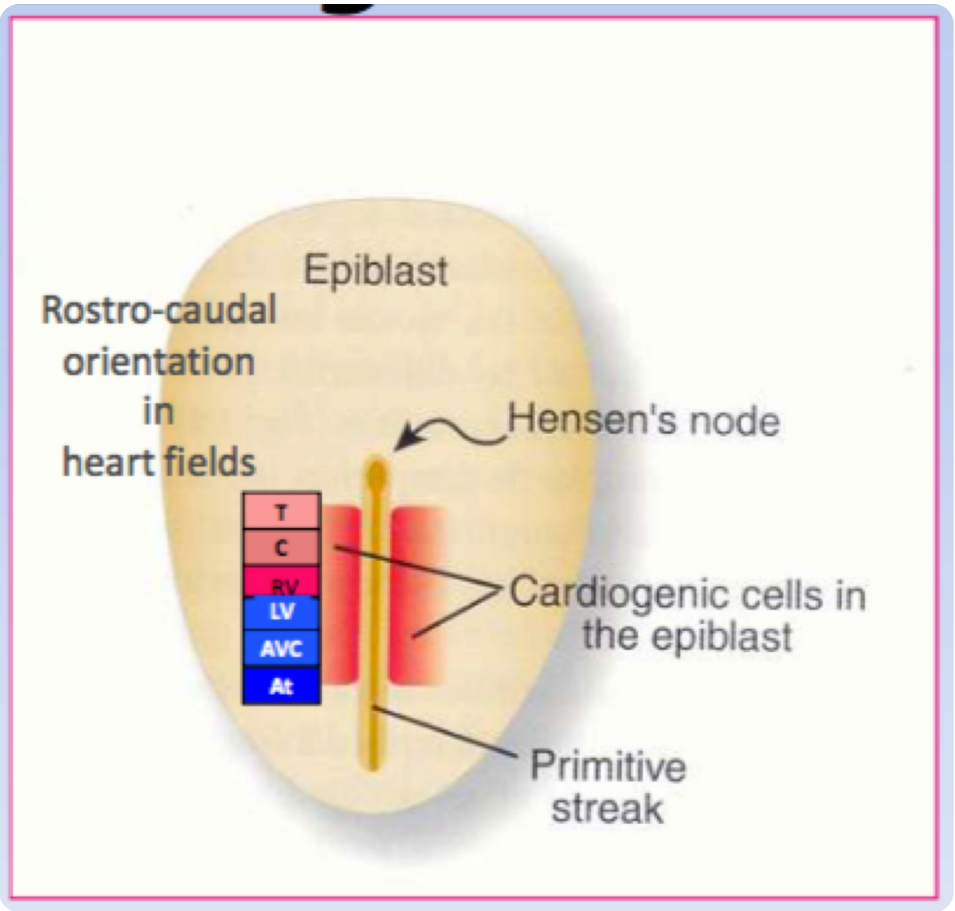


FIG. — Epiblast, streak, heart

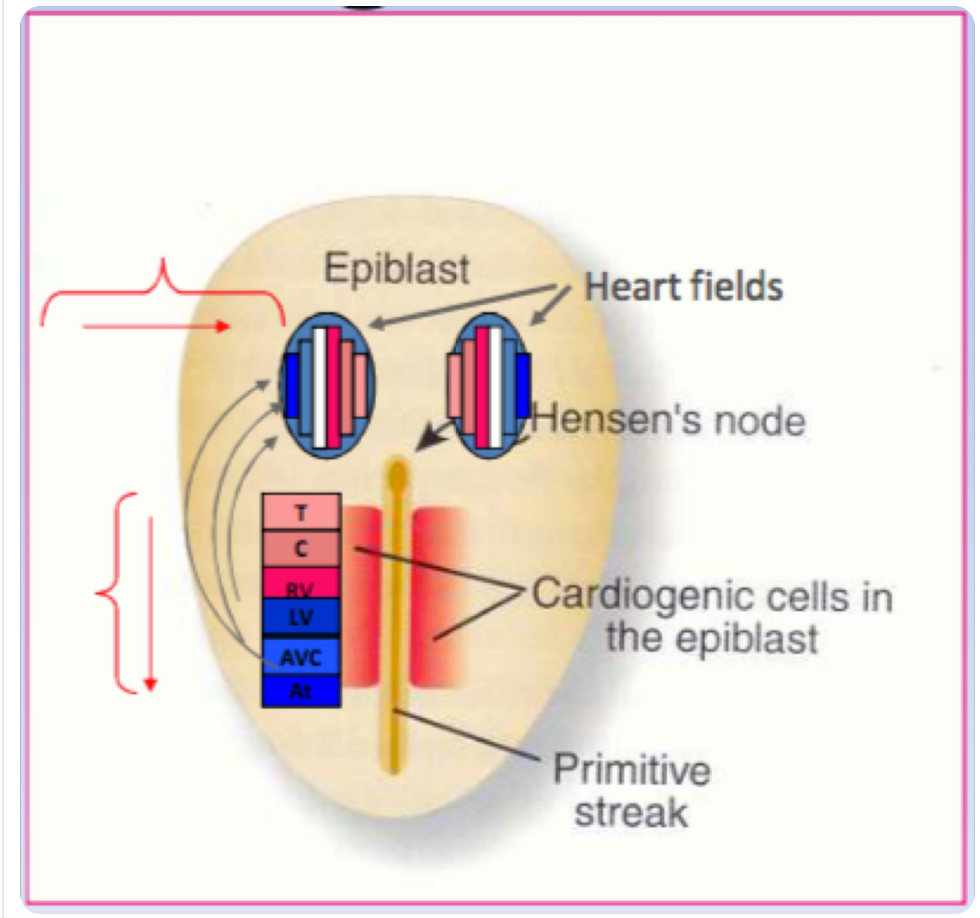


FIG. — Cardiac cell development

FIG. — Early neural tube

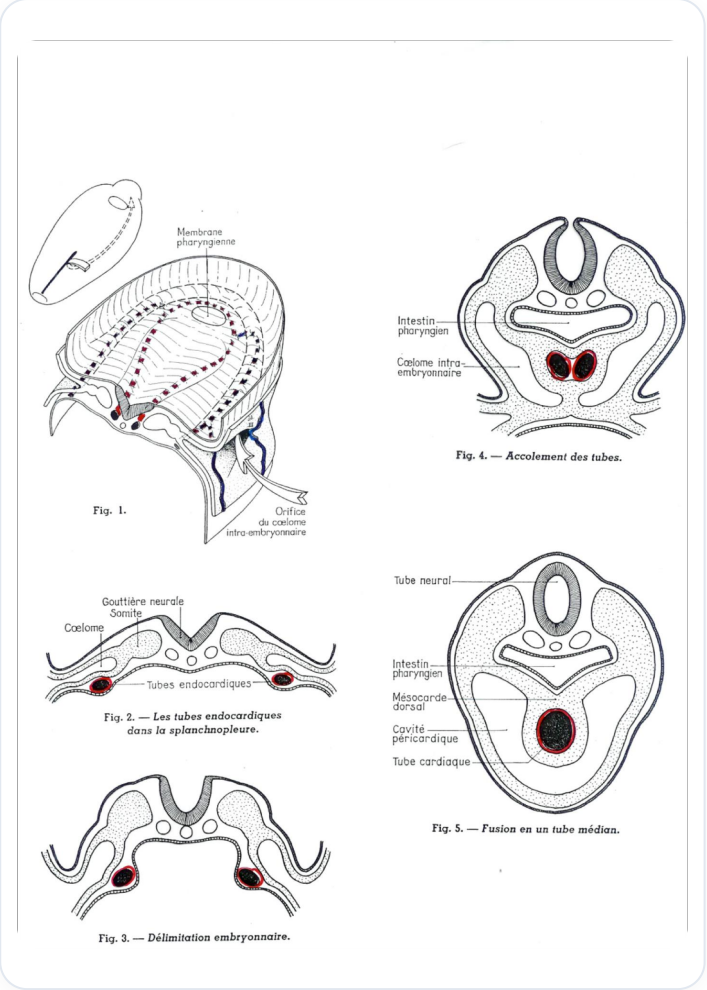


FIG. — Embryonic heart development

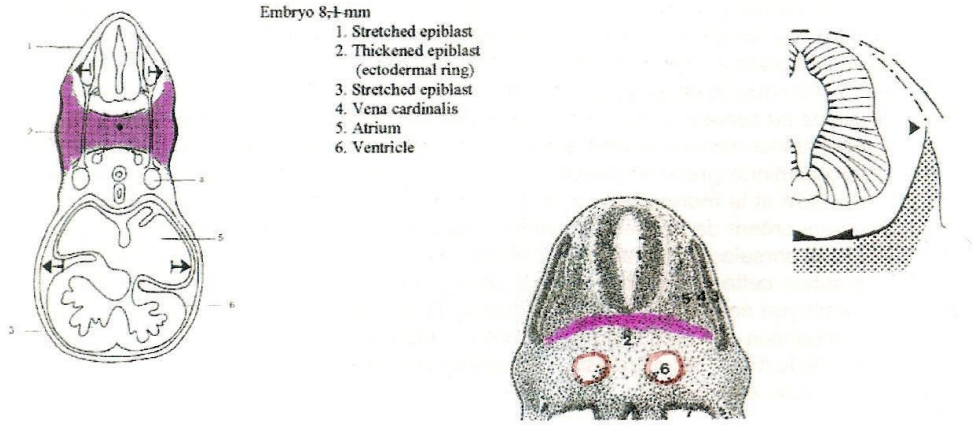


FIG. — Heart embryonic development

08. THE HEART : DYNAMICS OF DEVELOPMENTAL MOVEMENT

DURATION : 11:04 STUDY SHEET

COURSE SUMMARY

This video delves into the heart of cardiac embryology, exploring the fundamental stages of heart growth in early embryonic development. It addresses key concepts such as the approximation of the endocardial tubes, the 'looping' movement, and the inversion processes that transform the heart. The instructor emphasises the importance of understanding the dynamic between the heart and the liver, as well as the need to follow the embryonic blueprint to better support patients. By integrating this knowledge, practitioners will be able to refine their biodynamic approach, thus promoting harmonious development in their patients.

THE HEART: DYNAMICS OF DEVELOPMENTAL MOVEMENT

INTRODUCTION TO EMBRYONIC CARDIAC DEVELOPMENT

We will observe the **exponential growth** and **coiling** of the embryo.

Here, we only see the **vitelline cavity**. Above it, the **amniotic cavity** has been cut.

There is **longitudinal growth** and **flexion**. We can distinguish the **nervous system**, the **somites**, and the **vascular systems**.

Let's focus on an opening, the **future intraembryonic coelom orifice**. We must imagine it here. It will later transform this external coelom into an internal coelom.

The impressive growth of the system brings the two **primitive endocardial tubes** closer to the midline, due to a **differential growth rate**, influenced by the **notochord**.

This approximation of the endocardial tubes forms, above, the **intestine**, and all around, the **intraembryonic cavity**. At this stage, this will form the **pericardial cavity**, not the peritoneal cavity.

The **vitelline vesicle** moves closer to form the upper part of the intestine.

KEY MOVEMENTS OF CARDIAC DEVELOPMENT

The approximation of the two endocardial tubes forms the area of the developing **cardiac tube**.

We observe a movement of approximation towards the midline and, simultaneously, a movement where the brain "does this". It's a combination of both.

This process is synchronous with the **flexion** phase of the embryo. The meeting of the thumbs symbolises this first movement at the level of the heart.

THE HEART'S "LOOPING" MOVEMENT

The heart, in sagittal view, meets and seeks to develop. It lacks space behind it, and little space to the left or right. It only has space forwards.

Within the heart itself, it will find its own **fulcrum**. With the back blocked, it develops forwards, performing what is called the heart's **"looping" movement**. This is phase 2.

This continuous development forms a third level. The space between the **ventricular system** and the **atrium** becomes a fulcrum. The ventricle, initially above, ends up below in relation to the system. This is an **inversion** movement.

Here are the three main movements:

1. **Approximation** towards the midline and meeting.

2. **"Looping" movement** forwards.

3. **Inversion process** of the ventricle.

THE INFLUENCE OF THE PERITONEAL CAVITY AND SYMMETRY BREAKING

The third movement is linked to the inversion process, which is a developmental process of the **peritoneal cavity**.

The development of the **liver** causes a shift in movement. The lesser sac displaces the left bronchus higher than the right, linked to the **breaking of axial symmetry**.

We therefore observe a developmental movement forwards, but also to the left. This is a **double tilt**.

Let's mainly remember:

- The heart's **approximation** towards the midline.
- **Development** forwards.
- Simultaneous development to the left.

The **sinusoid system** of the liver will position itself upwards after this tilt.

ORGANISATION OF THE GREAT VESSELS

The first movement, then the second movement linked to the tilt, and finally a third movement that organises the **great vessels** above.

The heart is tilted, held by an axis, and balanced in relation to the **vascular plane**. There is greater development on one side.

It is a balance between the **thoracic** part, the **abdominal** part, and the **lower visceral** part. A very important movement occurs between the heart, stomach, and liver.

THE EMBRYONIC IMPRINT IN A BIODYNAMIC APPROACH

A very important reference point to rebalance is the **hepatic movement**. We observe whether the liver is well within its developmental movement or if it is receding.

If the liver stops, there is a small halt.

At the level of the heart, its development will do the opposite. There is a trajectory, like an **imprint**.

One must position oneself on the embryonic imprint, go where the system wishes to go; that is where the **stillness** is.

If the movement does not go far enough, the system must be helped. This organisation of the heart is strongly influenced by the liver. Hence the importance of understanding the role of the liver.

In biodynamic embryology, we do not impose. We reject neither **fascial restrictions**, nor the **fluid** aspect, nor **structural**, **visceral** or other techniques.

The approach is to adapt to what is felt in the present moment. The therapeutic force lies in the meeting of the two neutrals.

The regenerative processes in **biodynamic osteopathy** use these embryonic movements. It is not about direct therapy, but about recognising the embryonic developmental movements that appear in therapeutic processes guided by the **"tide"**.

It is crucial to recognise the frequency one is on.

THE HEART'S ENERGY AND ITS ENVIRONMENT

We rarely touch the heart directly first. The heart forms through the **echo** of the periphery.

The heart is extremely powerful because it receives very important information from the **neural crest**. It receives even more powerful electrical information on a neurological level.

This means that one can learn to feel the heart at a distance, or that the heart is felt at a distance. It has a power, a **radiation**. Measurements have shown radiation up to at least 5 metres from the person, probably linked to the neural crest.

It is on the **venous** plane that there will be development, called the **venous sinus** of the heart. In this venous sinus, the **cardinal veins**, **umbilical veins**, and **vitelline veins** will converge, undergoing many rearrangements.

The heart has a **fascial** environment and its own **electricity** for its beat. However, this electricity was not sufficient to maintain the heart in its proper function. It depends heavily on its entire fascial environment.

The **sterno-pericardial**, **phrenico-pericardial**, **vertebro-pericardial ligaments**, and the **superior aorta**, suspended up to the pharyngeal tubercle, support the heart's **rhythmicity**.

The heart's attachment system is a **closed oscillation system**. Its tissue and molecular structure is designed to undergo a stretching movement with each beat. It's like a system of **elastic bands** that constantly return the heart to its equilibrium position.

This structuring of the **pericardial fascial tissue** supports the heart, and the heart itself is supported in its rhythmicity by its fascial environment.

The entire fascial system around the heart can influence its rhythm. Hence the importance of the **structural freedom** of the cardiac environment. This is where direct action is possible.

The heart in **systole-diastole** should be seen as an **octopus**, with the **aortic arch** and the entire descending trunk down to the junctions.

It's not just the body, but also its "tentacles" that can be felt. In the belly, in the throat, in the feet, in the fingers.

This implies that a **destabilisation** of the hip can destabilise the heart. There are cases of **arrhythmias** that can be improved by releasing the hips.

There is a complex relationship through the **endothelium**, this tissue that connects the entire system.

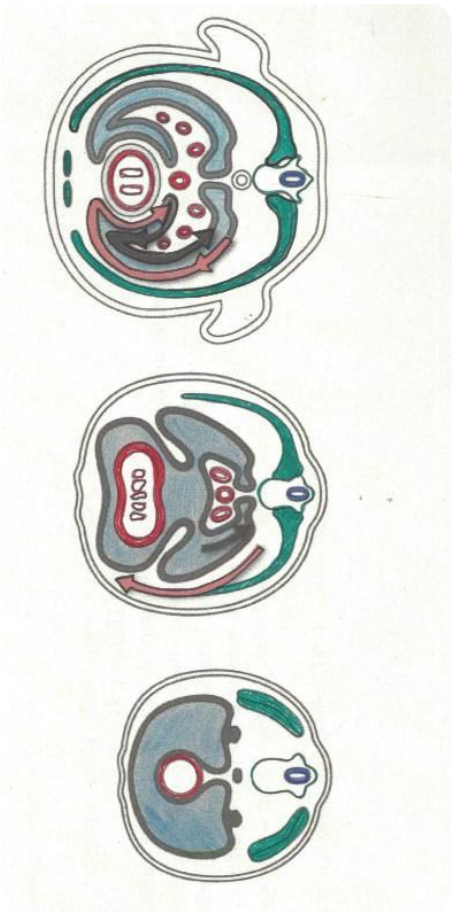


FIG. — Lateral embryonic folding

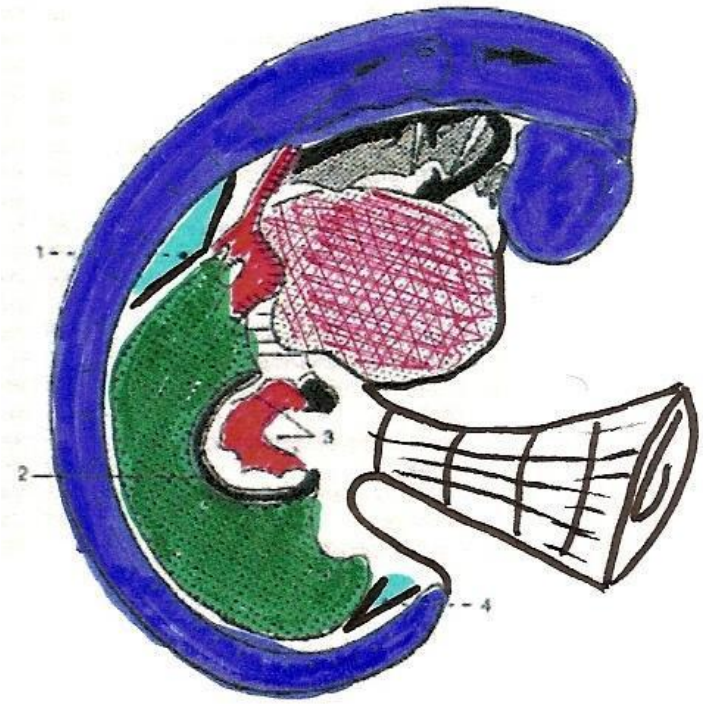


FIG. — Embryo with allantois

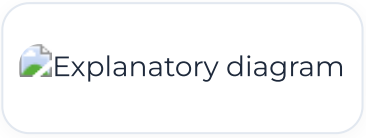


FIG. — Explanatory diagram

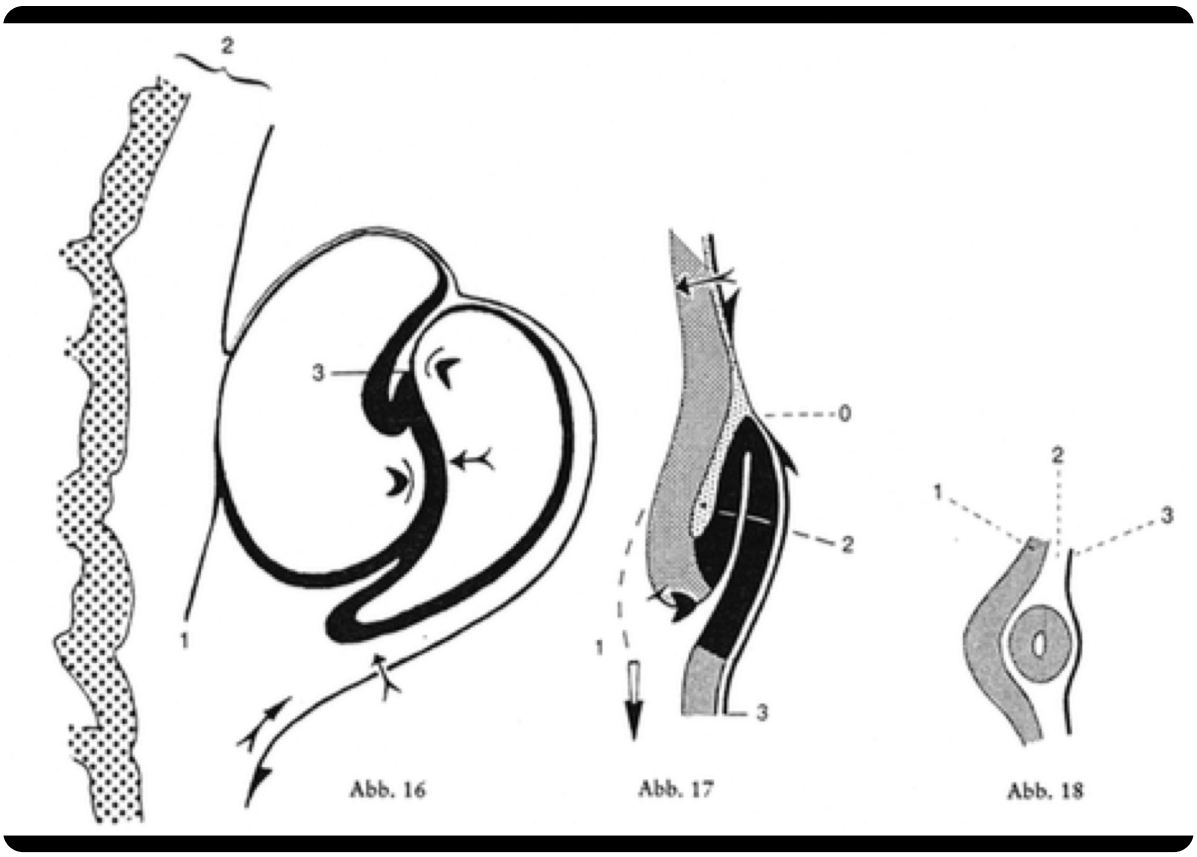


FIG. — Neural tube development

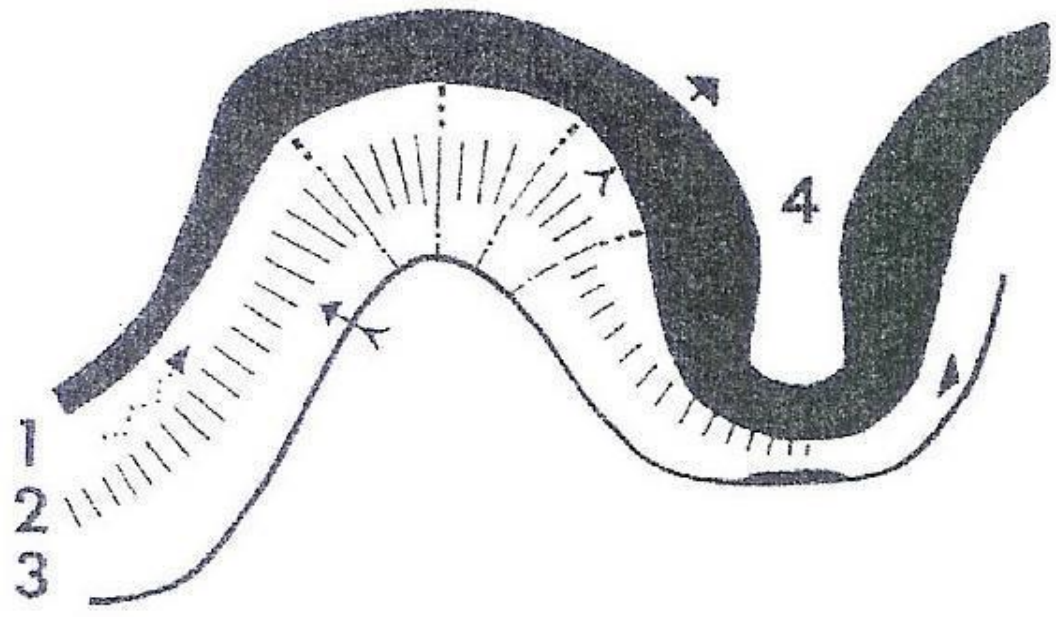


FIG. — Neural plate, groove

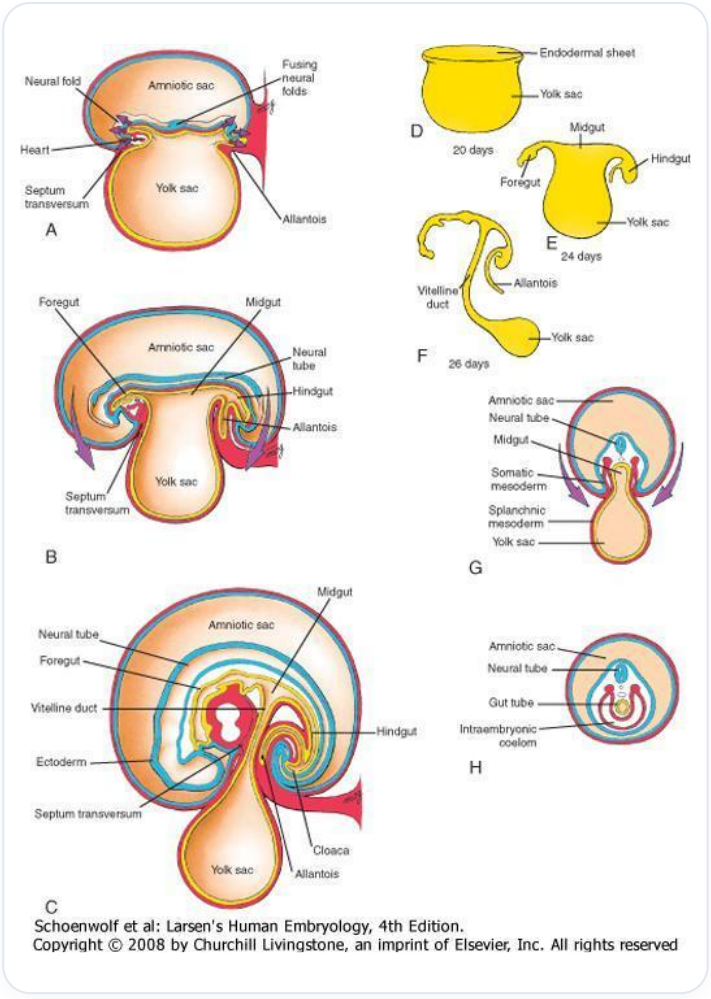


FIG. — Embryonic digestive tract development

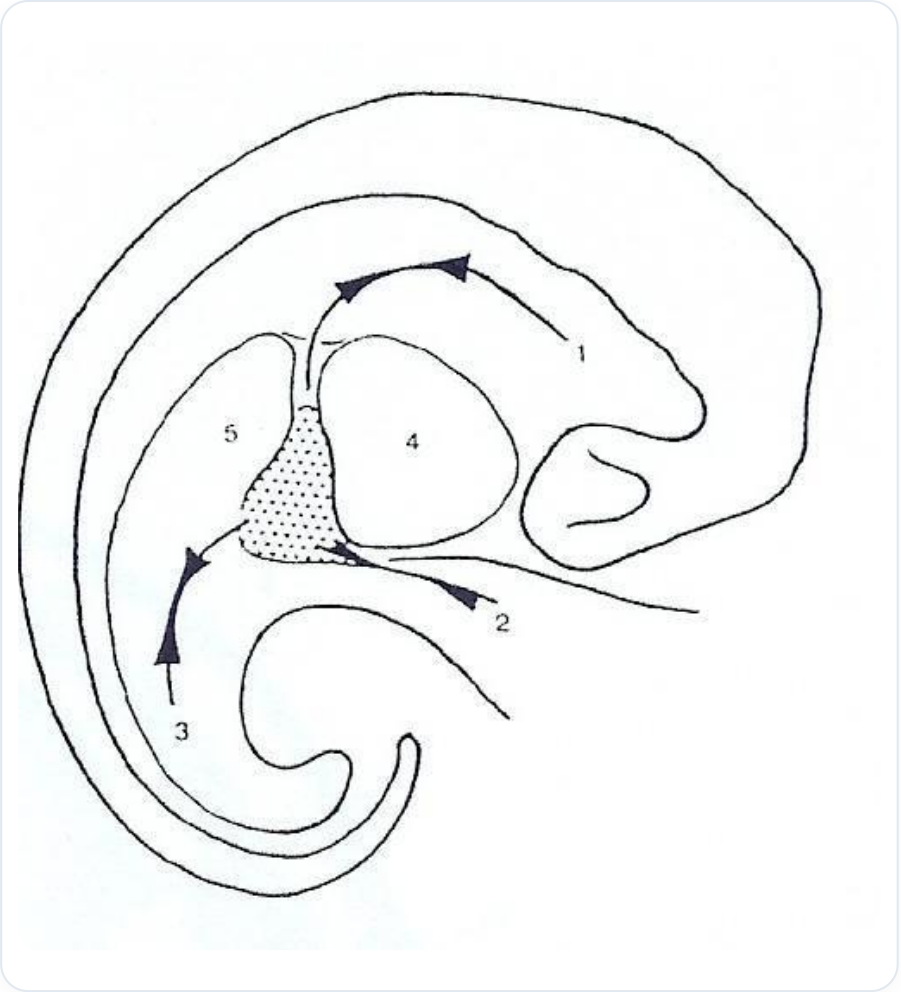


FIG. — Circulation embryo liquide céphalo-rachidien

09. THE HEART : 3 EXTERNAL INFORMATION TAKES

DURATION : 02:50 STUDY SHEET

COURSE SUMMARY

In this captivating video on biodynamic embryology, you will learn about three essential interactions between the heart and other vital systems: the eyes, the limbs, and the thyroid. Each interaction illustrates how these elements transmit crucial information that influences our perception and communication as therapists. By integrating these teachings, you will discover how touching and speaking with an open heart can transform not only your practice but also the relationship you build with your patients. Immerse yourself in this exploration that unifies body, heart, and mind for an even more holistic approach to healing.

THE HEART: 3 EXTERNAL INFORMATION INTAKES

This moment is crucial, as the **embryo** will accelerate its growth. We will observe a **complete coiling** of the brain, and, remarkably, three levels of **external information** will interact via this rhythmicity.

1. HEART - EYES INTERACTION

Firstly, the brain will coil around the heart. Significant information is transmitted at the level of the **optic placodes**, directly onto the heart. One could almost say that the **eyes** will "settle" on the heart.

2. HEART - HANDS INTERACTION

You will also see another interaction: the anlage of the **limbs** will come to settle on the heart. This completely informs the heart. Do not forget that the heart receives very important information at the level of the **neural crest**, an **electromagnetic** plane that allows us to feel the heart at a distance. This information will impregnate your hands on a rhythmic plane.

3. HEART - THYROID AND VOICE INTERACTION

There is another part that we do not see directly, but in this coiling, the entire **thyroid system** comes to settle on the heart. This is accompanied at some point by the opening of development upwards, which will form the **vocal cord system**. It is often said that the **tongue** is the bud of the heart. However, the tongue is linked to the thyroid and transmits information.

THE HEART-BODY CONNECTION: PERCEPTION AND COMMUNICATION

When I touch someone, I touch them with information received from my heart. I must maintain this connection. You will see that this will greatly change your perception.

- **The Gaze:** When I look at someone, I must look at them with my heart. The two are connected. The heart is informed on an embryonic plane, receiving purely electromagnetic information.
- **Speech:** Thirdly, when I speak, I must speak with my heart. I receive this thyroid information, an electrical information.

So I have three magnetic fields of information. When we receive this information, we must be very attentive to:

- Our thoughts.
- Our touch.
- Our gaze.

This is first designed to receive, then to give. But the primary function comes from the **heart's rhythm**, which gave your hand's **neurosensibility**. This heart acts in this way, then it opens. This is **love**, this is **joy**.

10. THE HEART : DEVELOPMENT OF ARTERIAL ANATOMY

DURATION : 04:14 STUDY SHEET

COURSE SUMMARY

This video delves into the heart of the embryonic development of blood cells, describing their origin from the umbilical vesicle to the bone marrow. It also explores the persistence of the mesenteric artery, a vital remnant of this process, and its role in intestinal development. The aortic arch is highlighted to illustrate the rhythmic movement of the heart and its impact on blood circulation. By associating cardiac dynamics with the fascial environment that maintains vital structures such as the oesophagus and bronchi, the video reveals how imbalances in this rhythmic zone can influence cardiological and musculoskeletal pathologies. An essential lesson for any therapist interested in bodily interconnections.

THE HEART: DEVELOPMENT OF ARTERIAL ANATOMY

ORIGIN AND DEVELOPMENT OF BLOOD CELLS

The first **blood cells** appear around the **18th day** of embryonic development. Their origin occurs in three important stages.

Initially, they come from the **umbilical vesicle**, where **primitive megaloblasts** are found. This phase lasts for the first two months.

From the beginning of the **second month**, a second movement of blood production emerges from the **liver**.

It is only around the **third month** that some of this production begins to occur in the **bone marrow**.

We can observe here a development from the periphery to the centre: first the **umbilico-pedicular** period, then the **hepatic** period, and finally the bone marrow for **red blood cells**.

THE VESTIGE OF VITELLINE DEVELOPMENT: THE MESENTERIC ARTERY

The only vestige that will remain of this vitelline development is the **mesenteric artery**.

It served as an axis of rotation, a **fulcrum**, for the rotation and development of the **intestinal loop**.

When the **amniotic cavity** forms all around it, the vitelline vesicle, which contains the original cells for circulation, reintegrates.

In part, with the vitelline vesicle, the **embryonic pedicle** will form the **umbilical colon** with the placenta. The vestige that will connect is the **mesenteric system** (the superior vesicle). This axis of rotation is crucial for embryonic development.

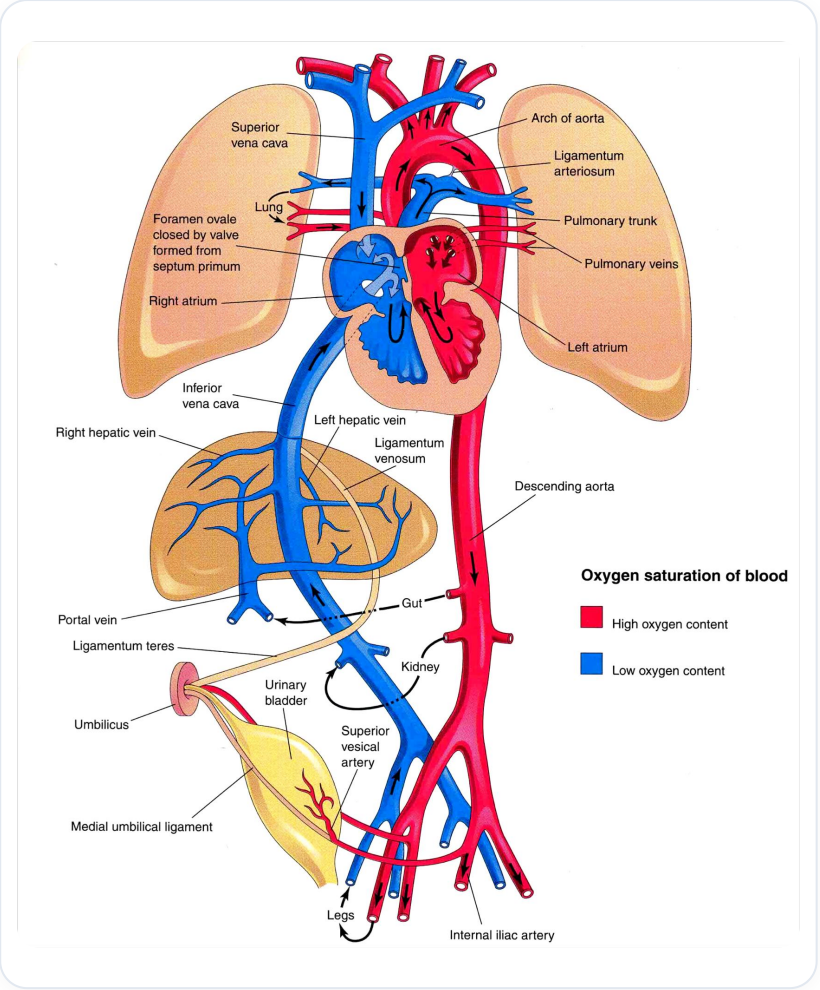


FIG. — Human fetal circulation

THE MOVEMENT OF THE HEART AND THE AORTIC ARCH

When we observe the heart, we see the **aortic arch**. Symbolically, we can say that it ascends or descends.

This movement of the aortic arch represents the **heartbeat**. It almost reproduces a developmental movement.

The aortic arch performs a **twisting** or **untwisting** movement during the cardiac cycle.

When the heart pushes, the aortic arch opens to absorb blood. When the heart empties, it closes, performing a particular movement.

It opens to take blood and closes to eject it, which intrinsically links the aortic arch to the function of the heart.

THE AORTIC ARCH AND THE CARDIAC CYCLE

It's a bit like a **basketball** player. When the heart rises, that's when the aorta ejects its blood volume.

It should be known that the aorta contains almost as much blood in its volume as the heart on the arterial level.

This means that the aorta actively participates in the **systolic ejection** of blood.

This is supported by the **fascial environment**, which could be compared to a **"fishnet stocking"**.

THE RHYTHMIC ZONE: A VITAL FASCIAL CROSSROADS

The aortic arch, the **bronchial bifurcation**, and the **oesophagus** are connected by a common **fascia**, which holds them together.

This area, characterised by the bronchial bifurcation, is what is called the **rhythmic zone point**.

Consider the vital rhythms: **100,000 heartbeats** in 24 hours, **23,000 breaths** in 24 hours, and **1,000 to 2,000 swallows**.

It is a large rhythmic zone that meets and attaches between **T3**, **T4**, and **T5**. These vertebrae form a fascial balance plane for the rhythmic zone, which is both **cardiac**, **pleural**, and **digestive**.

We have the aorta, the heart, and below, a very important bifurcation: the **iliac arteries**. Above, there is yet another level.

In fact, all of this forms a unitary and pure plane. We observe that certain pathologies of the **coxo-femoral** type or others can destabilise the heart. Conversely, what happens in these phases of rotation can manifest up to the level of the heart.

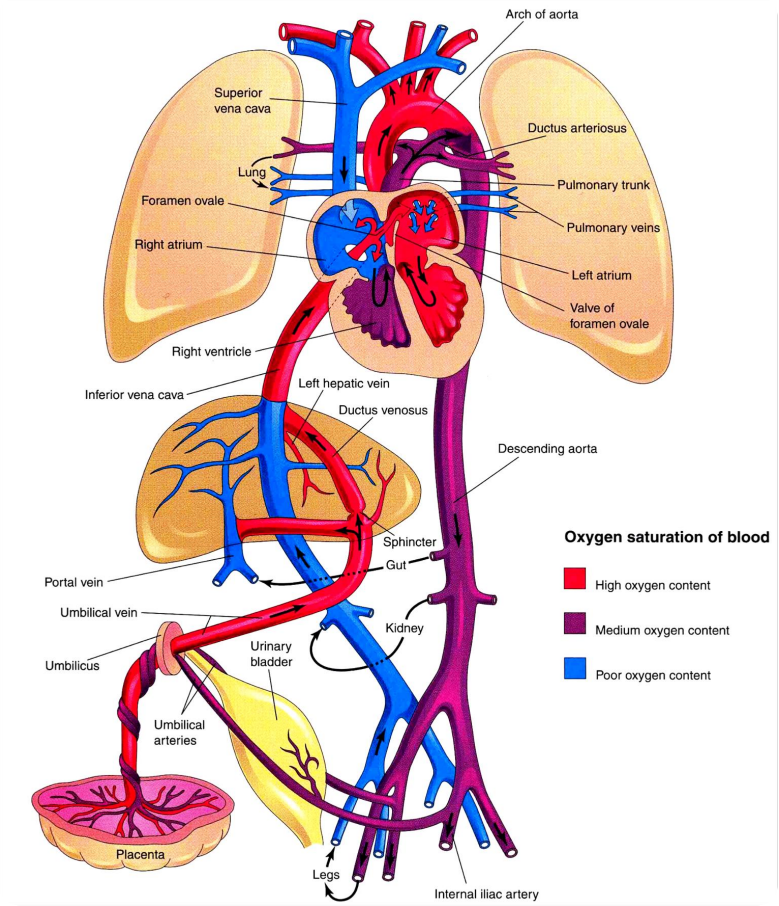


FIG. — Fetal circulation

11. THE HEART : SUMMARY AND PRACTICAL IMPLICATIONS

DURATION : 11:02 STUDY SHEET

COURSE SUMMARY

This video explores the importance of the heart in embryonic development, positioning it as the meeting point of vital forces between sperm and egg. It explains how the heart forms within the primitive streak, interacting with key structures such as the neural tube and mesoderm. The concepts of diastole and systole are also discussed, with particular attention to how these movements can be observed and integrated into osteopathic practice. Practitioners will learn to use embryonic fulcrums to revitalise and balance cardiac and cerebral systems, refining their approach through relevant points of support.

THE HEART: SUMMARY AND PRACTICAL IMPLICATIONS

THE HEART: SYMBOLIC MEETING POINT

The **heart** is considered the symbolic meeting point between the **spermatozoon** and the **ovum**. It is here that a **fundamental pattern** is established, manifesting even on a polarity plane.

FIG. — Two-cell embryo

EARLY APPEARANCE AND DEVELOPMENT

The emergence of the heart is particularly interesting at the level of the **primitive streak**.

It is the remnant below S2, the terminal end of the **notochord** and the site of sensation.

We approach it in its facial environment and its developmental movement. It will reintegrate **secondary neurulation** during the closure and establishment of the lesser pelvis.

Secondary neurulation is the space between the **filum terminale** and the **caudal bud**, involving differential growth of the **neural tube** relative to the anterior **endoblast**, with an opening of the lumen.

The anterior endoblast is the embryonic endoblast, but here, at the **mesoblastic** level and also at the heart level.

In this wave with the primitive streak, which ends at S2, a **primitive cardiac force** is already present. The presumed location in the epiblast is part of the heart's potential; it is the earliest identified location.

Cardiogenic cells are found at the epiblast level, in the presumptive layer of the epiblast, even before mesoderm formation.

We will find this presumptive lamina again. A growth movement is followed by congestion, then the integration and meeting of the two tubes, organised from the periphery by the **amniotic cavity**. The neural tube, the cardiac tube, and the integration of the diaphragm contribute to the establishment of the heart.

This occurs at the same time as the closure of the neural tube and the integration of the external **coelom** into the internal coelom.

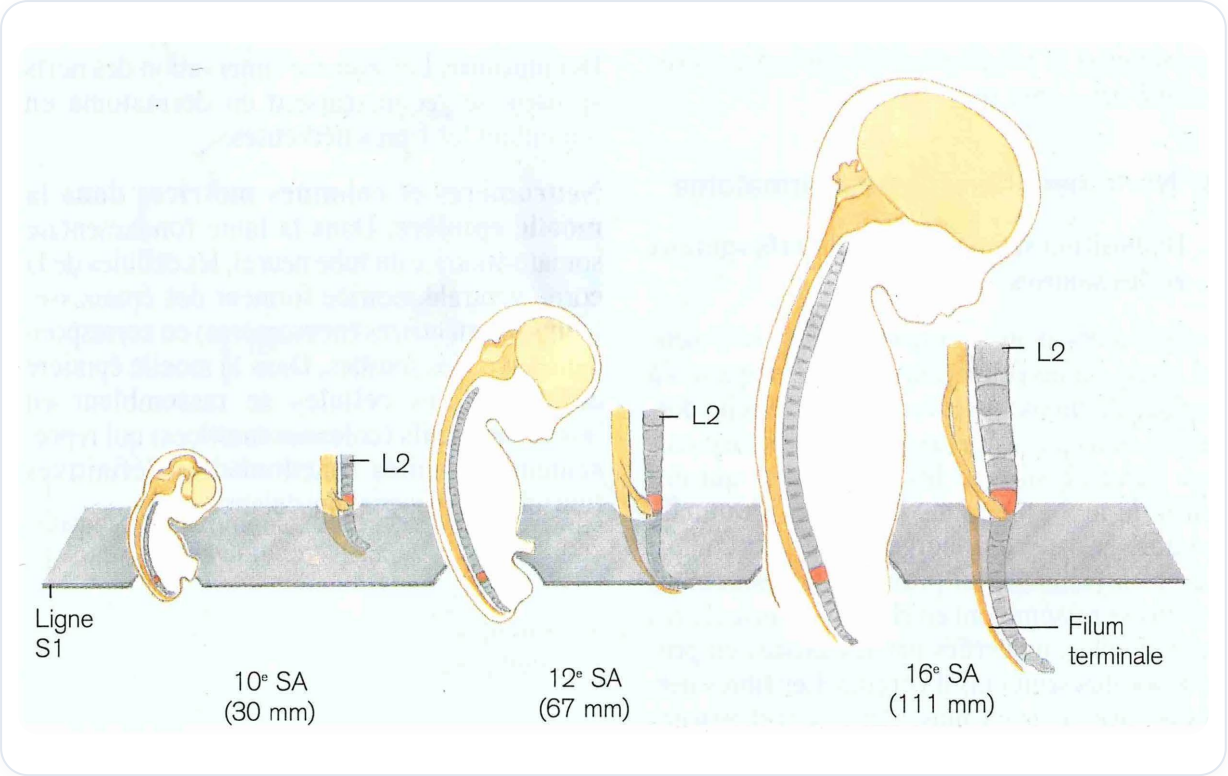


FIG. — Relative growth of spinal cord

THE PRIMITIVE STREAK AND THE CARDIAC FULCRUM

The earliest identifiable location is the primitive streak.

The notochord terminates at S2. The space between S2 and the end of the primitive streak contains the remnant of the primitive streak.

The **embryonic fulcrum** of the heart is on the coccyx.

The heart then adopts a rather **diastolic** posture, seeking energy and volume. It descends in its movement.

The heart also presents a fulcrum, i.e., a point of origin, which it can manifest differently.

CARDIAC MOVEMENTS AND STRUCTURES

A movement is established between the **brain**, with **cephalisation**, and the heart, **cardialisation**, allowing them to meet.

At the same time, the two endocardial tubes meet on the midline. This is followed by the **looping** of the heart, then the establishment of the **cardiac sinusoid** and the **aortic arch**, with the great vessels.

The **caval axis** serves as a landmark.

It is important to observe whether the heart expresses itself on a systolic or diastolic plane. Facially, it seeks more diastole or systole.

Relative to the iliac, it seeks much more **horizontality**, which indicates a diastolic movement. We can even go as far as to seek the **aortic pulse**.

CEREBRAL AND CARDIAC DEVELOPMENT

We will progressively integrate brain development in relation to developmental movement.

The first movement of brain development was **expansion**. This expansion depended on the **longitudinal axis** or the **mesenteric axis**. It depends on vitelline integration, the remnant of which will be found on the mesenteric axis.

Then, another important level for nutrition is the system between the **spleen** and the **liver**, with the return to the heart via the **vena cava**. Then, the aortic information will manifest through four or two entry points: carotid, above, below, vertebral.

We thus integrate brain development with the heart and the mesenteric system. By working on the brain, we observe which field needs to open. We can refine perceptions to determine the areas to work on, for example, on the carotid system in relation to the heart.

This can be done by providing simple points of support, whether they are embryonic fulcrums or points of support in the construction, to revitalise the regenerative force.

A fulcrum is a point of emergence. All the developmental force, such as the integration of the **vitelline vesicle** into an artery (a remnant of the developmental pathway and not an embryonic fulcrum), lies between the fulcrum and the finality.

We are beginning to perceive the game we are about to engage in, as we will be working on specific arteries. The goal is to restore **full homeostasis**, full vitality, full health.

ORIGIN OF THE FULCRUM AND MESODERMAL WAVE

I will touch upon the origin of its fulcrum, which is indirectly found at the coccygeal level, then which will gradually integrate into its system through the developmental movement of the **manotococcus**, this wave with this aspiration at the mesoderm level.

This is information that restructures, revitalises, restores strength to the tissue, restores function.

The development of embryonic forces allows the development of the system's main function.

THE CENTRAL ROLE OF THE DIAPHRAGM

The great master for all this will be the **diaphragm**. It is important to position oneself well on this diaphragm with three levels:

- **Amniotic cavity** on the transverse plane
- **Thoracic**
- **Abdominal**

This integration of the coelom, and at the same time the integration of the brain, the heart, the diaphragm, the posterior part of the peritoneum, to return to the **perineum** and, through this tilting of the liver, restore the complete closure of my linea alba, which will provide all the **urogenital** information.

THE LINEA ALBA AND THE UROGENITAL SYSTEM

This is how it will form. The **linea alba** represents the complete integration of the urogenital plane; it is its final story.

On this linea alba, we can look for concentrations of crystallised time, because it is a line of probability in relation to the **erythrocyte**, the formation of blood cells, thus a future, where the plane of the **copula** is also your future.

At the moment of exit, stretching occurs, and this is how the entire external and internal genital system will integrate on this plane of the anterior linea alba. It ends to rebuild.

On a blood cell plane, **erythrocytosis** or **gonocytosis** advance. At some point, the embryo makes a movement. There is a line of initiation in life. At birth, this line, the linea alba, must be opened, and one must be initiated into life.

EMBRYONIC FULCRUMS AND ARTERIAL SYSTEM

The idea will be to restore the correct **embryonic fulcrums**. The arterial system should be seen as an **octopus**.

Tomorrow, we will detail the arterial system at the facial level, as it is a trophic access point to the brain. We will need to understand the development of the carotid system and the vertebral system.

These two systems, with all the information on the **Circle of Willis** and the anterior, middle, posterior **cerebral arteries**, as well as the sinuses or valves (of Brechet or Thierry Goïdienne, etc.), are spaces of equilibrium.

CEREBRAL VASCULAR BALANCE

We need a balance between vascular supply and venous return. We now know that there is a **lymphatic system** in the brain.

To eliminate surpluses, some may have **lymphatic** or venous **congestion** in the brain. This is observed in people who are a little swollen, a little oedematous. The return is not happening.

A rebalancing work on the venous plane is then necessary, whether at the level of the **superior** or inferior **vena cava**. This is important to avoid congestion, especially in the ears, and anything that cannot be evacuated.

THERAPEUTIC POSTURE

I position myself well at the coccyx and the heart. One can visualise the **angle of Louis**, the vascular trunk of the aortic pulmonary type.

The heart is on the **longitudinal axis**. Only a part of it is dilated to the left, with a developmental phase on its apex to the left.

You will position yourself on the longitudinal plane of the heart. By giving a fulcrum here and a point of support on the heart, my attention is in space, neutral, and we let the system organise itself.

12. THE HEART : DIASTOLE & SYSTOLE

DURATION : 05:06 STUDY SHEET

COURSE SUMMARY

This video explores the essential dynamics of the heart's systolic and diastolic movements, focusing on their impact on bodily energy and emotions. You will learn to observe the heart's vertical and horizontal orientations, which can reveal crucial aspects of your patients' emotional health. The concepts of ventricular looping and interaction with other organ systems like the liver and lungs show how the heart maintains a regulatory pressure within its fascial environment. Practical techniques for manual assessment of the fascial skin and movements allow these notions to be applied in a therapeutic setting, while also revealing the link between the heart and the diaphragm for proper rhythmic and metabolic function.

13 - THE HEART: DIASTOLE & SYSTOLE

The **heart** stores **energy** in a **fascial** environment, thereby supporting its **electricity**, in both its ascending and descending phases.

There is an **innate predominance** in some people, who may manifest a **systolic** or **diastolic** function. A balance between these two functions is normally required. The heart repeats its embryonic movement quite directly.

HEART MOVEMENTS AND EMOTIONAL IMPLICATIONS

A specific movement can be felt. For example, people who are too oriented towards a **systolic** movement are often searching, but can also quickly become exhausted. Conversely, others start from this point and constantly seek a lower fulcrum.

VENTRICULAR LOOPING AND CARDIAC PHASES

The heart performs a characteristic **looping**, where the **ventricular** system, initially superior, becomes inferior, with a crucial fulcrum between the **atria** and the **ventricles**.

- **During systole:**

* The heart performs an **upward movement**.

* It performs a **supination** movement.

* It tends to **verticalise**.

* It performs an **untwisting**.

It is at this moment that the **aorta** opens to receive all the blood, causing **congestion**.

- **During diastole (descent):**

* All the blood from the aorta is ejected into the system.

It is sometimes easy to confuse the systolic movement with the descending movement. Systole is the moment of **ejection**. When the heart descends again, it **refills**.

When the volume is ejected, the pressure in the aortic arch system increases. At the moment of descent, the heart fills while ejecting the aorta.

HEART ORIENTATION: VERTICALISATION VS. HORIZONTALISATION

Do you have the impression that the heart tends to **verticalise** or **horizontalise**? Does it verticalise and ascend, or horizontalise and descend?

FULCRUMS FOR OBSERVATION

We will take three fulcrums:

1. At the level of the **aortic space** (second intercostal space).

2. At the level of the **pulmonary space**.

3. The **hypothenar eminence** will be placed on the apex of the heart.

THE HEART'S INFLUENTIAL SYSTEMS

The heart reacts according to two major systems that are very close to it and tends to help them as much as possible:

- The **liver**
- The **stomach**
- The **lungs**
- The **kidneys**

It always seeks to maintain this pressure in its environment. The heart is a **volume regulator**. The heart should be considered as a large artery extending with the aorta, like an **octopus**.

SUPINATION AND PRONATION MOVEMENTS

We will have a **supination** movement, then a **pronation** movement.

MANUAL ASSESSMENT AND FASCIAL REACTIVITY

We will observe the environment and how it is arranged. Then, we will place our hands at the level of the liver and observe its reaction on a **fascial** plane.

The aorta, if you observe it well anatomically, actually starts quite far back, which makes it difficult to visualise. However, it shares a common fascial environment with the **bronchi** and the **oesophagus**, which insert at the level of a vertebra often in adaptation.

RHYTHMIC ZONES AND DIAPHRAGMATIC RELATIONSHIP

There is:

- A **neurosensory** zone.
- A **rhythmic** zone.
- A **metabolic** zone.
- A **urogenital** zone.

For good movement, the heart must have a good relationship with the **diaphragm**. There is a direct relationship via the fascial continuity of the arch and the aorta.

Remember the very deep passage of the aorta, where the **diaphragmatic crura** insert on either side. To work on the heart, it is essential that the diaphragm is **open** and **supple**. Do not forget the other important passage at the level of the oesophagus.

PRACTICAL APPLICATION: FOLLOWING FASCIAL MOVEMENTS

We have studied the **embryonic pre-pattern fulcrum** which is positioned at the level of the primitive streak, at the level of the coccyx, with the arrival point.

Now, let's place the hand at the level of the heart and let it follow the fascial movement. The tissue, if you are attentive, will indicate whether the heart leans more towards systole or diastole.

Is it my liver reacting? Is it in relation to my stomach? Or is it in relation to my kidneys? (Three levels). Then, we will observe if this can rebalance in relation to this observation.

13. STRUCTURING THE VASCULAR SYSTEM : NOTES

DURATION : 03:24 STUDY SHEET

COURSE SUMMARY

In this video, we explore the structuring of the vascular system during embryonic development, focusing on the interconnection between the yolk sac and the amniotic cavity. The concepts of cephalisation, cardialisation, and diaphragmatic development are detailed, illustrating how the vascular system centralises as it develops. By analysing the cardinal systems and the formation of the venous vascular axis, we will discover how fluid movements and lateral germinations contribute to the formation of a 'protective aura', essential for the development of the cerebral system. This approach highlights the importance of reintegrating zone B when treating the nervous system, establishing a crucial link between the umbilicus and embryonic structures.

STRUCTURING OF THE VASCULAR SYSTEM

The **vascular system**, which develops at the periphery, is deeply integrated with the **vitelline vesicle**. **Wolff-Pander islands**, for example, form in the splitting of this vesicle.

Behind the vitelline vesicle is the **amniotic cavity**. Imagine it as a **balloon** containing this circulatory system, which is also found in the axis.

This "balloon" will deflate, and the entire peripheral system will centralise. This centralisation is orchestrated by the amniotic cavity.

Its differential growth rate, compared to the **primitive vascular field** and the development of the **brain**, generates a suction movement. Externally, this is accompanied by a **transverse thrust**, both thoracic and abdominal.

The longitudinal axis is more complex, integrating **cephalisation**, **cardialisation**, **diaphragmatisation**, and the entire **posterior peritoneum** down to the **perineum**.

This process is comparable to this integrating balloon. Simultaneously, at the periphery, there is a huge movement converging towards the **umbilical cord**. The cord, let us remember, is the meeting point between the **embryonic pedicle** and the vitelline vesicle, the latter gradually resorbing.

The secret here is **embryonic movement** and **developmental motility**: always bringing fluids towards the centre.

We will study the **cardinal systems** (subcardinal and supracardinal) in detail to finally arrive at a **venous vascular axis**. The latter presents specific vestiges.

The venous vascular system develops **transversely**, not longitudinally. It is composed of parts coming from different directions, resulting from a transverse union.

This contrasts with the arterial vascular system, which functions **longitudinally**, with two large conduits. On the venous side, these are **lateral germinations**, i.e., new trajectories.

When the fluids integrate towards the centre, **zone B** opens and takes shape, forming a kind of **protective aura**. This bubble contributes, among other things, to the construction of the **cerebral system**.

The first cavity to appear, or rather the second, is the amniotic cavity. It is at this moment that the **embryonic knob** becomes the future **epiblast**, the future nervous tissue.

The two must not be separated. If we want to treat the nervous system, we must always reintegrate zone B. This converges on the entire **median axis** and finally at the **umbilical** level.

The **umbilicus** is the final meeting point, the junction between the embryonic pedicle and the vitelline vesicle.

14. THE HEART : TAI CHI REMINDER & PRACTICE

DURATION : 03:38 STUDY SHEET

COURSE SUMMARY

This video teaches how the heart interacts with the brain and the circulatory system through embryonic movements, while reviewing the theoretical bases of the heart-brain system. You will learn specific palpatory techniques to assess the heart, liver, and kidneys, as well as their emotional impact, particularly how the kidney is linked to fear. By integrating these concepts with pulsology, you will develop a holistic therapeutic approach, enabling you to support the body in its healing journey.

THE HEART: TAI CHI RECALL & PRACTICE

INTRODUCTION TO THE HEART-BRAIN SYSTEM

We will explore the intrinsic link between the **brain**, the **heart**, and the **circulatory system**.

The **mesenteric system**, as the axis of rotation of peritoneal development, performs a downward movement. This movement of the brain is synchronised with the movements of the **kidneys** and the **liver**.

We will re-examine the **motility** of brain development and integrate it with our cardiac knowledge, to understand how the heart supports the brain.

Remember that the heart can be positioned in the **vitelline vesicle** from the earliest stages.

PALPATION ASSESSMENT: LANDMARKS AND OBSERVATIONS

To begin, I take a precise landmark: the **angle of Louis**. I know it is located at the level of my **second intercostal space**.

I place my middle finger on the angle of Louis, then I place my hand and align it with the axis of the heart.

CARDIAC OBSERVATION

Facing me, the heart is in **pronation**, which corresponds to **diastole**. This indicates that it is seeking, searching.

In the engagement of its life, it is somewhat in renunciation, more withdrawn than in its action.

HEPATIC ASSESSMENT

I then place my hand at the hepatic level, with very light pressure, as if an **ant** were placed there that one could not crush.

I observe the liver. If it is rather horizontal, it may mean that it is slightly **swollen**.

It tends to move in this direction.

RENAL ASSESSMENT

Now, let's examine the kidneys. I palpate a kidney and observe that it is also in pronation.

This indicates that it is more prone to its lesion, which means that it is the kidney that destabilises it here, while the liver provides support.

The kidney is emotionally linked to **fear**. The patient is looking for a point of support, but their history is blocked at this level.

For this patient, I will therefore work on the kidneys to support their heart.

EMBRYONIC MOVEMENTS AND THERAPEUTIC SUPPORT

I remind you that the **embryonic movement** of the lungs is distinct.

While that of the heart is different. Observe the difference in these movements.

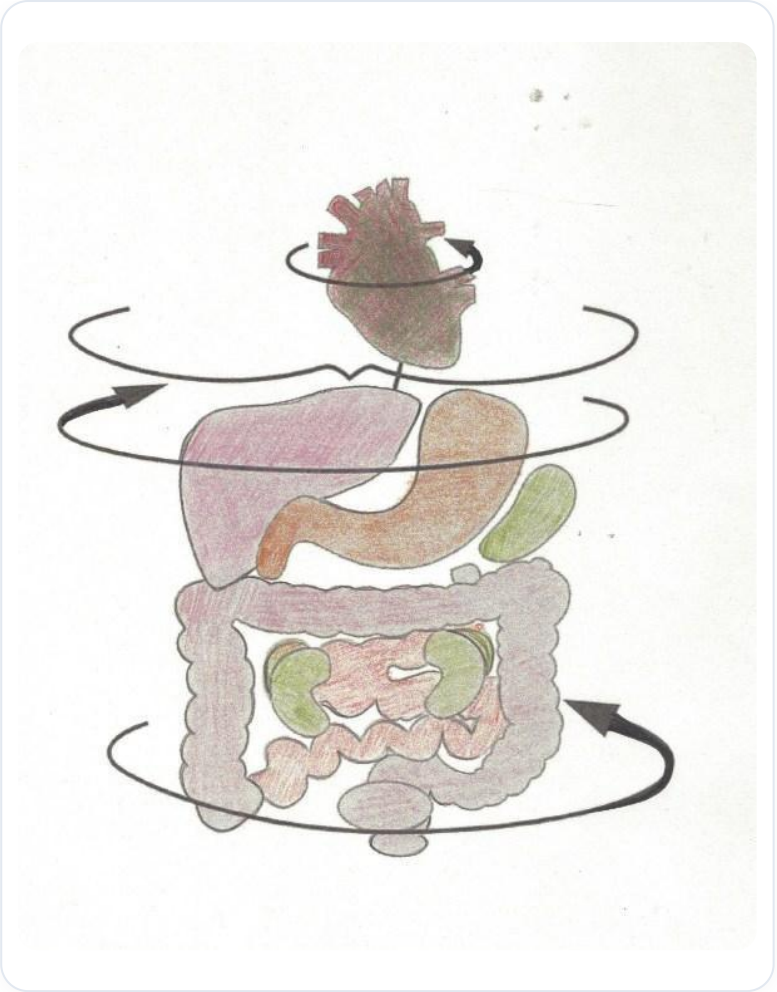


FIG. — Organ rotation

INDUCTION AND SUPPORT

After performing an induction towards its pronation, i.e., towards diastole, I support its kidneys.

Observe what happens. Do you feel the difference? The body seeks to reorganise itself.

Later, we will explore the motility of the development of the liver and kidneys, as well as their link with the **vascular system**.

THE HEART AND PULSOLOGY

It is essential to remember that the heart, in its form, is the reflection of the echo of its periphery.

This is why **pulsology** is such a fascinating field. It allows us to distinguish:

- A pure arterial type **pulse**.
- A venous type pulse, accompanied by **stases**.

For me, there are stases, probably located in the hepatic area.

An arterial pulse, when you take it, is characteristic, while a venous pulse manifests differently.

Despite this sensation on an artery, the question remains: am I more in **systole** or in **diastole** ultimately?

15. STRUCTURING OF THE VENOUS SYSTEM 1 : THE CARDINAL VEINS

DURATION : 12:46 STUDY SHEET

COURSE SUMMARY

This video offers a detailed teaching on the structuring of the venous system, with a particular emphasis on the cardinal veins, which play a fundamental role in embryology. The main venous systems, their embryological origins, and the transformation of venous structures, such as the umbilical and vitelline veins, are discussed. The exploration also includes the genesis of the heart and great vessels, as well as the importance of the subcardinal and supracardinal veins in the development of the venous system. This knowledge is crucial for practitioners and students in embryology and osteopathy, enriching their understanding of the interrelationships between embryological development and vascular anatomy.

STRUCTURING OF THE VENOUS SYSTEM: THE CARDINAL VEINS

THE THREE GENERAL VENOUS SYSTEMS

Three main **venous systems** are identified within the organism:

- **The neurocranial system:** Corresponds to the vertebral plexus.
- **The viscerocranial system:** Linked to the portal system.
- **The internal system (internal man):** Represented by the inferior and superior vena cava system.

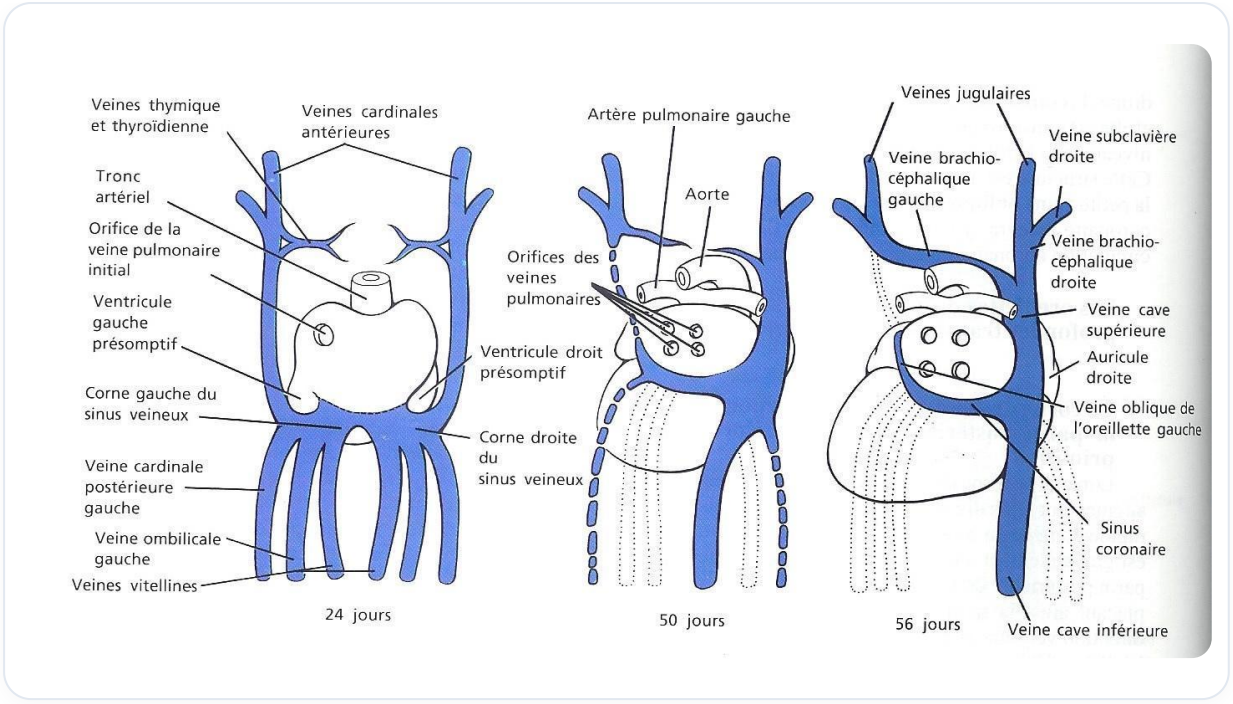


FIG. — Heart venous system development

EMBRYOLOGICAL ORIGINS OF THE VENOUS SYSTEMS

From an **embryological** point of view, the formation of these systems is linked to three precursors:

- **The common cardinal:** Gives rise to the neurocranium, the vertebral plexus, and the cava system (inferior and superior).
- **The vitelline:** Essentially linked to the viscerocranium.
- **The umbilical:** Largely disappears, leaving a terminal fibrous system. A persistent terminal vessel can form **Meckel's diverticulum**, a small pouch located just before the ileocecal valve, which can be the cause of diverticulitis.

THE CARDINAL SYSTEM

The cardinal system is divided into:

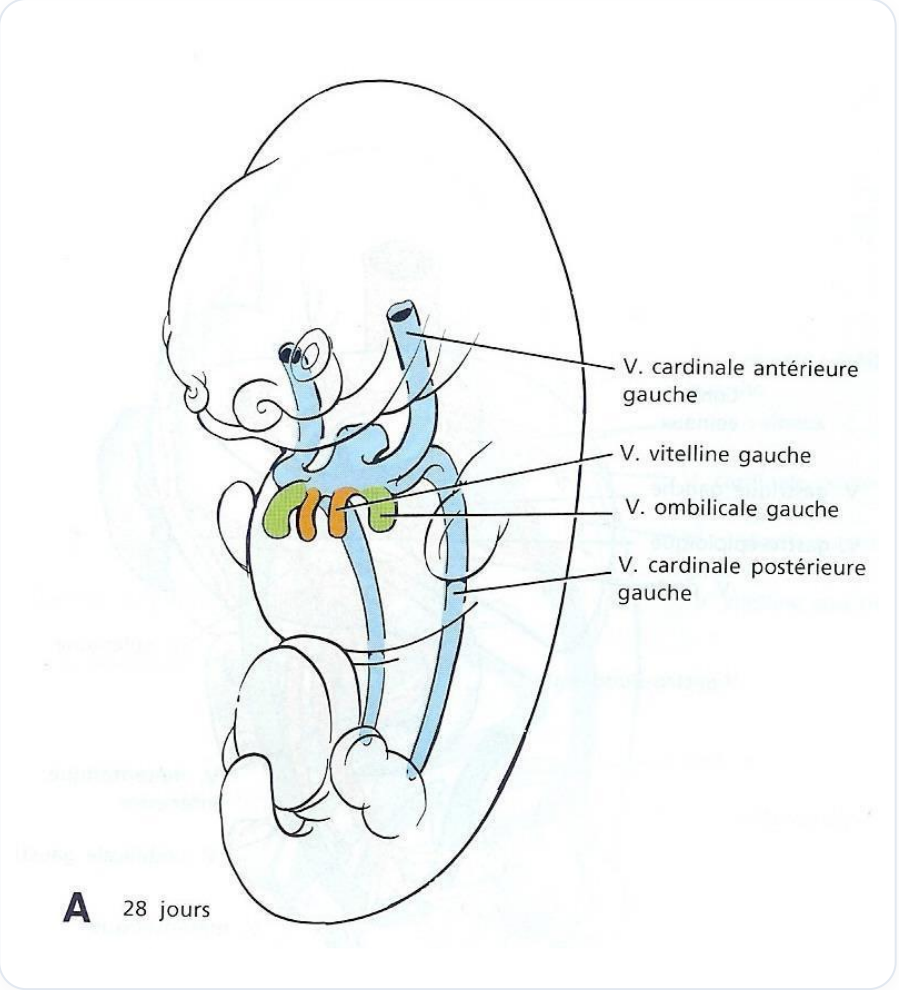


FIG. — Embryonic venous system

- **Superior or anterior cardinal vein:** Located laterally superiorly.
- **Posterior or inferior cardinal vein (or mesonephrotic):** Located laterally inferiorly.

These cardinal veins are also called **mesometric veins**. The whole corresponds to what is called the **venous sinus**. The umbilical vein and the vitelline vein are located in the middle.

FATE OF THE CARDINAL, UMBILICAL, AND VITELLINE VEINS

- **Umbilical veins:** Disappear, leaving only vestiges.
- **Right cardinal vein:** The part between the sinus and the thymic vein disappears. However, the right thymic vein anastomoses with the left thymic vein to form the **brachiocephalic vein**.
- **Vitelline vein:** The left part disappears. The right part remains and anastomoses with the right anterior cardinal vein to form the **inferior vena cava**.

THE COMPONENTS OF THE VESSEL AND CHANGES IN TRAJECTORY

For a vessel to exist, four components are necessary:

1. The **trajectory**.

2. The **metabolic concentration**.

3. The **vacuolisation**.

4. The **interstitial fluid**.

The disappearance of part of a vein indicates the loss of one of these components. A new trajectory is then drawn. Vascular development occurs laterally towards the centre and segmentally.

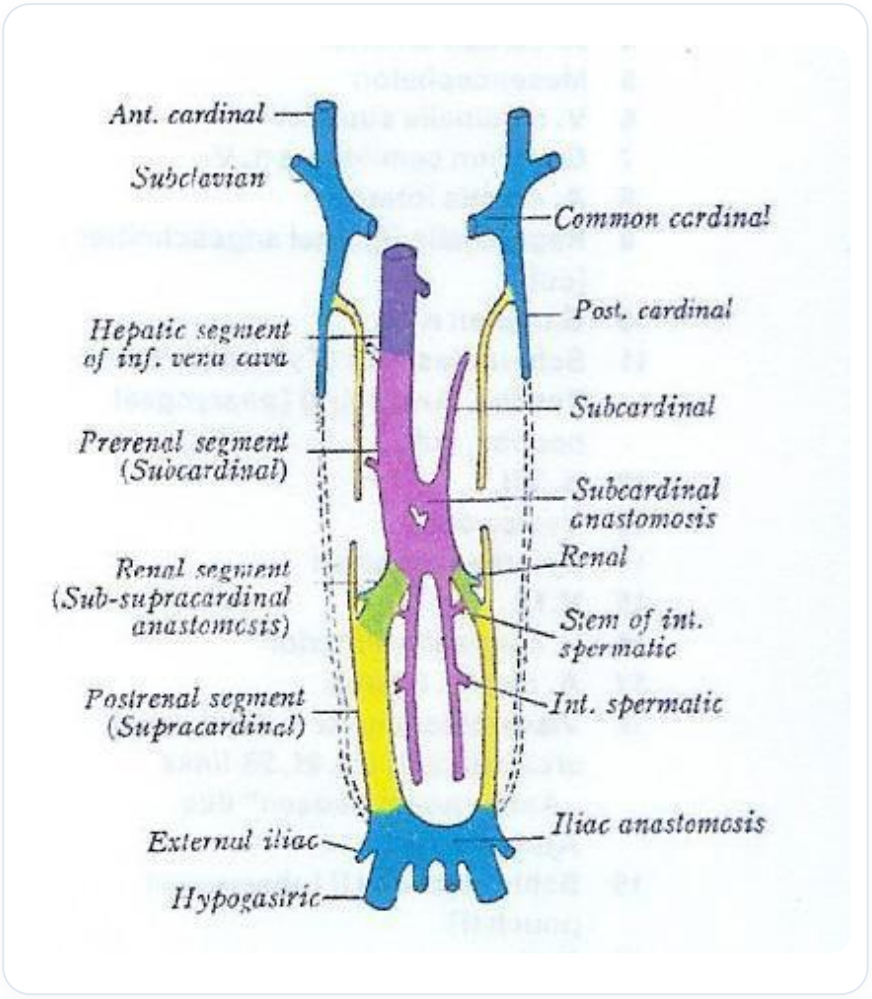


FIG. — Embryonic SVC and IVC

The vascular system of the internal man is characterised by the common cardinal vein and the posterior cardinal vein, resulting from phenomena of assimilation and disassimilation.

DEVELOPMENT OF THE HEART AND GREAT VESSELS

The heart develops from the **primitive streak**, relying on a movement of concentration towards the midline. This process includes the **corrosion field**, the **heart looping**, and its positioning towards the left. A longitudinal concentration axis is followed by the looping, creating a perpetual movement of the heart. Then follows the more horizontal positioning of the great vessels, particularly pulmonary and aortic.

The cardinal system develops more posteriorly, while the umbilical and vitelline systems are more anterior.

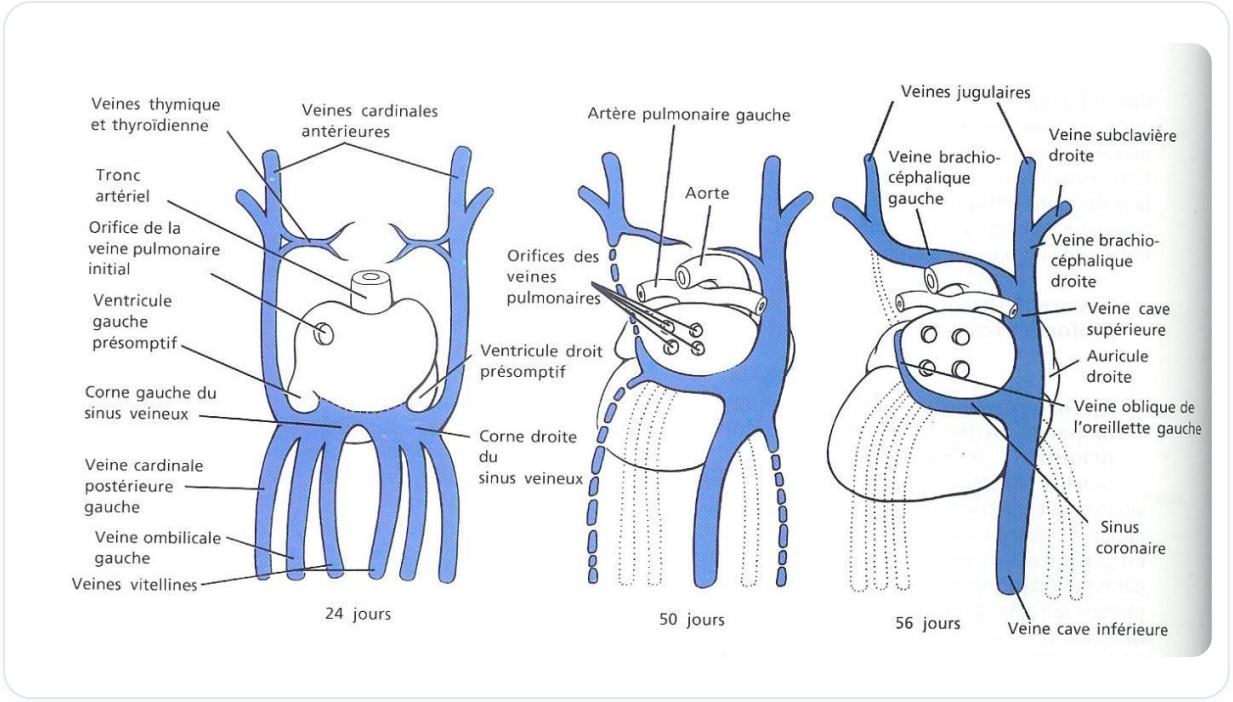


FIG. — Cardiac embryonic development

THE SUBCARDINAL AND SUPRACARDINAL VEINS

From the cardinal network, the **subcardinal veins** emerge at the base of the posterior cardinal veins.

- The major part of the left subcardinal vein regresses.
- The right subcardinal vein loses its connection with the posterior cardinal vein and develops a new anastomosis with the vitelline vein, participating in the formation of the cava system. This part will form the **inferior vena cava**, located between the liver and the kidney.

Derivatives of these subcardinals include the **suprarenal** vein, the **testicular** veins, the left renal part of the inferior vena cava, and the supracardinal.

This process is a "budding" by an aspiration field, creating new segmental trajectories while others disappear. The subcardinals re-bud to form a second network, and the process repeats with the supracardinals.

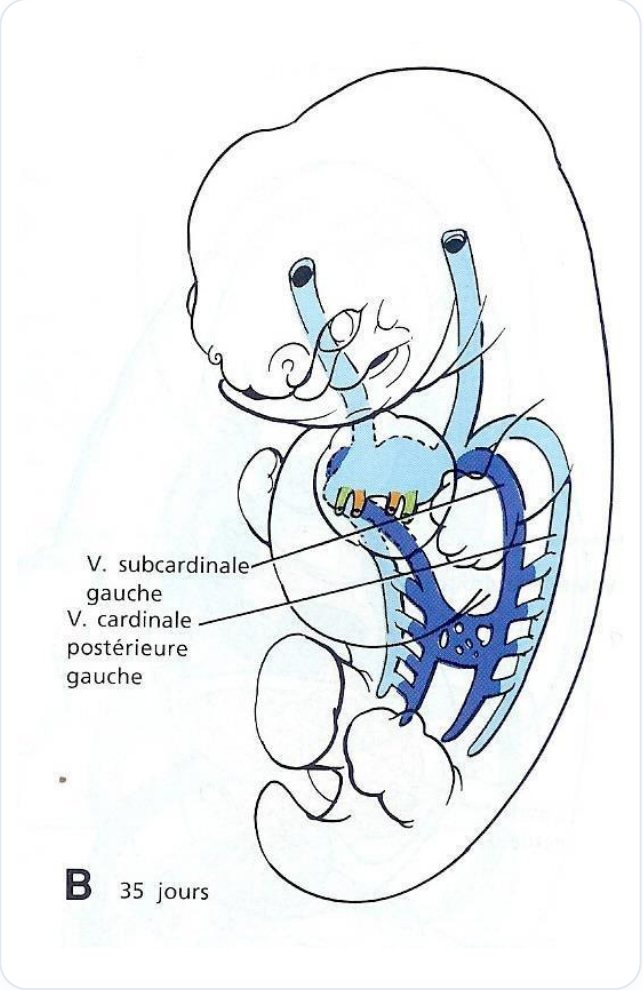


FIG. — Embryo 35 days vascularization

THE FURTHER FATE OF THE VENOUS NETWORKS

The only remaining vestige of the cardinal system will eventually be the **azygos system**. It will be replaced by the subcardinal, then supracardinal systems.

- Inferiorly, the iliac system is initially cardinal, but this system disappears.
- The subcardinal, partly disappeared, contributes to the purple segment that forms the inferior vena cava. This vein is a combination of part of the subcardinal and part of the supracardinal.

This development occurs according to a transverse movement. As a result, some segments of the inferior vena cava come from the supracardinal, with some vestiges of the subcardinal.

The developmental movement of the subcardinal is characterised by "spotting", a segmental budding that contains the entire network of the system.

DRAINAGE OF DIFFERENT AREAS

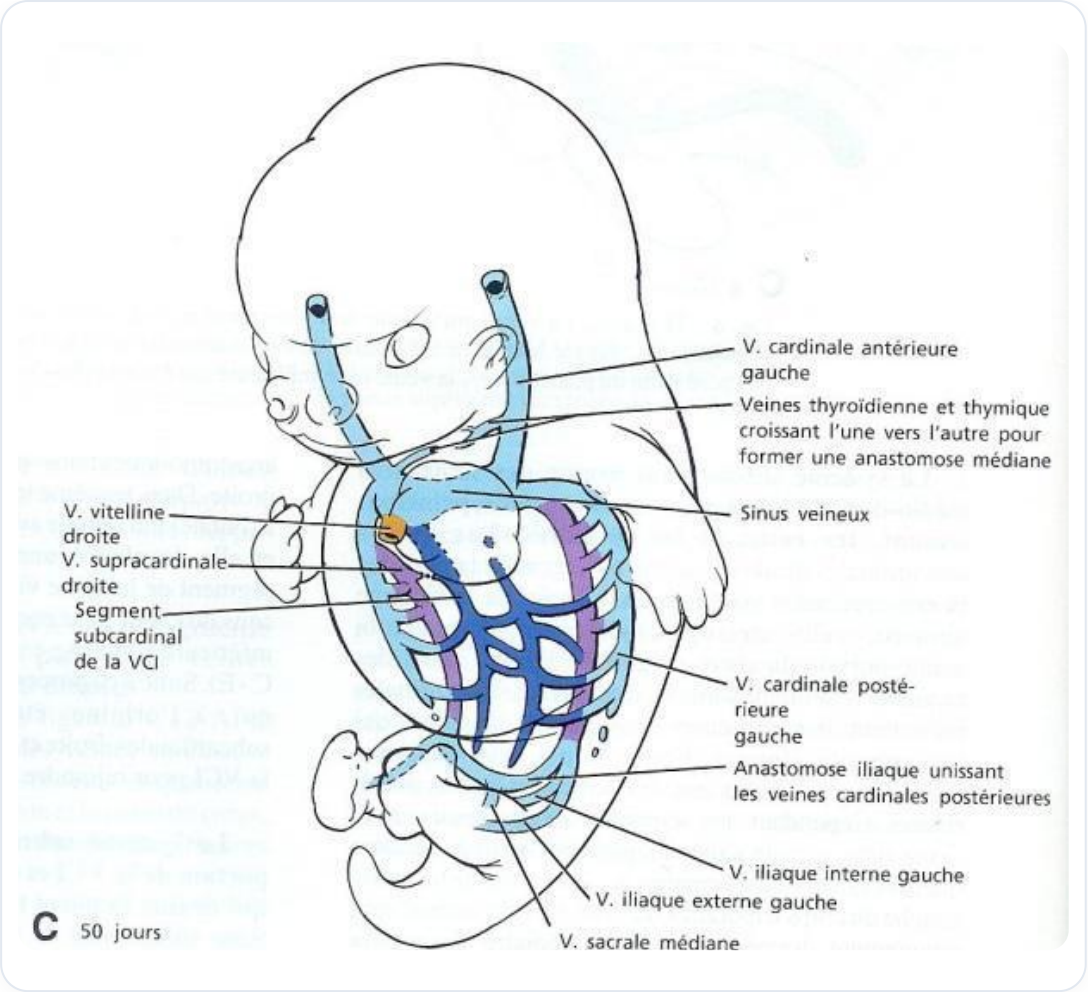


FIG. — Embryonic venous system

- **Azygos-hemiazygos system:** Ensures all thoracic drainage.
- **Mesodermal system (internal man):** Includes renal, spermatic, and suprarenal drainage.

The movement from the periphery to the centre is clearly visible. The supracardinal gives the entire thoracic and abdominal part, with a segmental difference for the left side (hemiazygos) and the right side (azygos).

The fusion of these networks in the abdominal region forms the **inferior vena cava**, which drains the entire system of the mesodermal man (support tissue).

KEY RESIDUAL VENOUS STRUCTURES

The only final vestiges are:

- **The brachiocephalic trunk.**
- **The superior vena cava.**
- **The iliac veins** below.
- **In the centre:** The meeting on the cava plane with renal, suprarenal (mesodermal type) and azygos structures.

- **To the left and right:** Hemiazygos and azygos, ensuring thoraco-abdominal drainage on the mesodermal plane.

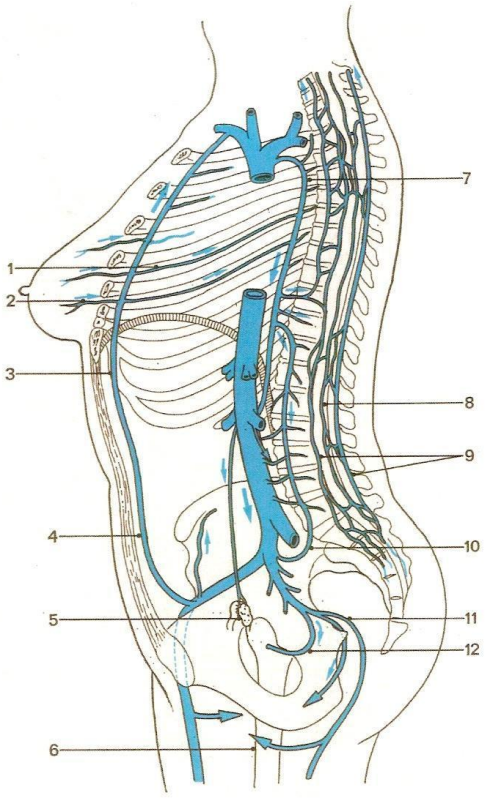


Figure 9. De la veine fémorale à l'abdomen. Référence (36), p 472.

FIG. — Deep venous system

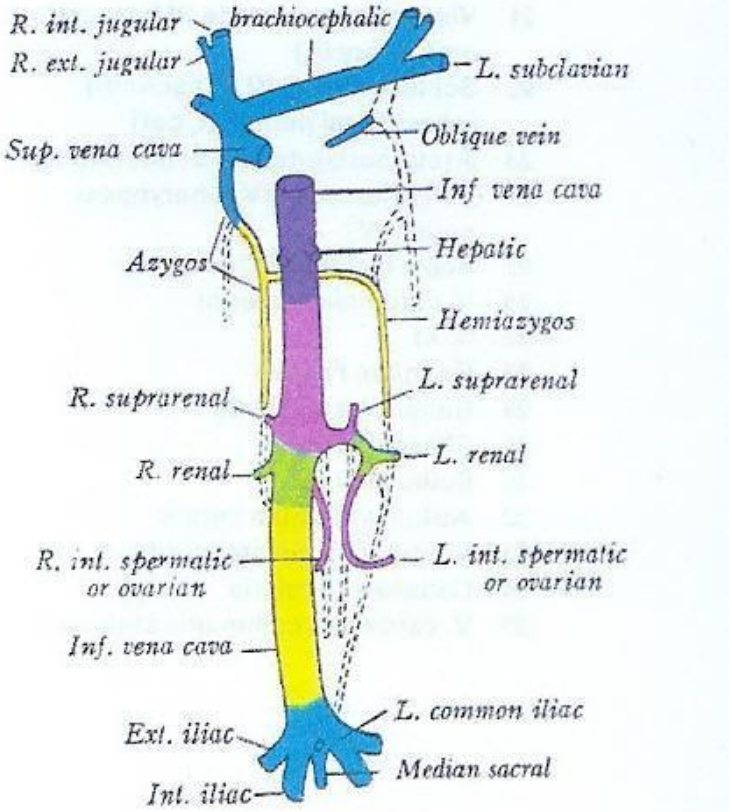


FIG. — Inferior vena cava development

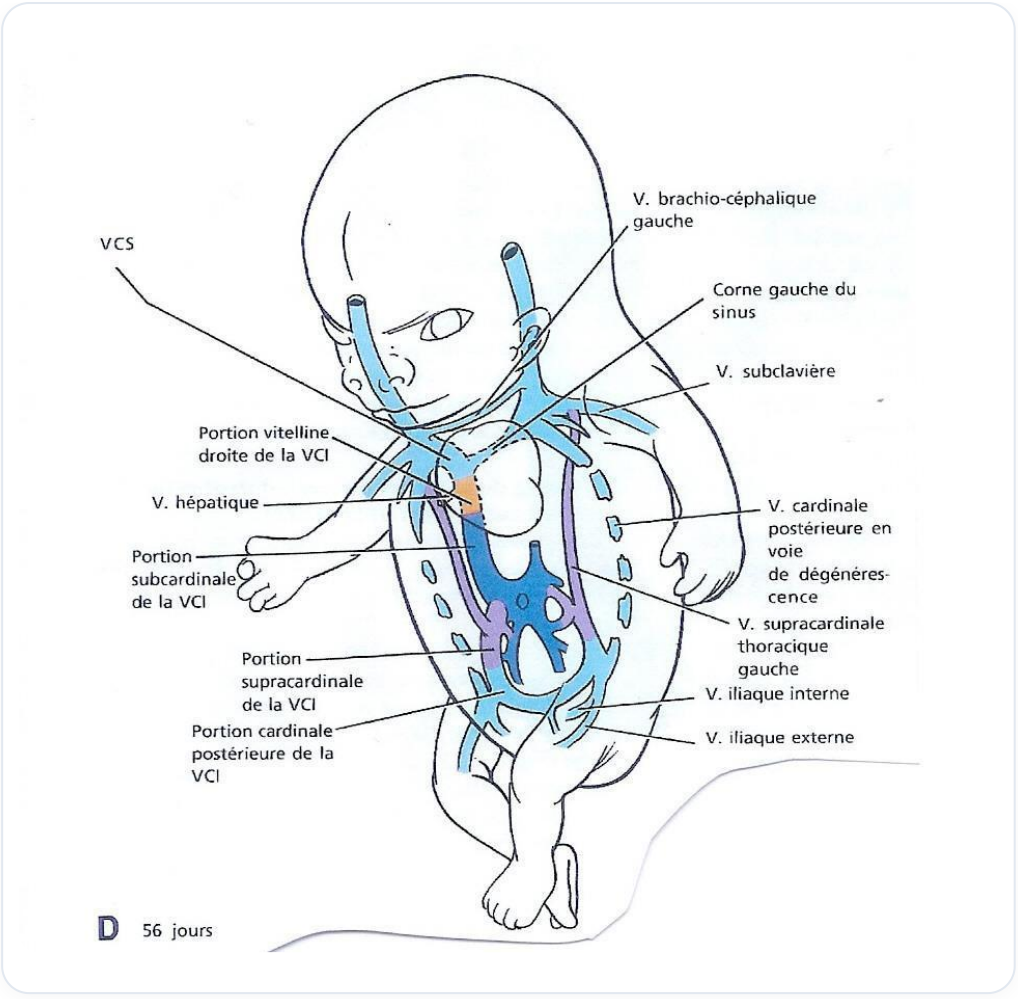


FIG. — Venous System Development

16. STRUCTURING OF THE VENOUS SYSTEM 2 : THE VITELLINE & UMBILICAL SYSTEM

DURATION : 08:22 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth exploration of the vitelline and umbilical systems, focusing on the development of the liver from mesodermal and endodermal tissues. It details the different stages of formation of the hepatic vascular network, including the vitelline and umbilical veins, as well as their complex interactions. Key concepts such as cephalisation, cardialisation, and the two currents of the hepatic portal system are explained, highlighting the importance of this network for blood drainage and overall metabolic health. The emotional impacts related to digestive organs are also addressed, thus enriching the biodynamic understanding of the human body.

17 - STRUCTURING THE VENOUS SYSTEM 2: THE VITELLINE & UMBILICAL SYSTEM

THE VITELLINE SYSTEM AND THE LIVER

The **vitelline system** is the second major network we will discuss, in connection with the digestive system. The **liver** is the main receptacle of this system.

It is essential to remember that the liver results from the combination of two tissues: a **mesodermal tissue** and an **endodermal tissue**.

My endodermal tube has an upper, middle, and lower intestine. The liver emerges at the junction between the upper and middle intestine, initially developing as a "field of aspiration," a subtle description of the digestive tissue.

The **20 vitelline veins**, mesodermal structures, are at the origin of liver formation.

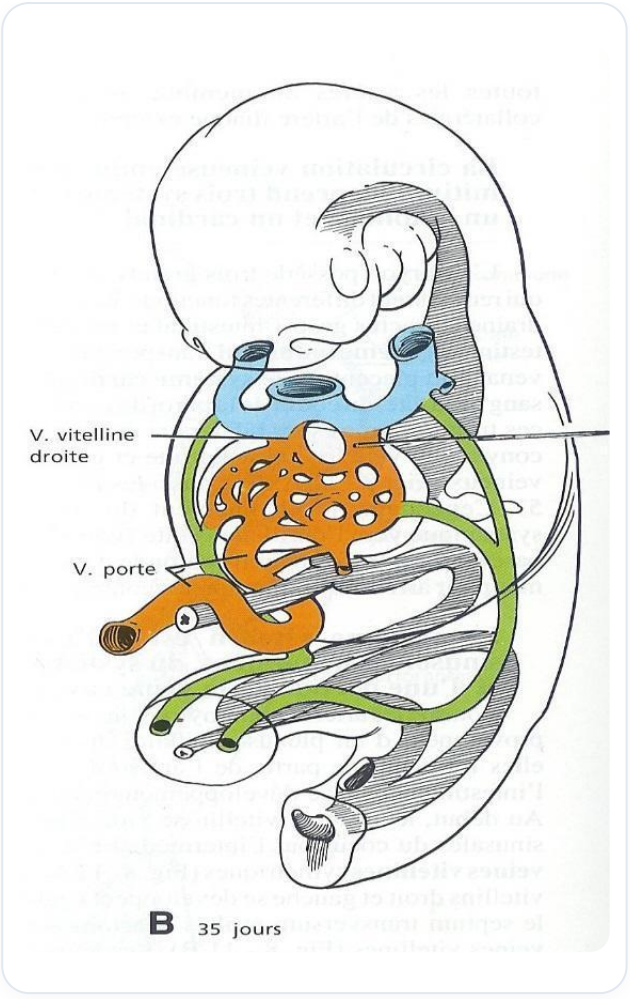


FIG. — Embryo hepatic vascular system

LIVER DEVELOPMENT SEQUENCE

Cephalisation leads to **cardialisation**, which causes **diaphragmatisation**. This process results in hepatic congestion at the mesodermal level, through vitelline integration.

The movement of cephalisation leads to cardialisation, then diaphragmatisation, leading to a phase of congestion below the diaphragm. This area of congestion corresponds to the mesodermal development of the liver. Initially, the liver receives no umbilical information, but is informed by disassimilation products from the brain, lower trunk, and vitelline vesicle.

Its first impulse is therefore a reception of **toxicity**. The liver, as an organ, has the primary function of receiving and transforming this toxicity, before having a regenerative impulse later.

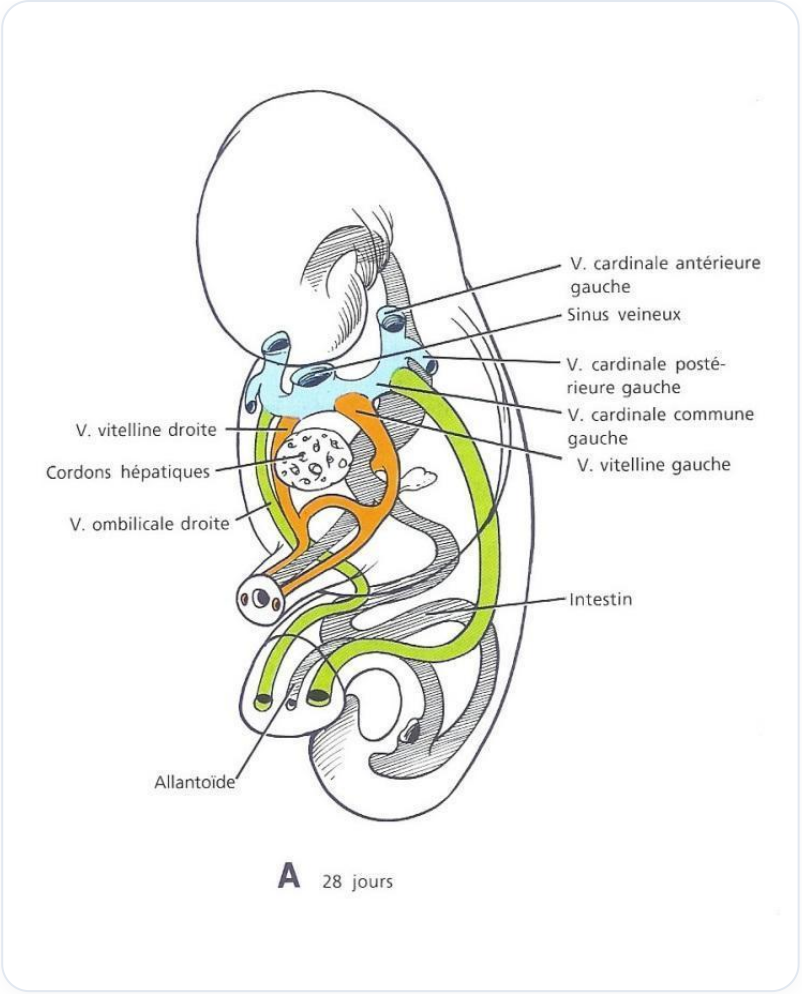


FIG. — Embryonic circulation 28 days

FORMATION OF THE HEPATIC VASCULAR NETWORK

We have the **right vitelline vein** and the **left vitelline vein**, with a small anastomosis already formed between the two. In parallel, the hepatic cord originating from the endoderm develops.

Between these two vitelline veins, an entire network forms, congests, and organises. This network will eventually form the venous duct and, later, structures such as the **gastric vein** and the **gastro-epiploic vein**. An important area will develop: the **portal vein**.

At another level, the **umbilical vein** develops, part of which will be integrated while another part will disappear. This network of vitelline systems meets to form a specific vascular network, acting as a drainage system.

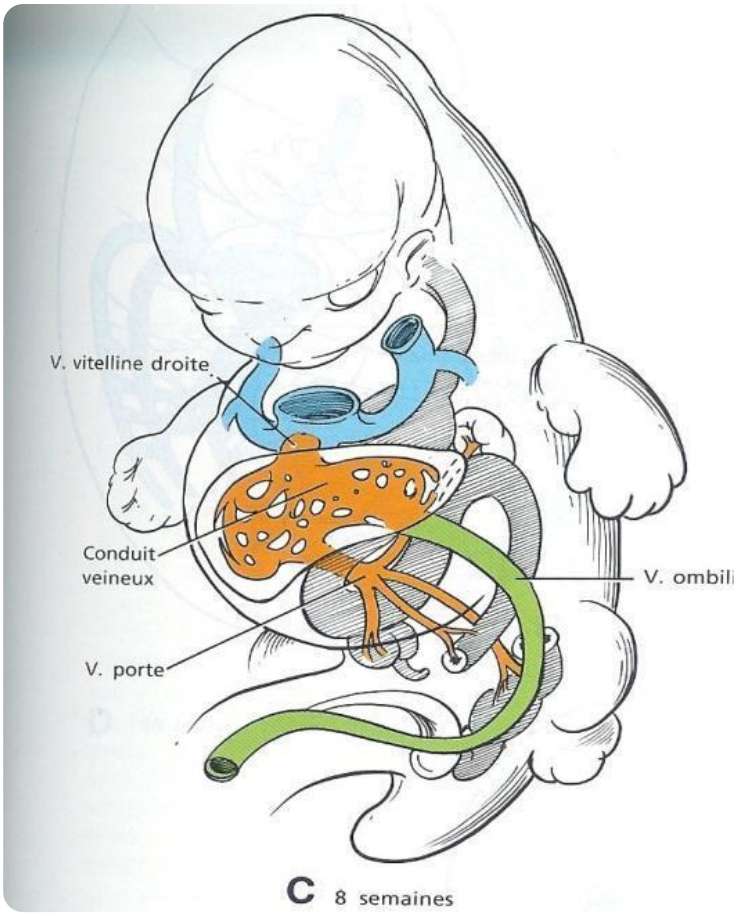


FIG. — Embryonic circulation 8 weeks

SEGMENTATION OF THE VITELLINE VEIN

Development takes place under the **septum transversum**. The vitelline vein, located under this septum, can be divided into three sections:

- **Suprahepatic section:** It will eventually form the hepatic duct, then the hepatic vein. The right vitelline vein becomes the hepatic vein, the region of flow towards the heart.
- **Hepatic section.**
- **Infrahepatic section.**

Another derivative of this vitelline part is the upper part of the **inferior vena cava**.

REGRESSIONS AND MODIFICATIONS OF DRAINAGE

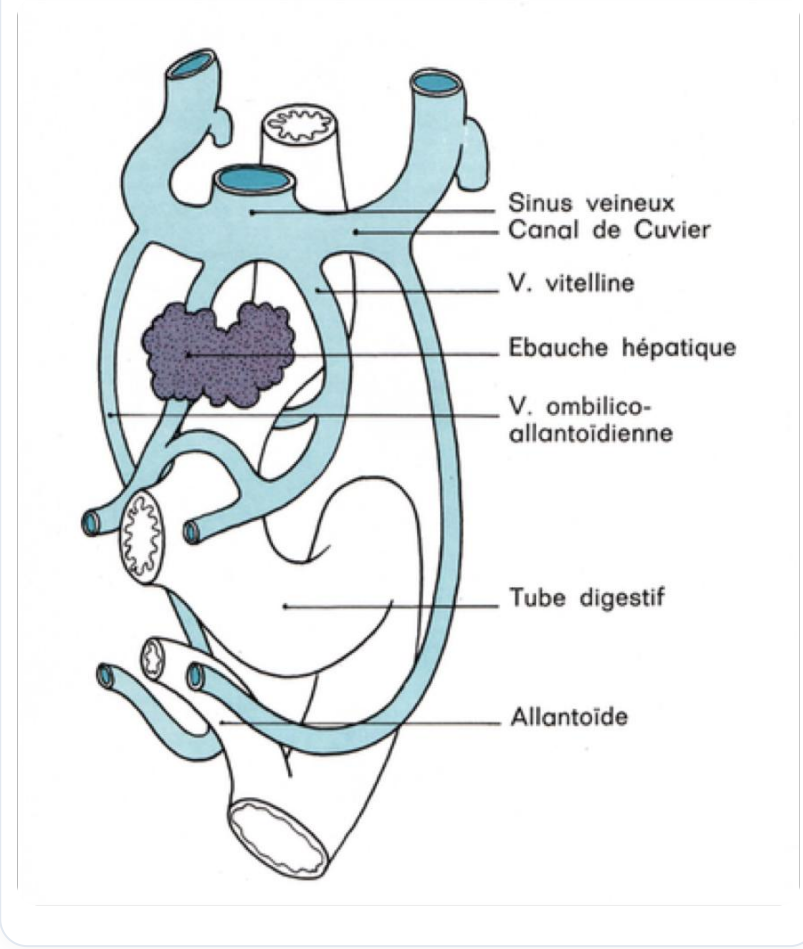


FIG. — Embryonic venous circulation

Regressions occur. The left umbilical vein regresses, but part of the right umbilical vein persists, forming a link for the **hepatic sinusoids**.

The umbilical vitelline vein is here. The left umbilical vein forms a link, then gradually disappears in its upper part. The **ductus venosus** will be formed with the portal trunk.

In the infrahepatic region, the left vitelline vein regresses, but develops an anastomosis with the right vitelline vein, leading to a change in blood drainage. Blood passes from the left side of the abdominal viscera to the right vitelline vein, creating a developing transverse network. This leads to a **switch** from the left side to the right side at the venous level, promoting growth on the right.

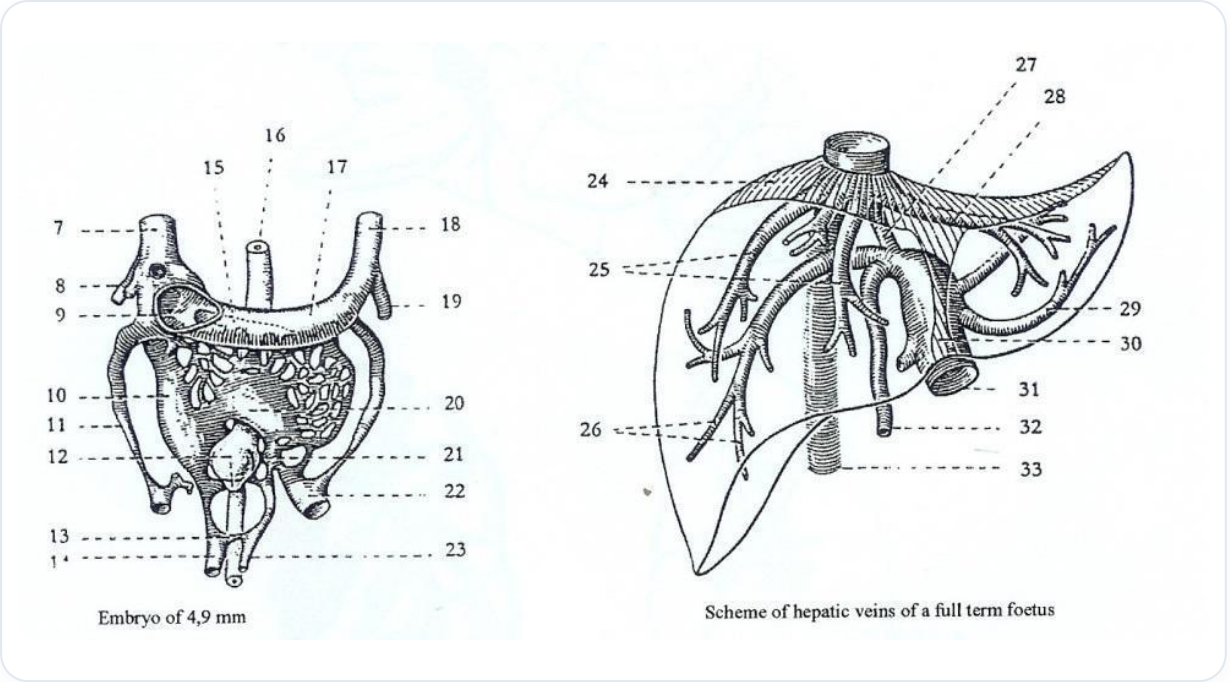


FIG. — Embryo, hepatic veins

THE TWO CURRENTS OF THE HEPATIC PORTAL SYSTEM

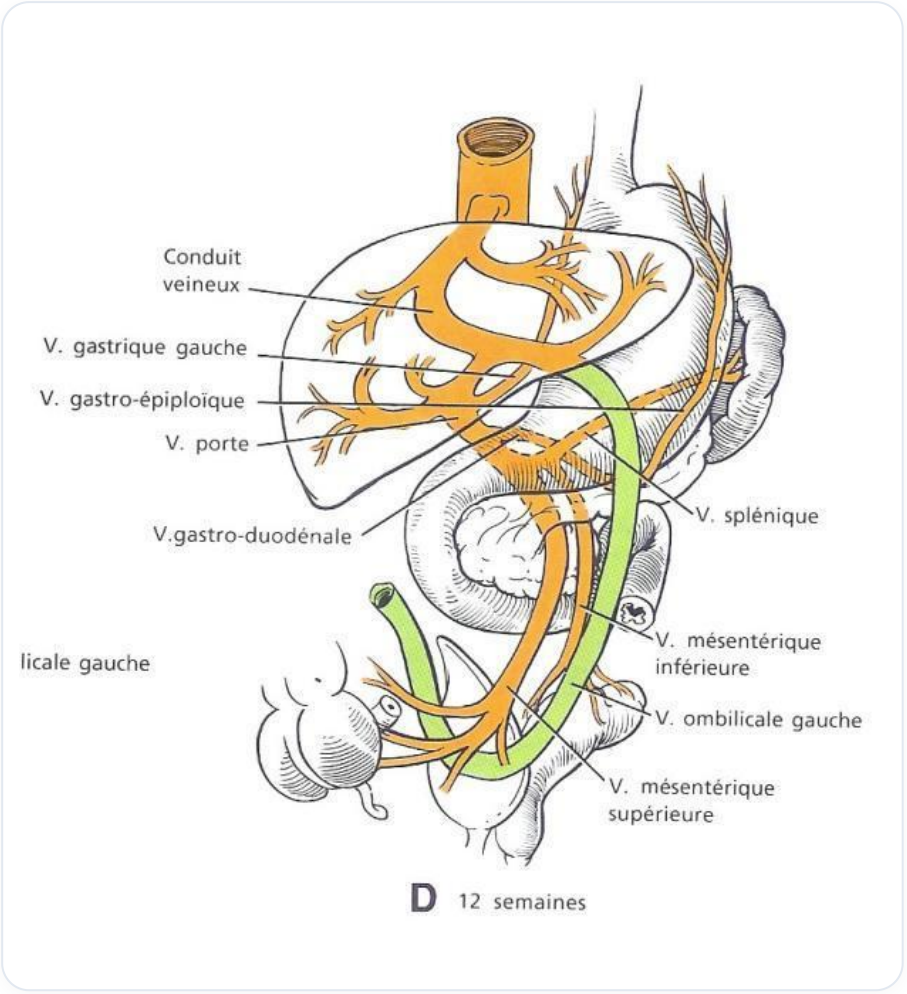


FIG. — Portal system 12 weeks

All blood coming from the **right mesenteric** side is drained by the portal system and is found mainly in the right liver. In contrast, everything inferior (stomach, spleen, and lower colon) drains more to the left liver.

Two currents can be observed within the portal system itself:

- One current oriented towards the left side.
- Another towards the right side.

The liver plays a digestive and metabolic role, but also a motor and emotional one. Organs like the stomach and spleen are related to the emotional system, dreams, and nightmares, leading to congestion.

The portal trunk is common here, but it is interesting to note that there is a flow that does not mix or can mix.

LIVER REGENERATION

The liver never regenerates at its endodermal level, but always regenerates at its mesodermal level. It can rebuild vessels, but not ducts.

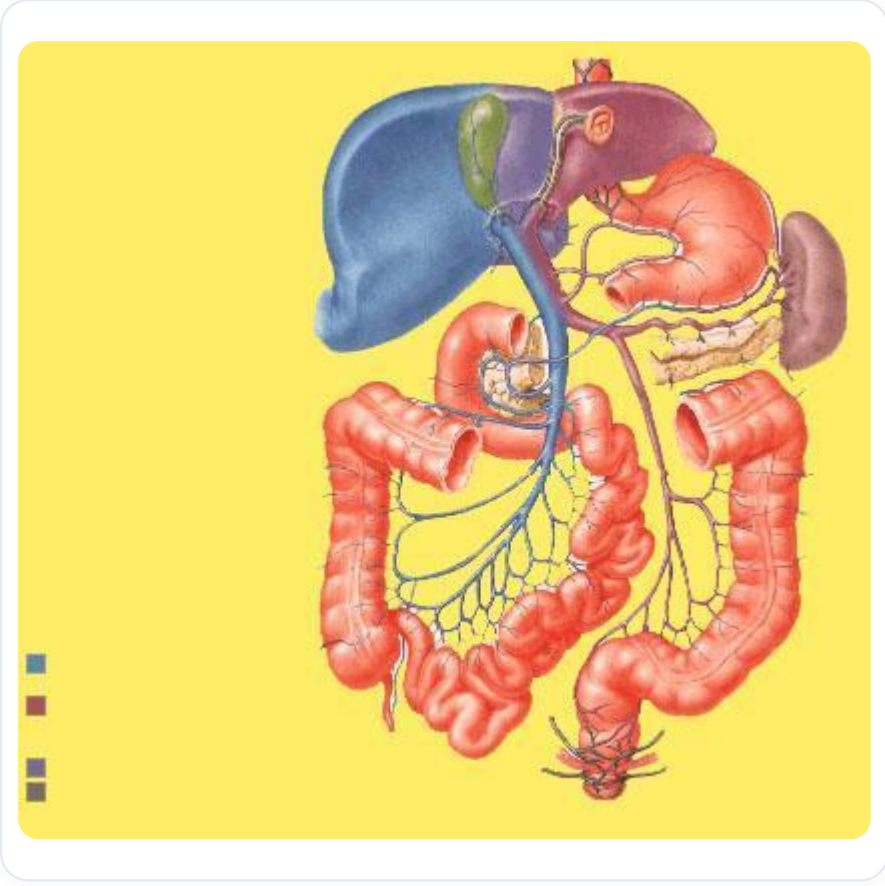


FIG. — Hepatic portal system

The liver is a set of **bile ducts** integrated with many vessels. It is important to remember that the liver only regenerates at night. There are three levels:

1. **When you eat:** A lot of venous blood and little arterial blood. The liver does not regenerate.

2. **Between meals:** Balance between arterial blood and venous blood. The liver regenerates a little.

3. **At night, when you sleep:** A lot of arterial blood and little venous blood.

Fasting is beneficial because it allows the liver to regenerate and cells to become stronger, releasing energy from the inside out.

THE HEART-LIVER-SPLEEN-SMALL INTESTINE FUNCTIONAL UNIT

A rebalancing occurs between the heart, liver, spleen, and mesenteric system (small intestine). This constitutes a functional unit.

The liver is a volume regulator, capable of increasing or decreasing **blood volume**. It plays a crucial role in embryonic development and at birth.

Bone and tissue compressions, as well as vascular changes, are observed. It is crucial to treat these aspects at a fluid level at birth, in addition to the compressive level.

IMPORTANCE OF CONFLUENCES

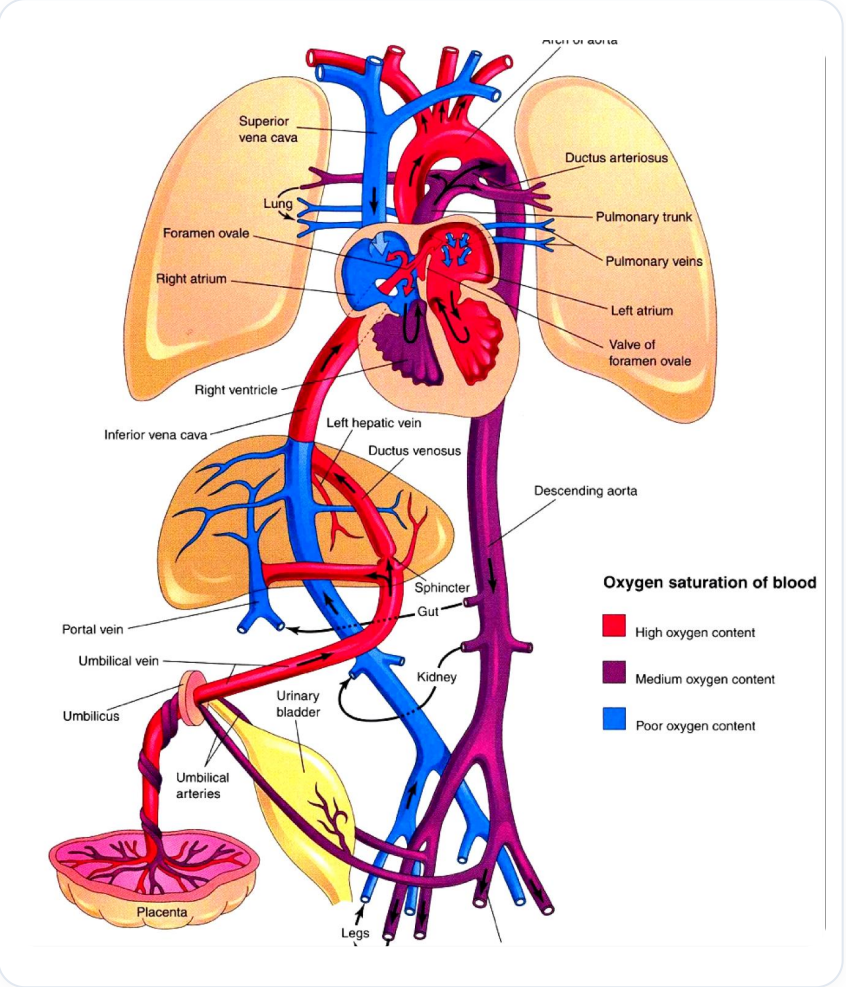


FIG. — Fetal circulation

Retrohepatic confluence is important for heart return. The **subhepatic confluence** and the **jugular confluence** represent two major keys.

It is essential not to overload the heart in terms of venous return. The heart must function correctly in systole, not in diastole. If the liver is systolic, it helps to resorb.

It is therefore necessary to release facial function to restore hepatic release. Do not forget the heart-liver-spleen connection.

VITELLINE VESICLE AND UMBILICAL CORD CONNECTION

The movement of the amniotic cavity shows the vitelline vesicle regressing and depositing on the umbilical cord. A loop will develop, integrating the intestinal loop inside.

In the umbilical cord, the spiralling vessels in the embryonic pedicle connect, with the remnant of the **yolk sac**. The intestinal tube and the embryonic pedicle join, integrating the allantois and the entire process with the large amniotic cavity.

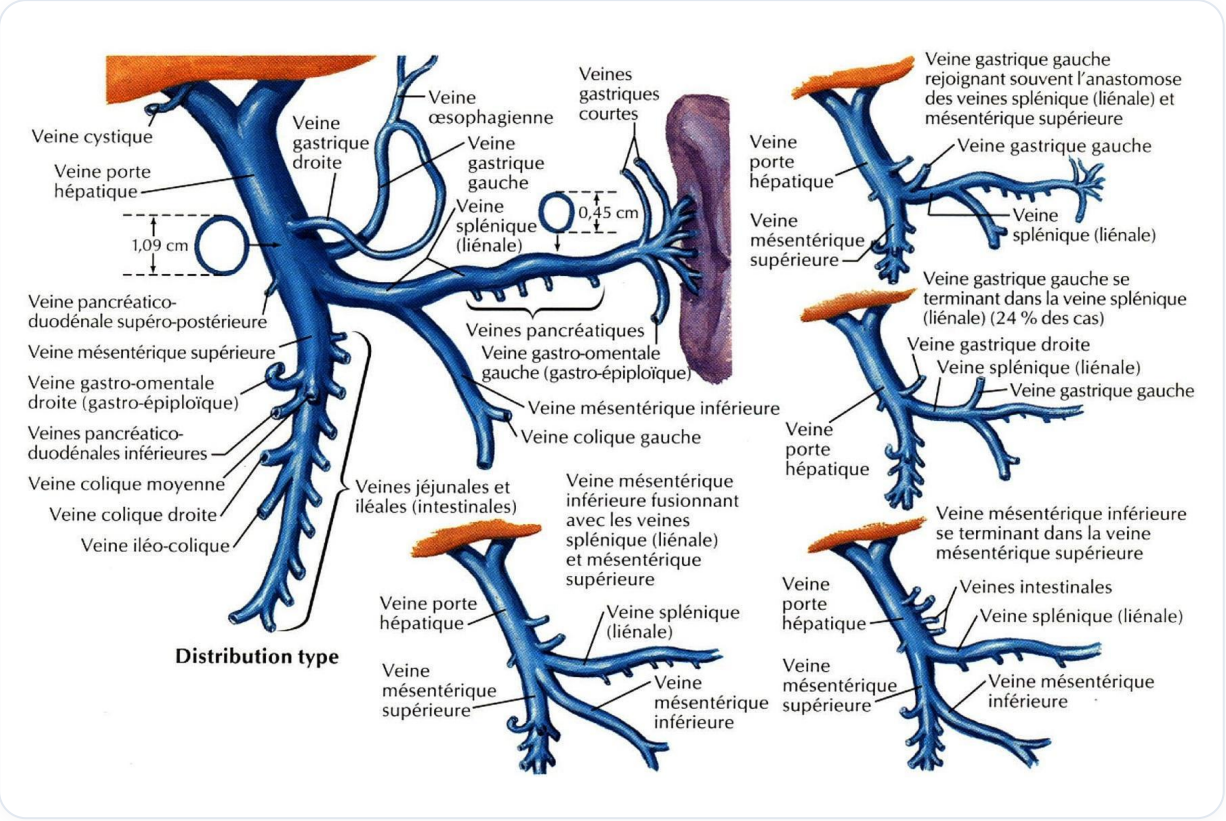


FIG. — Portal Vein Anatomy

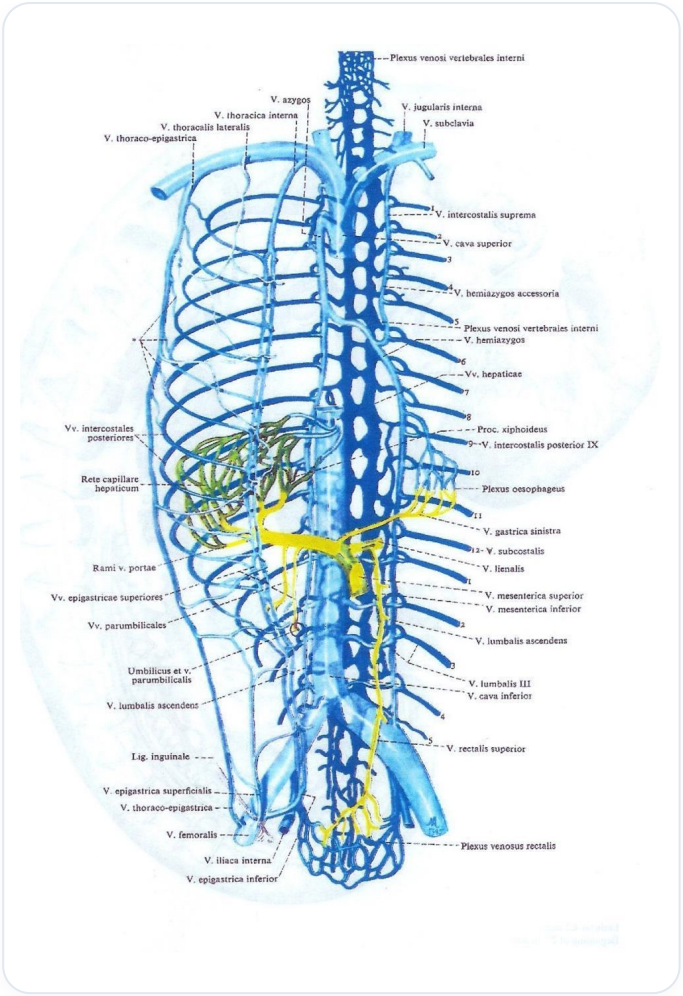


FIG. — Human venous system

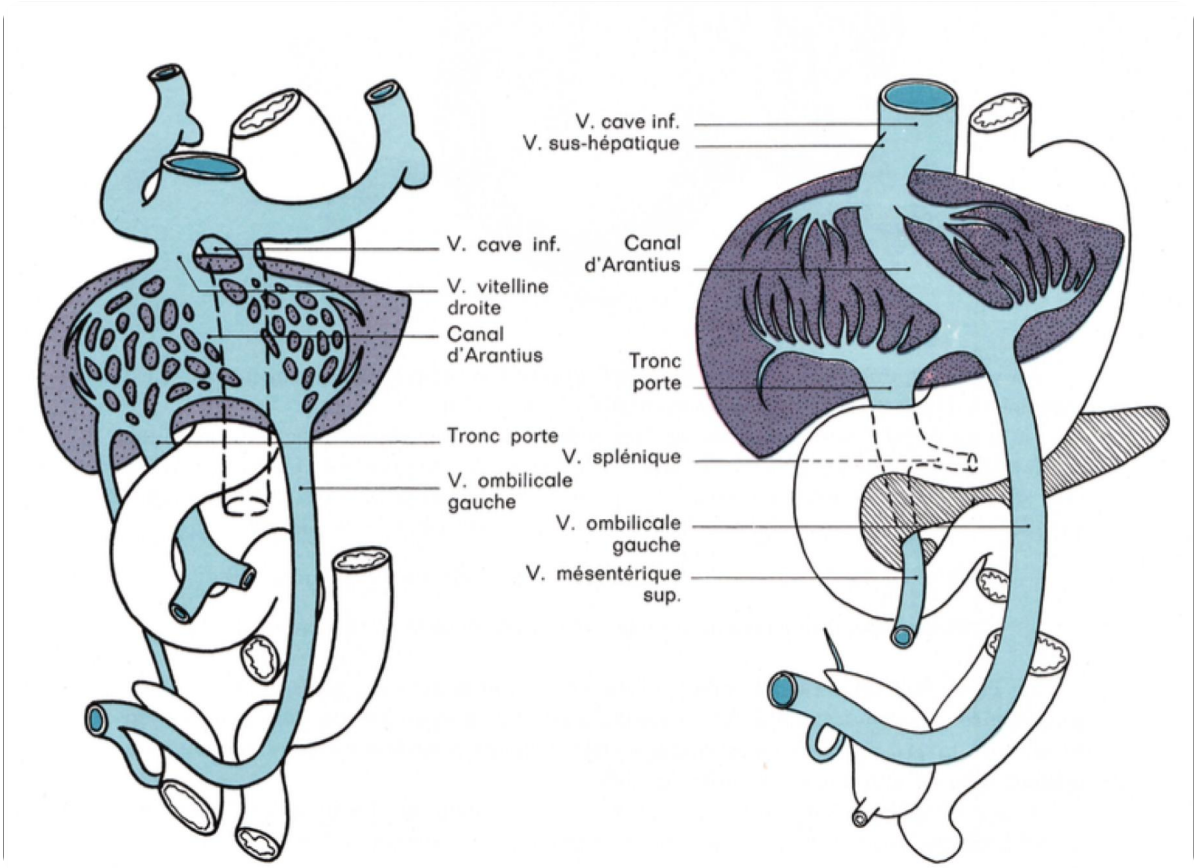


FIG. — Embryonic portal system

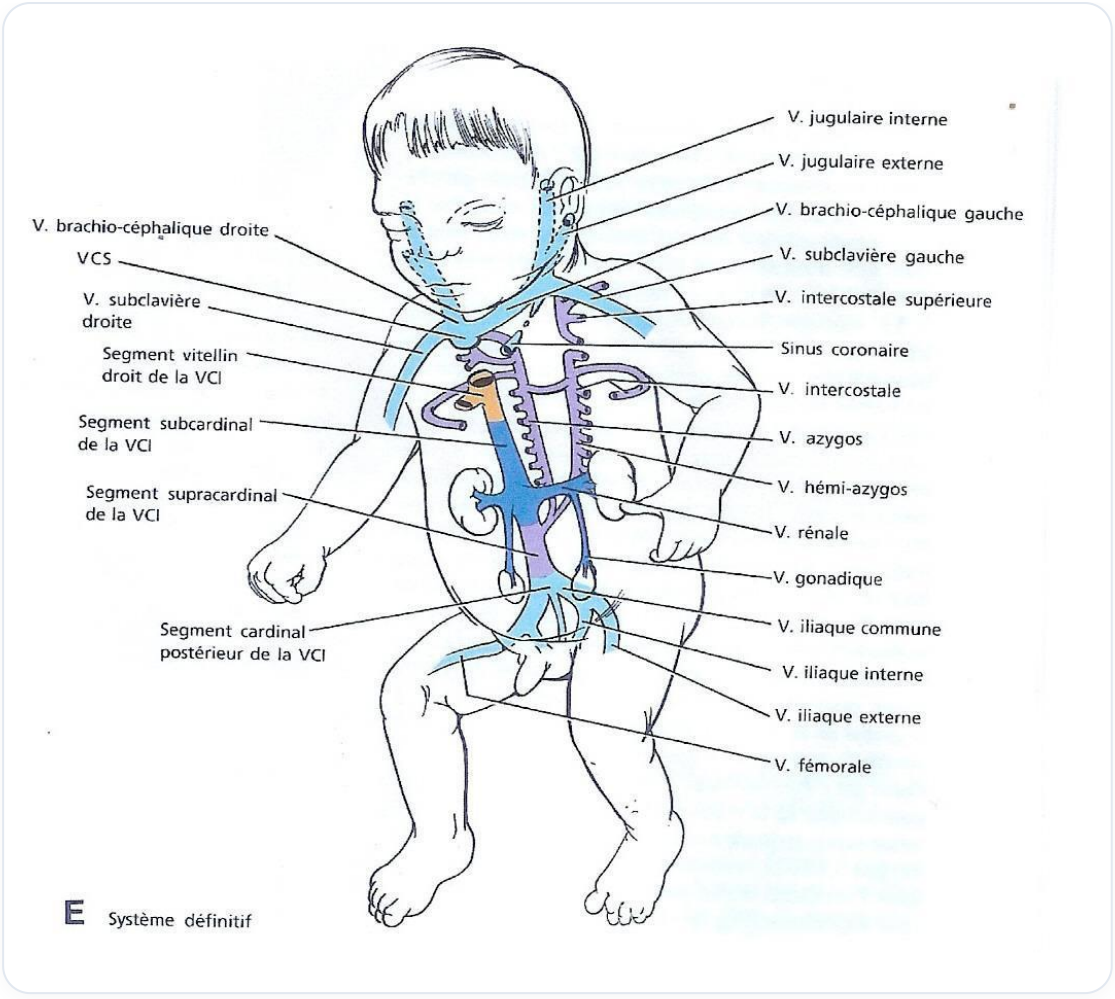


FIG. — Definitive venous system

17. VASCULAR SYSTEM : BRANCHIAL ARCHES

DURATION : 08:04 STUDY SHEET

COURSE SUMMARY

This video explores the development of the circulatory system from blood vascular islands, focusing on the evolution of the primitive aortic system and cardinal veins. The pharyngeal arches, which play a fundamental role in vascular and skeletal structure, are analysed in detail, including their derivatives and their influence on the skull and digestive tract. In addition to exploring vascular relationships, the video highlights the importance of balance in blood and lymphatic circulation, offering practical insights for osteopathy. The specific functions of the pharyngeal arches and their morphological contributions to structures such as the maxillary artery and the hyoid system are also discussed, providing a deep understanding necessary for effective therapeutic manipulations.

VASCULAR SYSTEM: BRANCHIAL ARCHES

We will explore the origin and development of the **circulatory system**, which begins within the blood vascular islands, located in the duplication of the **vitelline vesicle**. This process accelerates with the development of the **amniotic cavity** and the progressive integration of this vascular system.

THE PRIMITIVE AORTIC SYSTEM AND CARDINAL VEINS

We observe the appearance of a system called the **primitive aortic system**, including the future developing heart, as well as the **cardinal veins**. The latter, also called **mesonephric veins**, play a very important purifying role in connection with the first forming kidneys. The integration of these vessels into the aortic system becomes increasingly pronounced.

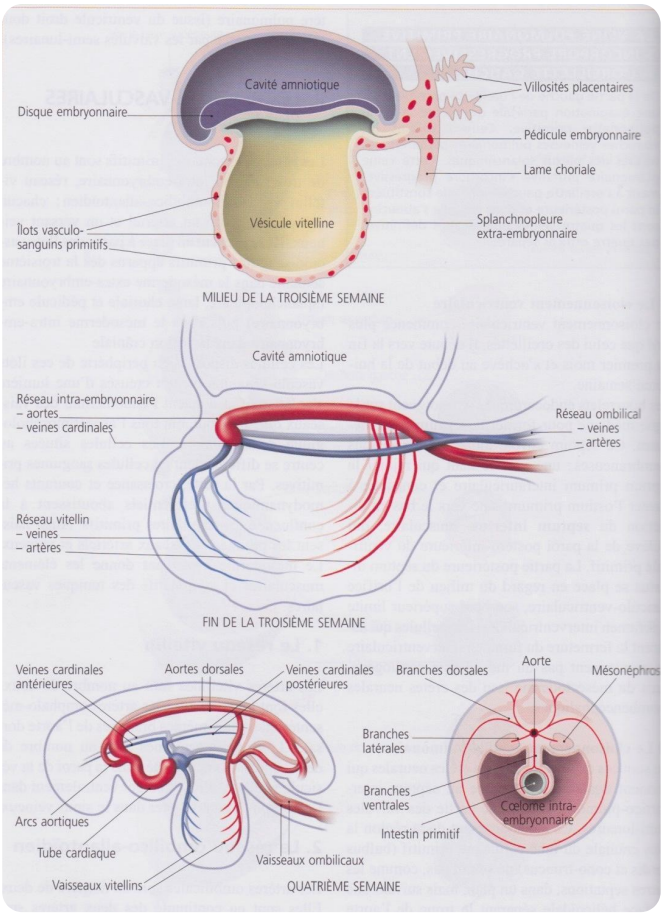


FIG. — Embryonic vascular development

IMPORTANCE OF VITELLINE AND UMBILICAL VEINS

The **vitelline veins** will initially form a plexus called the **vitelline artery plexus**. Its remnant will later become the **mesenteric system**. We also see the **umbilical arteries**, which will later join.

THE BRANCHIAL ARCHES

Our digestive tract is adjacent to structures called **branchial arches**. There are six of them, from pharyngeal pouch number 1 to 6. It is important to note that arch number 5 appears and disappears very quickly, which is why it is sometimes omitted in some descriptions, but it is indeed initially present.

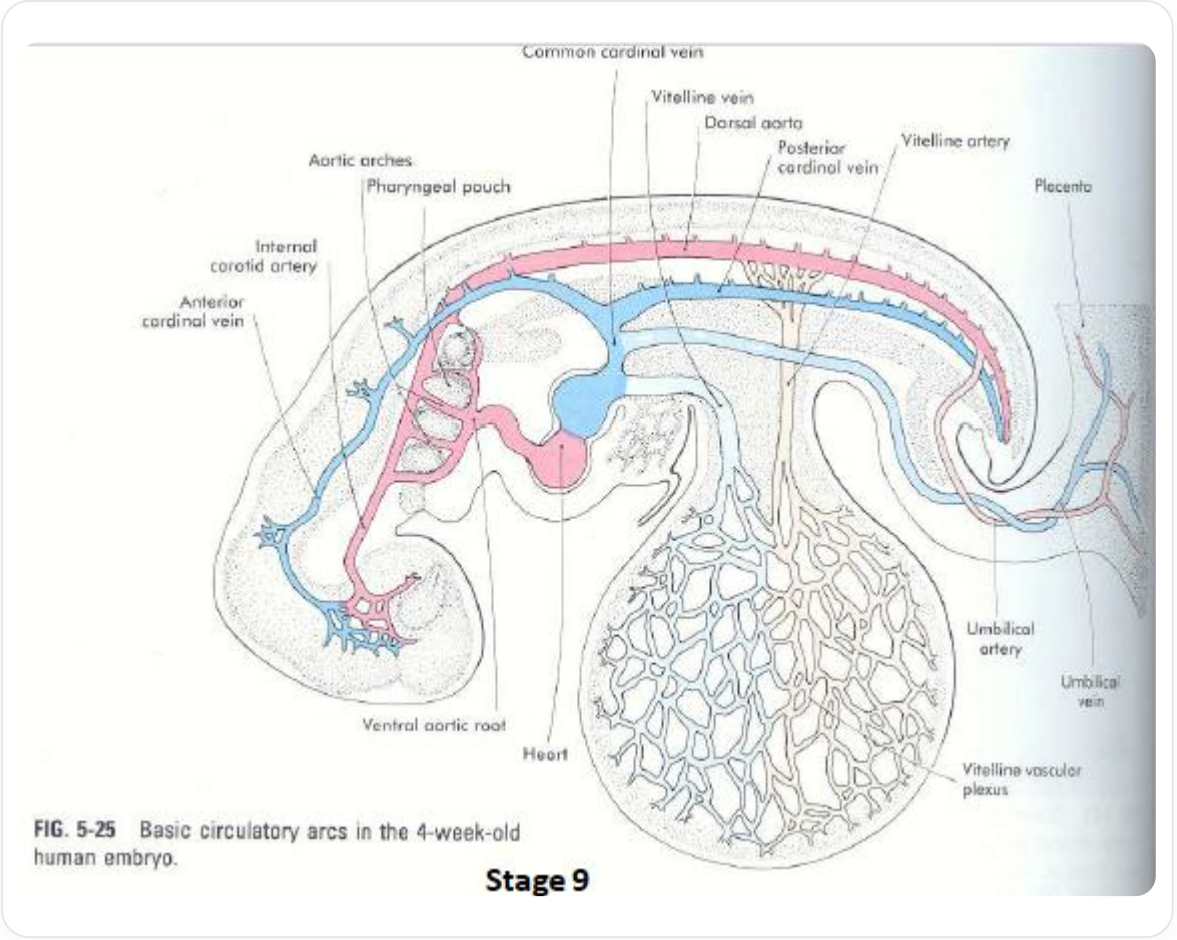


FIG. — Embryo circulation 4 weeks

VASCULAR EVOLUTION AND THE ROLE OF THE CRANIUM

It is fascinating to observe the transformations of this vascular system. One of the key ideas to remember is that the **aortic arch** derives from multiple segments. When working on the cranium, in relation to the brain and its vascular system, it is crucial to understand that the brain is primarily vascularised by the **carotid arteries** and the **posterior vertebral arteries**. These four vascular entry points represent essential areas to check and work on.

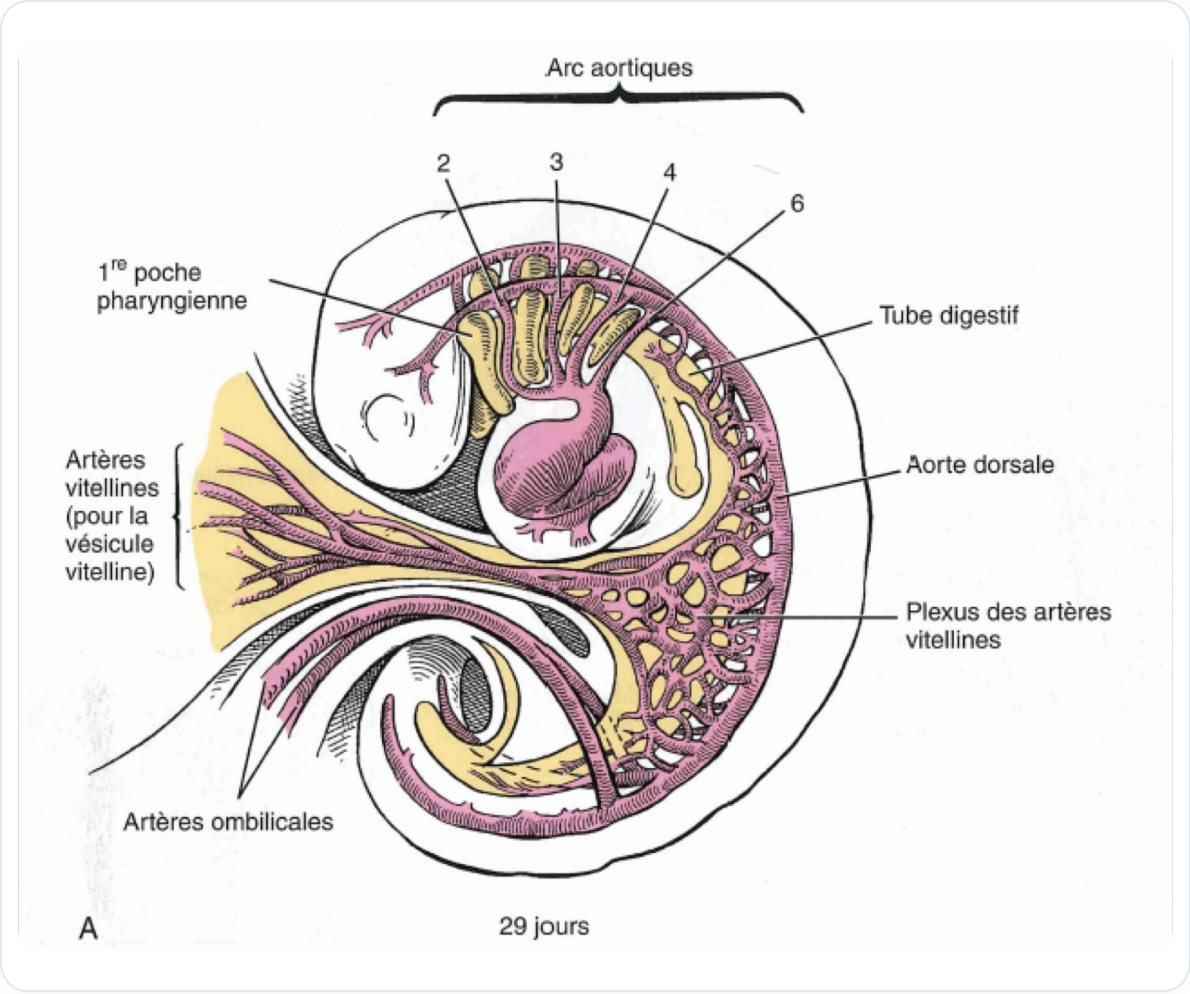


FIG. — Embryo 29 days

BALANCE AND ENERGY FLOWS

Circulation is a matter of **balance** at the capillary beds. This is where the exchange between the arterial system and the venous system occurs. These areas are places of energy transfer, where the balance between what enters and what leaves is paramount.

FORMATION OF THE LYMPHATIC SYSTEM

The arterial and venous system, combined with the resultant at the capillary beds, gives rise to the **lymphatic system**. This is a "transpiration" that creates a specific cleansing network. It is therefore essential to have areas for irrigation and for decongestion, in continuity with the **vertebral plexus**.

SPECIFIC ARTERIAL STRUCTURES OF THE BRANCHIAL ARCHES

Each branchial arch gives rise to a specific arterial structure.

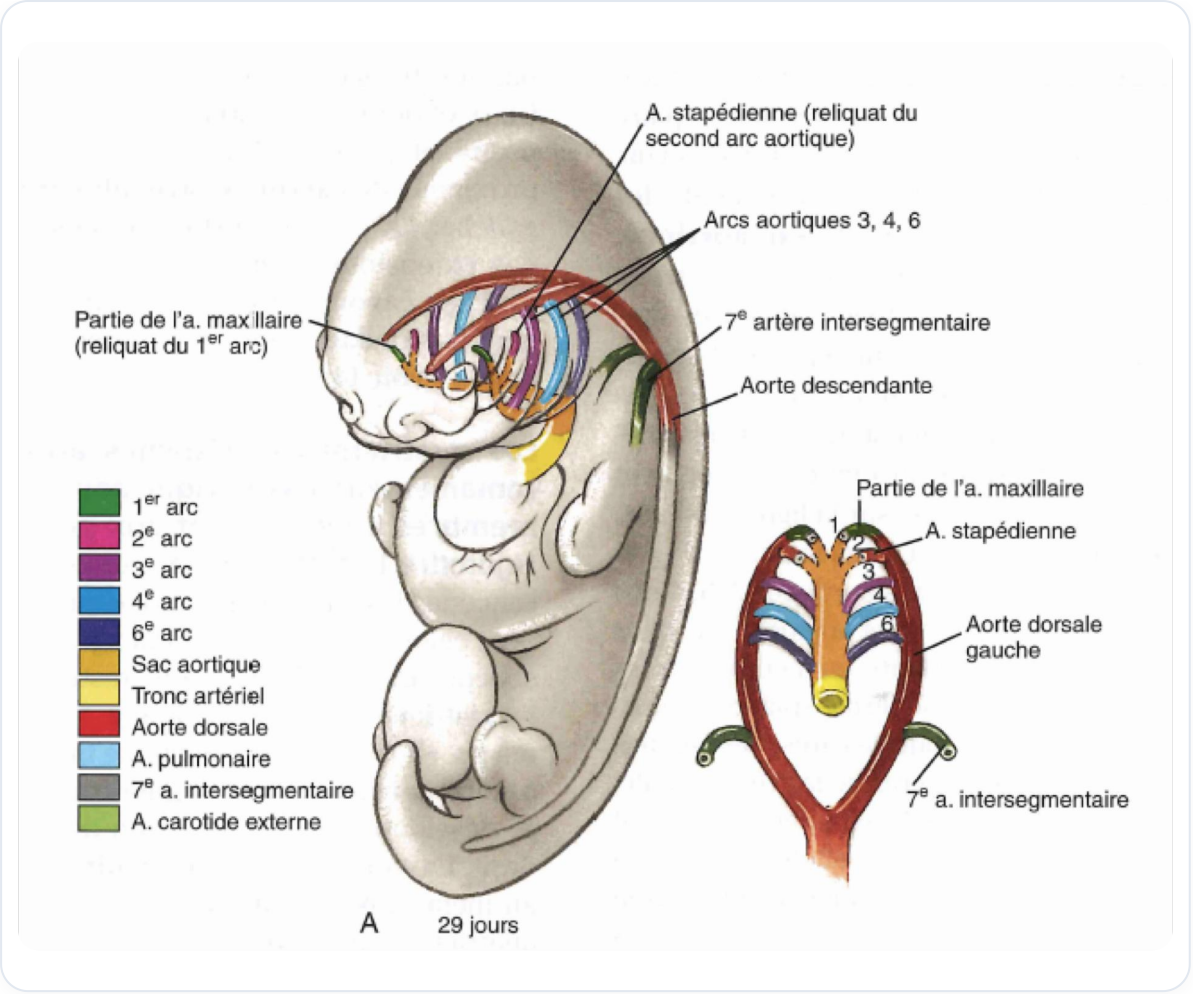


FIG. — Embryonic aortic arches

BRANCHIAL ARCH NUMBER 1

The remnant of branchial arch number 1 will form the **maxillary artery**. Skeletally, its cartilage will give rise to all derived structures up to the inner ears. For children, this area is particularly sensitive. They generally do not like their cranium to be touched, but are more receptive to manipulations of the viscerocranium.

CRANIUM-DIGESTIVE TRACT RELATIONSHIP

The brain forms the cranium, while the surrounding space influences the formation of the digestive tract. Remember the lines of force, or **dural sheaths** (anterior, middle, posterior), which orient towards the base of the cranium. All branchial arches also converge towards this base, which acts as an essential reference and support point (vascular, neurological, visceral). It is therefore important to balance these posterior, middle, and anterior parts.

DETAILS OF THE BRANCHIAL ARCHES AND THEIR DERIVATIVES

Here is an overview of the vascular and cartilaginous derivatives of the branchial arches:

- **Branchial arch number 1:**

* Maxillary artery.

* Meckel's cartilage, important for the stabilisation of the tympanic conduit.

* It is essential to reconnect this movement and rebalance it towards its fulcrum, the base of the cranium, allowing space to operate.

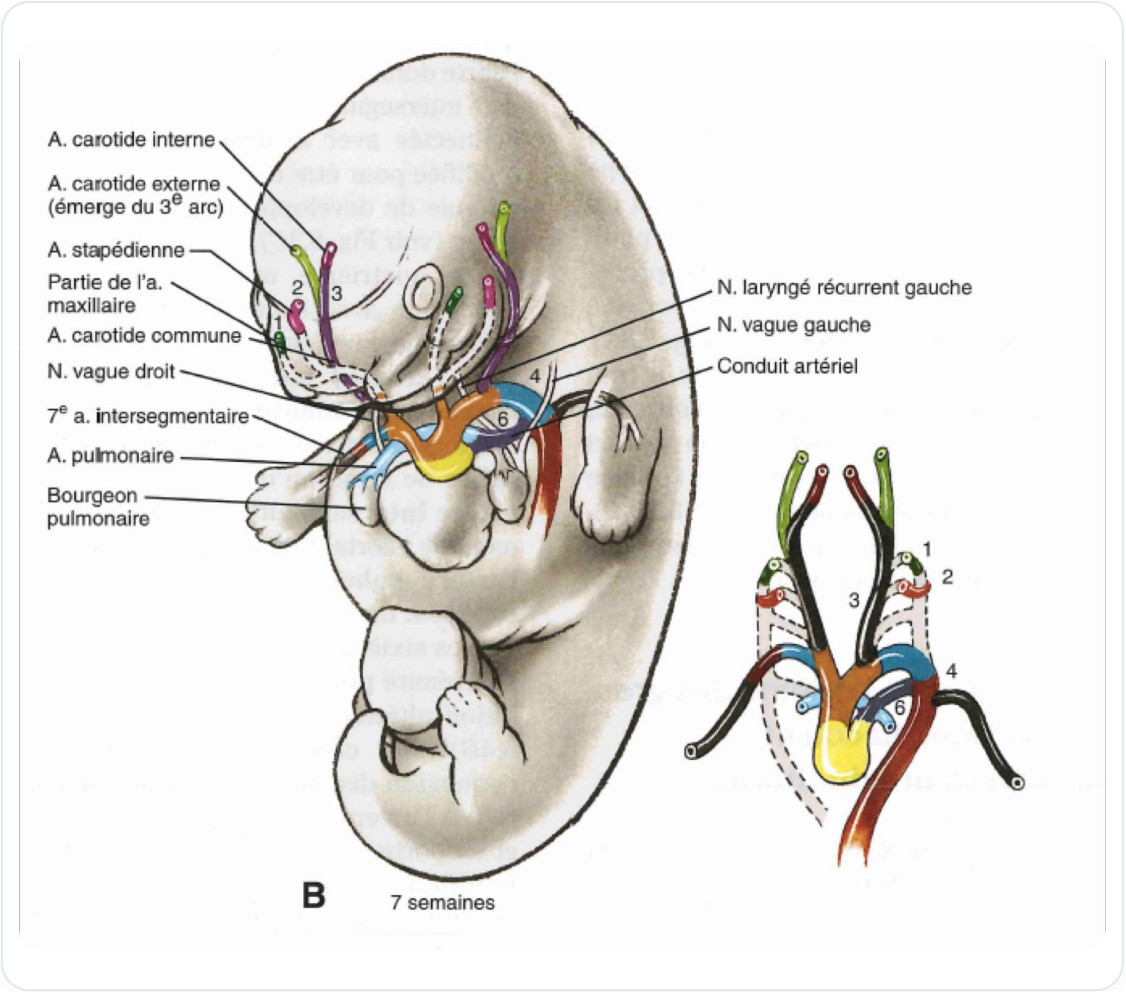


FIG. — Embryonic vascular system

- **Branchial arch number 2:**

* The hyoid system, from the stapedial and tympanic artery (which gives the stapedial artery), to the cartilages of the superior ring just at the thymic level.

* This system organises itself vascularly in a coherent manner.

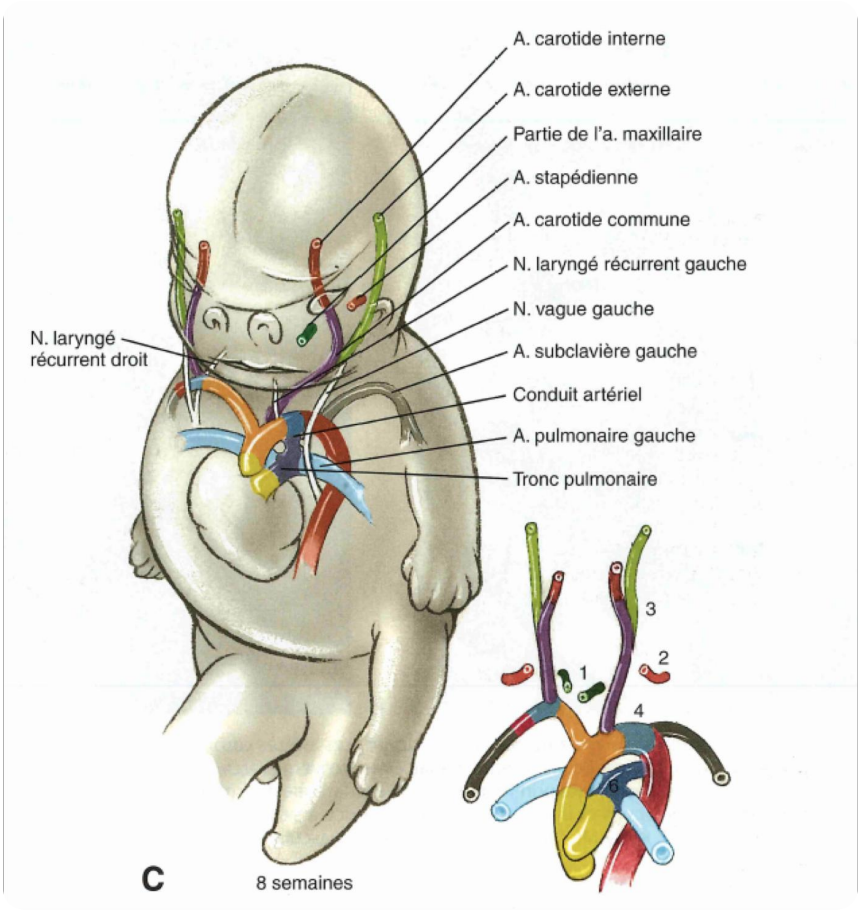


FIG. — Embryonic aortic arches

- **Branchial arch number 3:**

* The common carotid artery.

- **Branchial arch number 4:**

* The aorta and subclavian arteries.

- **Branchial arch number 5:**

* There is no persistent arterial derivative.

- **Branchial arch number 6:**

* The pulmonary artery.

Tableau 12 - 1. Dérivés des arcs pharyngiens et leurs tissus d'origine

ARC PHARYNGIEN	ARTERE DE L'ARC[a]	ELEMENTS SQUELETTIQUES	MUSCLES	NERF CRANIEN[b]
1	Branche terminale de l'artère maxillaire	Dérivés des cartilages (provenant de la crête neurale): du cartilage maxillaire: alisphénoïde, enclume cartilage mandibulaire (Meckel): marteau Dérivés par ossification directe du mésenchyme dermique de l'arc: maxillaire, zygomatique, portion squameuse de l'os temporal, mandibule	Muscles de la mastication (temporal, masséter et ptérygoïdiens), mylo-hyoïdien,ventre antérieur du digastrique, tenseur du tympan, tenseur du voile du palais (provenant du somitomère crânien 4)	Divisions mandibulaire et maxillaire du nerf trijumeau (V)
2	Artère stapédienne (de l'embryon), artère carotico-tympanique (de l'adulte)	Etrier, processus styloïde, ligament stylo-hyoïdien, petites cornes et bord supérieur de l'os hyoïde(dérivés du cartilage du second arc [Reichert]; provenant de la crête neurale)	Muscles de la mimique (orbiculaire de l'oeil, orbiculaire de la bouche, risorius, platysma, auriculaire, fronto-occipital et buccinateur), ventre postérieur du digastrique, stylo-hyoïdien, de l'étrier (provenant du somitomère crânien 6)	Nerf facial (VII)
3	Artère carotide commune, racine de l'artère carotide interne	Bord inférieur et grandes cornes de l'os hyoïde (provenant du cartilage du troisième arc;) issu de la crête neurale)	Stylo-pharyngien (provenant du somitomère crânien 7)	Nerf glosso-pharyngien (IX)
4	Arc de l'aorte, artère subclavière droite; bourgeons initiaux des artères pulmonaires	Cartilages du larynx (dérivés du cartilage du quatrième arc provenant du mésoblaste de la lame latérale)	Constricteurs du du pharynx, crico-thyroïdien, élévateur du voile du palais (issus des somites occipitaux 2 à 4)	Branche laryngée supérieure du nerf vague (X)
6	Conduit artériel, bourgeons des artères pulmonaires définitives	Cartilages du larynx (dérivés du cartilage du sixième arc; issus du mésoblaste de la lame latérale)	Muscles intrinsèques du larynx (issus des somites occipitaux 1 et 2)	Branche laryngée récurrente du nerf vague (X)

FIG. — Pharyngeal arch derivatives

All these systems form a functional unit, including, for example, the **laryngeal cartilages**, the **intracellular muscles**, and the **recurrent branch of the vagus nerve**. We work on these units by offering points of support, allowing the lines of motility to develop and the neutrals to meet, letting space contain the primary respiration.

THE FACE AND THE NEURAL CREST

The face has enormous **vascularisation**. Your facial bones, your skin, your facial cartilages are all derived from the **neural crest**, from the **mesoectoderm**. In addition to the points of support, we observe what is called the **neural crest stream**: prosencephalon, mesencephalon, rhombencephalon, followed by rhombomeres 1 to 7. This stream, this invasion of the neural crest into the mesoectoderm, is crucial developmental information.

ORGANISATION AROUND THE BASE OF THE CRANIUM

All the organisation of the branchial arches and the lines of force will tend towards this fundamental point of support which is the **base of the cranium**, corresponding to the tip of the **notochord**.

ROLES OF THE CIRCULATORY SYSTEMS

- **Arterial system:** It generates movement and ensures a homeostatic balance of volume and pressure.
- **Venous system:** It is a level of exchange, more osmotic.
- **Lymph:** It is responsible for elimination, particularly toxic elimination, and plays a role in immunity.

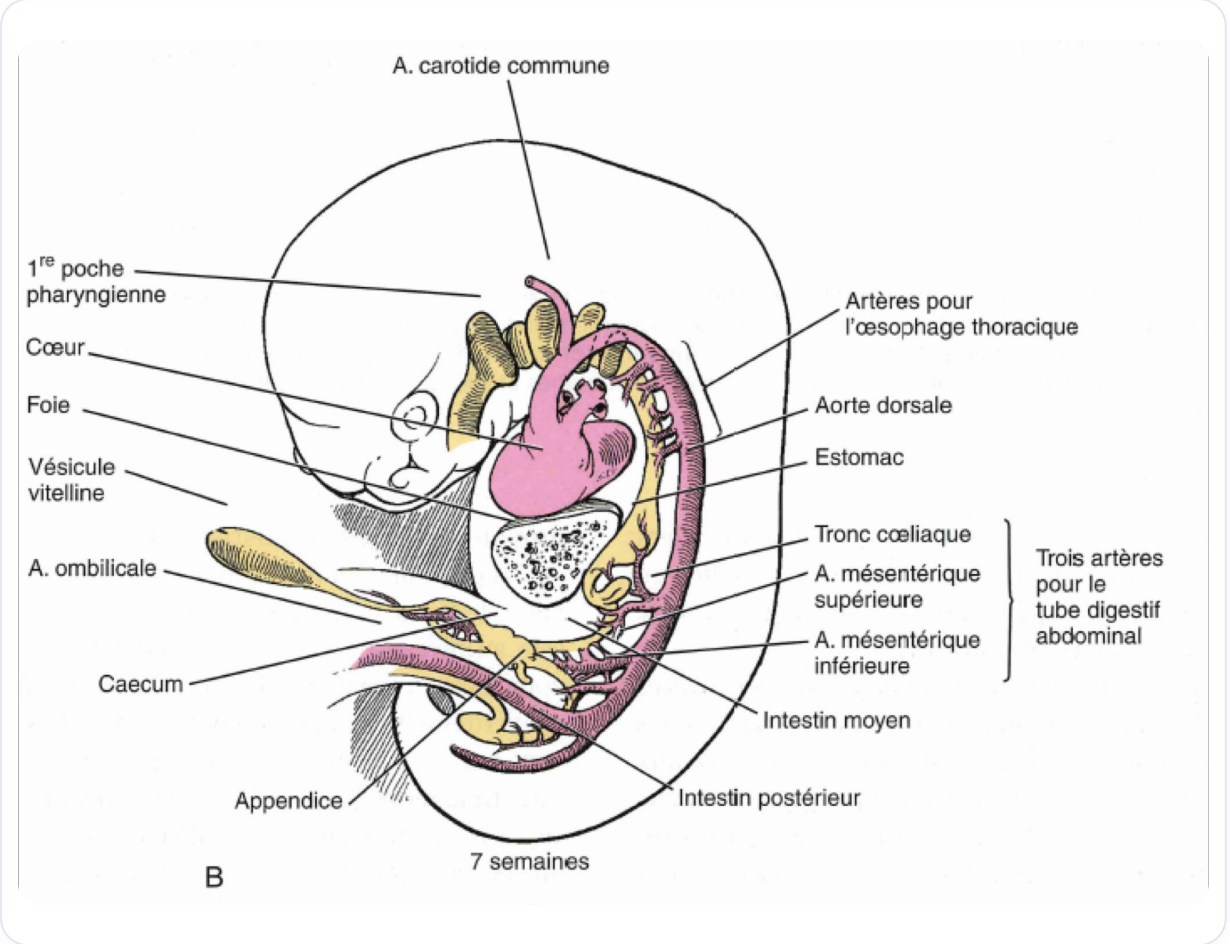


FIG. — Embryo 7 weeks, digestive system

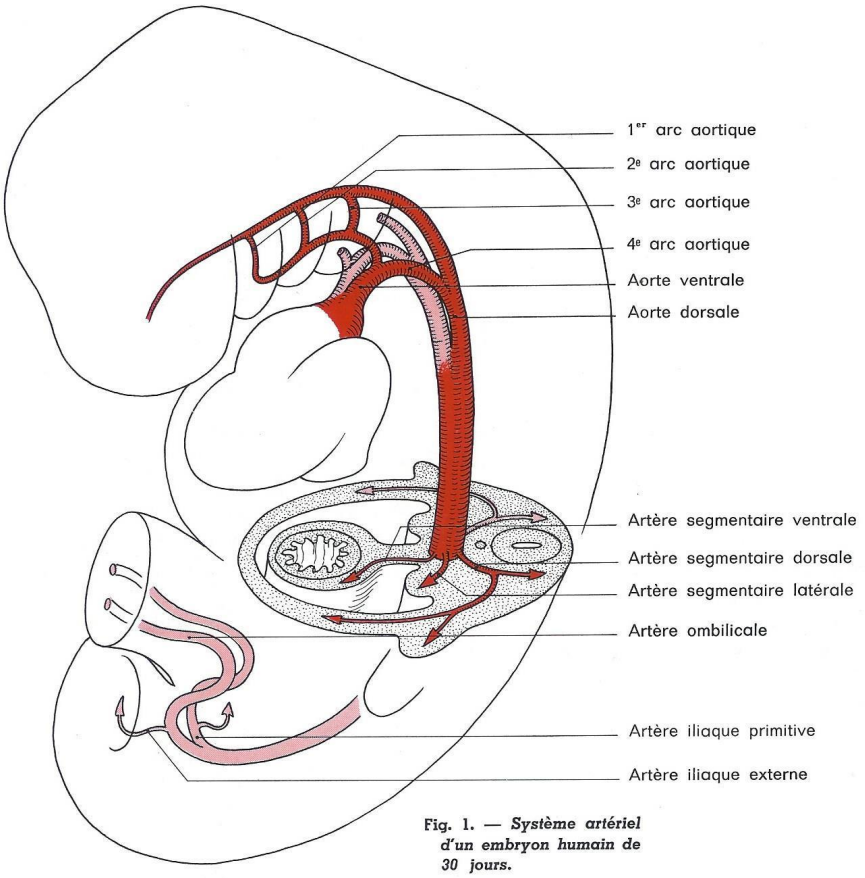


Fig. 1. — *Système artériel d'un embryon humain de 30 jours.*

FIG. — Embryonic arterial system

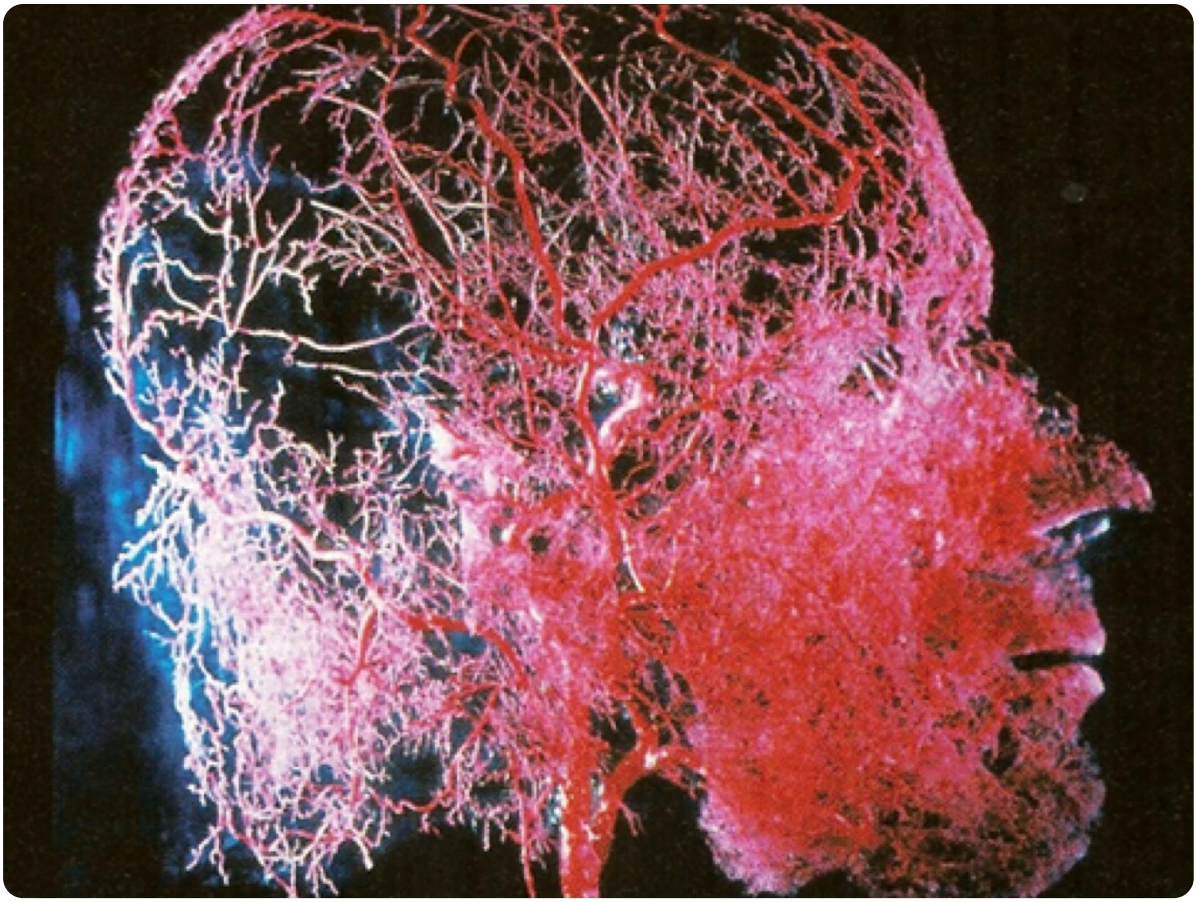


FIG. — Human head vascularization

The **cerebrospinal fluid** (CSF) is the controller. It ensures the specific molecular and hormonal balance of the body. Its return occurs according to the law of the venous system, with the **choroid plexuses** for its production and the **Pacchionian granulations** for its resorption. Lymph, let us remember, also returns to this system, participating in the overall balance.

THE CAPILLARY BEDS: MEETING AND TRANSFORMATION POINTS

The great plan of balance lies in the **capillary beds**, the meeting place between the arterial and venous systems. At this level, there is a transformation of energy, a rebalancing of rhythmic zones, and an adjustment of the **vasoregulation** of the different planes. These capillary beds are the "guardians of homeostatic balance" on the fluid plane, corresponding to areas of copula, specific rhythmic zones, essential for bodily balance.

18. THE VENOUS SYSTEM : NOTES

DURATION : 07:29 STUDY SHEET

COURSE SUMMARY

This video delves into the crucial role of the venous system, addressing topics such as the vertebral plexuses and their connection to the nervous system, as well as the importance of effective venous drainage for bodily health. You will learn to identify and release venous compressions that can lead to chronic pain, such as neck pain or pelvic disorders. Furthermore, it explains how abdominal pressure and diaphragmatic function impact circulatory balance, with direct implications for emotional and physical well-being. Practical techniques for improving lymphatic and venous drainage will also be described, allowing you to apply this knowledge in your therapeutic practice.

THE VENOUS SYSTEM

THE VERTEBRAL PLEXUSES AND THE NERVOUS SYSTEM

At a neurological level, the **vertebral plexuses** are composed of several essential parts:

- The **deep medullary system**.
- The **internal vertebral venous system**, located between the dura mater and the dural sinus.
- The **external vertebral venous plexus**.
- The **diploic systems** and the **emissary veins**, more specific to the skull.

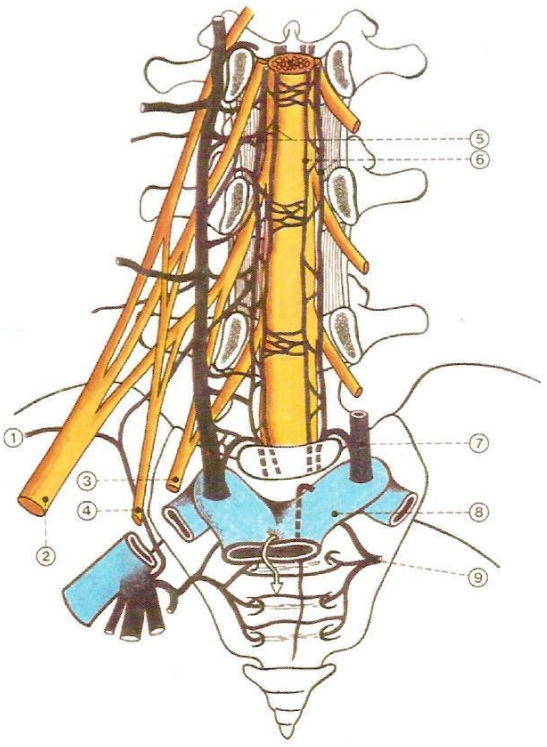


Figure 12. Rapports des plexus vertébraux. Référence (36), p 116.

FIG. — Vertebral plexuses

There is also a **medullary plexus** and an **internal plexus**, notably the **pterygoid sinus**, which can be a site of congestion. It is crucial to learn how to release this system, which acts as a safety valve in case of engorgement.

In cases of **chronic cervicalgia** or **acute torticollis**, these are often venous compressions. Inside a nerve (perineurium, endoneurium, epineurium), small vessels can become engorged with blood and compress the neuronal system. Our role is then to **decongest** these areas.

Discopathy occurs when **venous stasis** releases **cytokines** and creates acidity that destroys the system. Rebalancing pressures becomes paramount. The **dural sinus** is continuous with the vertebral plexus, and the mobility of the **sacrum** can influence headaches.

VENOUS DRAINAGE AND CRANIAL SINUSES

Venous blood flows towards the **cranial sinuses**: the **straight sinus**, the **superior sagittal sinus**, the **inferior sagittal sinus**, the **straight sinus**, and the **transverse sinus**.

These systems are connected to the **fourth ventricle**, particularly the **foramen of Magendie**, which is an opening involved in the pulsation of the **cerebrospinal fluid (CSF)** system. By releasing the vascular system from the periphery to the deep, we reopen the vascular fields and

redrain the entire system.

This drainage occurs from the **carotids**, the **vertebral arteries**, and all the sinuses via the vertebral plexus, directly involving a free sacrum.

THE FUNCTIONAL UNIT OF PELVIC VEINS AND LOWER LIMBS

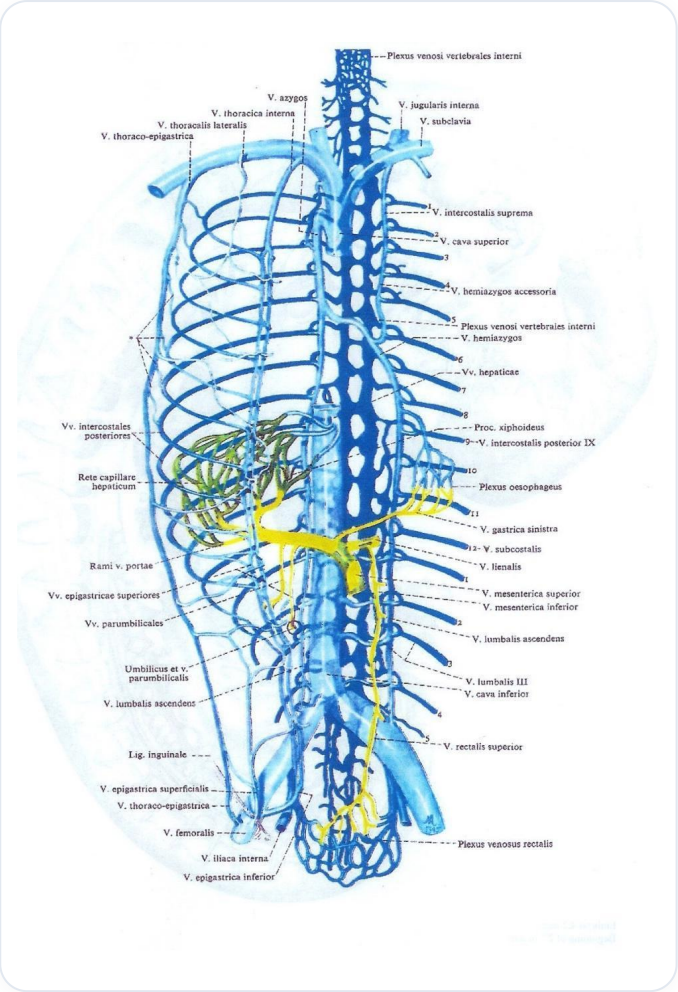


FIG. — Human venous system

The **pelvic veins** and those of the **lower limbs** constitute a functional unit. An imbalance in the lower limbs (feet, knees, pelvis) has repercussions on the **urogenital system**.

For example, walking activates drainage thanks to **Lejars' sole** and the **navicular-cuboid articulation**, which functions like a diaphragm. In cases of small pelvis disorders in women, it is essential to check and release the foot to restore this return force, otherwise stasis may occur.

CONGESTIONS AND COLLATERAL PATHWAYS

Figure 11. ...parmi beaucoup d'autres: les variances simples, doubles et plexiformes de la veine iliaque interne. Référence (37).

FIG. — Internal iliac vein

Lower limb pain can be related to **nerve compression** that is not of urogenital origin. Certain **circulatory collateral pathways** are the reappearance of a foetal state that regresses under physiological conditions.

Uterine retroversions can lead to frequent congestions. The paucity of **venous valves** promotes bidirectional circulation, allowing blood to change direction in case of congestion, but can also create areas of local stasis (prostate, etc.).

Abdominal pressure influences the path of venous blood. An increase in this pressure can direct blood towards the vertebral plexuses, leading to vertebral congestions and **low back pain** of digestive origin. It is often necessary to restore pressures to relieve the liver and restore dynamic balance.

IMPACT OF VENOUS CIRCULATORY DISORDERS

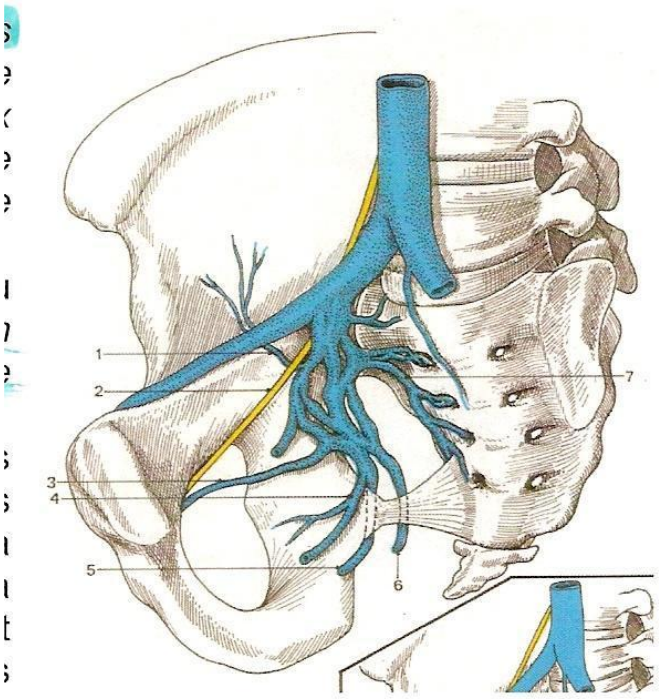


Figure 10. La veine iliaque interne: une disposition possible, mais ... Référence (36,) p 108

FIG. — Posterior view pelvic vascularization

Disorders of venous circulation can be the cause of multiple symptoms:

- Persistent deep **dyspareunia**.
- **Dysmenorrhoea**.
- **Constipation**.
- **Tenesmus**.
- General impairment (asthenia, depressive tendencies, emotional irritability).

THE DIAPHRAGMATIC PUMP AND LYMPHATIC DRAINAGE

The great key lies in the diaphragmatic pump, which acts on the **vascular**, **arterial**, and **lymphatic** systems simultaneously. The development of the vascular system generates blood pressure in the capillaries, creating exudates in the intercellular network. This network forms new pathways, particularly longitudinal ones.

The **cisterna chyli** (or Pécquet's cistern) is a very important network, responsible for a large amount of resorption at the level of the left supraclavicular fossa. The **psoas muscles**, thanks to the **perirenal fascia**, drain up to 1.5 litres of lymph per day through their movements.

The psoas is a thick organ that can retain toxins. **Toxemic overload** can block it, preventing the release of toxins from the peritoneum into the venous and lymphatic systems. A small muscle in the neck, the **levator scapulae muscle**, has a similar structure to the psoas and also participates in the uptake of toxicity.

THE SEVEN DIAPHRAGMS

We consider that there are **seven crucial diaphragms** for the body:

1. The feet

2. The knees

3. The small pelvis

4. The thoraco-abdominal diaphragm

5. The cerebellum and its balance with the base of the skull

6. The parietals

7. The most important drainage space for the CSF system, above the parietals.

The parietals, like **solar panels**, must breathe and capture light, just as the **radii** are "solar bones". The integration of these systems into the brain, in connection with the **vitelline** and **umbilical** systems and the **apical impulse**, allows the brain to relax.

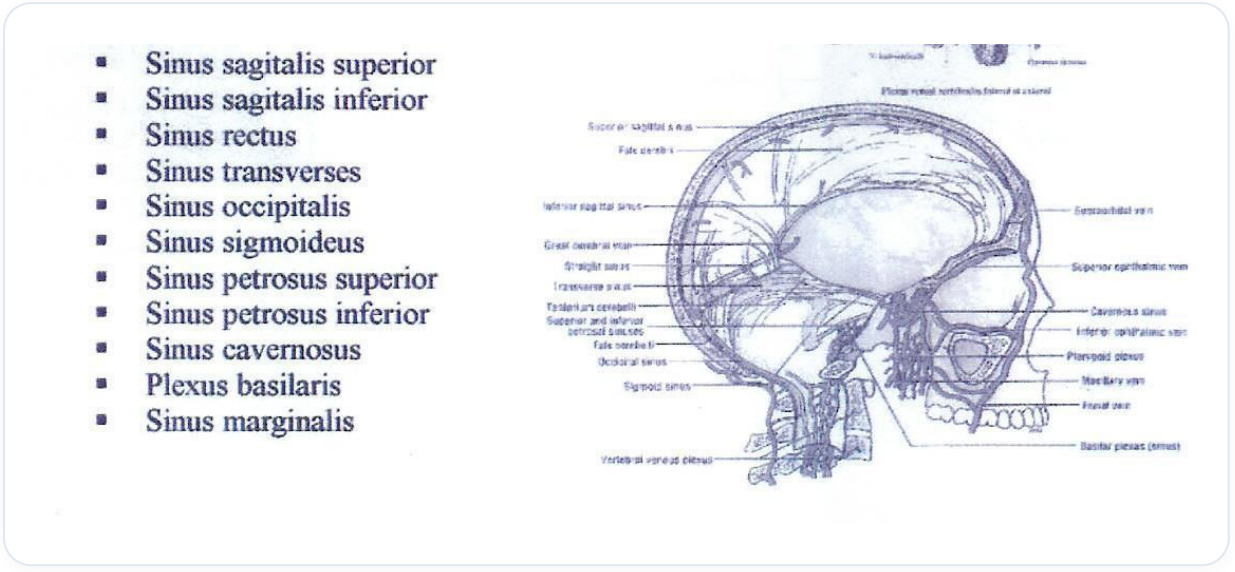


FIG. — Cranial venous sinuses

19. SUMMARY OF VASCULAR STRUCTURING BY THE AMNIOTIC CAVITY

DURATION : 05:54 STUDY SHEET

COURSE SUMMARY

This video deeply explores vascular structuring by the amniotic cavity, highlighting its integration with the vitelline system and its role in embryonic development. Key concepts discussed include the movement of resorption towards the centre of the embryo, arterial integration, and the importance of the amniotic cavity in redefining internal spaces. Essential movements such as cephalisation, embryonic coiling, and fluid repatriation are discussed, while emphasising the impact of the notochordal system and the neural tube. This knowledge is crucial for practitioners seeking to recognise the internal dynamics that influence embryonic health and overall well-being.

SUMMARY OF VASCULAR STRUCTURING BY THE AMNIOTIC CAVITY

THE ORIGIN AND INTEGRATION OF THE VASCULAR SYSTEM

The **vascular system**, at its periphery, is strongly integrated into the **vitelline system**. The **amniotic cavity**, initially the same size as the vitelline vesicle, plays a fundamental role.

The embryo elongates and grows considerably. The importance of this amniotic cavity lies in its ability to completely integrate the vitelline system. This movement is crucial and constitutes the central point of our study.

THE RESORPTION MOVEMENT AND ZONE B

In this movement, the entire vitelline system resorbs towards the centre of the embryo, reintegrating the whole. This amniotic cavity, which can be referred to as **Zone B**, in relation to the aortic system, draws everything towards the centre.

This process occurs on both a **transverse abdominal** and **thoracic** plane.

With the growth and establishment of the heart, as well as the significant development of the brain, the amniotic cavity takes over. It completely encompasses the entire space of the **external coelom**. Only a part of this internal coelom will be integrated into an internal coelom. The more the amniotic cavity grows, the more the system integrates.

ARTERIAL INTEGRATION AND EMBRYONIC COILING

Two key elements summarise the integration of the arterial system, from the periphery to the centre: **cephalisation** and **embryonic coiling**. This includes cardialisation, diaphragmatisation, and hepatisation.

The amniotic cavity acts as an integrating movement, redefining spaces. In this dynamic, both horizontally and vertically, Zone B is constructed. The vitelline vesicle, containing primitive information at the venous level, interacts with the exterior to reintegrate the body and the venous system.

THE ROLE OF FLUIDS AND THE VITELLINE VESICLE

The secret lies in the **repatriation of fluids** towards the centre, with the fascial system playing a predominant role in this release.

The vitelline vesicle is a great source of information. The integration movement includes:

- The transformation of the external coelom into an internal coelom.
- The closure of the neural tube.
- Integration and movement towards the centre.
- Construction, orchestrated by the cavity.

RESTORING THE CAVITY AND ZONE B

The objective is to restore this cavity and Zone B to reintegrate the internal circulation.

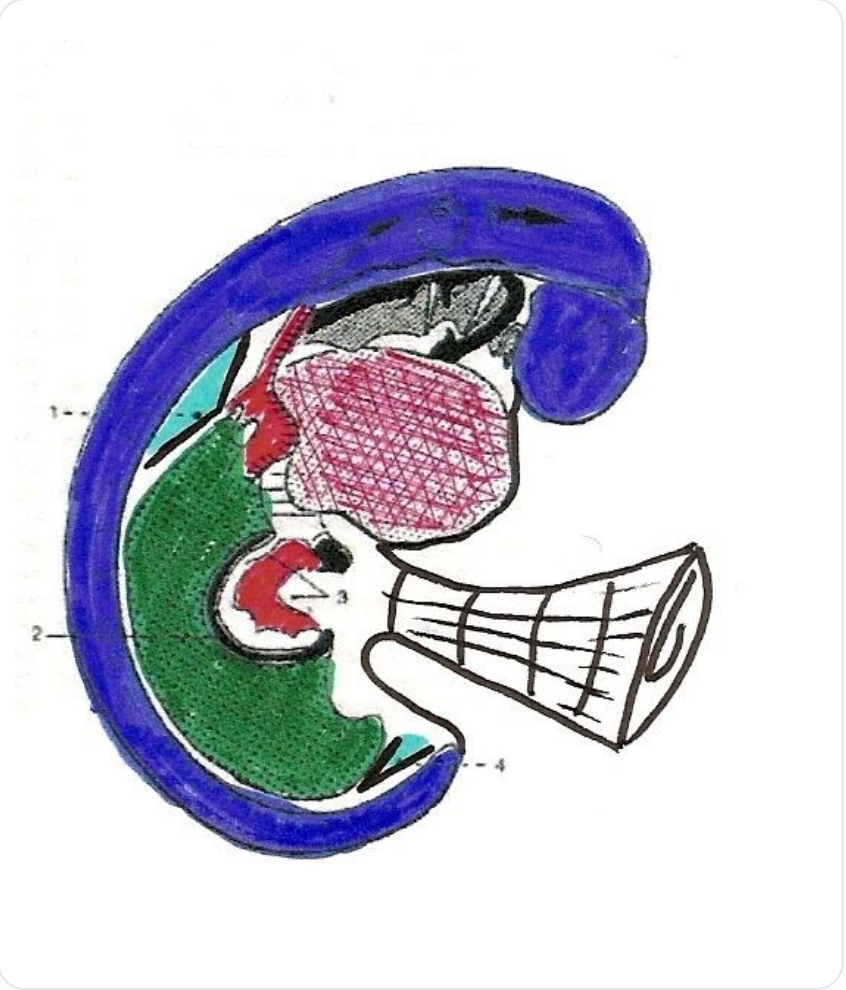


FIG. — This image appears to be an overview, so inserting it after a general contextualizing sentence about the reintegration of the circulatory system seems appropriate

Visualising the diagram as a person lying down:

- In blue, integration into a neural tube.
- The rest forming the skin, an **ectodermal** layer.

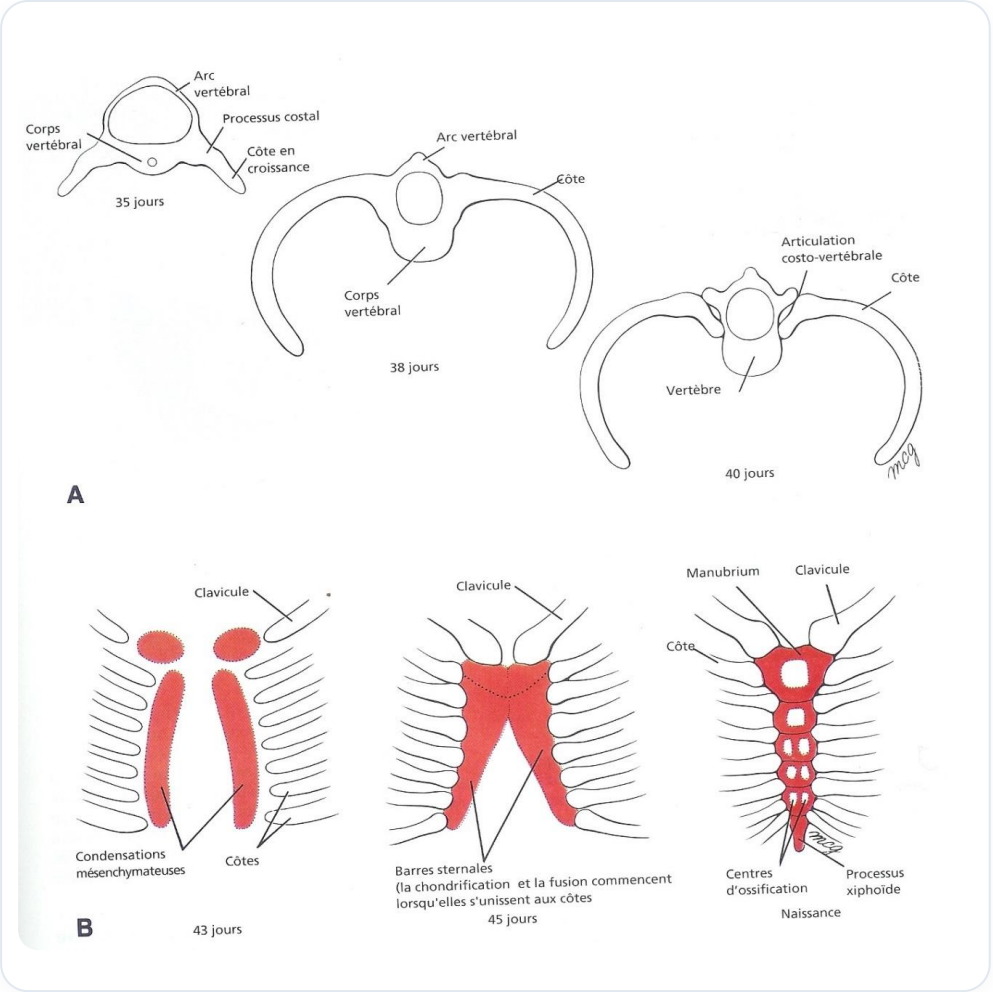


FIG. — The image seems to detail the integration of the nervous system or an aspect of embryonic structuring

Simultaneously, the integration of the vitelline vesicle contributes to forming part of the digestive system, while the entire venous system integrates. The movement to follow is that of the growing amniotic cavity, leading to a fusion to form the longitudinal part, the digestive tube.

The integration of the external coelom into a future internal coelom is visualised in yellow, where everything converges on the midline.

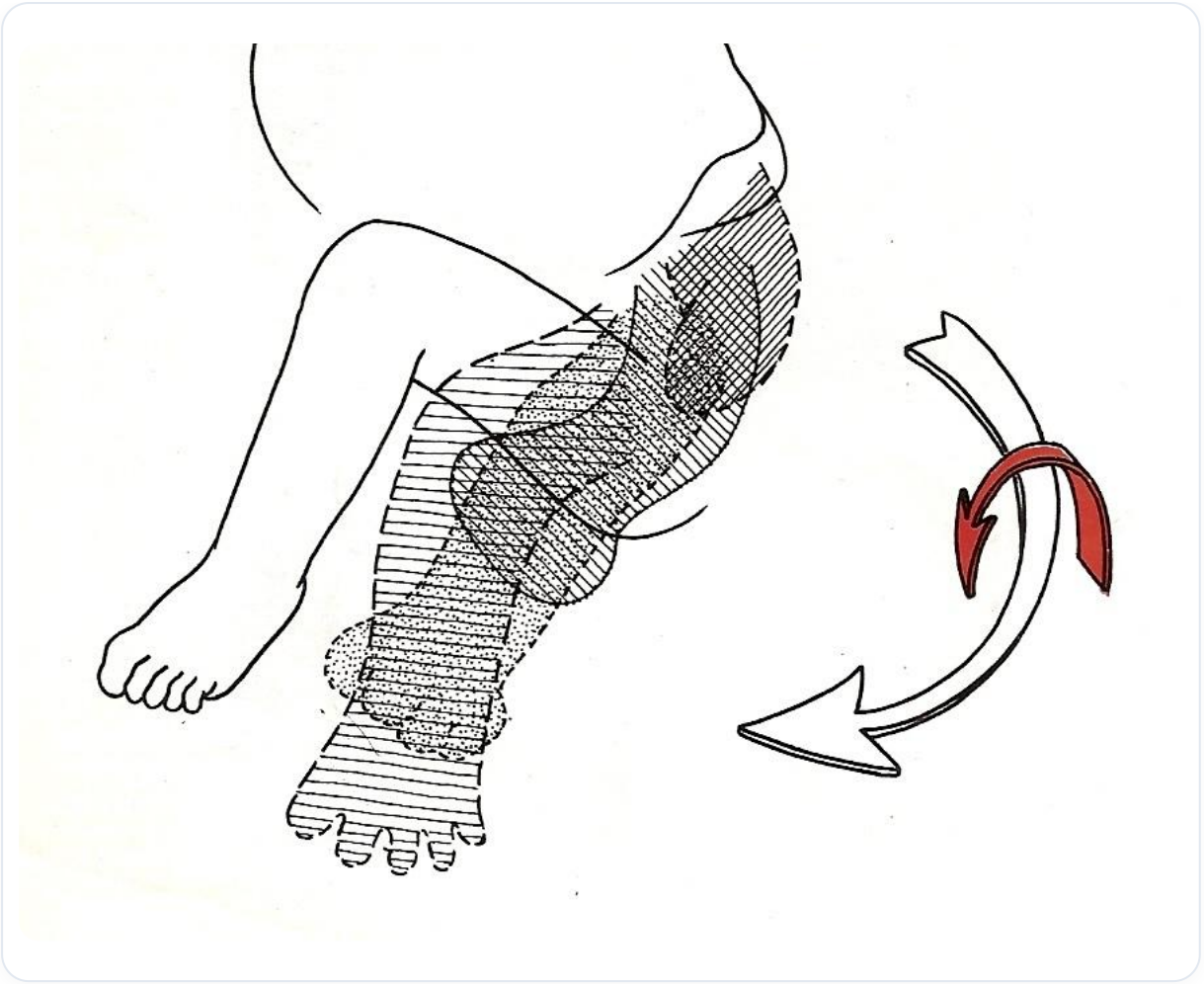


FIG. — Probably a stage of digestive tract formation or a longitudinal structure

THE GREAT ENGINE: NOTOCHORDAL SYSTEM AND NEURAL TUBE

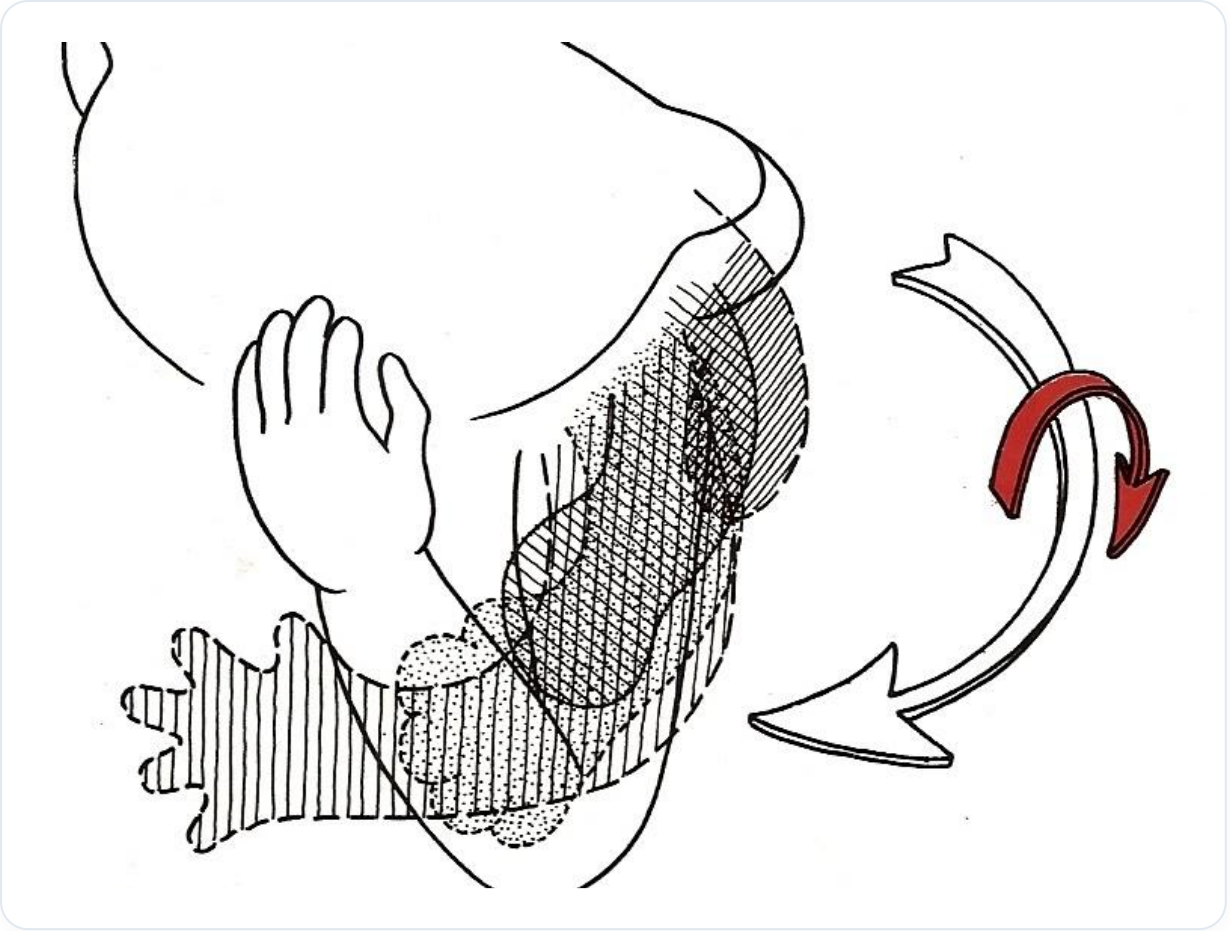


FIG. — Probably the convergence of structures towards the midline, consistent with the sentence

The main engine of this process is the **notochordal system**, which influences the neural tube and organises the primitive aortic vascular system. It redirects information with the progressive integration of the entire transverse system, from the periphery to the centre.

We will follow this movement step by step. It is essential to "allow" the primary respiration to unfold.

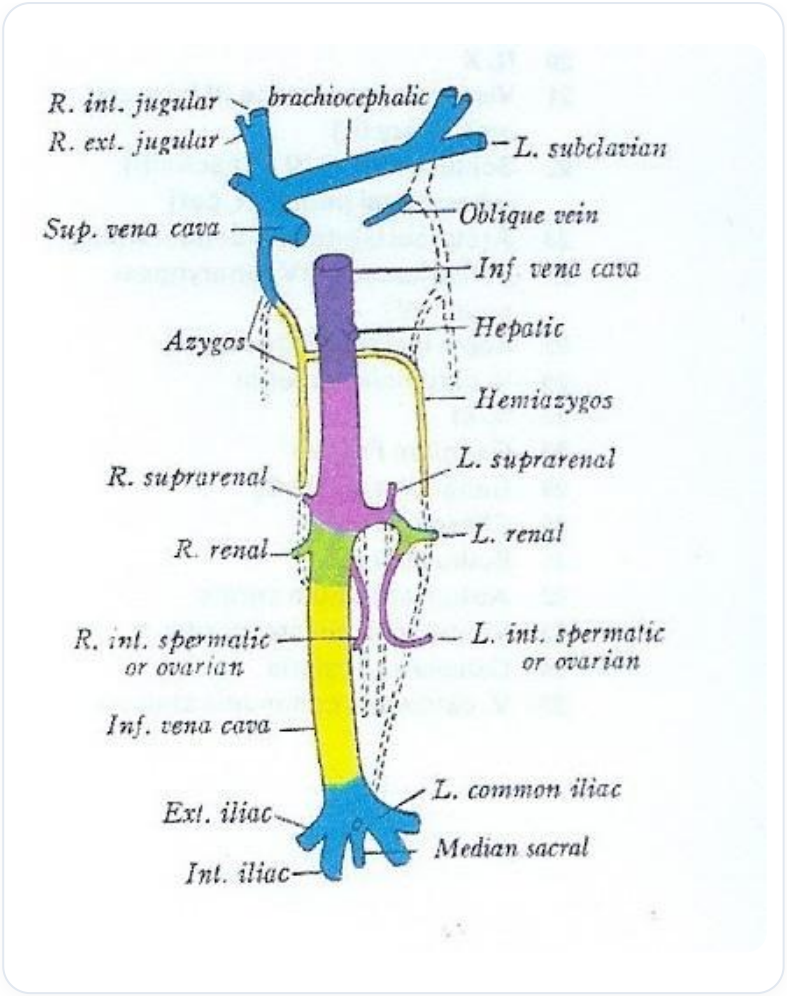


FIG. — Notochordal system or a transverse view of integration, corresponding to the text

ENDOVOLUNTARY MOVEMENT OF THE LUNGS AND AWARENESS OF THE UMBILICAL PEDICLE

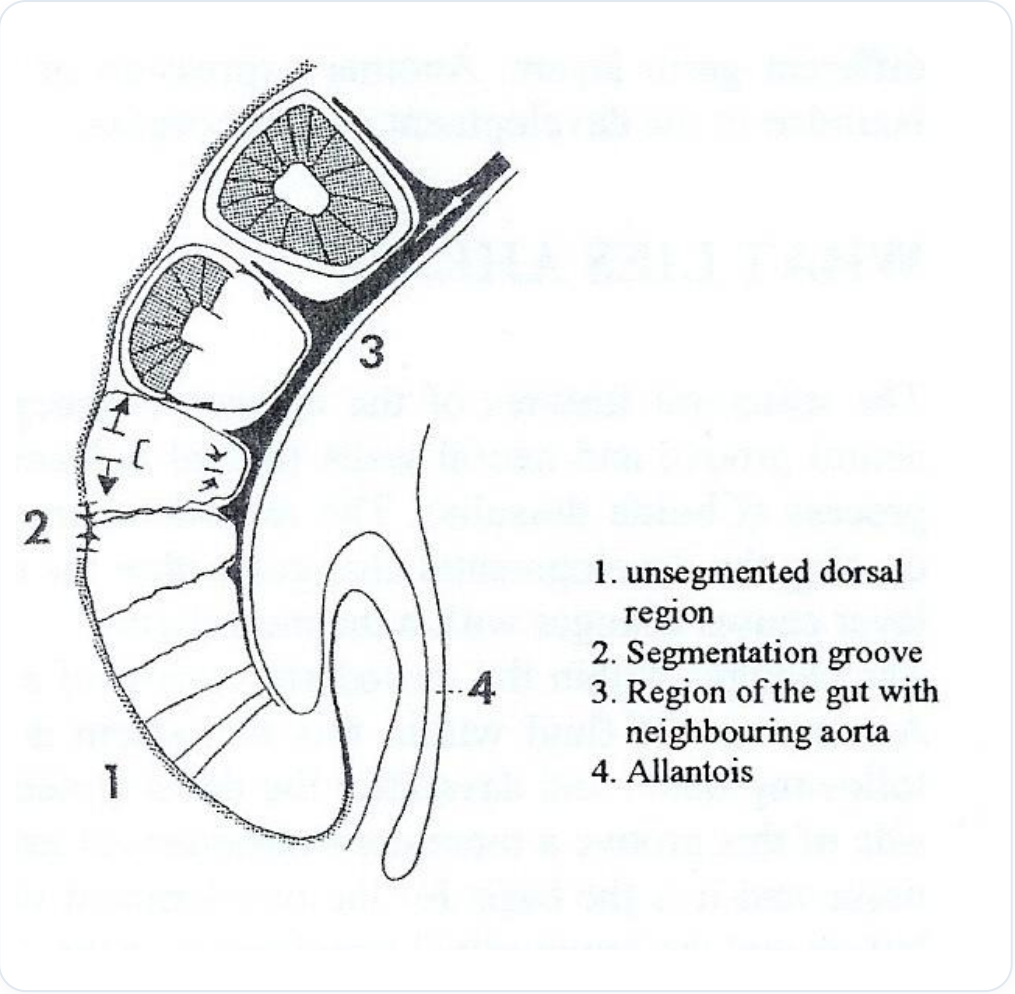


FIG. — The image corresponds to the description of the integration of systems such as the vitelline and hepatic, and this sentence marks a transition to the notion of 'movement'

The **endovoluntary movement of the lungs** originates from the endoderm and is guided by the vascular system. Awareness of the **umbilical pedicle** must return to this space, allowing one to perceive the force of the growing amniotic cavity, while integrating the processes of cephalisation, cardialisation, diaphragmatisation, and hepatisation.

The return then occurs. The vitelline vesicle, propelled by differential growth speed, will be reincorporated into the scheme. It will push the pedicle to join the vitelline vesicle and later form the umbilical cord.

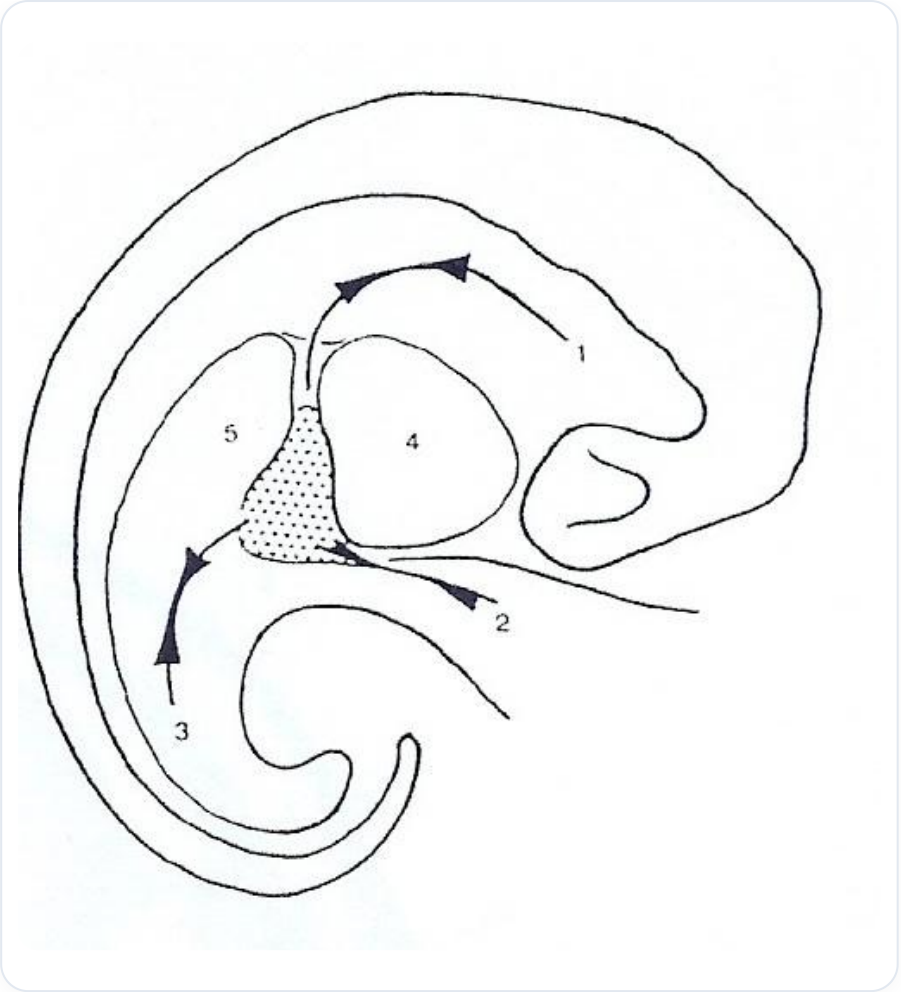


FIG. — The image corresponds to the description of the integration of systems like the vitelline and hepatic

Everything converges here, starting from a pedicle to return to a cord.

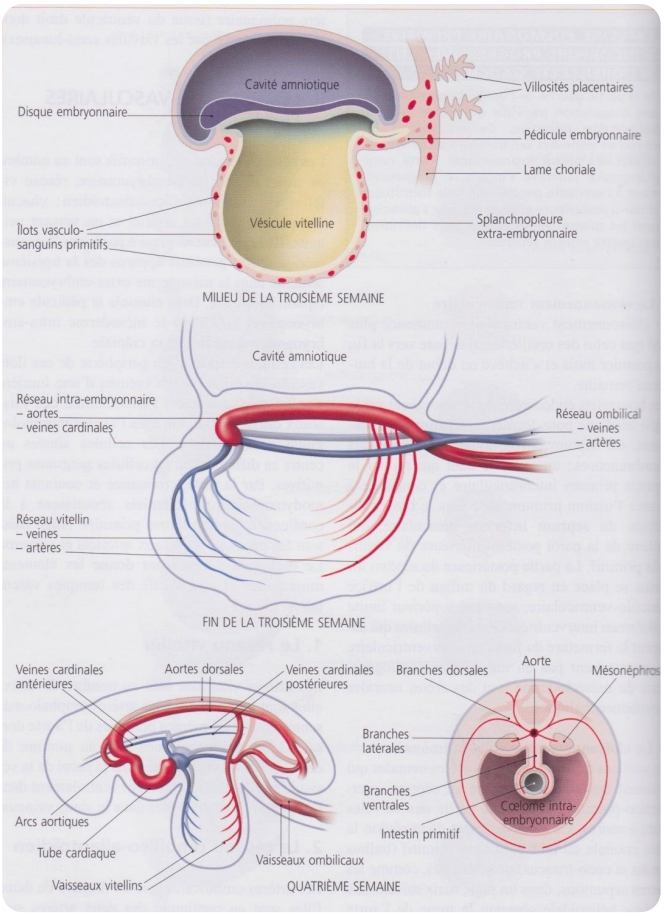


FIG. — Regression of the vitelline vesicle and umbilical cord formation, which is logical after mentioning the 'return'

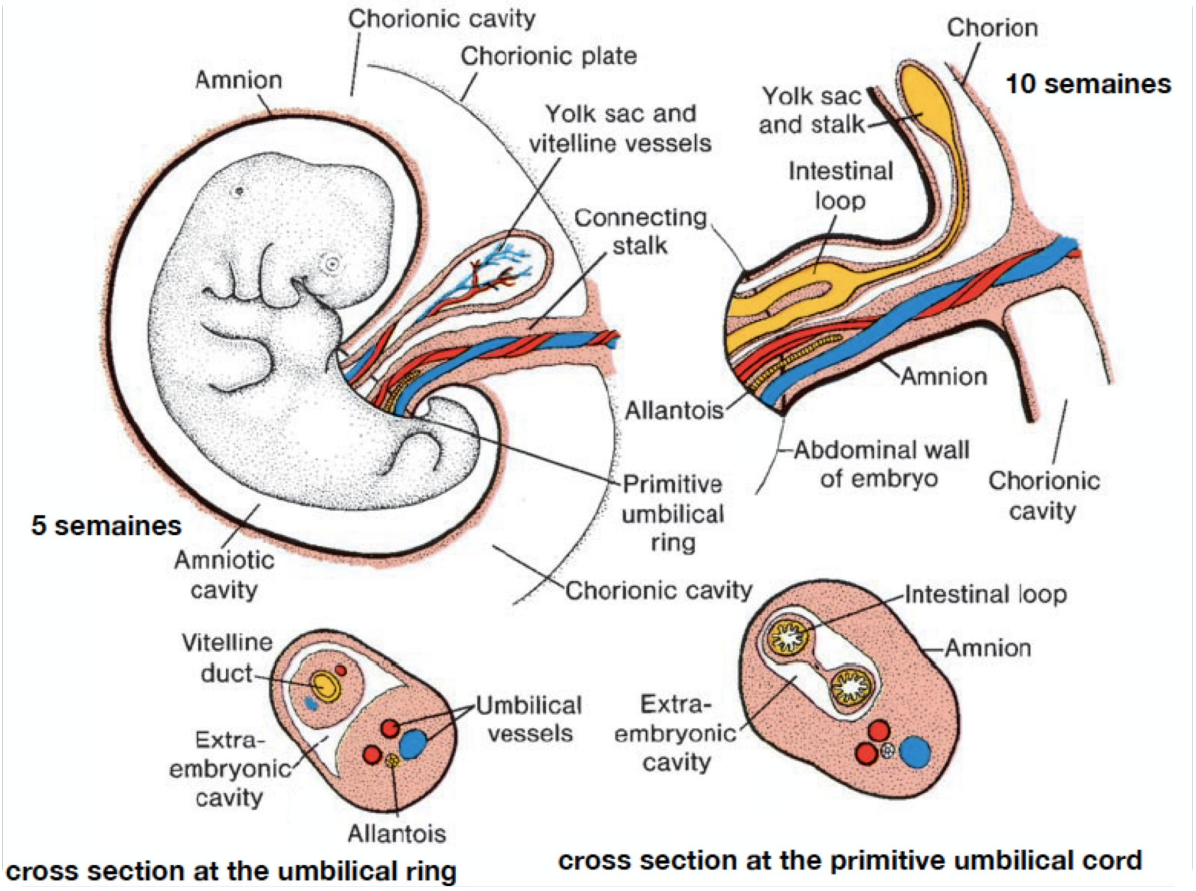


FIG. — Umbilical cord formation

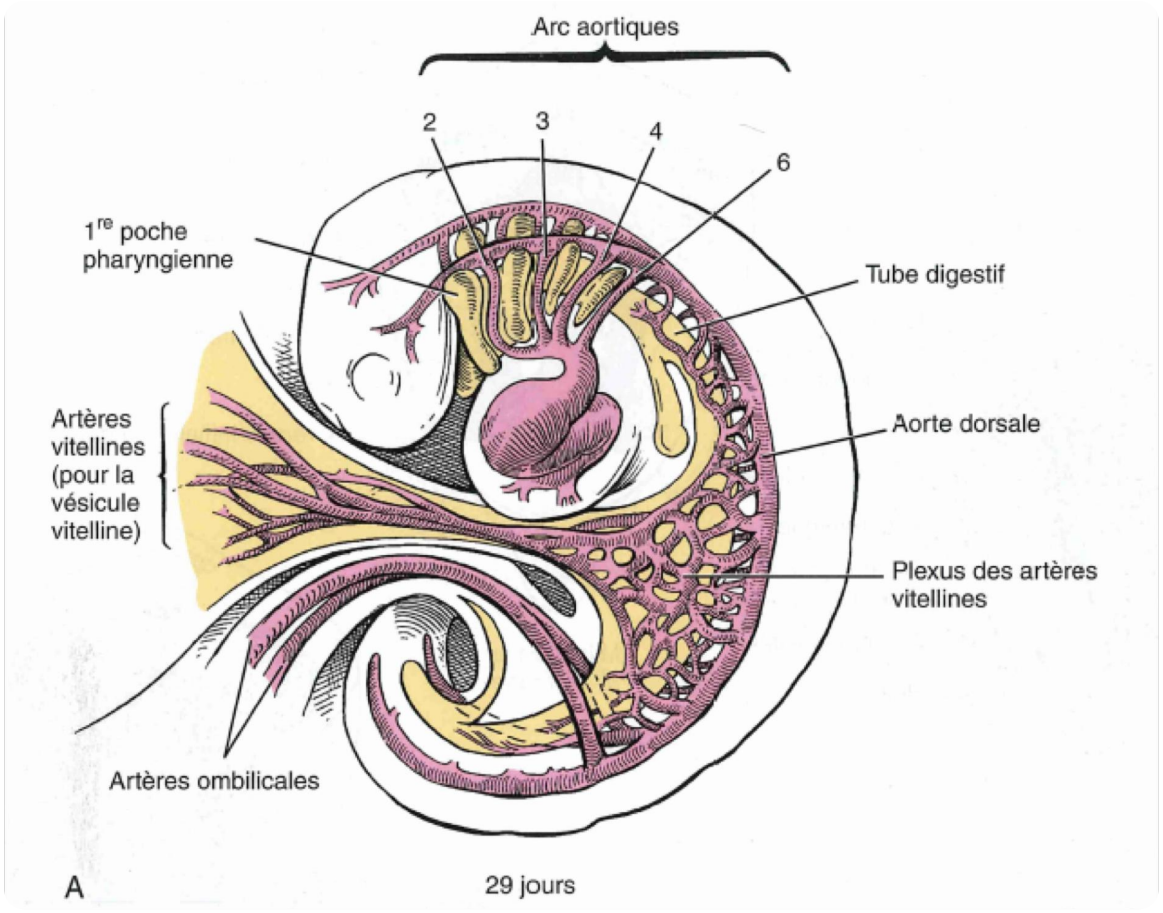


FIG. — This image appears to be a recapitulative view or a final visualization of the integration of the vitelline system and its remnants, consistent with the end of the paragraph

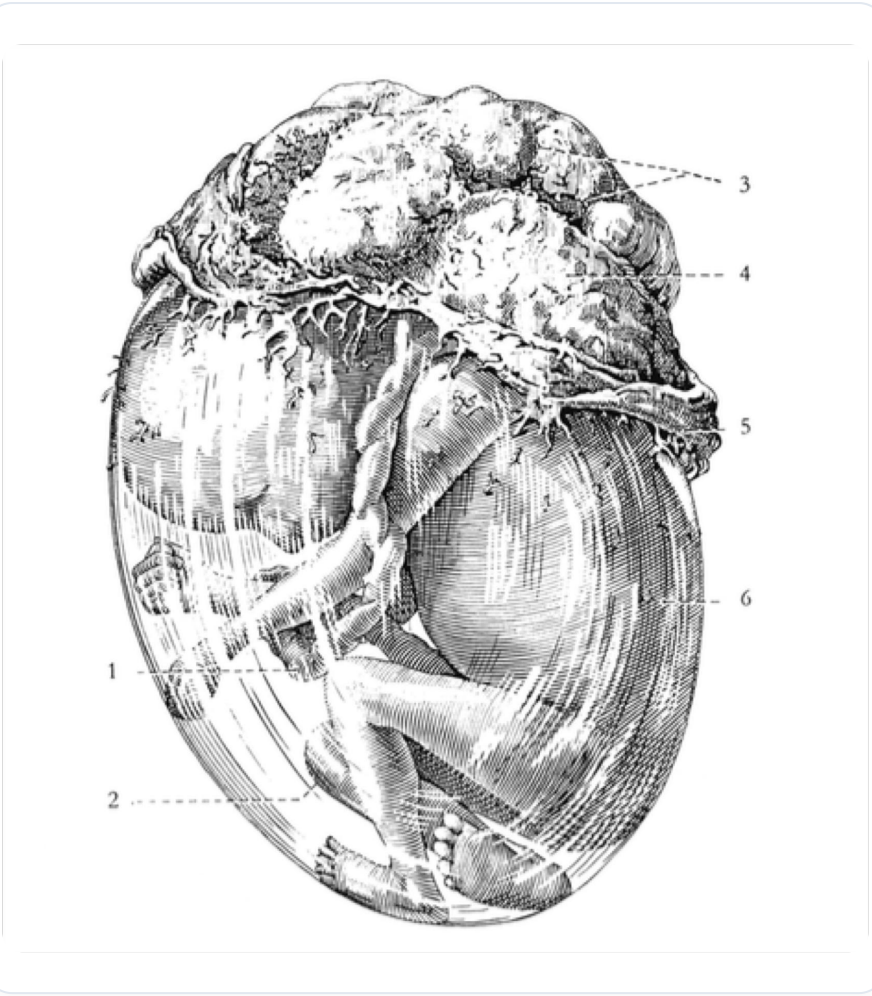


FIG. — Fetus in uterus

20. PRACTICE OF STRUCTURING THE VASCULAR SYSTEM

DURATION : 02:50 STUDY SHEET

COURSE SUMMARY

This video immerses you in the practice of structuring the vascular system, emphasising the importance of finding and emerging the neutral in the patient. Through techniques such as the micro-crystalline field and dynamic stillness, you will learn to create an environment conducive to healing by integrating the concepts of silence in the room and mobility at the diaphragmatic pillars. The session also focuses on the amniotic cavity and the importance of primary respiration as the breath of life, essential for a holistic therapeutic approach. This approach offers an opportunity to learn not only how to intervene, but also how to allow the body's natural processes to flourish.

PRACTICE OF STRUCTURING THE VASCULAR SYSTEM

We will seek the **stillness** of zone C in relation to zone A on the diaphragmatic plane. There is a point of stillness in the room.

I will guide you at a given moment. We will start from **neutral** and gradually allow the patient's neutral to develop. We will focus on the **micro-crystalline field**, taking up the entire space.

To achieve the most effective therapeutic action possible, it is essential to reach a neutral. Then, it is important to allow this neutral to emerge in the patient. This is a process that needs to **mature**.

For this, we need the **meeting point**. Once I have established this meeting point, the person, in trust, can progressively disengage, let go, and move towards their neutral.

Little by little, these two neutrals will begin to blend. Once these two neutrals blend, a more visible space, an **emergence**, can appear: what is called the **Primary Respiration**.

Primary Respiration is not only present in the body; it is omnipresent, much like the **Breath of Life**.

We will focus on the **amniotic cavity**, this zone B. At the same time, we will dissociate our attention to focus on the **diaphragmatic crura**.

The most fixed point of my diaphragm is formed by the crura. We will find that, ultimately, we will reach a neutral. These are enormous fulcrums, almost a silent point.

With a little patience, we will allow the neutral to emerge more broadly, also using **crystalline stillness**.

We will speak rather of the **crystalline immobility** of the room. There is a form of immobility in zone C that we must perceive. We will allow the body to centre itself in relation to this zone C.

Although it may seem subtle, this will happen naturally on a **fluidic**, almost electrical, plane.

Then, we will begin to blend this stillness of the room, or this dynamic immobility of the room, with the dynamic immobility at the level of the diaphragmatic crura.

While being aware of the amniotic cavity and the integration of my **vitelline vesicle** to form my venous system, we will blend this stillness of the room with the dynamic immobility.

It is crucial to surrender completely to the **Primary Respiration**, to the present motion, to the tide, that is, to the **therapeutic force**, to the potency.

Little by little, let's see what this can generate. The goal, in **non-doing**, is not to induce, but to trust the osteopath.

21. GUIDED PRACTICE : WORKING WITH ZONE B

DURATION : 06:20 STUDY SHEET

COURSE SUMMARY

In this guided practice dedicated to working with Zone B, you will learn to establish delicate communication with your patient, promoting self-regulation and a deep connection with the primary respiration. The teaching emphasises the perception of the body as a dynamic field, where the synchronisation of thoracic and primary respirations plays a central role. By engaging in this delicate process of meeting the neutrals, you will promote regeneration and dynamic stillness, thus allowing the fluids and the amniotic cavity to rebalance. This therapeutic approach paves the way for a better understanding of bodily function and harmonious cooperation between practitioner and patient to promote healing.

GUIDED PRACTICE: ZONE B WORK

In this practice, the objective is to find **health**, to work with it, and to fully trust the process of **self-regulation** generated by the **primary respiration**, when the two still points merge.

You will make contact with the patient in the gentlest way possible. It's like asking for **communication**, for **permission**. The movement present is health.

BODY PERCEPTION

We touch as if we had **ants** that could leave, but cannot, and that we cannot crush. Take your time, let your still point mature further, position yourself perfectly.

Allow **thoracic breathing**, and gradually primary respiration, to synchronise. Your hands are placed on the **micro-crystalline field**, meaning you encompass the entirety of the person, even their **aura**, their **amniotic cavity**. You hold the entirety of the person in your hand. The breathing moves, the breath of life moves, primary respiration, and the awareness of the space around you.

This respiration itself has a fulcrum located in **Zone A**, within the body. Respiration is like a door opening and a door closing. If there is a door opening and a door closing, there is like a **hinge**, a fulcrum for this respiration. You have silence or **dynamic stillness** in the room.

You have the breathing moving and the breath of life, primary respiration. Gradually, your still point deepens.

SYNCHRONISATION AND STILLNESS

You synchronise with primary respiration, the breath of life, and your thoracic breathing. Simply synchronise.

Become aware of the **crystalline stillness** of the room. The fulcrum, the immobile point of this pendulum, represents the point of stillness of thoracic breathing.

You remain in the space, in the presence of the awareness of space. You have in mind all this movement, the stillness that is there, and its transverse and longitudinal organisation.

Somewhere, you will reconstruct this layer. At the same time, you will restore all the strength of the **fluids** towards the centre.

You have the breathing, the breath of life, and the awareness of the room. The stillness.

THE MEETING OF STILL POINTS

Gradually, with a little calm, when the still point deepens, these two points of silence will merge. This will give a new quality: **silence**.

For this, you have observed the breathing, the end of exhalation. Just before the patient inhales again, there is an awareness, there is a door. Place yourself in this quality.

The breathing, the crystalline stillness of the room, the breath of life, and the meeting, gradually, of this **diaphragm**, of these **diaphragmatic pillars**, of this fulcrum of respiration, and of this silence of the room, of this stillness of the room.

Gradually, the still points must meet. They will merge, they will unify.

RECONSTRUCTION AND REGENERATION

You have the **amniotic cavity** expanding laterally, like an egg completely surrounding you and reconstructing itself. The entire fluid system recentres.

- Zone A calms down.
- Zone B reconstructs itself.
- Zone C merges with stillness.

With the dynamic stillness of the fulcrum of respiration. Let the still point deepen.

This **midline**, as a function, is a centre of health. It is an opening to space; there is no real distance. Let yourself be placed by it. Open your perceptual, peripheral field.

THE THERAPEUTIC FUNCTION

The meeting of the two still points will create, open the **therapeutic function**. Now, the small tide is present.

Let yourself be repositioned by it. It cannot be injured. This is what will gradually allow the correct spatio-temporal reorientation of form and its function. **Unity** with nature.

It converts therapeutic forces. Let this process continue for a few more minutes. Surrender yourself.

The amniotic cavity, Zone B, reconstructs itself. The body finds its fulcrums, its points of silence to be reborn.

There is a full force of **regeneration** that is setting in, that is activating.

ALIGNMENT AND AWARENESS

Carry your awareness to the **horizon**. From your midline to the horizon. From the centre of your being to the horizon. From the horizon to your midline.

The breathing, the breath of life. Gradually, a **tight** may arrive. Let yourself be surprised.

It's like a warmth that will begin to enter. A **metabolic increase**. A **rebalancing**.

It's a mantra that cuts off the mind: Oum! O yā ā O vénā Vēnā Sī Nē O.

22. THE LIVER : GENERALITIES

DURATION : 11:53 STUDY SHEET

COURSE SUMMARY

This video explores the complex embryological origin of the liver, composed of tissues from the upper, middle, and lower intestine, as well as essential mesodermal structures. It details the liver's functional duality, acting both as an exocrine gland for bile secretion and an endocrine gland thanks to its mesodermal component. It also highlights the importance of embryonic growth and haemodynamics in the development of this vital organ. Furthermore, the video emphasises the functional integrity and anatomical relationships between the liver and the diaphragm, illustrates how imbalances can lead to dysfunctions, while stressing the diaphragm's role as the 'gate of life' directly influencing liver health.

THE LIVER: GENERALITIES

1. EMBRYOLOGICAL ORIGIN OF THE LIVER

The **liver** has a complex embryological origin, formed from three parts:

- The foregut
- The midgut
- The hindgut

The junction between the foregut and the midgut is the site of an aspiration phenomenon, known as a metabolic field of **loosing fields**. This aspiration field initiates the development of the liver.

Observing a histological section, we see that this tissue, initially a collection of cells, undergoes an aspiration movement. Below this primordial tissue is a **mesodermal** tissue, which will later include the arterial system of the liver, of vitelline origin.

This process of aspiration and fractionation is common to the development of all glands. Each small budding recreates a new aspiration field, thus forming a set of small channels, the **hepatic ducts**.

2. DISTINCTION BETWEEN EXOCRINE AND ENDOCRINE GLANDS

There are two types of glands: **exocrine** and **endocrine**.

- An endocrine gland loses its lumen and secretes its substance directly into the blood.
- The liver, on the other hand, is an exocrine gland. It secretes its biliary substances inside its lumen, returning the sugars necessary for digestion to the digestive system.

In summary, the liver is an aspiration field that associates with the mesoderm, particularly with vessels of vitelline origin, to form its structure. It originates from the intestine but integrates mesodermal components.

3. DUAL ENDODERMAL AND MESODERMAL FUNCTION OF THE LIVER

The liver has a **dual function**:

- **Endodermal function:** of a digestive and specialised cannobiliary type, characterised by the **enterohepatic cycle**, which is very important for the secretion of bile, which is then reabsorbed at the caecal level.
- **Mesodermal function:** an endocrine function, linked to its mesodermal component.

This duality explains how the liver, even after partial ablation, can regenerate its mesodermal structure.

4. ROLE OF THE LIVER IN GROWTH AND HAEMODYNAMICS

For an aspiration field to exist, there must be a phase of **longitudinal growth** of the embryo. This aspiration field is therefore linked to growth, participating in the processes of **cephalisation**, **cardialisation**, and **hepatisation**.

The primary function of the liver is essentially **haemodynamic** and enables a potential for life. Its first impulse is to receive the products of disassimilation, i.e., the catabolic products of the entire body.

This means that the liver is fundamentally designed to participate in **regeneration**. It is the main organ of bodily regeneration, whether at the tendinous, muscular, or other structural levels.

The liver receives multiple inputs: portal, lymphatic, and cardiac.

5. FUNCTIONAL INTEGRITY AND ANATOMICAL RELATIONSHIPS

In osteopathy, the loss of **functional integrity** of the liver with the diaphragm is crucial. The **triangular ligament**, the **falciform ligament**, and the capsule of the liver are in tissue, cellular, and even molecular continuity with the diaphragm and pericardium.

This origin extends beyond the precordial plate. Undifferentiated **mesenchymal** cells, originating from this area, give rise to the **septum transversum** and the pericardium.

The septum transversum, among other things, forms the triangular ligament of the liver, which extends into the hepatic falciform ligament. It also contributes to the **greater omentum**.

The heart, diaphragm, and part of the liver are intimately linked. The liver acts as a fulcrum that indirectly contributes to the formation of the diaphragmatic crura.

The septum transversum, in association with the pericardium, remains stable, while the surrounding structures grow around it, forming this integration through the movement of cerebral development.

A simple imbalance, such as a costal displacement, can affect the position of the diaphragm and the liver.

6. THE DIAPHRAGM: GATEWAY TO LIFE

The diaphragm is often referred to as the **"gateway to life"** due to its orifices and essential role. The interaction with the liver is central.

The rhythmic movement of the diaphragm, with approximately **23,000 breaths per day**, makes the work of the liver dependent on diaphragmatic and pulmonary movement.

We observe differential pressures in the body:

- Negative pressure at the pulmonary level.
- Positive pressure at the abdominal level.
- Negative pressure at the pelvic level.
- Positive pressure in the upper body.

7. IMPORTANCE OF ABDOMINAL NORMOTENSION

Abdominal normotension is an important reference: epigastric tension is slightly higher than hypogastric tension.

- Frequent abdominal hypertension can increase tension in the upper area and decrease strength in the lower areas.
- Treating this area often involves considering the depressive zone.
- Treating the lungs amounts to treating the peritoneum.

The destabilisation of one part leads to the destabilisation of the other. It is therefore crucial to work on the spaces of **volaemia** to support the diaphragmatic process. The liver, being voluminous, and its proximity to the heart and lungs, creates a vital exchange space.

It is necessary to determine the type of abdominal tension and to work on the volaemic patterns. Supporting this area is essential because a loss of function here will decrease overall vitality. The diaphragm will then strive to maintain balance.

8. LIVER, FLUIDS, AND CARDIAC RELATIONSHIPS

The liver manages the qualities of the **fluids** exchanged, particularly for digestion and volume.

It quantifies and modifies the portal tension it transmits to the heart. In reality, the liver is a specialisation of the **duodenum**, which receives a multitude of information (autocrine, paracrine, neurocrine, endocrine).

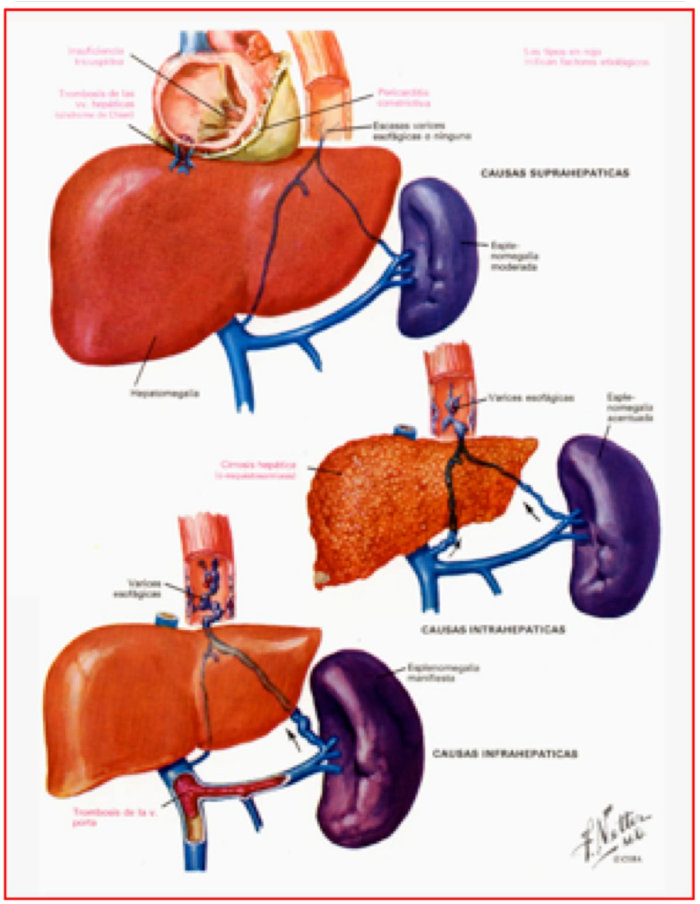


FIG. — Portal hypertension

The duodenum is a crossroads of information for the pancreas and the liver, providing digestive information based on inputs. A good digestive state is therefore paramount.

When the liver loses its movements and the quality of its exchanges, particularly at the lymphatic level, this can destabilise the heart and make it ill. There is a very important relationship between the heart and the liver.

- If the liver begins to destabilise, the heart can become ill.

- Conversely, relieving the liver can help a suffering heart.

Many cardiac affections are actually **metabolic dysfunctions** of the liver.

9. LIVER AND LYMPHATIC SYSTEM

The liver can overload the **lymphatic system**, which in turn overloads the main lymphatic duct and the lateral neurological plexus of the sympathetic chain, leading to distant repercussions.

For example, a painful left shoulder can be a sign of **hepatic congestion**. These are early indicators of hepatic dysregulation, the large primary conduit being first on the left.

One must not forget the **"trinity"** heart-liver-spleen, to which the mesenteric system is added. These "sponges" or states of volaemia work in synergy.

The axis between the liver and the heart is the **vena cava**, whose construction is complex.

10. LIVER AND IMMUNE SYSTEM

There is an important immune relationship involving the **thymic system**. The liver is involved in the passage of **T lymphocytes** with the thymus, and the return of immune information via the spleen. This process contributes to thymic balance and the maturation of T lymphocytes.

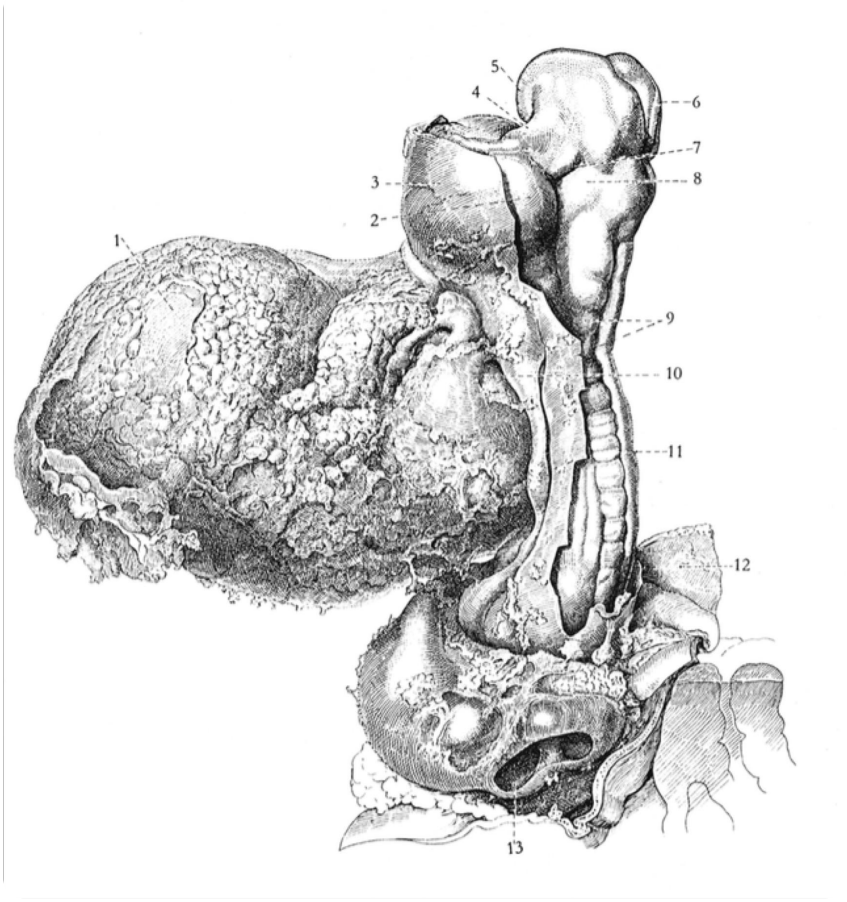


FIG. — Human embryo dissection

23. PRACTICAL NOTES BEFORE APPROACHING THE LIVER

DURATION : 06:13 STUDY SHEET

COURSE SUMMARY

In this session, practitioners will learn to explore the liver, taking into account its integration into the whole body and its relationships with the heart and surrounding structures. Concepts of embryonic motility are at the heart of the practice, where the technique will begin with neutral contact and progress towards a deeper approach, thus promoting primary respiration and the emergence of a therapeutic process. Participants will be guided on the importance of working in connection with embryonic fulcrums, particularly the midline and the heart, in order to energise the vascular and neurological systems, and thereby improve the effectiveness of their treatment.

PRACTICE BEFORE APPROACHING THE LIVER

In this session, we will explore the **liver** in relation to its **global environment**. It is essential not to adopt an overly **selective** or **local** approach.

The liver plays a central role in the organisation of the **omental bursa**, the positioning of the **stomach**, the axis of the **spleen**, and the **greater omentum**. We will practice on the liver while considering its relationship with the **heart** and the omental bursa, using an appropriate handhold.

The objective is to allow the liver to regain its **motility**, as it is this motility that creates its function. We will begin with a **neutral** and superficial contact, which will progressively deepen as the **primary respiration** manifests. The meeting of these two neutrals is what brings forth the **therapeutic process**.

The liver is crucial in the organisation of the embryo, particularly in the **transverse** axis and the establishment of the **gastro-duodeno-pancreatico-splenic** axis, thus managing all movement. This tilting, which we previously discussed with the **venous** and **arterial** networks, is an essential developmental movement for growth. The liver moves upon itself, not by rotation, but by **displacement**, which gives it a developmental force.

This motility of embryonic origin is fundamental. When we place our hands, we must position ourselves in a **fulcrum** where the entire system can develop. If we limit ourselves to a local approach, we lose sight of the original fulcrum.

The closer we get to the embryonic fulcrum, the more we access this force, and the more the body can provide us with information, because it knows it. A high frequency promotes **regenerative power**. Consider the word "fulcrum" as a **function** rather than a mere point of support.

The **midline**, for example, is a fulcrum for the totality. We have identified the heart area as an embryonic fulcrum, which cannot be reduced to a simple point.

Working on motility related to **brain** development can serve as a tool. Let's recall the curvatures, this "tai chi" of the brain:

- **Expansion**
- **Cephalic flexion**
- **Cervical flexion**
- **Pontine flexion**
- **Telencephalisation**

The telencephalisation phase is crucial, as it induces the straightening and establishment of the **anterior white line**. This involves the formation of the **star**.

We can work solely on motility or use **specific points** such as:

- The tip of the **monotocorte**
- The **supraocciput**
- The base of the **occiput**
- The **presphenoid**

These points are related to these developments. For example, the tip of the monotocorte is an important reference point in **submesencephalic flexion**.

I can thus search for the **still point** in the developing system, depending on the points of support on which I position myself. Certain phases will occur simultaneously, illustrating the **synchronicity** of development.

Regarding brain development, an important vestige is found at the level of the **mesenteric system**. The **vitelline** or **mesenteric** axis is crucial for re-energising a developing brain.

From a vascular perspective, it is important to note that the brain is entirely dependent on the heart for its function and formation. The heart continuously informs the brain, and a **pulsation** is present within it.

Keep in mind the image of the **septum transversum** coiling with the establishment of the liver, integrated with the **vitelline vesicle**. The establishment of the **amniotic cavity** is the result of our previous work.

There is a movement of developmental motility generated by the liver, which organises the entire **hepato-gastro-duodeno-pancreatico-splenic** system. We observe the tilting of the heart on its left axis, an inverse movement at the global hepatic level, and a counter-rotation on the digestive plane.

We will first work on this large **subdiaphragmatic** organisational centre, by establishing the omental bursa. Then, we will examine the relationship between the liver and the heart, which will be our two working points.

We have the **cardiac imprint**, the **pulmonary imprint**, and the enormous **hepatic imprint**. We will perform work on the liver in relation to the establishment of the heart and the intestine, adopting a **biodynamic** approach.

This means we will place our hands and allow the movement to occur. We may have a movement that seeks or a movement that expresses its originality. We will provide this point so that the system can reorganise itself.

The liver is a **specialisation** to be considered as a **fractal expansion** of the intestine. It consists of a set of **bile ducts** with two types of information:

- **Vascular** type information, corresponding to the **portal** type liver.
- A very red face may indicate a **mesodermal** type hepatic problem.
- A very yellow face may indicate an **endodermal** type hepatic problem.

These two approaches allow for a complementary understanding of the system.

24. LIVER PRACTICE

DURATION : 03:58 STUDY SHEET

COURSE SUMMARY

This video explores the interaction between hepatic growth movement and surrounding organs, emphasising the importance of the liver's position beneath the diaphragm and the aortic axis. Through a delicate approach, the teacher guides therapists on how to connect with the embryonic dynamics of the liver, cultivating respectful listening and a neutral state, while encouraging the use of gentle gestures to promote its regeneration. The concepts of spatial awareness and primary respiration are also addressed, highlighting the importance of patience in the therapeutic process.

LIVER PRACTICE: MOVEMENT AND THERAPEUTIC APPROACH

THE MOVEMENT OF HEPATIC GROWTH AND ITS ANATOMICAL RELATIONSHIPS

The **liver**, through its **growth movement**, exerts a major influence on the arrangement of neighbouring organs. It initiates a **sub-diaphragmatic tilt**.

This growth impacts the positioning of the **stomach**, as well as the **retro-gastric pouch**. The **duodenum** and the **spleen** are also dependent on this considerable expansion of the liver.

From a **fascial** perspective, this entire system is suspended from the **aorta**. This is a crucial reference axis, as it constitutes the main pathway of the **trophic gate**, around which the system appears to pivot.

SUPPORTING THE LIVER'S DEVELOPMENTAL MOVEMENT

To support this movement, I position myself slightly above the **costal arch**. My goal is to exaggerate the movement to visualise it, but it is a **pure developmental** movement.

I place myself in **embryonic listening**, first establishing my **stillness**. My hands are placed, gently encompassing, without exerting excessive pressure.

The image to remember is that of **ants**: one must neither crush them nor let them escape. In my mind, I visualise the aortic vascular axis.

At the end of the exhalation, I connect to this specific quality of the liver, which seeks to return to its source, to move slightly backwards. I let it do so.

FIRST SENSATION AND FULCRUM

My first sensation is that it is not completely free in its movement. It seeks to **regenerate**.

I place a **fulcrum** with my elbow, without too much force, and I provide this fulcrum with my **primary respiration**. I let the stillness settle and the system manifest itself.

We will take the axis of the **heart** and the **vena cava**. The movement is a little more delicate, difficult to mimic.

THE AWARENESS OF SPACE AND THERAPEUTIC STILLNESS

The second step is the **awareness of space**, the opening and perception of this space. Then comes the notion of **stillness**, which is essential.

When this stillness begins to mature, little by little, the two stillnesses meet. It is at this moment that the process becomes **therapeutic**.

Impatience is an obstacle; if we create a space of impatience, this meeting cannot take place. The process is there, and it is at this stage that one can dare to let the primary respiration act.

I know there is a great desire to intervene, but this practice can also be done on other levels. I am teaching an **energetic**, embryonic approach here, but one can very well work on fascial planes, etc. My observation is that other approaches do not hold as well.

25. THE RENAL SYSTEM : GLOBAL EMBRYONIC KINETICS

DURATION : 06:56 STUDY SHEET

COURSE SUMMARY

This video explores in depth the embryonic kinetics of the renal system, starting with its mesodermal origin, then illustrating how flexion and endorotation forces influence kidney development. Key concepts include the cranio-caudal development of the different stages of the kidney – Pronephros, Mesonephros, and Metanephros – and the dynamics of compression and rotation movements within the mesoderm. Therapists will learn to recognise how these embryonic mechanisms establish the foundations of organ systems, thus offering a better understanding of human anatomy and physiology.

THE RENAL SYSTEM: GLOBAL EMBRYONIC KINETICS

1. THE MESODERMAL ORIGIN OF THE RENAL SYSTEM

The lesser pelvis and the **renal system** are intrinsically linked in their development.

The establishment of the renal system begins at the **mesodermal** level, from the primitive streak. We find there:

- **The somatic plate**: This will give rise to the somites, which then differentiate into **sclerotome**, **myotome**, and **dermatome**.
- **The lateral plate**: This will later form the **somatopleure** and the **splanchnopleure**.
- **The intermediate plate**: It is in this tissue that the renal system develops. It will give rise to the **excretory** and **urogenital tissue**. From this stage, all the potential is present.

The kidney is a tissue formed by compression resulting from flexion, which is a concentration of **intermediate mesoderm**.

2. FLEXION AND ENDOROTATION: THE SCULPTING FORCES

At the embryonic stage, we observe:

- The **neural tube**.
- The **notochord**.

- The **somites** (dermatome, sclerotome).
- The **aortas**, which are very important.

The intermediate plate, initially open, closes on its longitudinal axis.

The growth of the neural tube and the approximation of the aortas lead to:

- A movement of **longitudinal compression**.
- A movement of **endorotation** in the longitudinal axis (a torsion).

The tissue, in its embryonic memory, receives compression and a rotational movement. It is a tissue that revitalises through compression.

This first biodynamic information of developmental kinetics is a concentration of mesoderm generated by flexion and this movement of approximation.

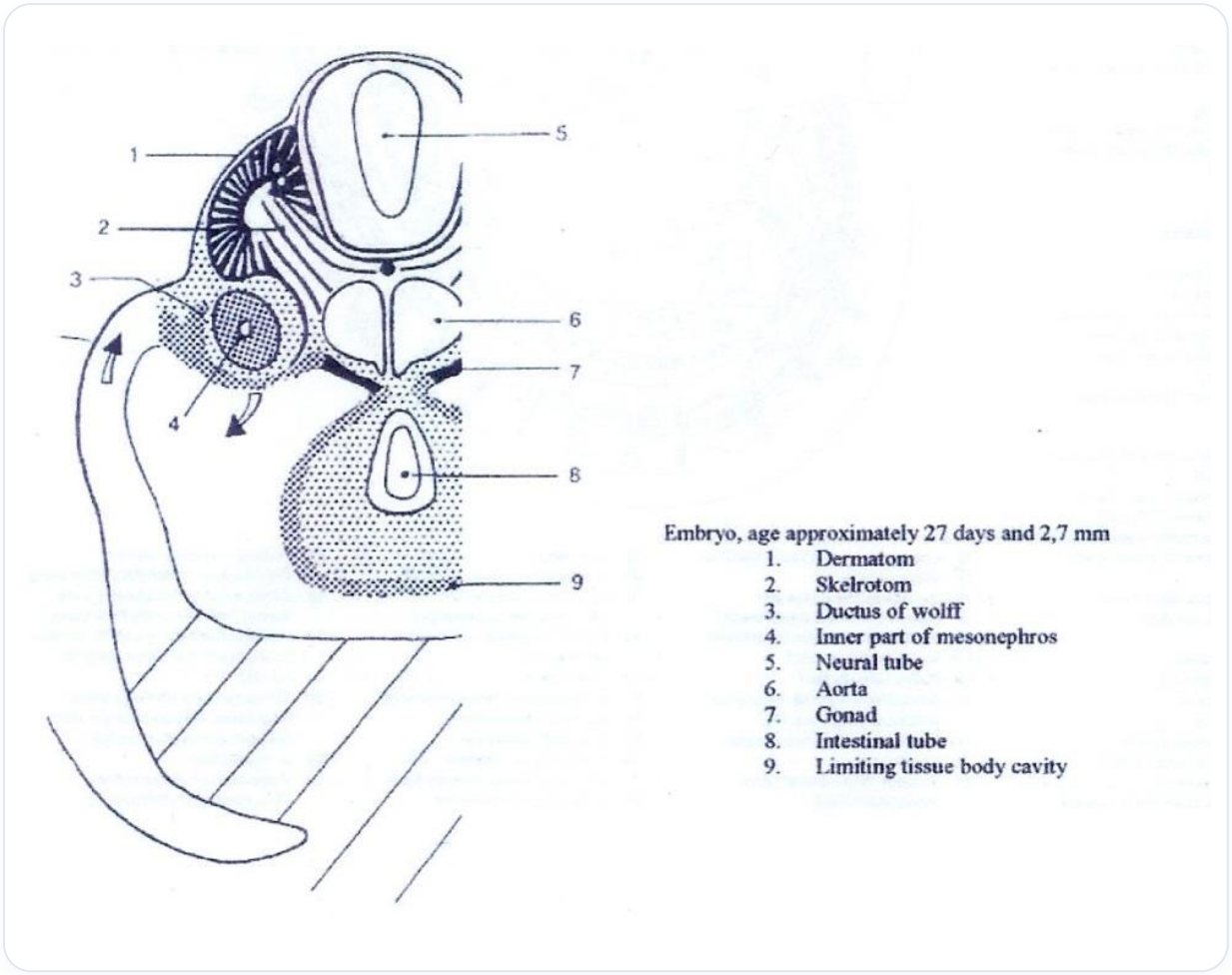


FIG. — Concept of kidney formation by compression and flexion

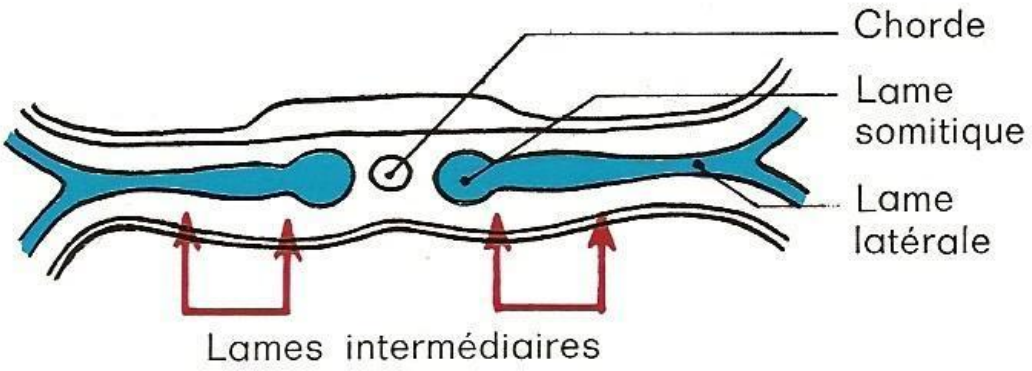


Fig. 1. — *Schéma de l'embryon, plan tridermique.*

FIG. — Tridermic plan embryo

The posterior intermediate plate of the posterior parietal peritoneum also undergoes this movement.

In this ridge, a canal appears: the **mesonephric duct**, a collecting duct in formation.

3. EMBRYONIC ELONGATION AND MIDLINE ROTATION

At the beginning of the fifth week, the embryo elongates. A larger fold is observed at the level of the **coelom**.

From a frontal view, these two large folds reveal a **rotation** organised by the approximation of the mesoaortas on the midline.

There is therefore **compression and rotation**, this is the first major information in the mesoderm. This indicates a repositioning of the system that we will detail.

4. THE CRANIO-CAUDAL DEVELOPMENT OF THE KIDNEY

The kidney, in its developmental phase, builds from **cranial** to **caudal**. Three main stages are distinguished:

1. **The Pronephros**:

* This is the first part of the kidney.

* It develops at an almost cervical level.

* It quickly emits a small collecting duct and then disappears.

2. **The Mesonephros**:

* This is the second part that develops.

* It becomes a functional kidney, forming a connection with the aorta and the first **glomeruli**.

* It also disappears after a certain time.

3. **The Metanephros**:

* This is the third part that develops.

* It is induced by a **ureteric bud**.

* It will form the definitive kidney.

Each part allows the emergence of elements that will be incorporated into the definitive kidney.

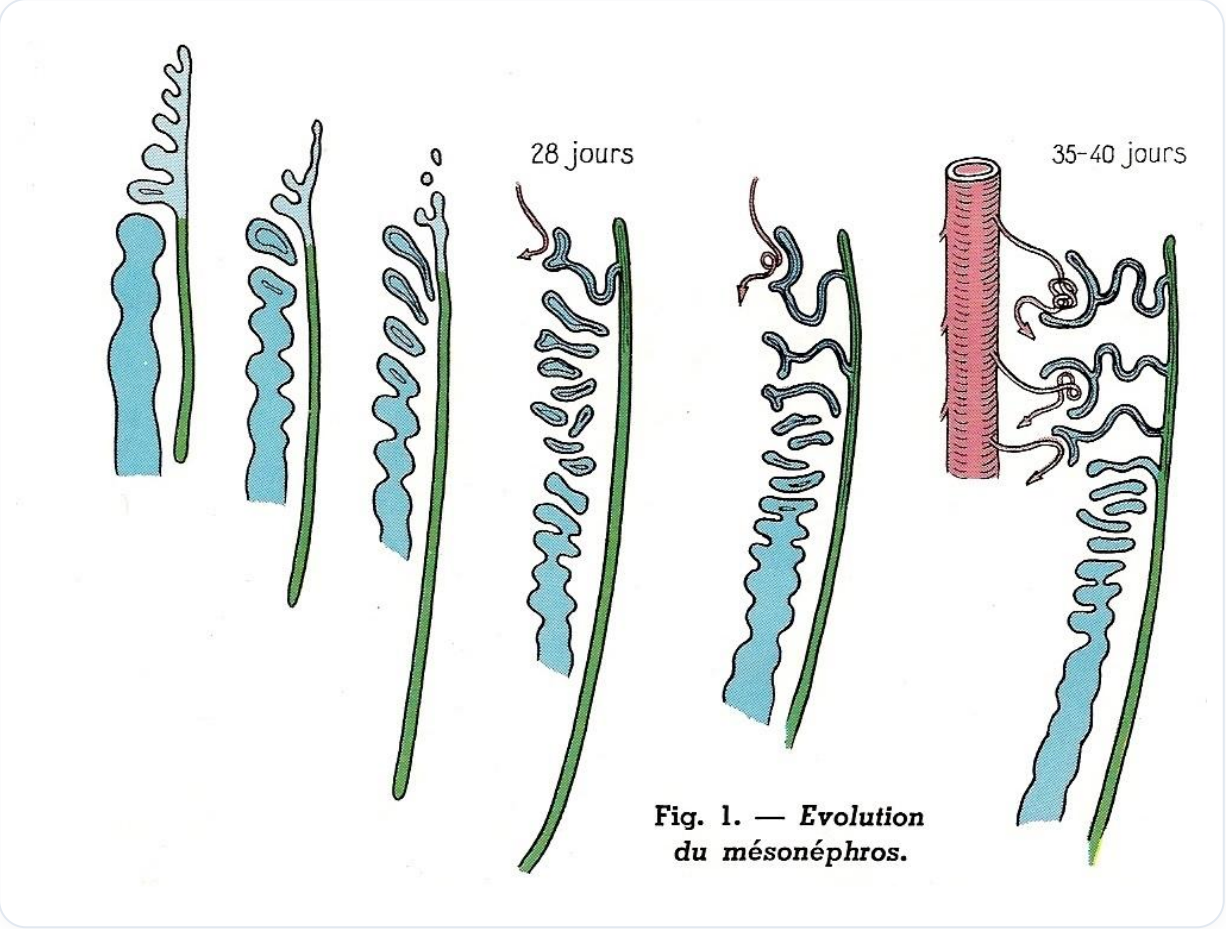


FIG. — Développement crânio-caudal du rein (Pronéphron, Mésonéphron, Métanéphron) qui est le sujet du paragraphe

This is a **cranio-caudal** developmental sequence. The first impulse movement of the intermediate plate is compression, due to the differential growth between the neural tube and the aorta, which compresses the system.

5. REPOSITIONING AND DIFFERENTIAL GROWTH

Initially, the **adrenal glands** are enormous and follow this phase.

However, the kidneys do not ascend. It is the posterior part that descends faster.

The **sacrum** and the kidney become reference points. The kidney is the **fulcrum** for the development of the posterior part.

There is a cranio-caudal developmental sequence: Pronephros, Mesonephros, Metanephros. They do not appear at the same time. Compression occurs from top to bottom.

The primitive kidney gives rise to a small primitive collecting duct, which will be incorporated into the second kidney, the mesonephros, with a mesonephric duct.

Then, the third part will be the definitive kidney, called the metanephric kidney.

Initially, the mesodermal plate is undifferentiated. But to have the definitive kidney, this first small, cervical system is needed, which almost follows the path of a **meridian**. The **bladder** system will also develop from a small canal.

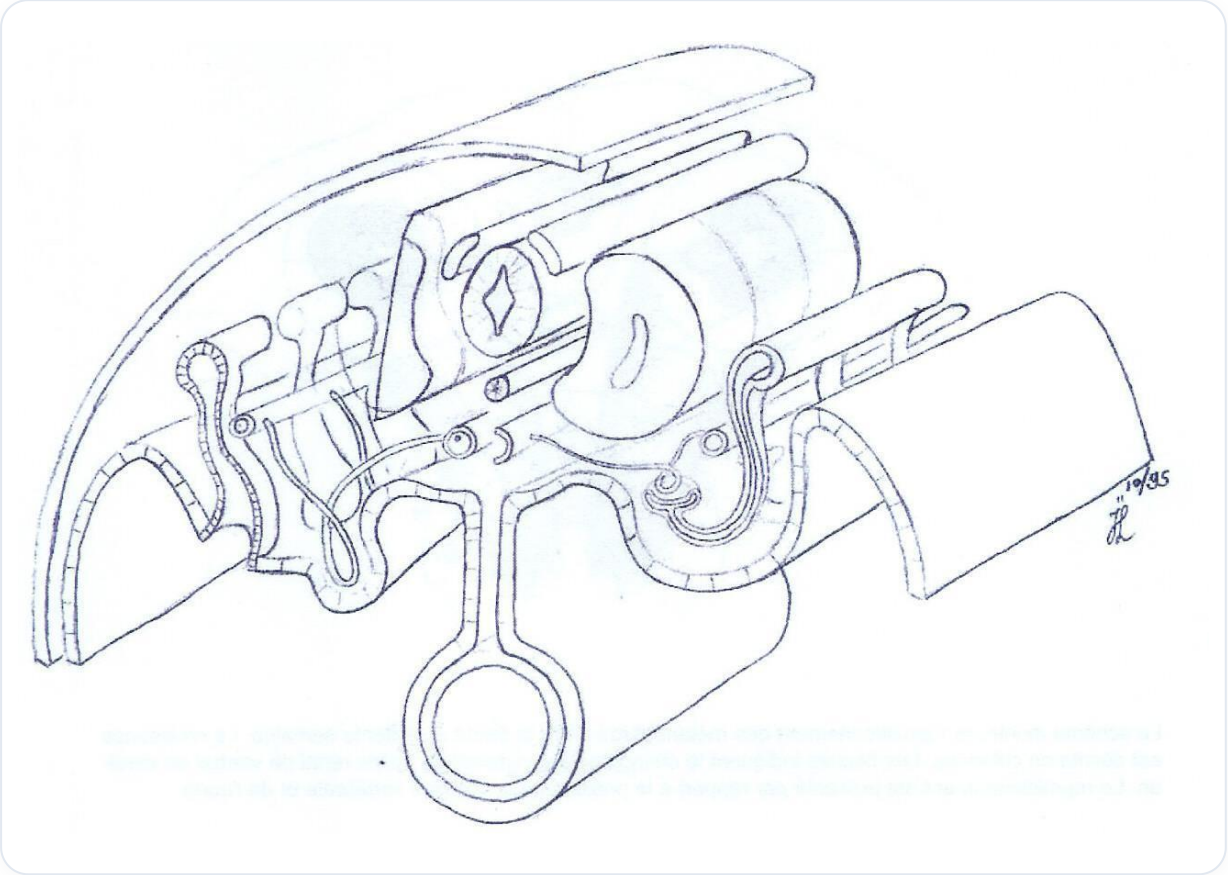


FIG. — Vertebrate embryonic anatomy

26. THE RENAL SYSTEM : EMBRYONIC KINETICS IN DETAIL

DURATION : 06:06 STUDY SHEET

COURSE SUMMARY

This video deeply explores the development of the renal system, focusing on the embryonic sequence from the pronephros to the metanephros. Starting from day 28, it addresses the key stages of glomerulus formation and the preparation of pro-urine, while highlighting the significant impact of the liver on renal development. In parallel, it discusses the interactions between the Wolffian and Müllerian ducts and the importance of the mesodermal layer in renal constitution. This cutting-edge approach aims to enrich the theoretical and practical understanding of embryonic development, offering students and therapists valuable tools to integrate this knowledge into their osteopathic practice.

THE RENAL SYSTEM: EMBRYONIC KINETICS IN DETAIL

THE PRONEPHROS: FIRST RENAL OUTLINE

The **rapid growth** of the liver leads to the compression and disappearance of the **pronephros**. Despite its regression, the pronephros gives rise to the **mesonephric duct**, a small, primordial collecting duct.

Around **day 28**, this duct forms a primary pronephros, characterised by a small **glomerulus**, the first image of a functional kidney. This is the time of **neural tube closure**, the integration of the **external coelom** into the internal coelom, and the delimitation of the embryo.

PREPARATION OF PRO-URINE

At this stage, the embryo begins to prepare the first **pro-urine**, which will contribute to the increase in **amniotic fluid**.

Initially an exudate from amniotic cells, the amniotic fluid then receives contributions from the **lungs**. This link strengthens around **days 35-40**. It is truly around **day 45**, with the complete closure of the anterior white line, that a first pro-urine appears. An **autocrine exchange** then establishes with the embryonic system, where the embryo is "constellated" by what it will eject and drink.

Day 28 thus marks a first functional connection with the appearance of several **glomeruli** at the level of the mesonephros.

THE MESONEPHROS AND HEPATIC INFLUENCE

The embryo continues to grow, the liver develops more and more, leading to the progressive regression of the **mesonephros**. The upper part of the mesonephric structures is compressed and disappears under the effect of this hepatic growth.

This highlights the importance of the liver in renal development, even for definitive function. This hepatic compression is necessary to subsequently induce the **metanephros**, the definitive renal structure.

The metanephros is already present in the structure of the **intermediate mesoderm**, but not yet induced.

THE WOLFFIAN DUCT AND THE MÜLLERIAN DUCT

The duct called **"Wolffian duct"**, **"mesonephric duct"** or **"ductus mesonephricus"** are synonyms.

In the urogenital system, two major ducts are essential: the Wolffian duct and the Müllerian duct. These two ducts will integrate to form the urogenital system.

The Wolffian duct or ductus mesonephricus induces the metanephros, which corresponds to the terminal part of the intermediate mesoderm, the latter being the entire posterior mesoderm of the embryo.

POSTERIOR DEVELOPMENT AND SWELLING

This posterior mesoderm, corresponding to the **posterior peritoneal cavity**, is the site of a swelling movement. Behind the aorta, a bulge occurs.

This is where another structure appears: the **adrenal glands** (or medullary adrenal glands initially). A small duct is induced here, the **bud** or the **ureteric duct**, which will induce the **metanephric blastema**.

THE SEQUENCE OF RENAL DEVELOPMENT

The embryo possesses an intermediate mesoderm containing all the **mesodermal** potential for the kidney. The growth and delimitation of the embryo organise a flexion movement.

1. **Pronephros**: The posterior part of this mesoderm, by compression of its anterior part, forms a tube, a mesodermal concentration. This process, associated with the approximation of the aortas, induces a rotational movement. From the cervical level, flexion induces a pronephros that segments into small pouches, leaving a small descending duct called the mesonephric duct.

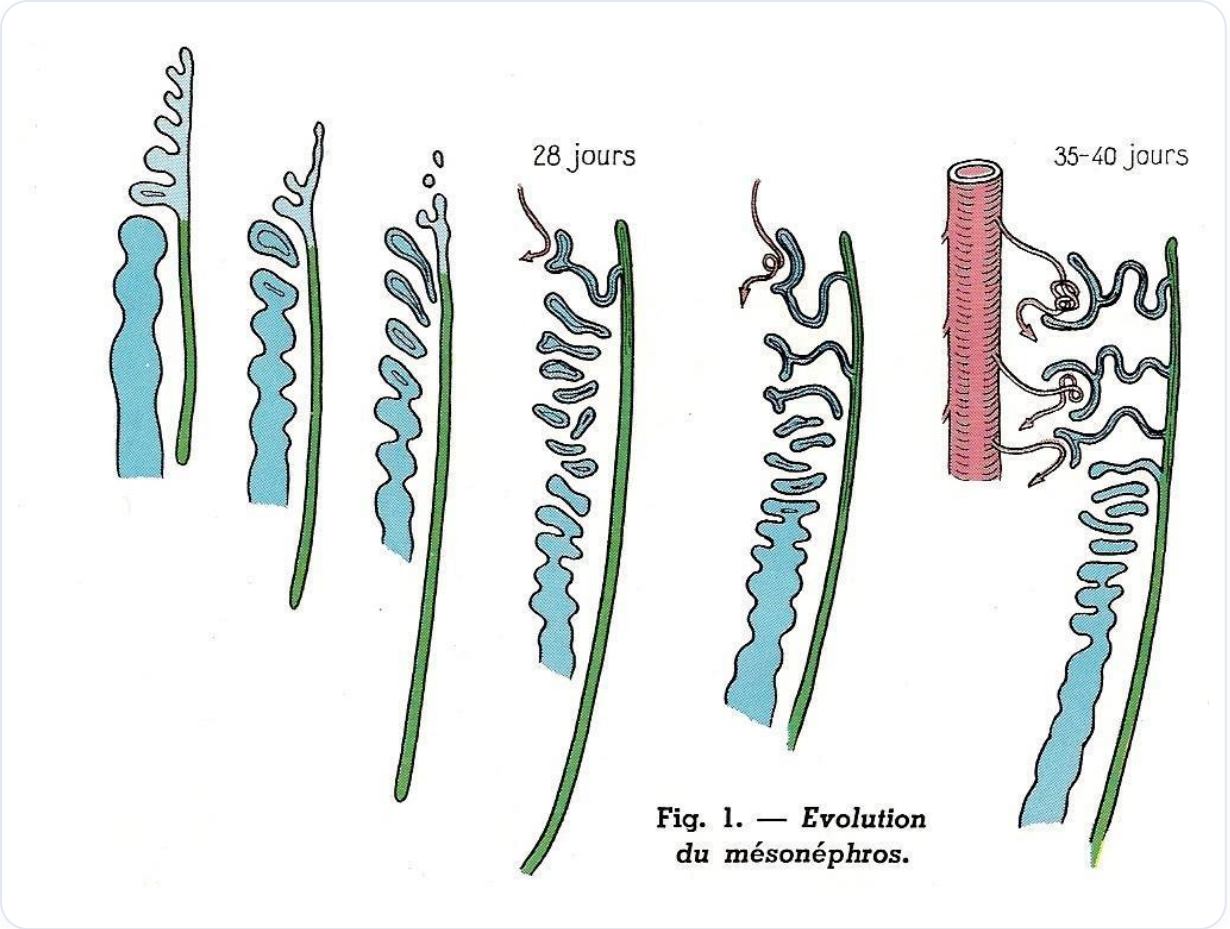


Fig. 1. — *Evolution du mésonéphros.*

FIG. — Evolution of the mesonephros

2. **Mesonephros**: The pronephros disappears very quickly, giving way to a second embryonic renal structure: the mesonephros. A connection is established with the mesonephric duct, and the mesonephros segments into small vesicles to form a first small pronephros. The latter comes into contact with the vascular system around day 28 and begins to function around day 45.

3. **Metanephros**: Then, the growth of the liver and the lengthening of the embryo lead to the disappearance of the mesonephros. Only the collecting duct remains. From this duct, a small opening forms, creating a **ureteric bud** by a suction movement. This bud induces the terminal part of the intermediate mesoderm into a **meso-metanephric blastema**, which will give rise to the definitive kidney.

27. THE RENAL SYSTEM : GROWTH AND DIFFERENTIATION

DURATION : 05:58 STUDY SHEET

COURSE SUMMARY

This video deeply explores the growth and differentiation of the renal system during embryogenesis. It highlights the central role of the liver as a major organiser, whose dynamics and growth influence the development of surrounding structures. By analysing the process of renal 'ascent', which is actually the result of the expansion of the rest of the body, as well as the phenomenon of separation between the bladder and the rectum, this session offers valuable insights into embryological reference points. Narrations on renal motility and retroperitoneal articulation demonstrate how cooperation between ectoderm and endoderm, under the influence of the mesoderm, is essential for the harmonious formation of the urinary organs.

THE RENAL SYSTEM: GROWTH AND DIFFERENTIATION

THE DYNAMIC OF POSTERIOR GROWTH AND ITS ROLE

The **posterior** part of the embryo experiences much faster **growth**, creating a kind of **"tearing"** in the lower intestine. This separation between the lower intestine and the allantoic process changes the dynamics of the posterior part.

This phenomenon is caused by a **differential growth rate** of the posterior part compared to the anterior part, maintained by the allantois and organised by the liver.

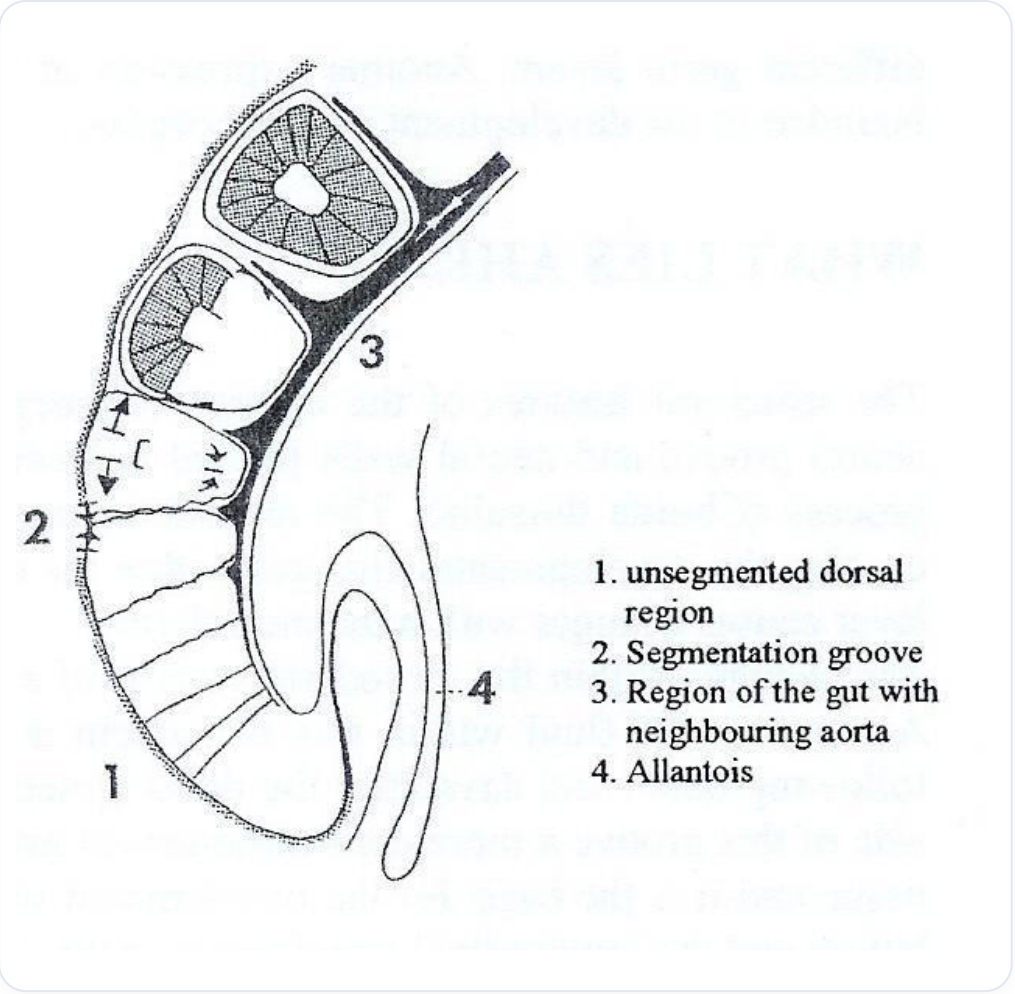


FIG. — Embryo sagittal section

THE LIVER: SYSTEM DRIVER AND ORGANISER

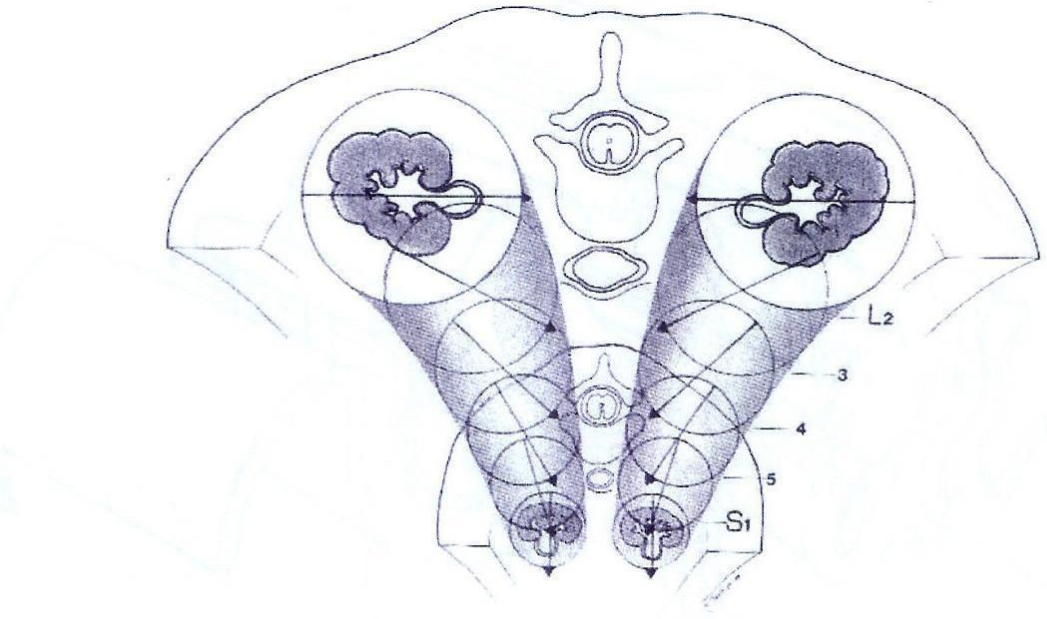
It is crucial to understand this dynamic, this **diaphragmatic** relationship, and this overall growth. The liver, through its dynamics, becomes a **major organiser**.

As an enormous organ, the liver plays an essential role in the development of the entire system through its growth and activity.

THE POSITIONING OF THE KIDNEYS AND ADRENAL GLANDS

The kidneys appear to **"ascend"** from their sacral origin to the lumbar region, but this **"ascension"** is an illusion. The adrenal glands, induced by the tilting of the liver, are initially enormous and descend with it. They give the impression of regressing, but it is actually the environment that grows around them.

It is important to think about embryonic development in terms of **changing references**. The growth of posterior structures positions the kidneys. It is not that they ascend, but that the rest descends and grows.



;chéma montre le repositionnement des métanéphros entre la 6ème et la 9ème semaine. La croissance décrite en colonnes. Les flèches indiquent le changement progressif du pelvis rénal de ventral en médi- Le repositionnement est présenté par rapport à la position de la colonne vertébrale et de l'aorte.

FIG. — Embryonic renal ascent

THE ROLE OF THE LIVER AND THE TRANSVERSE SEPTUM

The liver creates a **compression** from the upper to the lower part, essential for evolution. It is connected to the **transverse septum**, which, although it ascends high (embryonic origin at C3), does not move. It is the entire surrounding system that grows, giving the appearance of a **descent**.

The transverse septum, which does not move, becomes a **fulcrum** and finishes its descent at L2-L3. In reality, it is the entire environment that has moved.

THE DIFFERENTIATION OF THE URINARY TRACT

The **bladder** differentiates from the endoderm of the hindgut. The posterior growth rate of the neural tube is such that it **"tears"** the space between the future rectum and the future bladder, provided by the allantois.

Initially, the bladder and rectum formed a single structure. This separation of the cloacal space from the rectum establishes the different structures. The kidneys (metanephros) remain in place and become a **fulcrum** in relation to the posterior part of the pelvis.

THE RETROPERITONEAL ARTICULATION

The path at the kidney level reveals a **longitudinal axis** of the embryo. At the sacrum, there is a fulcrum, then the renal lodge.

The essential work is the opening of **motility** at the kidney level. The anterior compression explains their shape.

The retroperitoneal region becomes an **articulation** created by two movements:

1. The longitudinal growth of the neural tube.

2. The endodermal movement.

The integrity of these two functions depends on this intermediate articulation, which is an articulation between the **ectoderm** and the **endoderm**. The parietal system is dependent on the **mesodermal** system, which makes the mesoderm a royal pathway in embryonic development.

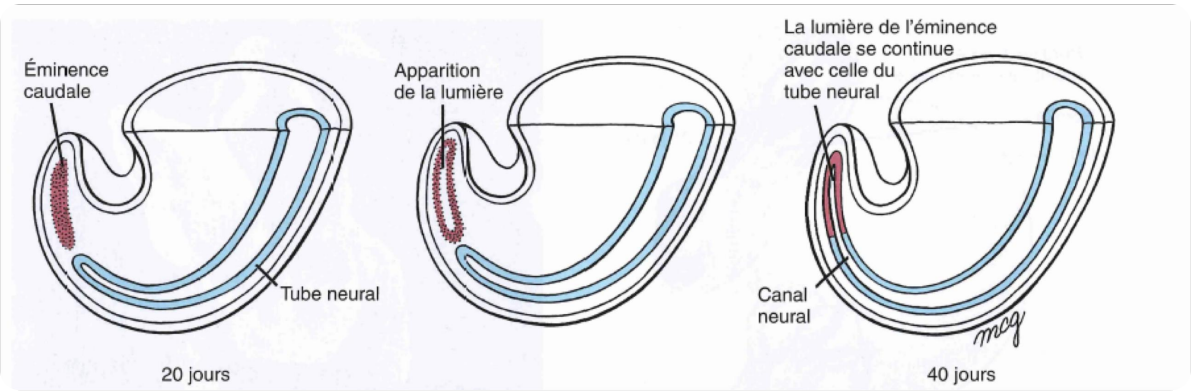


FIG. — Eminence/neural canal development

28. PRACTICE : KIDNEYS BLADDER

DURATION : 05:27 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth exploration of kidney and bladder practice through the lens of biodynamic embryology. You will learn the importance of the embryonic pathway of the kidneys and how primary respiration influences bodily balance. Practical techniques include sacrum manipulation and recognising connections between the kidneys and the uterus, revealing potential tensions and imbalances. You will also be guided through rebalancing exercises between ectoderm and endoderm, while becoming aware of unfinished healing processes related to kidney torsion movements. An enriching sensory approach for any therapist seeking techniques to optimise their patients' health.

BIODYNAMIC EMBRYOLOGY: KIDNEY AND BLADDER PRACTICE

THE EMBRYONIC PATHWAY OF THE KIDNEY AND PRIMARY RESPIRATION

The **embryonic pathway** of kidney development is crucial in a transgenerational approach. **Primary respiration** plays an essential role, where a left-right differential shift and compression manifest, leading to a perceptible movement.

SUPINE PRACTICE: SACRUM AND KIDNEY

In this practice, we focus on the **sacrum** and the **coccyx**. By holding the sacrum in my hand, I can imagine manipulating both kidneys, but I focus on just one. In the **motility** movement, the kidney becomes a fulcrum, allowing the sacrum to open and descend.

I feel the sacrum joining the kidney, which indicates a potential imbalance, either digestive or neuronal. Work on the **peritoneum** might be necessary to release tensions, particularly in the magnetic field of the uterus.

THE EXTERNAL ROTATION OF THE KIDNEYS

The **external rotation** of the kidneys is powerful information to feel at a distance. The aortas participate in this movement, which manifests as **dorsorotation**. This movement is more pronounced at the end of the process, while the aortas are not yet fully developed.

CONNECTION WITH A VOID AND THE UTERUS

I feel an attraction towards a **void**, a recognition that could indicate a generational problem. By connecting to this sensation, I allow primary respiration to act. This reveals that something is unfinished, linked to the uterus, even in its physical absence. This concentration of energy can manifest elsewhere, often in the neck, where people transfer their energy.

ECTO/ENDO REBALANCING PRACTICE

We will perform a simple and rebalancing practice, which acts as an articulation between the **ecto** and the **endo**. By manipulating the intermediate tissue, I focus on the **sacrum** and the renal lodge. I keep in mind the **unity of the body**, remembering to maintain this globality and the surrounding space. I wait for the **neutral** to gradually emerge.

THE UNFINISHED HEALING PROCESS

There is a **health** that expresses itself, but the process seems unfinished. This evokes a person on a journey, moving step by step, sometimes with a certain laziness. This awareness can motivate the kidney to continue its process. A brake persists, especially on the left, and I integrate it into primary respiration, placing myself in the patient's position.

TORSIONAL MOVEMENT OF THE KIDNEYS

The movement of the kidneys involves **torsion**. By taking one kidney in a clockwise direction, the other moves counter-clockwise, opening forwards. The anterior face moves towards the back, and although the **renal hilum** is not yet developed, this movement is essential. It is initiated by the **thoracic resin**, which plays a key role in the formation of the renal system.

29. THE GENITAL SYSTEM : IMPLANTATION

DURATION : 05:37 STUDY SHEET

COURSE SUMMARY

This video deeply explores the establishment of the genital system, focusing on the influence of the notochord and the nervous system on the development of genital structures. Key concepts include diaphragmatisation, the development of the adrenal glands and gonads, as well as the dynamic interaction between these organs, which guides the formation of the fallopian tubes, uterus, and Wolffian ducts. Through a biodynamic approach, practitioners will discover how these embryonic processes influence morphology and ligamentous structure, thus offering essential keys for an integrative understanding of the evolution of the urogenital systems in men and women.

THE GENITAL SYSTEM: ESTABLISHMENT

INFLUENCE OF THE NOTOCHORD AND NERVOUS SYSTEM

The **notochordal process** influences the **neural system**, which in turn impacts the **cardiac system**.

The cardiac system plays a crucial role in the establishment at the diaphragmatic level, more precisely at the level of the **transverse septum**, a phenomenon known as **diaphragmatisation**.

This diaphragmatisation process is also responsible for the development of the **adrenal glands**.

DEVELOPMENT OF ADRENALS AND GONADS

The growth of the **liver** generates a suction field, creating a posterior space in the **posterior parietal peritoneum**. This suction field contributes to the formation of the adrenals, which become very voluminous.

In parallel, the **urogenital system** is characterised by the migration of **gonocytes** from the **vitelline vesicle**. These gonocytes colonise the **genital ridge**, located at the posterior wall, thus stimulating the thickening of the epithelium and forming the **gonadal ridge**.

INTERACTION BETWEEN ADRENALS AND GONADS

The adrenals, pushed by the development of the liver, push the gonads laterally.

This developmental movement generates a **lateral fold** in the tissue, caused by the pressure of the adrenals on the gonads. This fold, visible in the posterior parietal peritoneum, is referred to as the **Müllerian duct**.

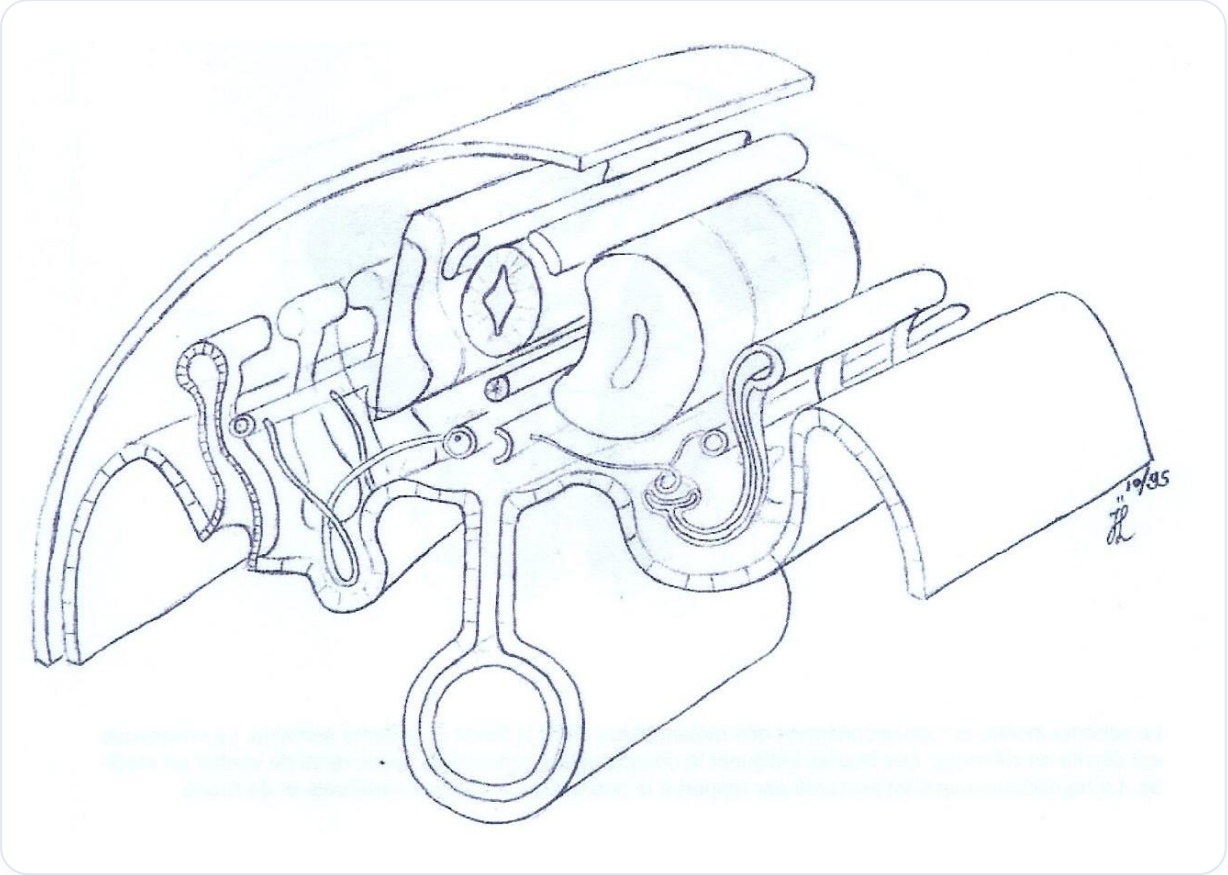


FIG. — This image seems to depict the effect of adrenals pushing gonads laterally, forming the Mullerian duct, as described

SEXUAL DIFFERENCES AND FORMATION OF GENITAL ORGANS

Initially, there is no visible structural difference between the sexes. However, genetically, females have more developed adrenals, accentuating the lateral movement.

This movement creates a fold in the tissue, which will give rise to the **Fallopian tubes** (left and right).

In the centre, the moving liver brings the elements closer together, while the lower limbs begin to form, creating a midline.

The two tubes of the Müllerian duct join, and under the influence of the liver and adrenals pushing the gonads laterally, this midline is structured.

Superiorly, the liver continues to exert traction, and the joining of the two tubes will lead to the formation of the **uterus** in females.

In males, this canal regresses and does not develop in the same way. The developing gonad will transform into an **ovary** in females, with the uterus originating from the posterior parietal peritoneum.

LIGAMENTOUS STRUCTURES AND DUCTS

The **broad ligament**, with its internal lines, will give rise to the **ovarian ligament**, the **uterosacral ligament**, and the **loose broad ligament**, all derived from the posterior parietal peritoneum.

The genital ridge, informed and developed, is linked to the development of the liver, which creates a new **retroperitoneal** space.

This suction field, typical of glands, pushes the gonads and kidneys, leading to the disappearance of the **mesonephros**.

The gonadal ridge itself moves laterally, forming a fold in the posterior parietal peritoneum, which will become the **Müllerian tubercle**.

The first level, originating from the collecting tubule, is the **mesonephric duct**, also known as the **Wolffian duct**. This duct will later serve as a conduit for the gonadal ridge (future testis or ovary).

Finally, the **diaphragmatic suspensory ligament** superiorly and the **gubernaculum** inferiorly will guide the ovaries into place in females or allow the descent of the testes through the **inguinal canal** in males.

30. THE MALE GENITAL SYSTEM

DURATION : 06:40 STUDY SHEET

COURSE SUMMARY

This video deeply explores the formation of the male reproductive system through different embryonic stages. You will learn how the genital ridge, influenced by the migration of gonocytes, gives rise to the gonads, as well as the critical role of the Wolffian ducts, which transform into structures such as the vas deferens and epididymis, while the Müllerian duct regresses. The video also details the mechanism of testicular descent, highlighting the importance of the gubernaculum ligament and its interaction with liver development. Finally, it addresses the prostate, its embryological origins, and its connection to the urogenital sinus, providing a holistic understanding necessary for practitioners and students in embryology and osteopathy.

THE MALE GENITAL SYSTEM

1. THE GENITAL RIDGE AND GONAD FORMATION

The **genital ridge** is a structure influenced by the migration of **gonocytes**, which originate from the **yolk sac**.

These gonocytes thicken the posterior coelomic epithelium, thus forming the anlage of the gonad, the future **ovary** or **testis**. This tissue is swollen by the proliferation of the gonocytes that reach it.

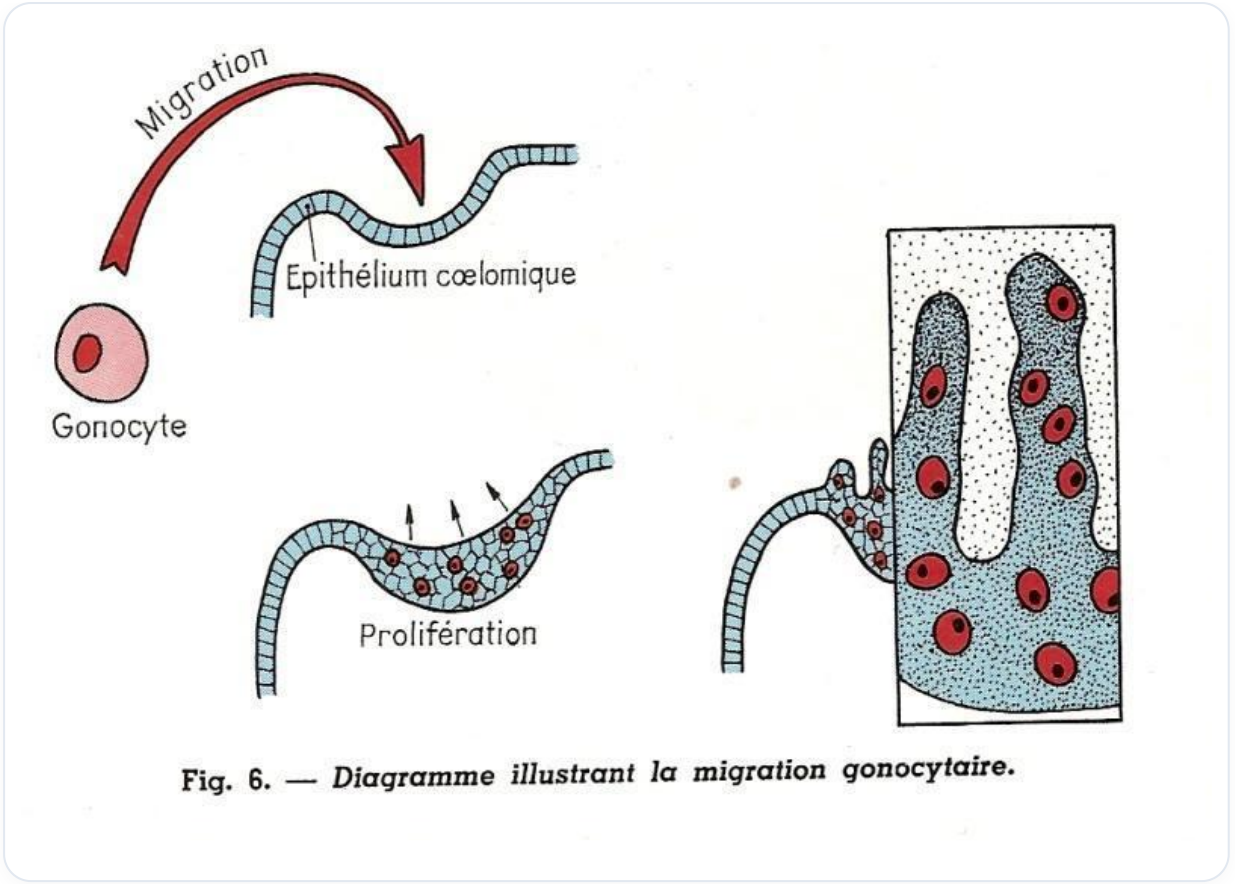


Fig. 6. — *Diagramme illustrant la migration gonocytaire.*

FIG. — Gonocyte migration

2. THE ROLE OF THE WOLFFIAN AND MÜLLERIAN DUCTS

The **Wolffian duct** is a collecting duct, already formed and then disappeared since the **pronephros**. It is pushed laterally and a small duct, the **Müllerian duct**, appears.

These two ducts, Müllerian and Wolffian, play a crucial role in the development of the **urogenital** system, in both males and females.

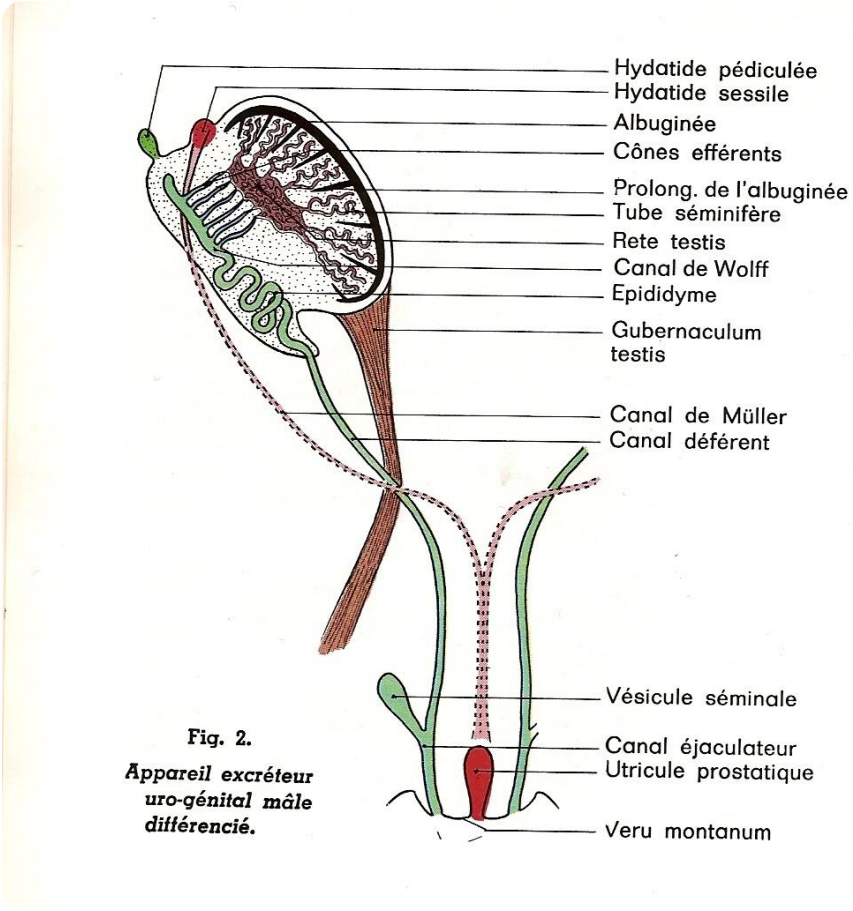


FIG. — General role of Wolffian and Müllerian ducts, in connection with the section introducing them

2.1. DEVELOPMENT IN MALES

In males, the Wolffian duct differentiates into the **vas deferens** and, by budding, into the **ureter**. It connects to the developing testicular area, forming connections that will give rise to the **epididymis**.

The Müllerian duct, on the other hand, regresses in males, leaving behind a small vestige, the **sessile hydatid**, which has no known function. However, it gives rise to the **prostatic utricle**, which will later be engulfed by endodermal tissue to form the **prostate**. This prostate will be connected to the system of vasa deferentia, originating from the Wolffian duct.

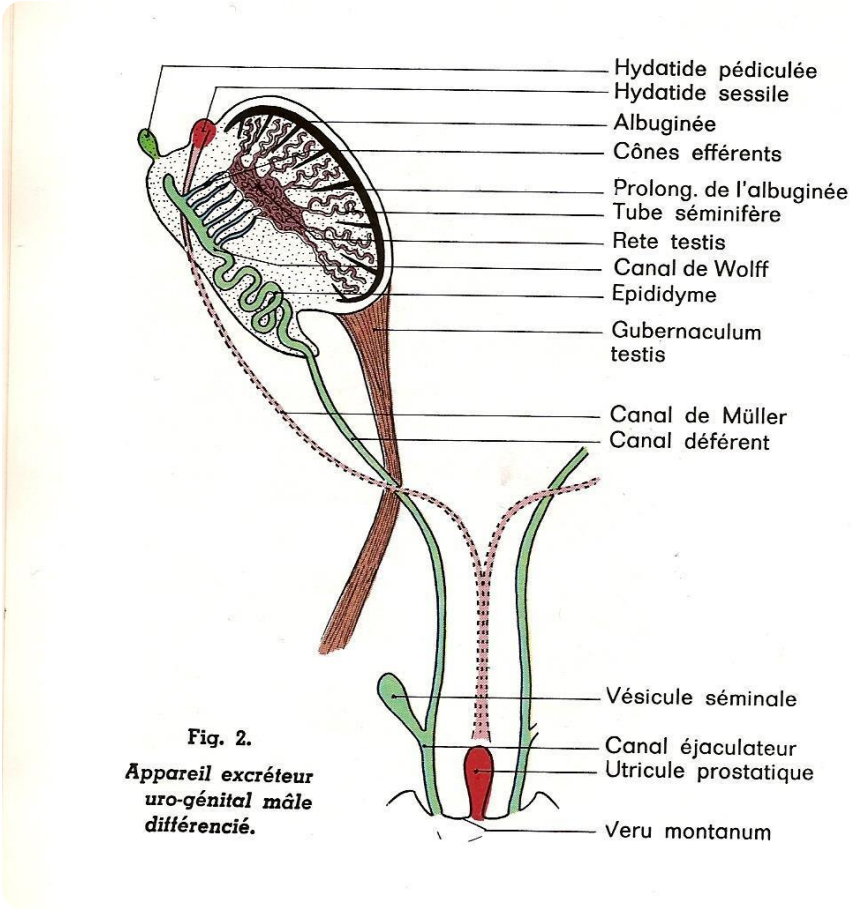


FIG. — Differentiated male genitalia

2.2. DEVELOPMENT IN FEMALES

In females, the Müllerian duct forms the **fallopian tubes** and the **uterus**.

3. THE GUBERNACULUM LIGAMENT AND TESTICULAR DESCENT

In males, a crucial ligament is the **gubernaculum testis**. It shortens and pulls the testes downwards.

This process is linked to the elongation of the rest of the system and the differential growth rates of the posterior part. Thus, the gonad, initially in a high position, descends. The gubernaculum, associated with the **inguinal ligament**, draws the testicular system downwards.

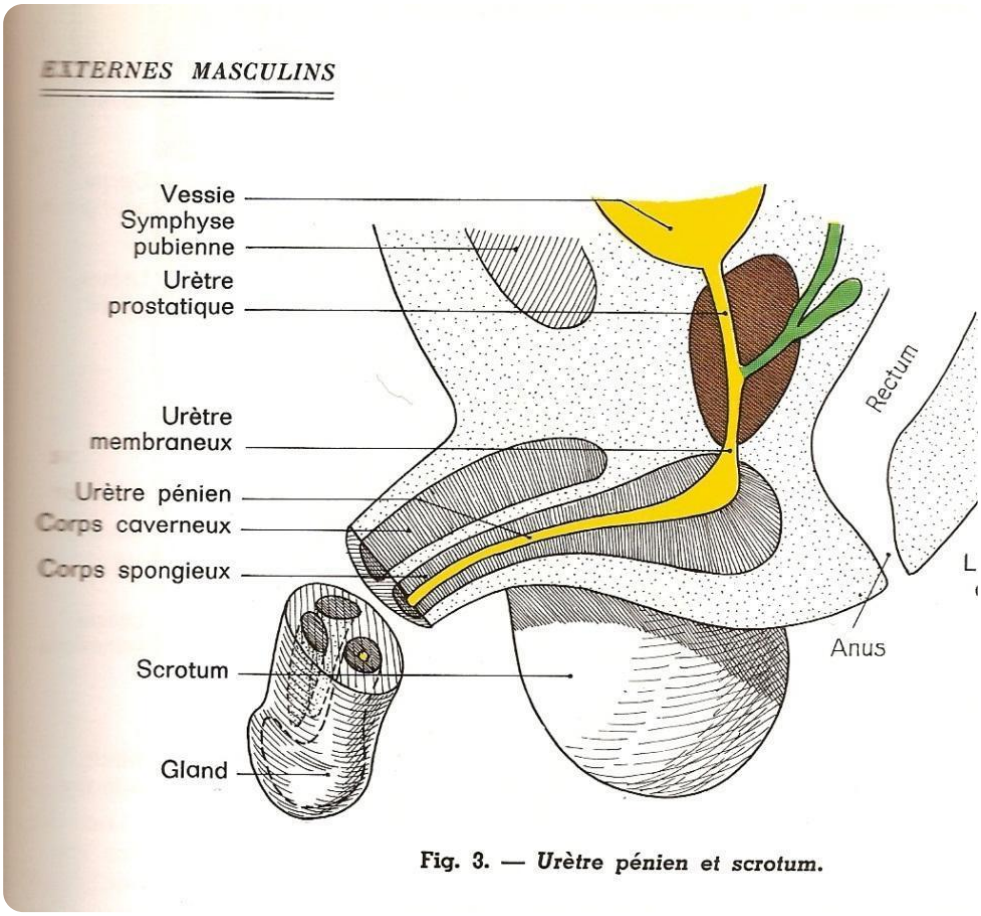


FIG. — Male reproductive system anatomy

4. THE INFLUENCE OF HEPATIC MOVEMENT

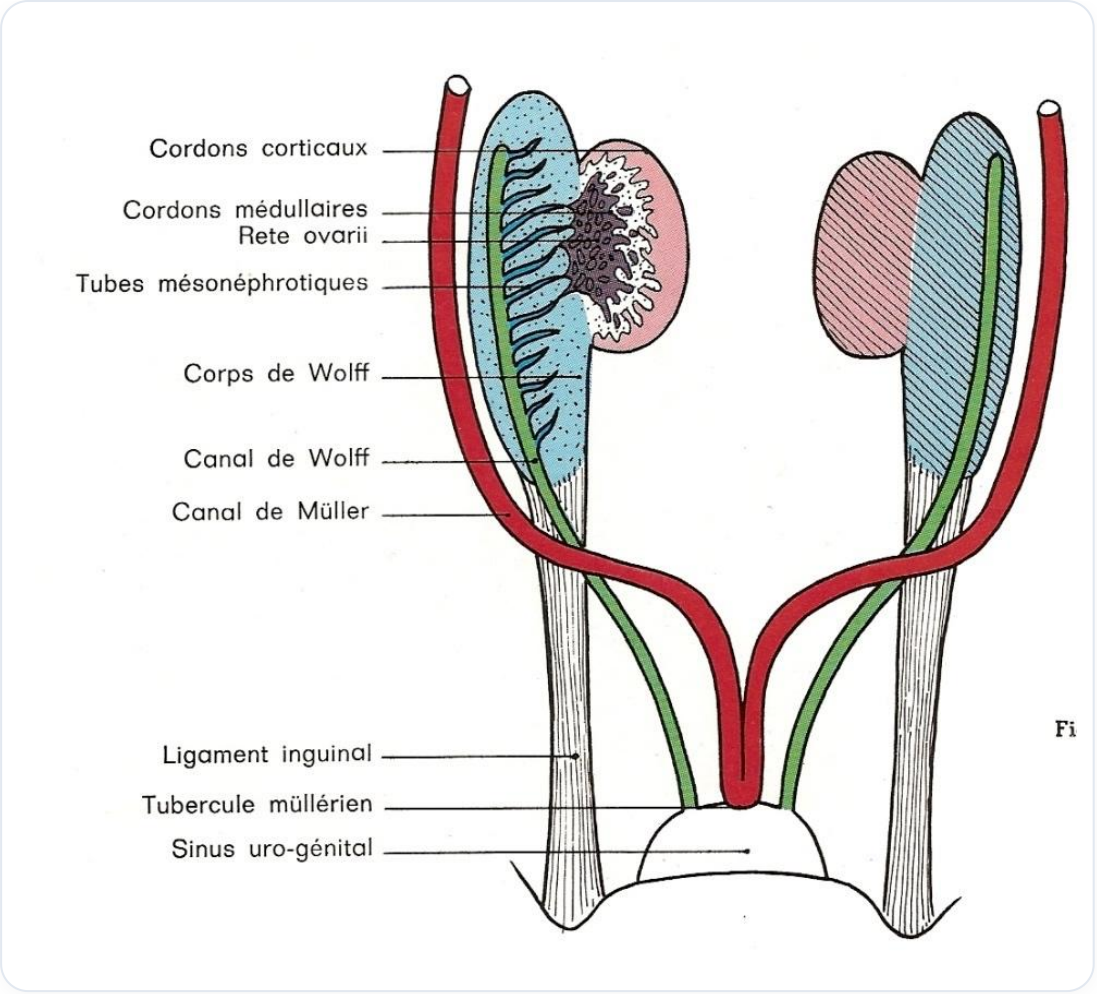


FIG. — Testicular descent, a key concept explained in this paragraph

The initial **hepatic movement** pushes the whole system back and organises the peritoneal cavity as well as urogenital development.

The liver therefore has major importance, both in its metabolic and embryonic functions.

5. THE PROSTATE AND THE UROGENITAL SINUS

The **prostatic utricle** is a remnant of the Müllerian duct, composed of urogenital epithelial tissue of endodermal origin.

The prostate develops from this tissue. The core of the prostate (the prostatic utricle) is **mesenchymal**, while its capsule is endodermal, originating from the **urogenital sinus**.

The urogenital sinus is the area where the **cloaca**, the **allantois** and, further back, the **rectum** were located. It is also from here that the **ureteric bud** originates, which induces the definitive kidney, itself derived from the Wolffian duct.

6. DETAILED TESTICULAR DESCENT

The initial position of the testes is very high. Testicular descent is due to the non-growth of the gubernaculum ligament, which pulls the system by the surrounding growth.

Meanwhile, the other ligament, the **gonadal suspensory ligament** (formerly called the ovarian suspensory ligament, but more precisely the **testicular suspensory** or **diaphragmatic** ligament), lengthens.

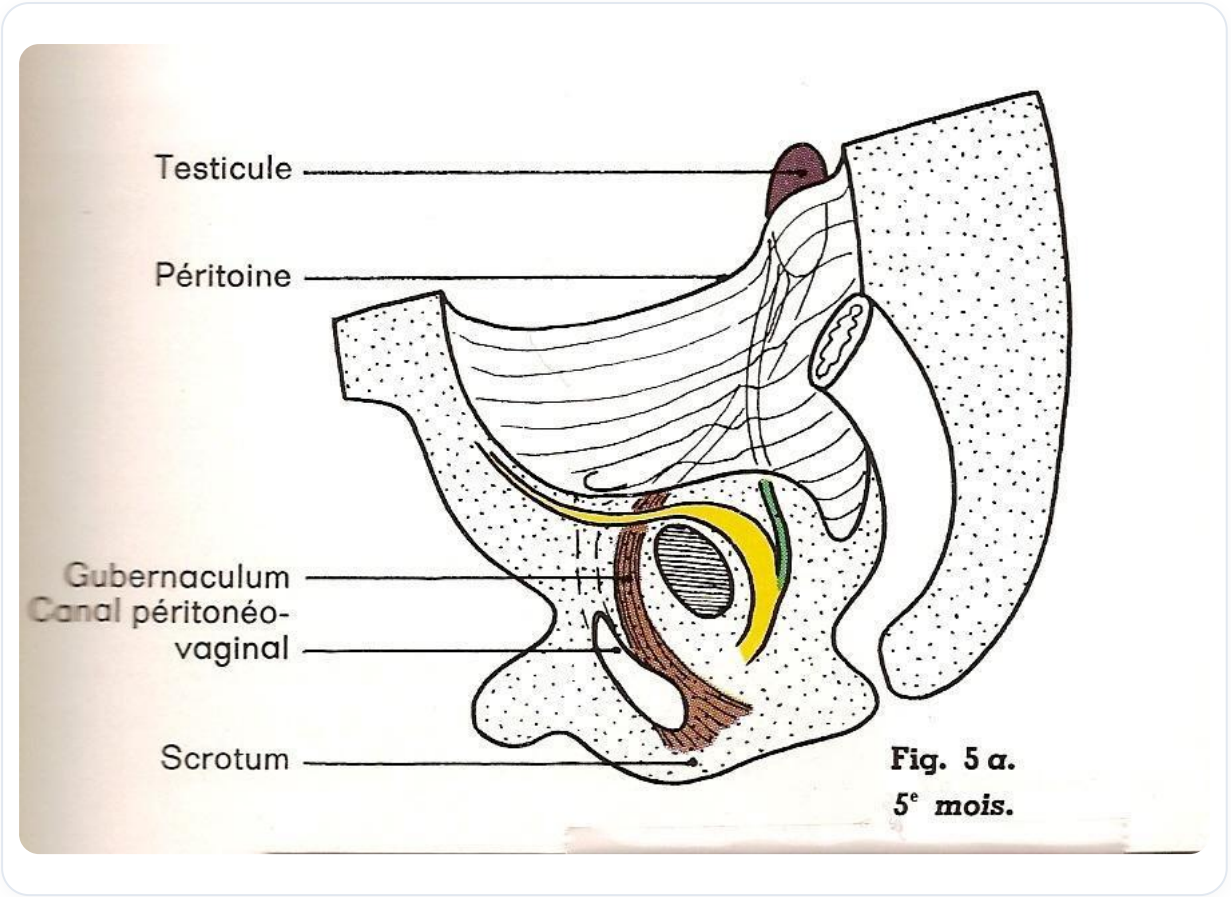


FIG. — Fetal testicular descent

This attachment between the gubernaculum ligament and the lower layers, combined with differential growth, causes testicular descent. It is in fact an overall growth that gives the impression of a reduction in the ligament, whereas it is the rest that grows.

The initial suspension of the testes up to the diaphragmatic plane, and more precisely to the **duodeno-pancreatic** junction, is a crucial point to review.

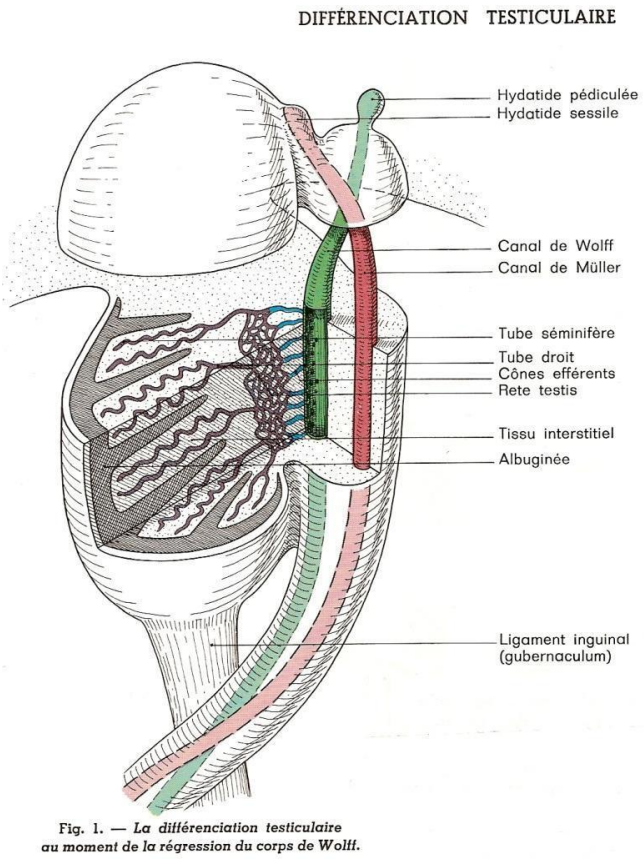
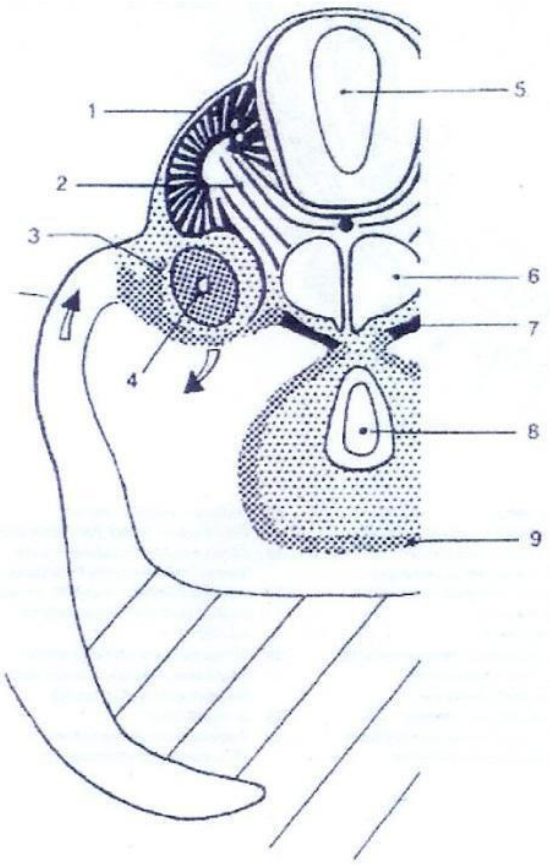


Fig. 1. — *La différenciation testiculaire au moment de la régression du corps de Wolff.*

FIG. — Testicular differentiation



Embryo, age approximately 27 days and 2,7 mm

1. Dermatom
2. Skelrotom
3. Ductus of wolff
4. Inner part of mesonephros
5. Neural tube
6. Aorta
7. Gonad
8. Intestinal tube
9. Limiting tissue body cavity

FIG. — Embryo 27 days

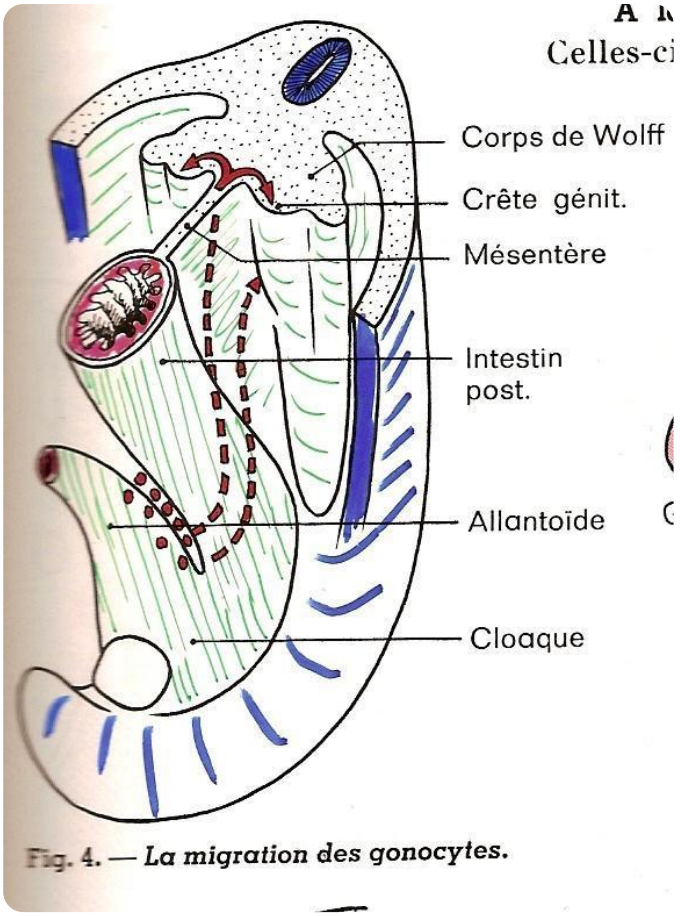


Fig. 4. — *La migration des gonocytes.*

FIG. — Migration of gonocytes

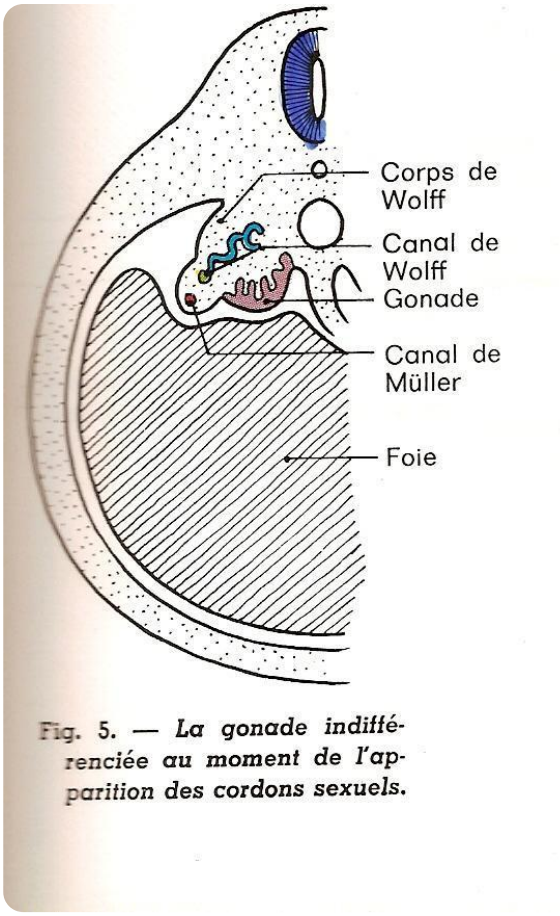


Fig. 5. — *La gonade indifférenciée au moment de l'apparition des cordons sexuels.*

FIG. — Undifferentiated gonad

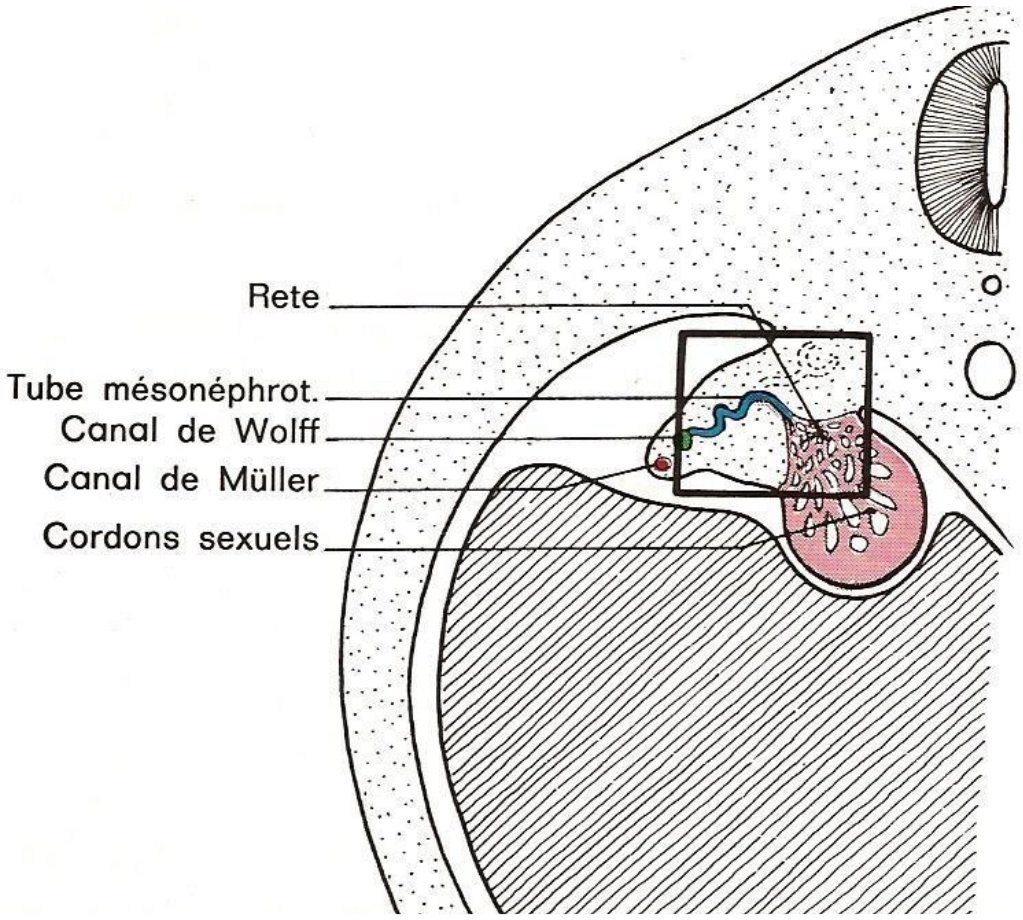


FIG. — Embryonic gonadal differentiation

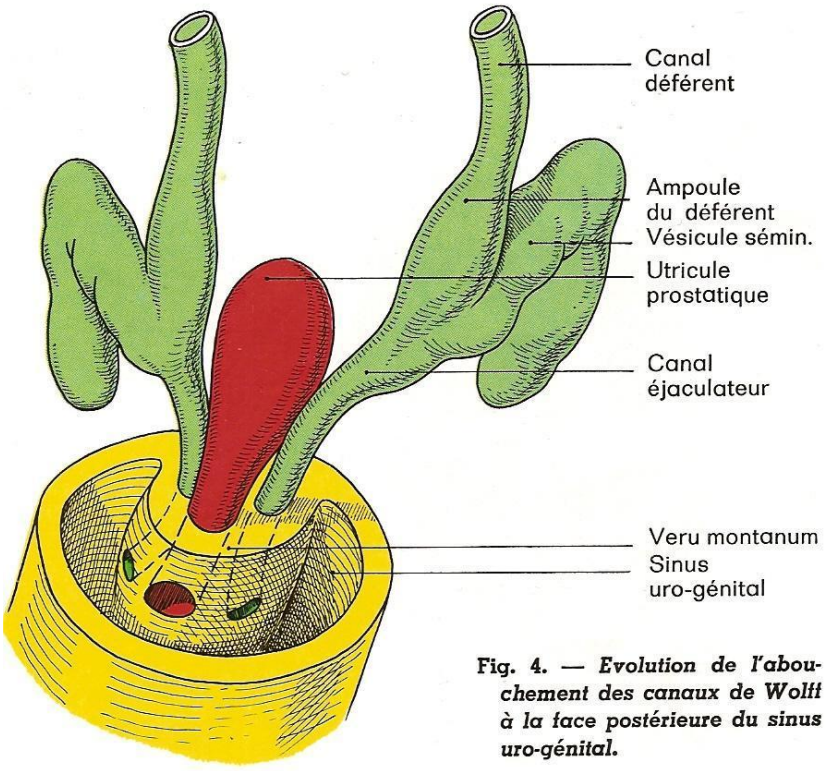


Fig. 4. — *Evolution de l'abouchement des canaux de Wolff à la face postérieure du sinus uro-génital.*

FIG. — Wolffian ducts

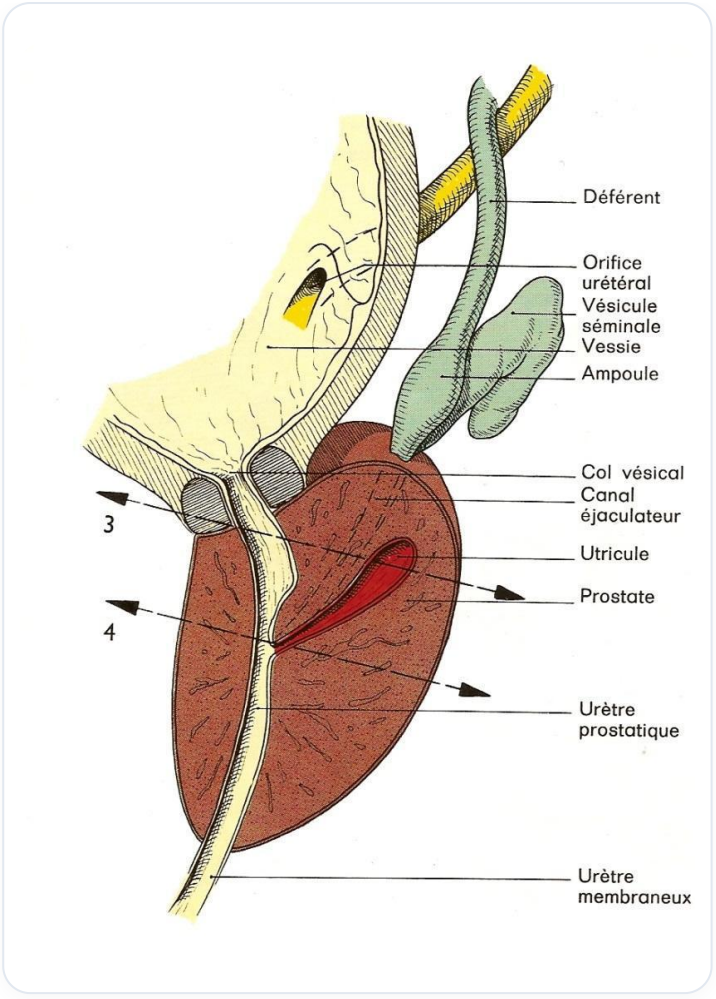


FIG. — Male prostate anatomy

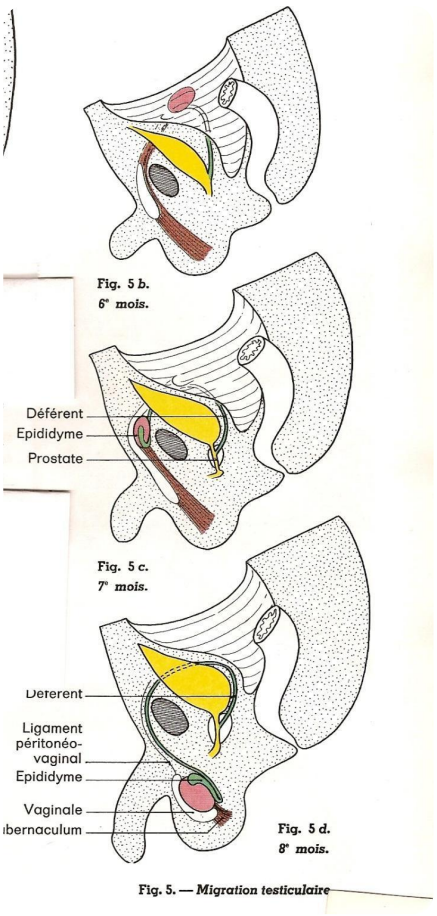


FIG. — Testicular migration

31. THE FEMALE GENITAL SYSTEM

DURATION : 06:15 STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the development of the female reproductive system, starting with the formation of the Wolffian and Müllerian ducts, and their impact on reproductive structure. The focus is on the uro-genital relationship, which differs between men and women, highlighting the creation of the uterus and fallopian tubes. We also explore the role of the broad and ovarian ligaments, as well as the importance of kidney health in hormonal regulation and fertility. The video addresses fascial continuity in the pelvis and how anomalies such as fibroids can influence overall body dynamics, providing crucial information for osteopathic practitioners and those interested in embryonic development.

THE FEMALE GENITAL SYSTEM

1. MESONEPHRIC DUCT (WOLFFIAN DUCT) AND PARAMESONEPHRIC DUCT (MÜLLERIAN DUCT)

The **Wolffian duct**, also known as the **mesonephric duct**, is the first duct to appear in the embryo.

In males, this duct descends and is integrated between the testes to form the **ejaculatory duct**. In contrast, the Müllerian duct remains as a vestige and contributes to the formation of the core of the prostate.

In females, the situation is reversed:

- The Wolffian duct differentiates into **Gardner's organs**, which are vestiges.
- The Müllerian duct, through fusion, becomes the **uterus** and **fallopian tubes**.

UROGENITAL DEVELOPMENT

The **urogenital** relationship is fundamental. In males, the epididymal duct, vas deferens, and ejaculatory duct form from this relationship.

In females, this development does not occur in the same way. The Müllerian duct meets at the midline to form the uterus.

Gardner's organs represent the vestiges of the Wolffian duct in females. The ejaculatory duct is oriented towards the base, and the further away it is, the more it will form the uterus. The closer it is, the more the vestige will be the **prostatic utricle**.

2. THE BROAD LIGAMENT AND OVARIAN LIGAMENTS

In both males and females, the **posterior parietal peritoneum** concentrates to form the **broad ligament**, as well as the small ovarian ligaments and round ligaments.

It is the same tissue, but with different concentrations. This posterior parietal peritoneum will form the uterus in females.

In males, this movement does not exist, and only a concentration of peritoneum remains.

3. THE KIDNEY AND ADRENAL GLAND

The kidney is located below the **adrenal gland**, separated by a capsule. This means that a **renal ptosis** does not lead to adrenal ptosis.

They are separated by a fascia called **Glisson's capsule**. The kidney requires a fatty space.

In very thin women, kidney ptosis can lead to menstrual irregularities and reduced fertility. **Revitalising a kidney** also means revitalising its fertility.

EMBRYONIC DEVELOPMENT OF THE KIDNEY

Initially, in a small embryo, the kidneys are enormous, as is the liver. This is reminiscent of brain development, which is very developed compared to the trunk at first.

These proportions evolve during development. The liver plays an important role in the development of the adrenal glands.

This is the first time that a **structural differentiation** between male and female is observed, linked to the greater development of the glands that create the **Müllerian tubercle**.

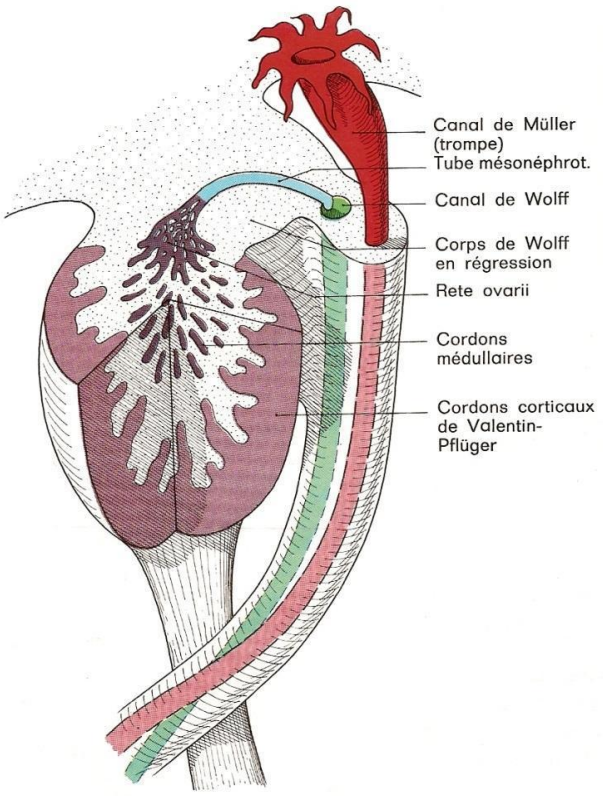


Fig. 1. — *Vue schématique de la différenciation ovarienne au moment de la régression du corps de Wolff.*

FIG. — Ovarian differentiation

4. DEVELOPMENT OF THE MÜLLERIAN SYSTEM (5TH WEEK)

By the fifth week, the system on the side begins to develop, restoring full power to the kidney.

The brain influences the establishment of the heart, which in turn establishes **diaphragmatisation**, the liver, adrenal glands, gonads, and the ridge.

The Müllerian duct develops much more strongly in females to create a pure canal, which joins to form the uterus.

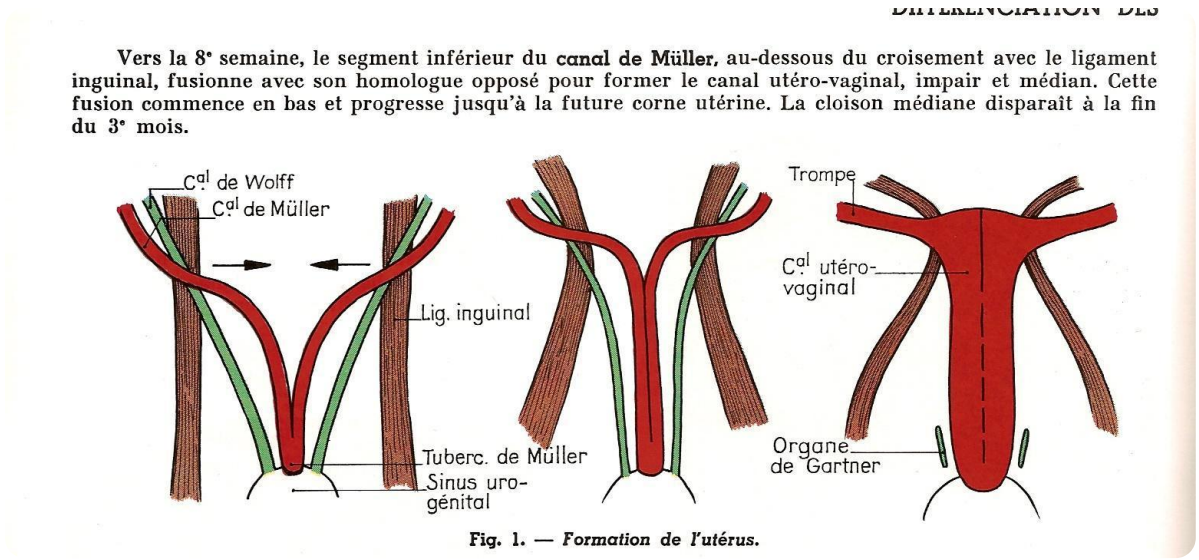


FIG. — Formation of embryonic uterus

In males, development is less pronounced, and what remains at the bottom of the Müllerian duct constitutes the core of the prostate.

TESTICULAR DESCENT

Testicular descent is due to the differentiated growth rate at the **gubernaculum**, which slows down and pulls the entire system.

This fascial continuity between the uterus and the entire part of the fascia, skin, and small pelvis is essential.

5. FASCIAL CONTINUITY AND INFLUENCE OF STRUCTURES

The uterus, ligaments, ovaries, and **peritoneal cavity** are clearly visible.

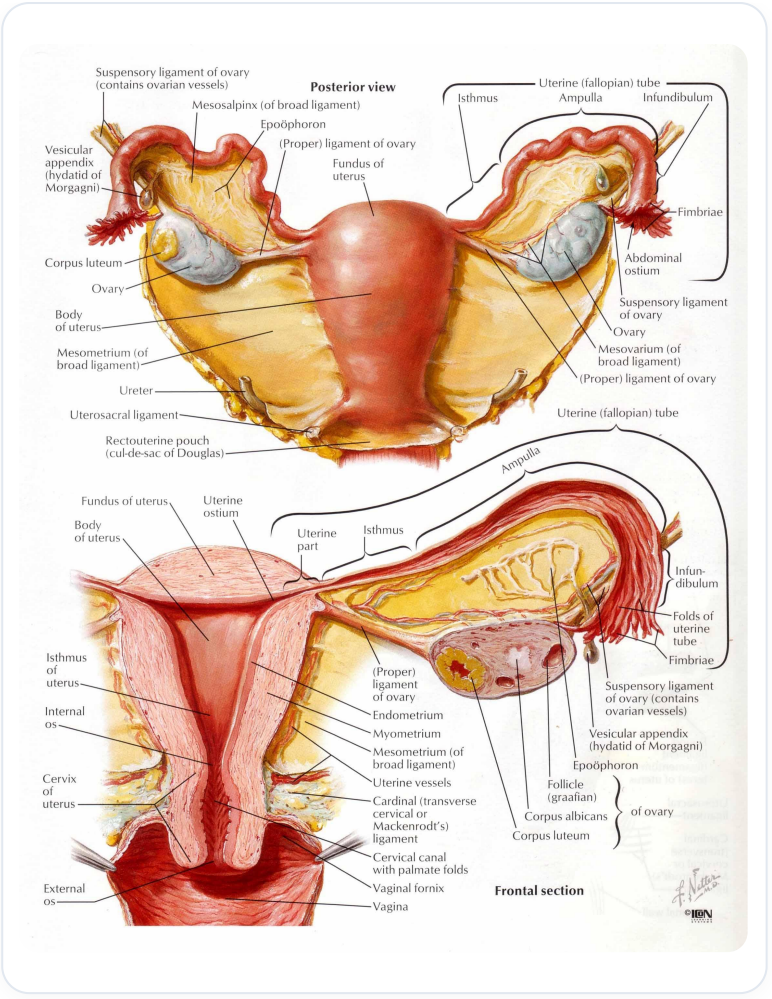


FIG. — Female reproductive system

By lifting the uterus with forceps, one can see that everything moves.

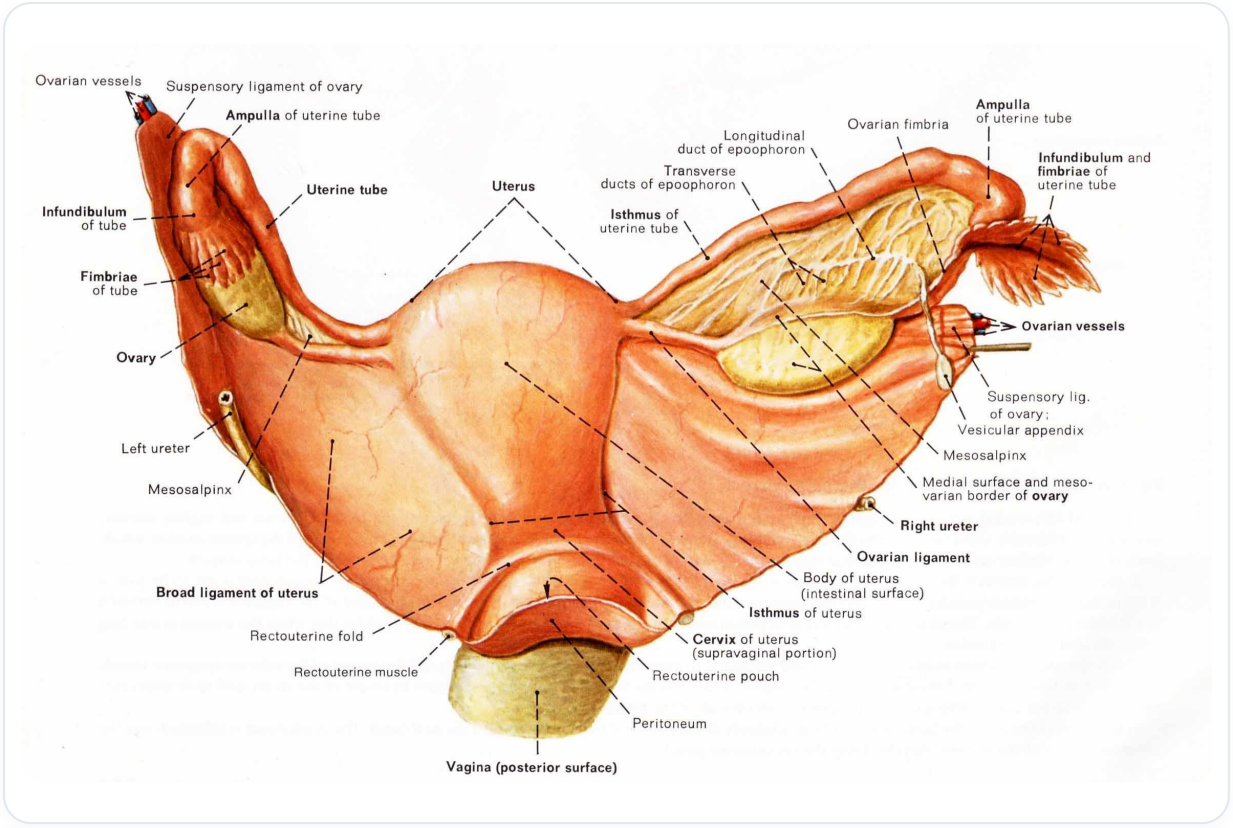


FIG. — Female reproductive system

The impact of a **fibroid** or a **cyst** is significant. Continuity with the ligaments is paramount.

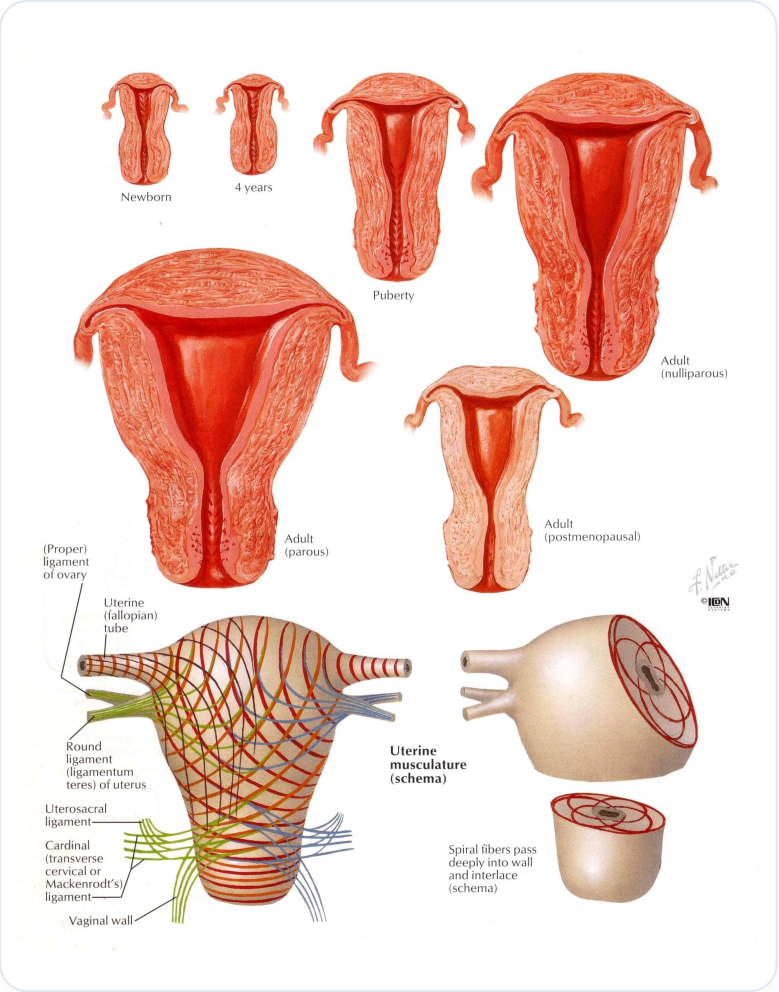


FIG. — Uterus and musculature

When pulled harder, one can see how the entire system reacts, including the round ligaments.

The peritoneum is a very intelligent tissue. The influence of a fibroid, for example, can modify the entire downstream system through a simple traction.

A large fibroid can lead to a cascade of consequences throughout the system.

ial de Müller ouvert dans la cavité péritonéale devient le pavillon de la trompe (fig. 1 et 2).

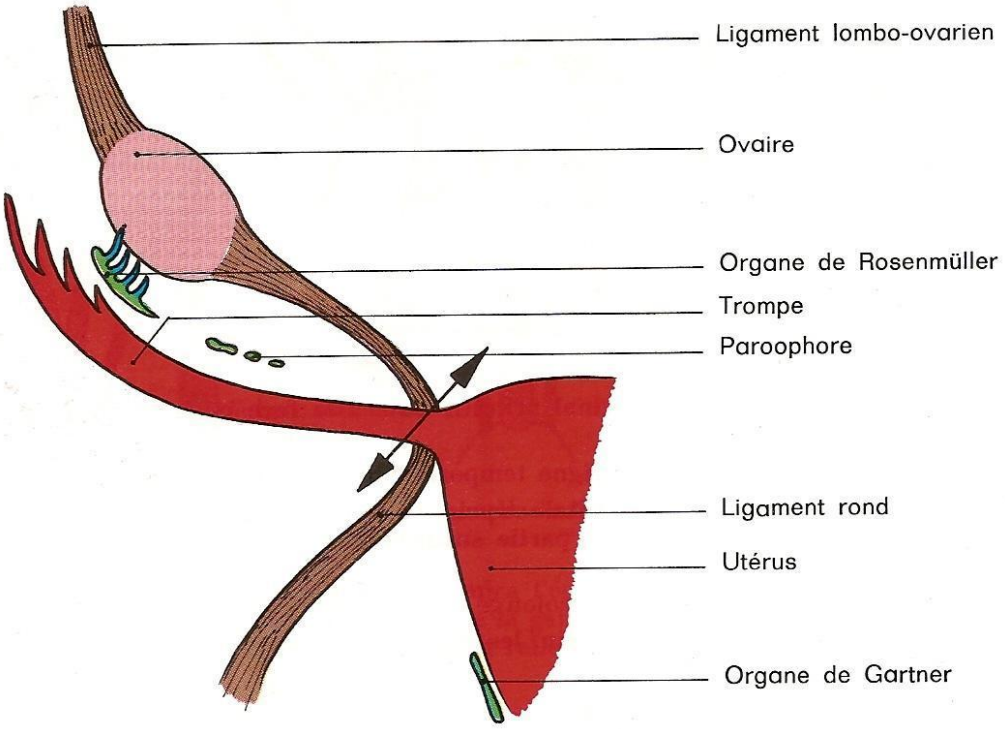
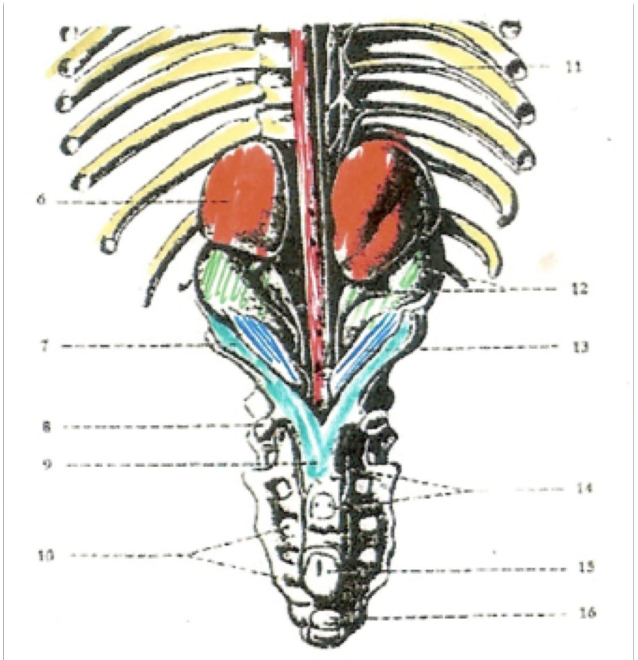


Fig. 1. — *Formation de la trompe et résidus wolffiens.*

FIG. — Female genital organs



6. Glandula suprarenalis
7. Müllerian duct
8. Lig.teres uteri
9. Uterus
10. For.sacralia ant.
11. Ramus communicans
12. Ren. Pelvis renalis
13. Ovarium
14. Urether
15. Anus
16. Anlage Os coccygis

FIG. — Fetal genital system

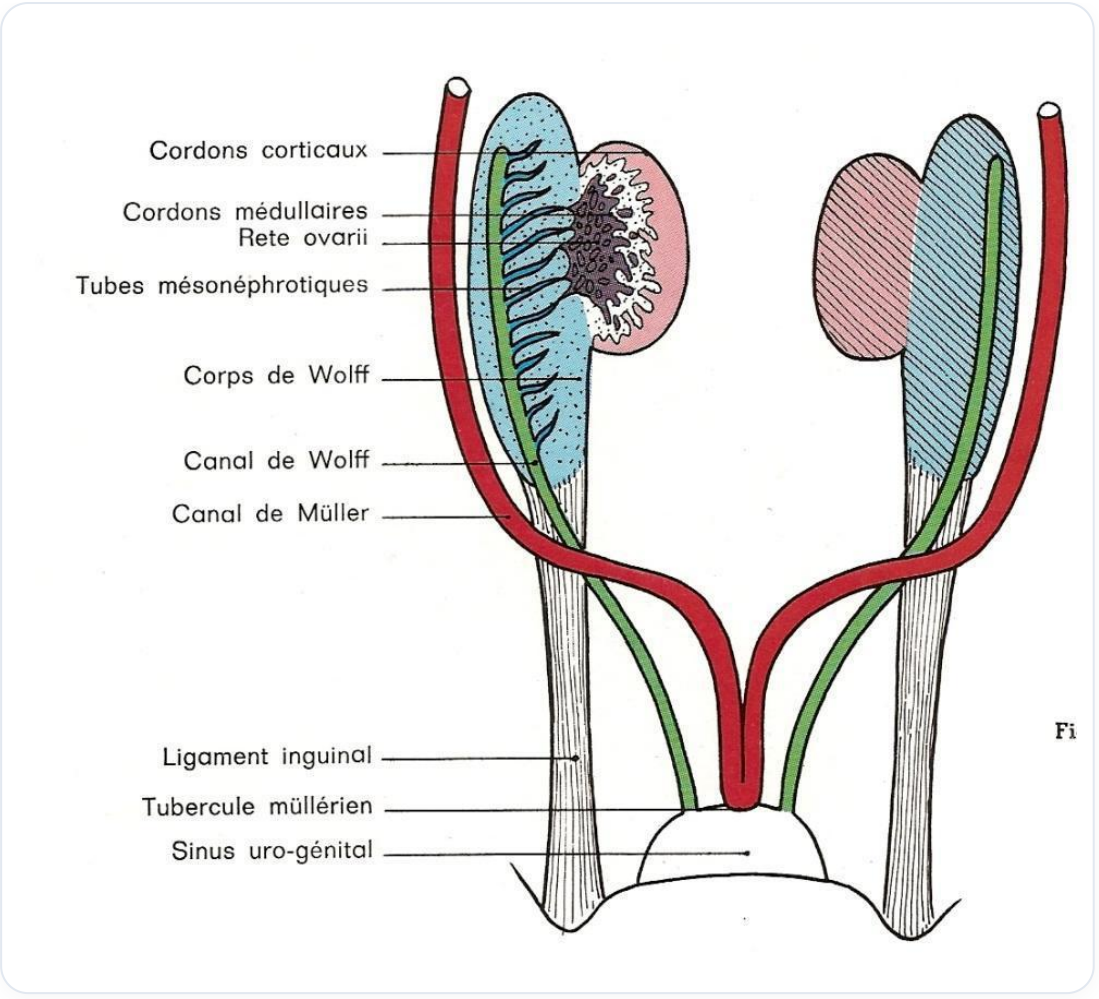


FIG. — Undifferentiated genital system

32. THE LOWER GENITAL APPARATUS

DURATION : 08:54 STUDY SHEET

COURSE SUMMARY

In this immersive video on the Lower Genital Tract, we explore the embryology of the vagina and uterus, highlighting the posterior growth dynamics that form these essential structures. Gynaecological balance is approached through the analysis of movements between the sacrum and the peritoneum, and how these interactions affect the health of the reproductive organs. Focusing on imbalances such as miscarriages or endometriosis, the video proposes fascinating therapeutic techniques based on concepts of energetic cleansing, fascial balance, and transgenerational links. This approach allows practitioners to learn how to

THE LOWER GENITAL TRACT: A BIODYNAMIC APPROACH

EMBRYOLOGY OF THE VAGINA AND UTERUS

The insertion of the **bladder** at the **Müllerian tubercle**, located in the urogenital sinus, is essential. This area includes the bladder, urethra, and the attachment to the vagina.

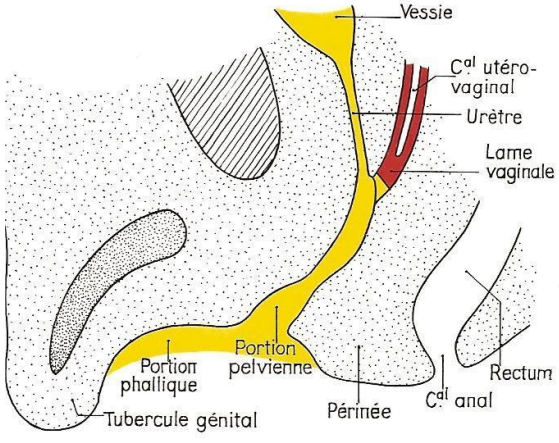


Fig. 2. — *Ouverture de la membrane uro-génitale.*

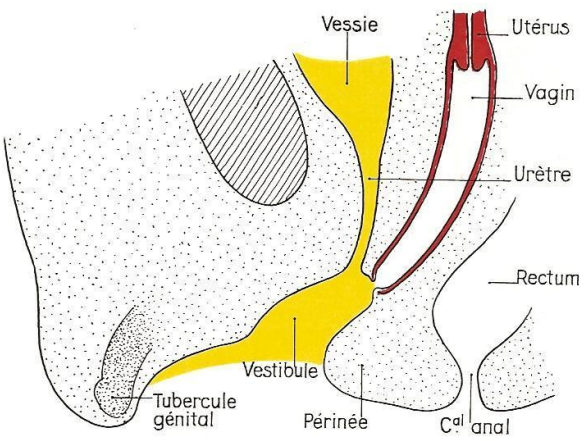


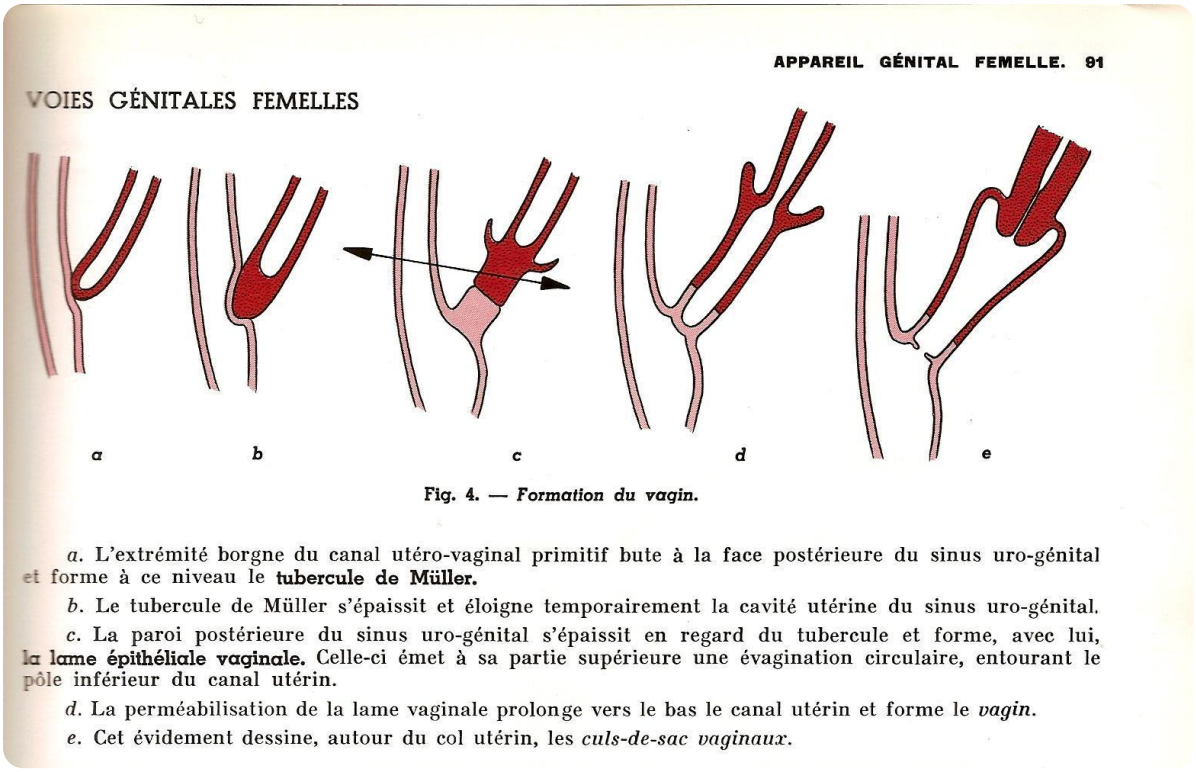
Fig. 3. — *Le vestibule définitif.*

FIG. — Development of female urogenital tract

The vagina derives from the **mesoderm**, specifically the Müllerian duct. The formation of the **vaginal plates** results from the posterior growth of the sacrum and the neural tube.

The differential growth rate of this posterior part creates the opening space of the vagina. It is the **postero-anterior** movement, between the kidney and the sacrum, that allows this extension.

The Müllerian tubercle thickens and is then stretched, not by superficial traction, but by the faster push of the posterior part. This phenomenon leads to the lengthening and formation of the vagina and cervix.



APPAREIL GÉNITAL FEMELLE. 91

VOIES GÉNITALES FEMELLES

a b c d e

Fig. 4. — *Formation du vagin.*

a. L'extrémité borgne du canal utéro-vaginal primitif bute à la face postérieure du sinus uro-génital et forme à ce niveau le **tubercule de Müller.**

b. Le tubercule de Müller s'épaissit et éloigne temporairement la cavité utérine du sinus uro-génital.

c. La paroi postérieure du sinus uro-génital s'épaissit en regard du tubercule et forme, avec lui, **la lame épithéliale vaginale.** Celle-ci émet à sa partie supérieure une évagination circulaire, entourant le pôle inférieur du canal utérin.

d. La perméabilisation de la lame vaginale prolonge vers le bas le canal utérin et forme le *vagin*.

e. Cet évidement dessine, autour du col utérin, les *culs-de-sac vaginaux*.

FIG. — Vaginogenesis

The posterior wall of the urogenital sinus thickens, giving rise to the vaginal plates which open and stretch. This dynamic is similar, with the posterior part growing faster.

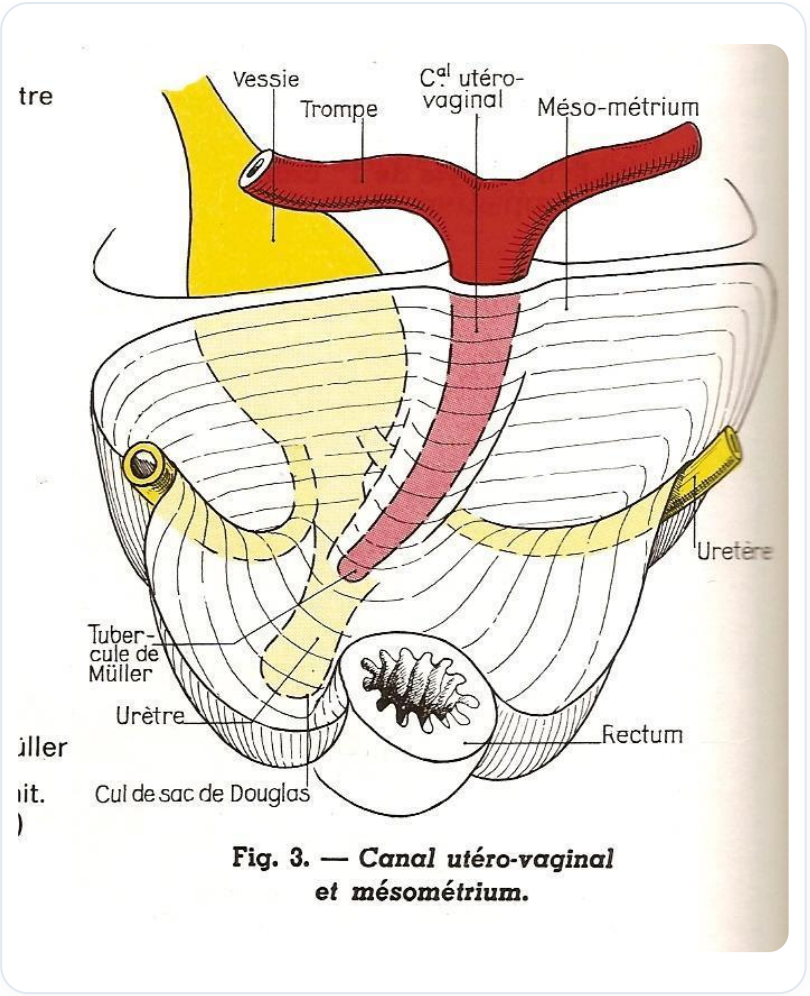


Fig. 3. — *Canal utéro-vaginal et mésométrium.*

FIG. — Uterovaginal canal

GYNAECOLOGICAL BALANCE AND BIODYNAMIC MOVEMENT

During the examination of the uterus, a movement between the **sacrum** and the **peritoneum** is perceived. It is in the interaction of these two poles that vaginal and uterine balance is found.

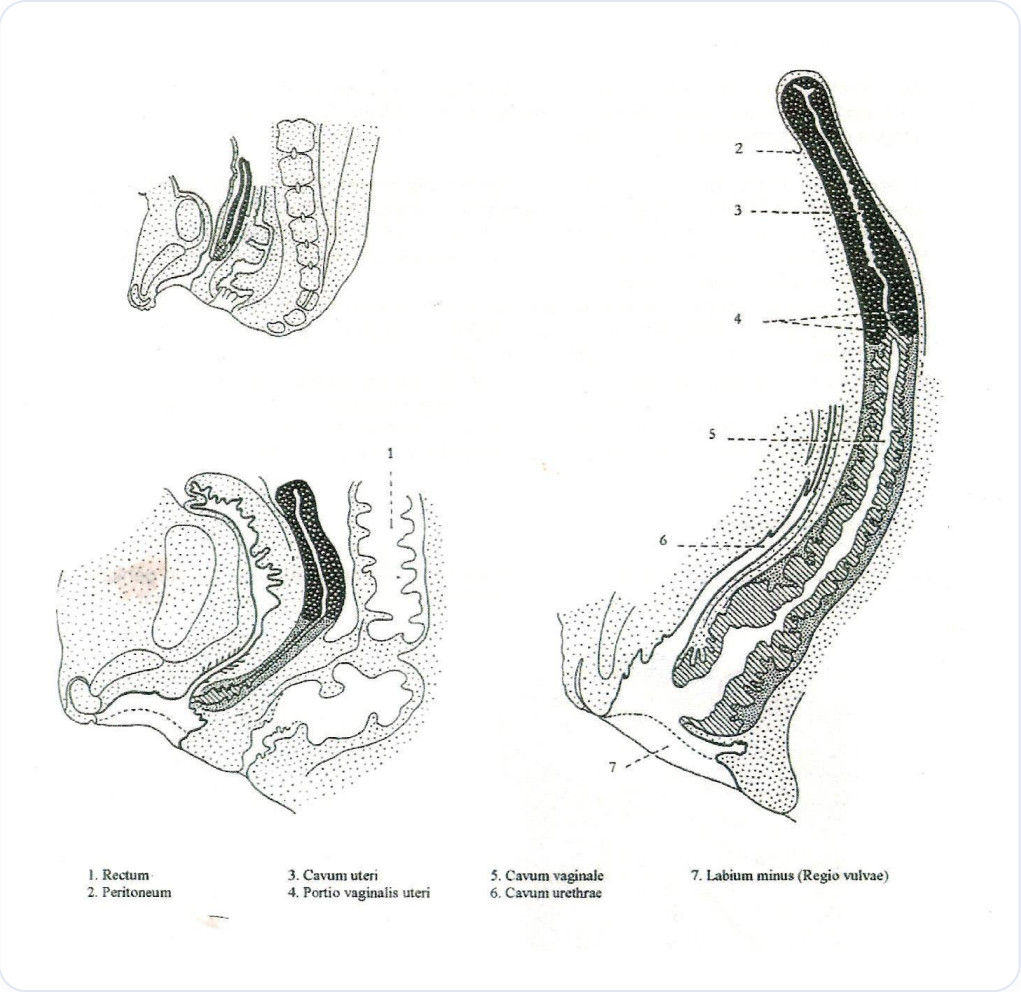


FIG. — Female uterine anatomy

At the gonadal level, an inverse movement is felt in the abdomen, following the ligament towards the pubis, like a descending wave.

The descent of the sacrum is influenced by the neural tube. Another crucial fulcrum is the combination of the pubis and the sacrum, forming an essential fascial balance of the female lesser pelvis.

This balance involves a descending line of force relative to the sacrum, influencing various functions of the female body.

THERAPEUTIC APPROACH TO UTERINE IMBALANCES

Situations such as **miscarriages** leave an energetic imprint in the uterus. Energetic cleansing of the uterus then becomes necessary.

We can approach the uterus with a **magnetic force**, learning to feel the energetic field at a distance. The practitioner perceives a frequency, a zone of heat that densifies, a sign of the energy present.

My approach consists of "scanning" and identifying the areas requiring cleansing. This involves restoring freedom to the tissues and energies.

A **retroverted uterus**, for example, can lead to congestive phases. The remote work aims to rebalance the gynaecological system, in close connection with **gastrin**.

Endometriosis, characterised by a proliferation of tissue ascending into the peritoneum, is often associated with lower back and sacroiliac pain.

THE INTRAPERITONEAL CAVITY AND THE TRANSGENERATIONAL DIMENSION

In women, there is a direct link between the cervix, the fallopian tubes, and the **intraperitoneal cavity**. This cavity is of great importance, perceived as a "sky" or a primordial space.

Each state of the body represents the best possible functional response at a given moment. This response is influenced by our lives, but also by those of our ancestors.

A retroverted uterus, for example, may be the best response the body has found to resolve a deep **transgenerational** or personal problem.

It is essential to understand why a transgenerational pattern manifests in this way. Therapy aims to decode this "best response" and offer the necessary release, if requested.

The embryo perfectly illustrates this notion: at each stage of development, it presents the best possible physiological response.

SECONDARY NEURULATION AND THE FORMATION OF THE FILUM TERMINALE

The **neural tube** is associated with a posterior caudal eminence, creating a space of light and connection. The caudal bud and the terminal part of the **notochord** are observed.

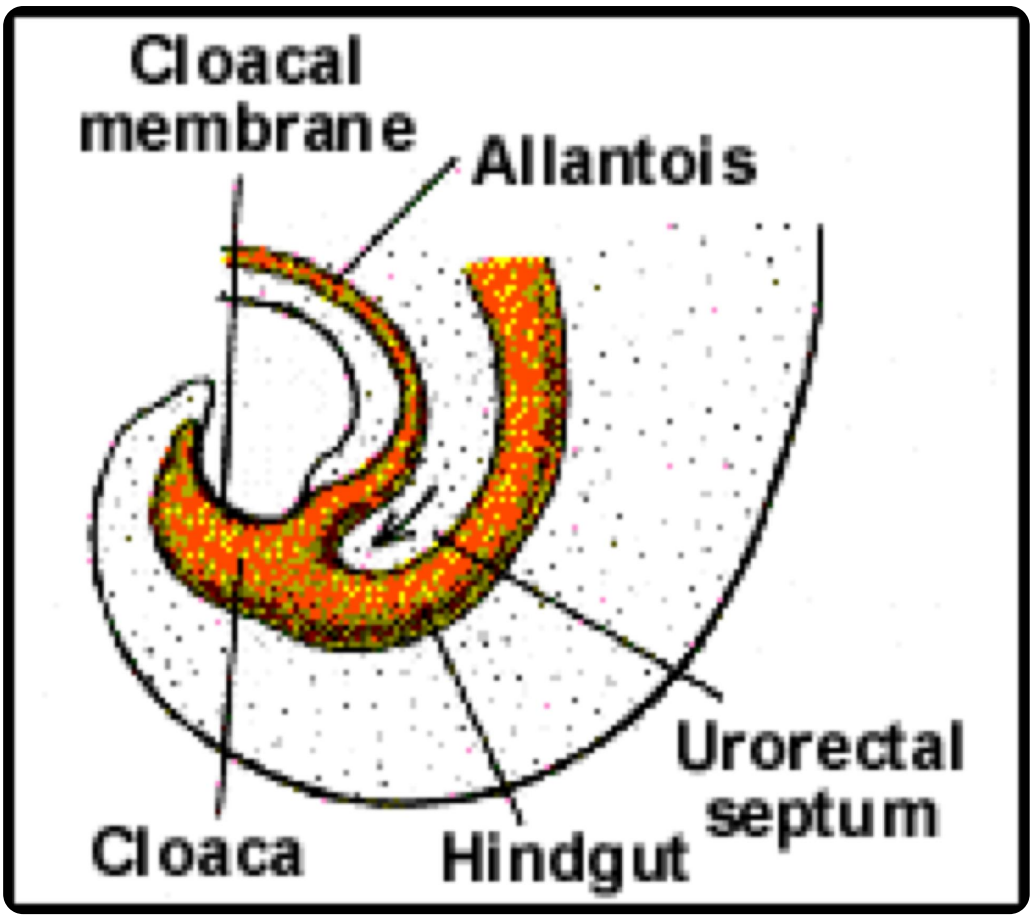


FIG. — Embryonic cloacal development

The space of the rectum and the neural tube with the posterior neuropore are in close relation. A connection forms thanks to a more pronounced developmental movement in the posterior part.

This involves a closure then a reopening to establish this connection. This process leads to the emergence of the posterior neuropore and the lengthening to form the **filum terminale**.

This is called **secondary neurulation**. Manipulation of the sacrum reveals these differential growth rates, where the initial junction lengthens posteriorly.

The integration of the caudal eminence ensures continuity with the neural tube, forming this ascending secondary neurulation.

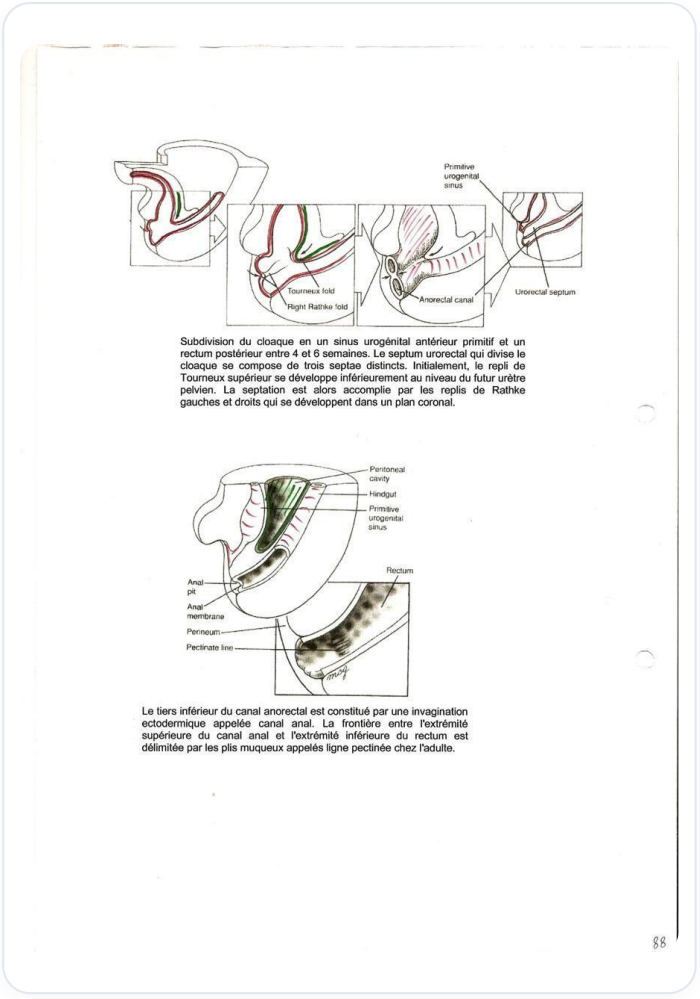


FIG. — Cloacal septation

DEVELOPMENT OF THE PERINEUM AND HEPATIC INFLUENCE

At the level of the **perineum**, we observe the emergence of the lower limbs and a traction that is exerted. This movement is deeply linked to the uterine system.

The tilting of the **liver**, in synergy with the descent of the neural tube and the sacrum, influences the developmental movement that pulls towards the linea alba, via the falciform ligament.

This hepatic development causes a tilt that affects the entire developing uterine area. It could explain phenomena of migration or differential traction related to uterine position.

A ligament develops between the pubis and the umbilicus, encapsulating the history of uterine descent and formation.

Just as the sternum contains the history of our flexion, this ventromedial ligament (on the linea alba) concentrates the history of all posterior formation, of the **gonadisation** and **uterisation** of the pelvis.

The linea alba is the meeting point of all **urogenital** history.

33. SYNTHESIS OF THE GENITAL SYSTEM

DURATION : 03:26 STUDY SHEET

COURSE SUMMARY

This video offers an in-depth synthesis of genital development, highlighting embryological analogies and the differentiation of male and female sexual organs. You will learn how undifferentiated structures transform, under hormonal influence, to form distinct organs such as the penis and clitoris, while integrating the processes of midline closure. The concepts of gonadal sex, embryonic ducts, and somatic implications of psychic conflicts are also addressed, allowing for an enriching therapeutic approach for practitioners.

SYNTHESIS OF THE GENITAL SYSTEM

The development of the **genital system** shows remarkable correspondences between male and female structures, originating from the same undifferentiated embryo. These structures evolve under hormonal influence, modulating their progressive closure along the **midline**.

DIFFERENTIATION OF EXTERNAL GENITAL ORGANS

Initially, the structures are undifferentiated and then specialise into **male** or **female sexual organs**.

- **Genital tubercle:**

- Male: Glans and shaft of the penis

- Female: Glans and body of the clitoris

- **Urogenital sinus:**

- Male: Penile urethra

- Female: Vestibule of the vagina

- **Urethral fold:**

- Male: Penis and penile urethra

- Female: Labia minora

- **Labioscrotal fold:**

- Male: Scrotum

- Female: Labia majora

The progressive closure of these structures along the midline is a key process. This begins with the separation between the **cloaca** and the **rectum**.

The differentiation of the external genital organs is clearly visible:

- The clitoris develops into the labia minora and majora.
- In males, the embryonic clitoris becomes the glans and penis.
- The embryonic labia minora close to form the scrotum.

This differentiation begins between the **fifth** and **eighth week** of pregnancy, concurrently with hepatic and adrenal development.

MIDLINE AND EMBRYOLOGICAL ANALOGIES

A posterior view of the embryo reveals the **falciform ligament**, the **teres ligament**, and the **median fold**. The linea alba is the continuity of this tissue, resulting from closure along the midline. This is the culmination of all this fusion movement.

There is an analogy between the formation of the genital organs and that of other median structures of the body:

- The nose develops like a uterus.
- The sternum develops like a uterine system.

All these systems converge towards the midline, whether it is the sternum, the face, or the genital organs. This is the closure of the anterior midline.

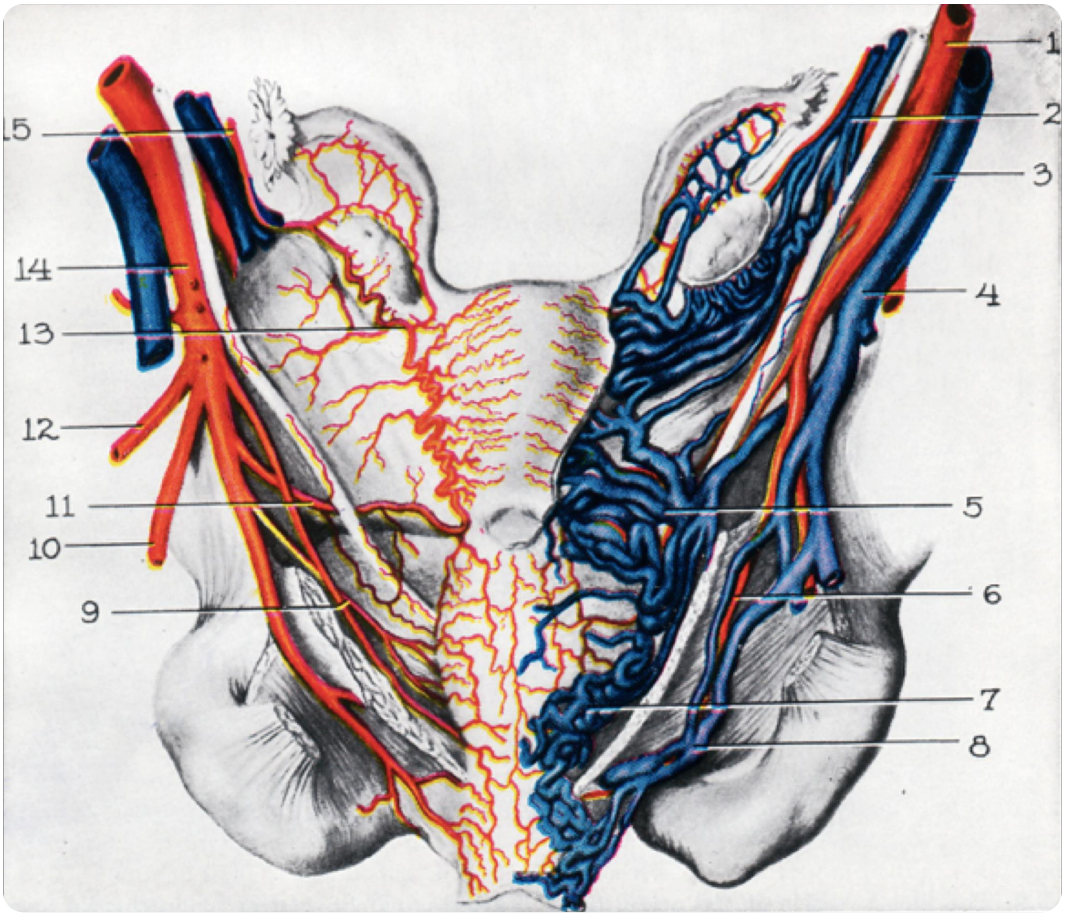


FIG. — Vascularization of cleft lip/palate

GONADAL SEX AND EMBRYONIC DUCTS

The development of the **genital ducts** depends on gonadal sex:

- **In the absence of androgen secretion (female):**

- The Wolffian duct regresses.

- The Müllerian duct develops to form the fallopian tube, uterus, and vagina.

- The urogenital sinus remains open (vestibule, labia minora). These structures, derived from the endoderm, will give rise to the urethra and the surrounding prostatic part.

- **In the presence of androgen secretion (male):**

- The Wolffian duct becomes the epididymis, vas deferens, seminal vesicle, and ejaculatory duct.

- The Müllerian duct regresses.

- The urogenital sinus closes.

The genital tubercle in males becomes the penis, and in females, the clitoris. The genital swellings form the labia minora and majora in females.

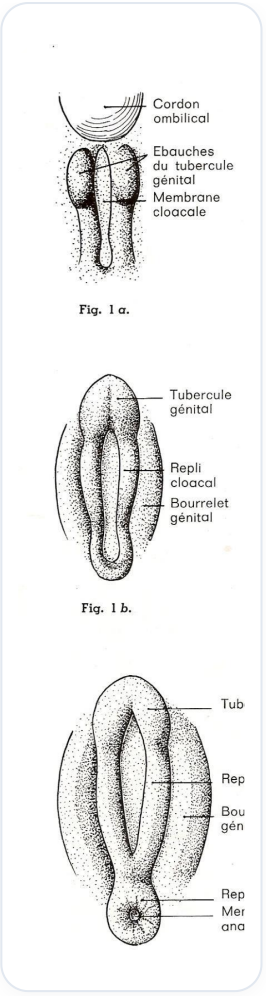


FIG. — Development of genital organs

SOMATISATION OF PSYCHIC CONFLICTS

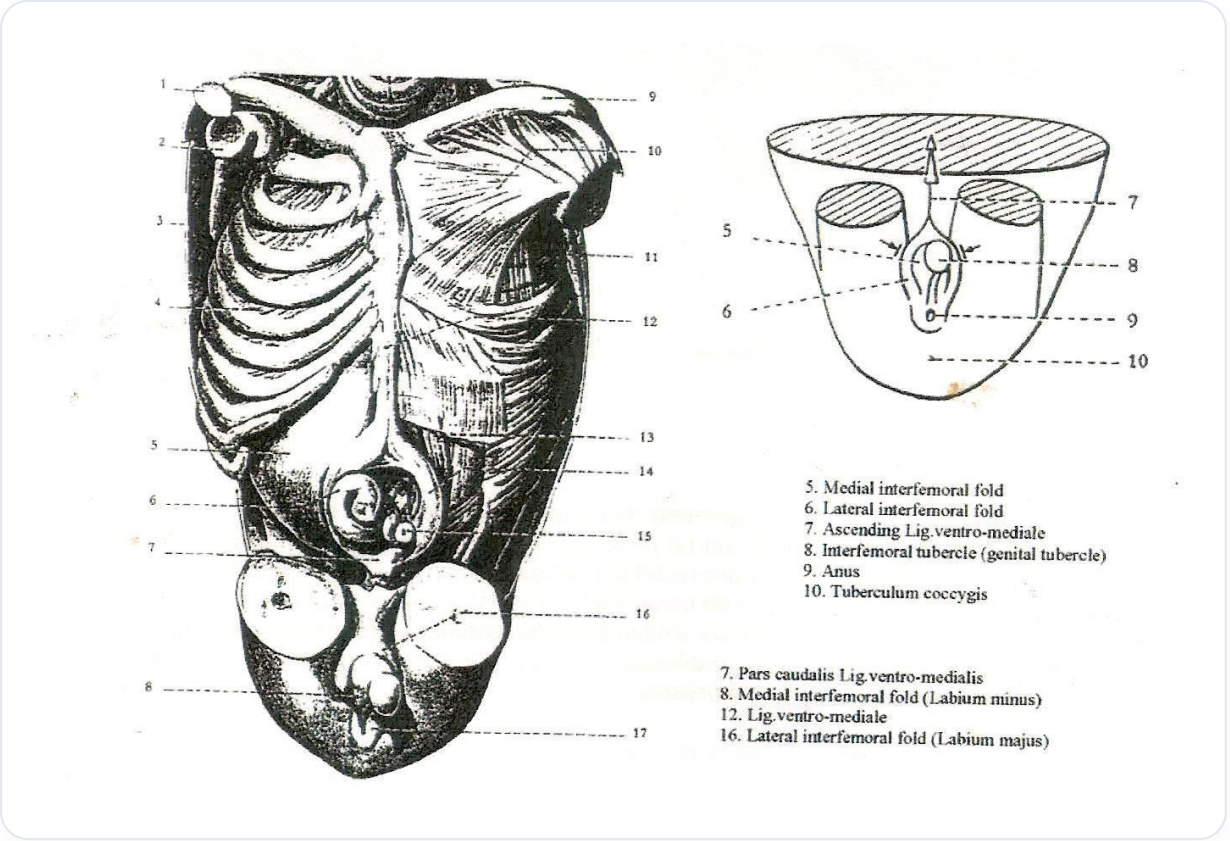


FIG. — External embryonic development

It is important to note that our body **somatises** psychic conflicts. If a conflict is not resolved, it can manifest physically according to its symbolism:

- **Separation:** Can affect the skin.
- **Urogenital, metabolic, rhythmic, or neurosensory conflict:** Will be expressed in the corresponding system.

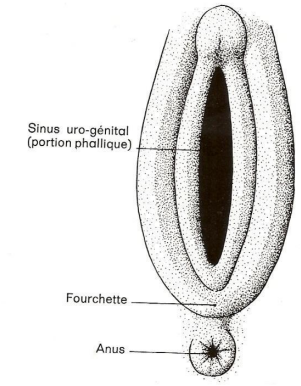


Fig. 3.

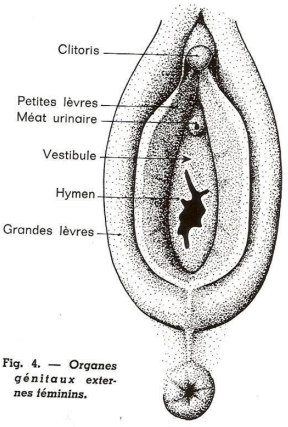


Fig. 4. — *Organes génitaux externes féminins.*

FIG. — Female genital organs

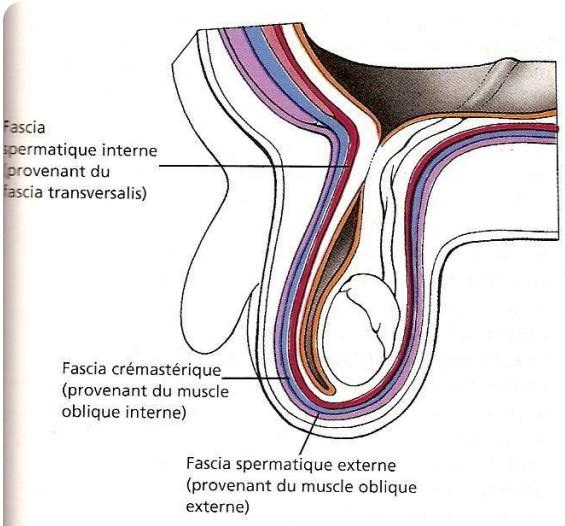


Fig. 10 - 21. Les trois feuillets évaginés de la paroi abdominale sont poussés dans le scrotum par le développement du processus vaginal et ils deviennent les trois enveloppes du cordon spermatique . Ces trois feuillets isolent le testicule et la vaginale dans un compartiment commun.

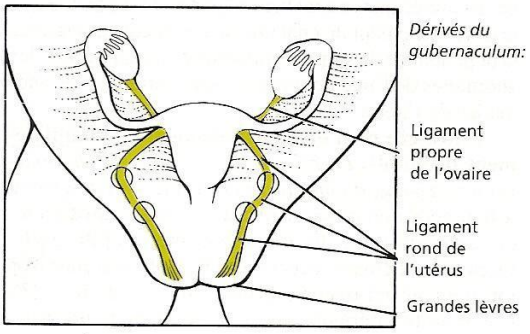


Fig. 10 - 22. Les ovaires descendent quelque peu au cours du développement et ils sont poussés vers les ligaments larges cependant que les conduits paramésonéphrotiques s'unissent pour former l'utérus. Dans le sexe féminin, le gubernaculum grandit à l'allure de celle du corps et il est attaché au conduit paramésonéphrotique au point de croisement avec la paroi postérieure du tronc. En conséquence, dans le sexe féminin, le vestige du gubernaculum unit les grandes lèvres de la vulve à la paroi de l'utérus et il est ensuite rabattu latéralement pour attacher l'ovaire.

FIG. — Descent of genital organs

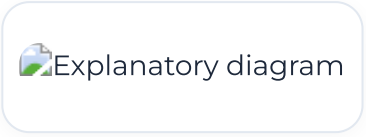


FIG. — Explanatory diagram

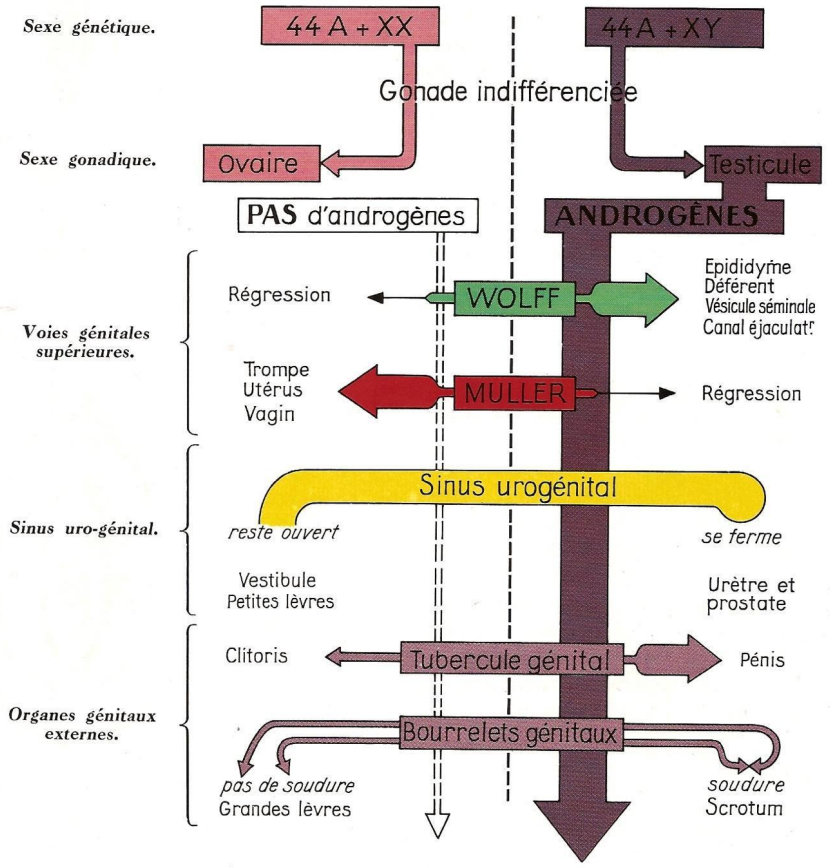


FIG. — Human sexual differentiation

Tableau 10 - 1. Développement des organes génitaux externes masculins et féminins

EBAUCHE PRESOMPTIVE	STRUCTURE MALE	STRUCTURE FEMELLE
Tubercule génital	Gland et corps du pénis	Gland et corps du clitoris
Sinus uro-génital définitif	Urètre pénien	Vestibule du vagin
Pli urétral	Pénis entourant l'urètre pénien	Petites lèvres de la vulve
Pli labio-scrotal	Scrotum	Grandes lèvres de la vulve

FIG. — External genital development

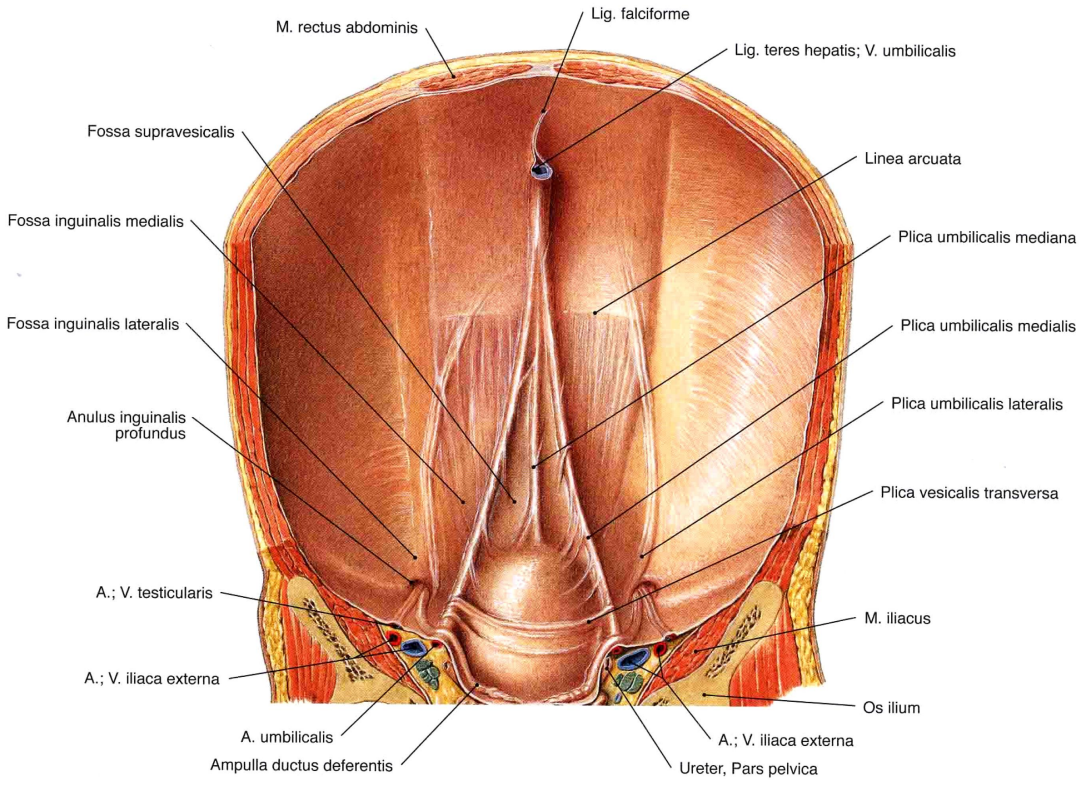


FIG. — Anterior abdominal wall

34. PRACTICE : THE GENITAL SYSTEM

DURATION : 03:16 STUDY SHEET

COURSE SUMMARY

In this video, you will learn two practical methods for balancing the female pelvis, essential for optimising the birth axis. The techniques focus on identifying the adductor tubercle and using support points to support the pelvis without excessive pressure. By exploring alternative approaches to repositioning the pelvis, the practitioner will also promote decongestion and improve circulation, taking into account the specific needs of pregnant women. You will discover the importance of fascia during childbirth, highlighting the risks of an unbalanced pelvis that could hinder birth.

THE GENITAL SYSTEM: BIODYNAMIC PRACTICE

I'm going to show you two ways to prepare the **female pelvis** to rebalance it and work on the **birth axis**. This approach is also valid for the **uterine** or **prostatic system**. However, there is a fundamental difference: men do not have to give birth or carry a child.

BALANCING THE FEMALE PELVIS

IDENTIFYING THE AXIS AND IMBALANCE

We will identify the **tubercle of the third adductor**, present on both sides. By referring to these two tubercles, I can determine the **pelvic axis** in space. If the pelvis is unbalanced, for example, if one side is slightly depressed and too short, this indicates an imbalance.

BIODYNAMIC REBALANCING TECHNIQUE

To balance this structure, I position myself **biodynamically**. I hold the woman's pelvis, supporting it at different points. The objective is to support the entire pelvis while having the axis in front of me. I then allow biodynamics to operate. I relate to my central line and take a precise fulcrum.

I work with a **support triangle**, but without applying excessive pressure. This awareness of support allows the pelvis to reposition itself in relation to the fulcrum.

ALTERNATIVE IN CASE OF DIFFICULTY

If this method is not sufficient, I can use another fulcrum. By palpating, I can already determine the birth axis, which is crucial. If the pelvis remains unbalanced, I position myself behind the patient and lean on the **ischium**. I then let the process unfold naturally, with the patient standing relatively upright. With a good fulcrum from my hands, I let the body work to rebalance problems in the lesser pelvis.

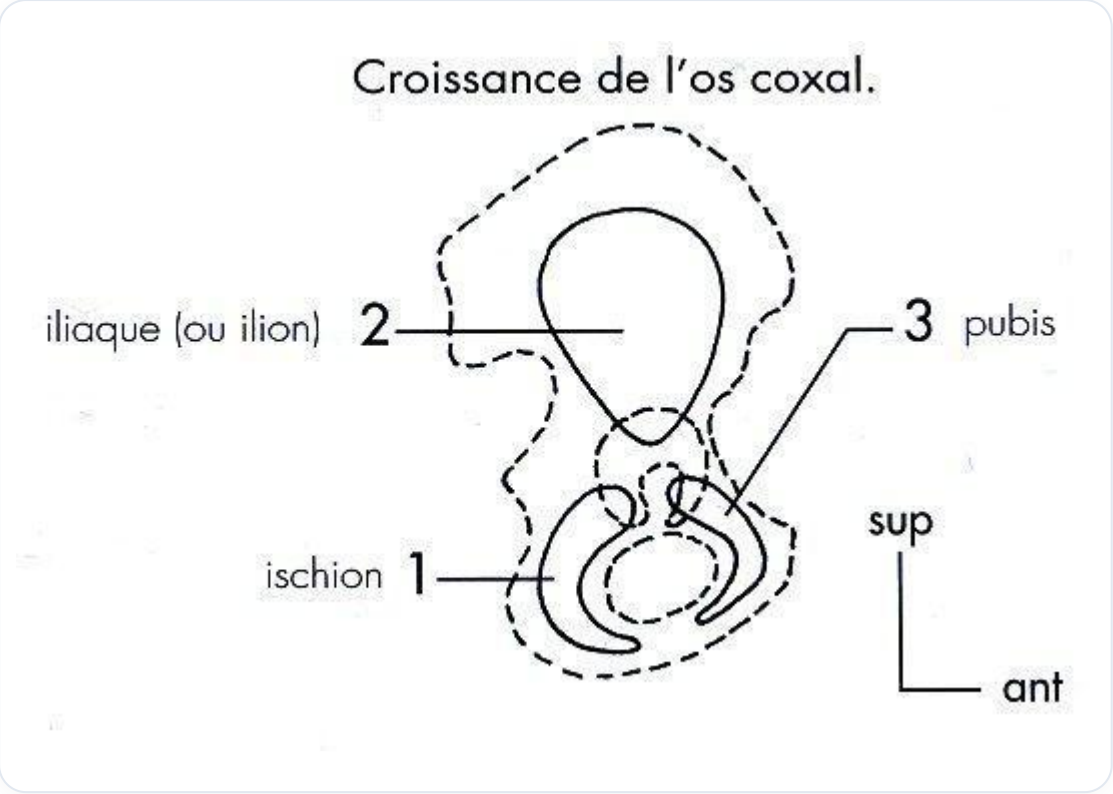


FIG. — Growth of pelvic bone

BENEFITS AND SPECIFIC CONSIDERATIONS

DECONGESTION AND CIRCULATION

This technique promotes **decongestion**. A good recentring of the pelvis helps to free the **vascular system**, thus facilitating the evacuation of congestion.

PREGNANCY AND POSITIONING

When a woman is pregnant, it is not always easy to position her on her back, especially if she suffers from severe lower back pain. In this case, it is preferable to avoid lying her down so as not to undo the work done during straightening. I prefer to let the space and **primary respiration** work.

CONTROL AND VERIFICATION

During pregnancy, I systematically check the birth axis and the freedom of the feet. I position myself behind if necessary and let the body rebalance itself; it is the surrounding space that will guide this process.

ROLE OF FASCIA IN CHILDBIRTH

It is fundamental to know that during birth, the child is guided by the **fascia of the pelvis**. If the pelvis is unbalanced, the child could become stuck during the birth movement.

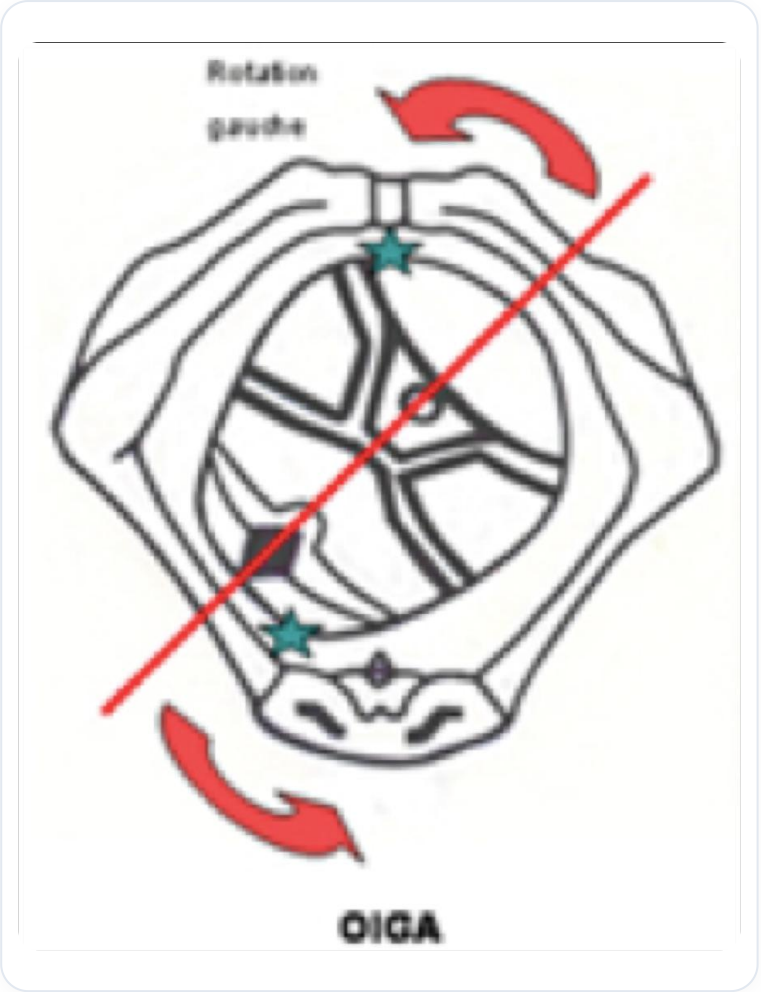


FIG. — OIGA engagement

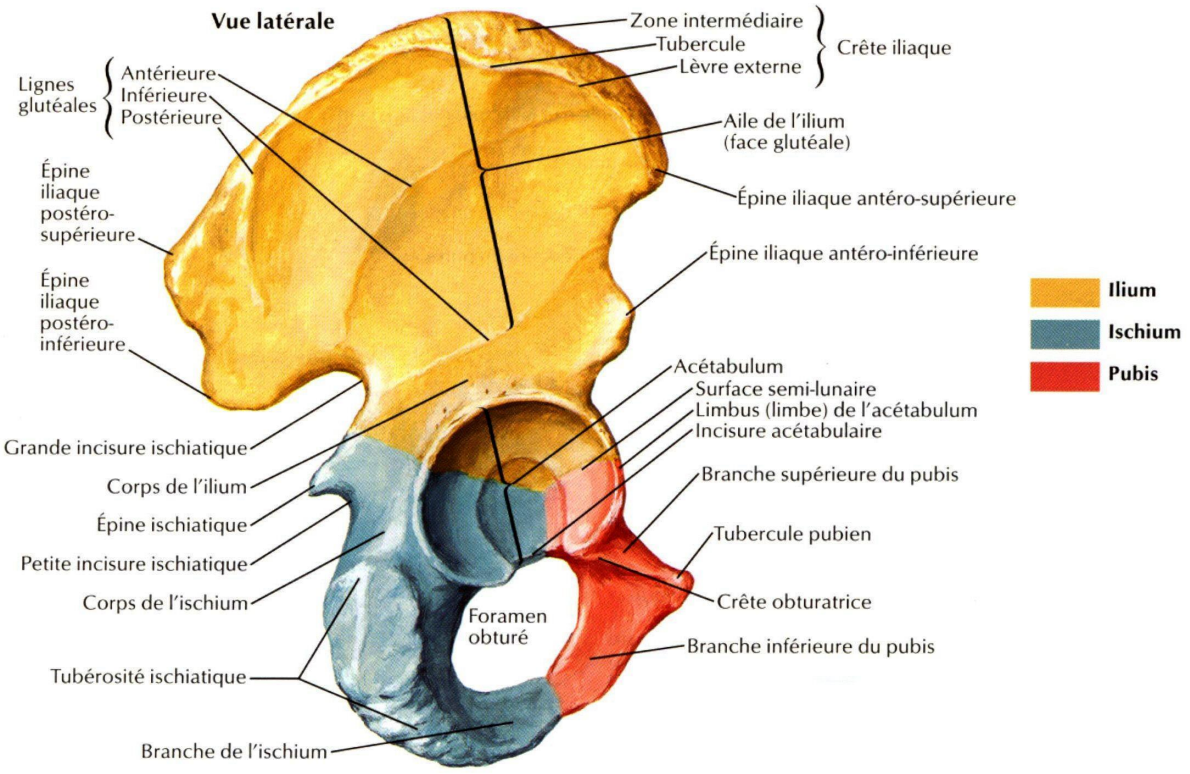


FIG. — Lateral hip bone

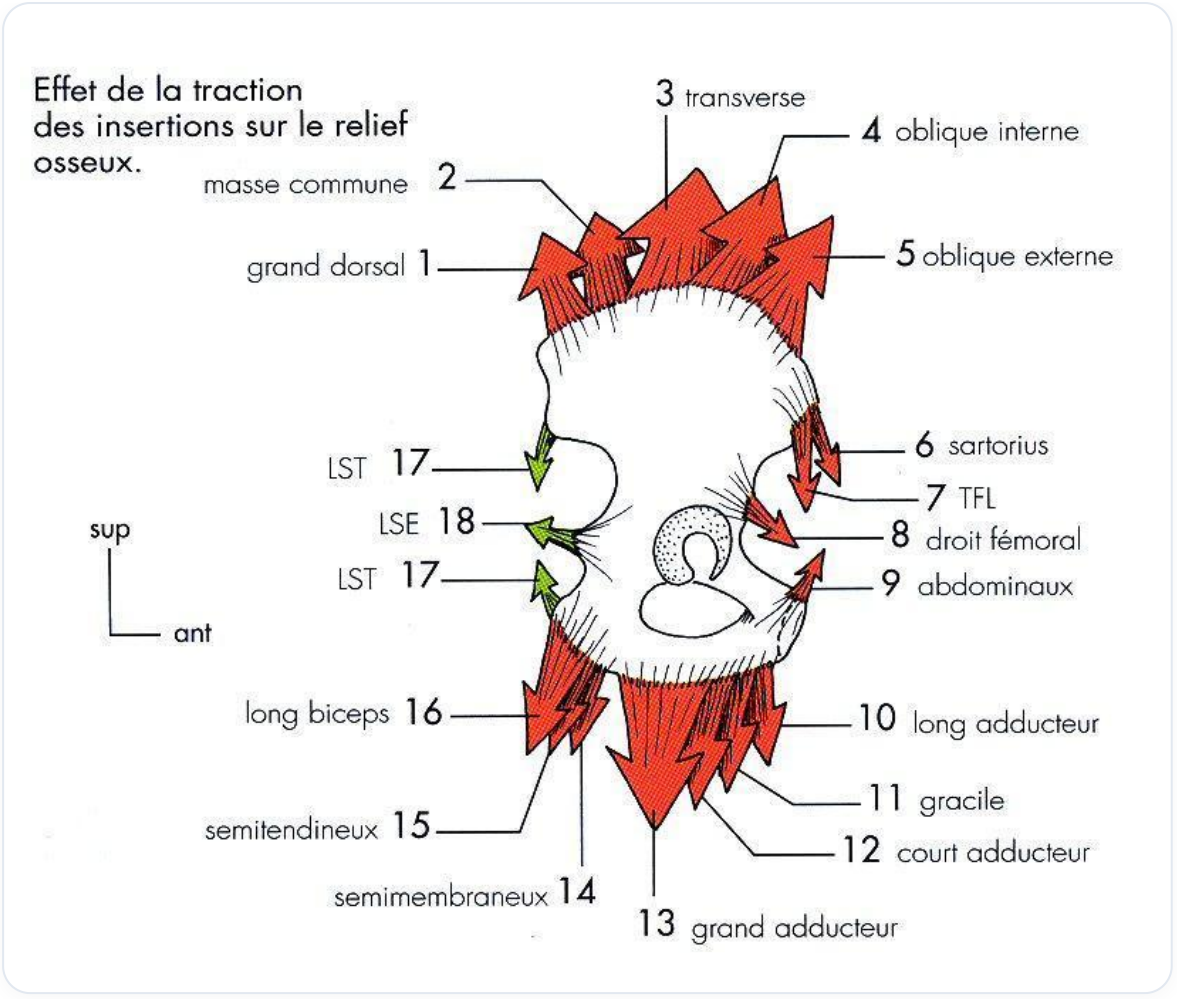


FIG. — Pelvic muscles

35. MEDITATION : 4 PRACTICES

DURATION : 09:57 STUDY SHEET

COURSE SUMMARY

This video offers a set of meditation practices adapted to biodynamic embryology, with a focus on breathing, energy activation, and opening the central channel. Participants will explore breathing techniques such as Nadi Shodhana to balance prana, while learning to create a meditative connection conducive to healing. Key concepts covered include harmony, active listening to the patient's needs, and the importance of increasing one's own energetic vibration to promote well-being. The video invites therapists to move beyond the desire to help, fostering a non-judgmental and complementary approach towards their patients.

MEDITATION AND PRACTICES IN BIODYNAMIC EMBRYOLOGY

POSTURAL AND RESPIRATORY PREPARATION

To begin, sit in a relatively upright position. Take your thumbs and place them under your armpits. Straighten your body slightly, as we will be working with the **central canal**.

Breathe in deeply through your nose, then breathe out through your mouth, releasing notions of the **"self"**, as well as **mental** and egoistic **toxins**. Let them leave you.

Breathe in again, then breathe out what is unnecessary. Breathe in one last time to gain **clear awareness**.

NADI SHODHANA BREATHING (ALTERNATE NOSTRIL BREATHING)

Place your longest finger at **Nasion**. Close your right nostril and breathe in through your left nostril. Focus on inhaling **prana**.

Focus fully on this level of prana. Breathe in through the left nostril, then close the other side and breathe out through the right nostril.

Breathe in again with the right nostril. Close and breathe out through the left nostril. Repeat a cycle.

Gently breathe in prana, air, light, vibration, through the left nostril. Breathe out through the right nostril.

Breathe in once more through the right nostril. Breathe out through the left nostril. Perform this cycle one last time.

Gently breathe in through the left nostril, breathe out through the right nostril, then breathe out through the left nostril.

OPENING THE CENTRAL CANAL

Breathe in, inflate your abdomen, then contract the **perineum**. Pull in your stomach and raise your diaphragm to lift the air pocket.

Tuck your chin, close your glottis, and gradually bring a sensation of pressure to open this area.

Breathe in, bring the pranic air pocket into your belly. Pull in, make a small pelvic tilt. Close the glottis. Tilt and raise the diaphragm.

Make a small tilt from the bottom of the glottis to the top, then release. At this moment, open your central canal in the form of a symbol.

Breathe in once more. One last time, breathe in. Now, completely empty the air.

Then, through the void, raise your diaphragm, then lower it. This will warm the body a little and open things up.

ENERGY ACTIVATION

Take your hands and open them, stretch all the **meridians** to prepare them well, as well as your body. Bring the **chi** into your hands.

You enter a meditative state. Close your eyes and become aware of prana. Prana is a means, a vehicle.

Remember that prana is air, light, and energy in the form of vibration. In a way, you are nourished by prana.

If you become aware of this quality, you will become aware of what is called the **All**. If you become aware of your breath in this three-dimensionality, it is so simple that we don't do it.

HEALING AND VIBRATION

In a way, healing from illness is an **awakening**. At some point, you will no longer speak of illness.

If you know how to increase your **vibration**, if you know how to increase this frequency, you will experience healing. That is why, as a therapist, you must always increase your vibration.

Increase it to become like **untouchable**. One should not perceive this detail specifically, otherwise one falls into **pride**.

One must give everything, give everything back. This is where one must be in **non-willing**. In a way, whether we are aware of it or not, we pass through prana.

LISTENING AND HARMONY

Be in a time of **non-judgment**. And there, the patient will ask you to place yourself at the level that suits them best.

This can be visceral, cranial, vertebral, anything. You must listen and follow the present movement.

In a way, the only thing that exists is **harmony**. Harmony is obtained by the complementarity of **duality** in the extension of the equality of the other.

Often, it is pride to give an answer before the question. Often, without realizing it, we create points of support, we create question marks for the patient.

Man does not create his illness, he suffers it. It is an answer to the question that, more or less consciously, he has asked himself. Do not breathe by your will.

The will to help others can ultimately harm more than it helps. Increase your vibration.

THE POWER OF WORDS AND THOUGHTS

One way to increase your vibration is to try to become aware that every **word** is a vibration.

Eliminate as many **low-energy** words from your vocabulary as possible. Thus, you will cleanse your **inner diamond**.

This is where you need to become conscious. Every word, every gesture, every look, is a vibration. It's as if the Earth needs it.

Be like **weavers of light**. Every time you have a thought, it goes into the **Akasha** and is distributed in the great web. You are entitled to this consciousness.

There, you can feel a vibration in your body, like small **tingles**.

36. EMBRYO KINETICS : SUMMARY

DURATION : 04:48 STUDY SHEET

COURSE SUMMARY

This video deeply explores the dynamics of embryonic kinetics, highlighting fundamental movements such as peritoneal coiling, the formation of the auditory angle, and the development of the peritoneal framework. You will learn how these movements influence the development of vital organs and their integration into the body, focusing on the interactions between the diaphragm, the urogenital system, and the etheric force. Concepts of inner peace and wisdom are also addressed, emphasising the importance of emotional balance for effective practice. Finally, the video offers applicable techniques for integrating pranic force into your practice, providing a harmonious approach to body and mind.

BIODYNAMIC EMBRYONIC KINETICS (SUMMARY)

THE PERITONEAL COILING MOVEMENT AND THE MIDLINE

At the peritoneal level, in the closure of the **anterior midline**, a **coiling movement** occurs, moving from the posterior to the anterior part.

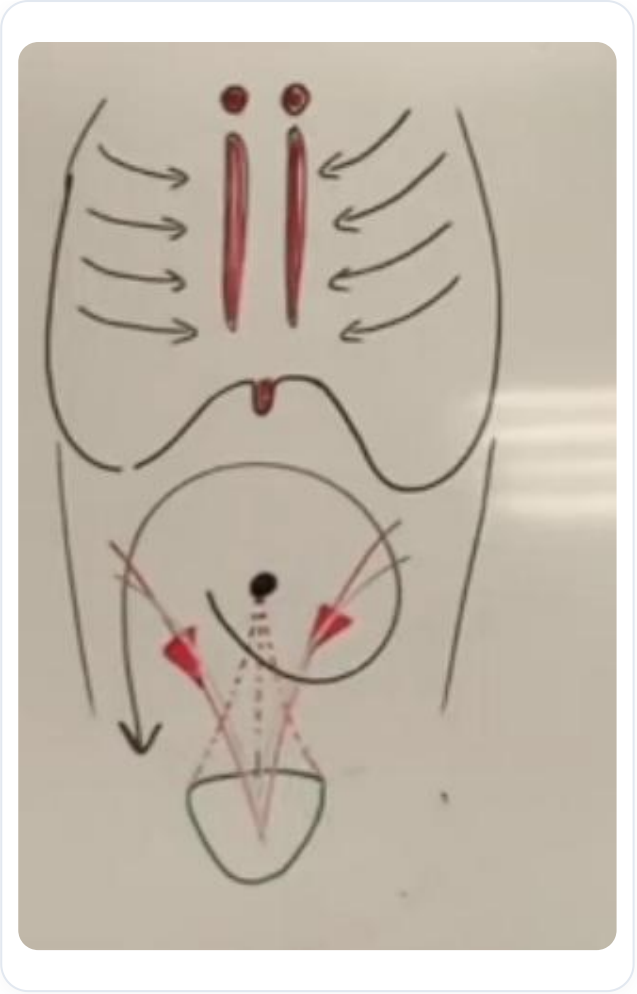


FIG. — The general schematic image fits at the beginning of the text, after the initial description of peritoneal coiling movement

This dynamic leads to a closure along the entire anterior axis, specifically the **anterior linea alba**.

FORMATION OF THE ANGLE OF THE EAR

This movement first forms a primary level: the **angle of the ear**, resulting from the establishment of the **mediastinum**.

When the mediastinum reaches its first plane of equilibrium, the angle of the ear is closed. To touch this angle is to touch the entire rhythmic zone of the lung and heart.

THE DEVELOPMENT OF THE PERITONEAL FRAME AND THE XIPHOID PROCESS

Another level develops in relation to the **developmental spiral** of the peritoneal frame.

When the **caecum** completes its final descent into the pelvis through rotation, it adopts a 270-degree anti-clockwise position, carried by a developmental force. The **xiphoid process** then finds its position, signifying that it is the result of the establishment of the **peritoneal cavity**, connecting the mediastinum and the peritoneum.

THE DIAPHRAGMATIC LEVEL AND CEPHALISATION

An important level is the **diaphragmatic level**.

A significant force develops for the establishment of the lesser pelvis, leading to **cephalisation**. This leads to **cardialisation**, which then leads to **diaphragmatisation**, and then to **hepatisation**.

ESTABLISHMENT OF THE POSTERIOR WALL OF THE PERITONEAL CAVITY

We observe the establishment of the **posterior wall** of the peritoneal cavity, including the **renal lodge** which develops from top to bottom in a descending movement.

The entire renal system is concerned, as well as the liver, the **adrenal glands**, and the **gonads**.

Progressively, the **uterine** or **prostatic force** manifests, integrating the **Wolffian duct** or the **Müllerian duct** towards the midline.

This is a concentration of energy that descends, contributing to the equilibrium of the peritoneum.

THE ALLANTOID PROCESS AND THE SEPARATION OF THE CLOACA

The **allantoid process** is established with the posterior wall of the sacrum.

There is a separation due to differential **growth rates** between the posterior part and the inferior part, with the posterior part developing more rapidly.

This arrangement separates the cloaca into:

- The posterior **rectal system**
- The anterior **bladder system**

UROGENITAL INTEGRATION AND THE UMBILICUS

This entire mechanism integrates towards the **anterior linea alba** through the tilting of the liver, returning forwards.

Within this linea alba lies the entire integration of the **urogenital system**, whose story concludes here.

In this tissue, we find the entire finality, both in the descending linea alba and in the global integration, ultimately reaching the level of the **umbilicus**.

We start from the pedicle and return to this level.

THE ETHERIC FORCE AND INNER PEACE

We integrate a symbol: the **etheric force** which must join the **astral force** to find its power at the level of the heart.

The first step is to find **peace** in our body. If we have peace in the body, we can properly develop **wisdom**.

These two elements go hand in hand: wisdom and peace meet in the heart, revealing the true meaning of **love**. One becomes a being in service, finding what is called **devotion**.

THE EMBRYO, THE PRANIC FORCE, AND TECHNIQUES

The entire embryonic system illustrates this **pranic force** which will be integrated through our **breathing** and our **nutrition**.

The techniques discussed should not be perceived as techniques, but rather as a **frequency**.

37. EMBRYOLOGICAL TAI CHI : REMINDER

DURATION : 01:38 STUDY SHEET

COURSE SUMMARY

This video offers a clear and structured reminder of the embryological Tai Chi, focusing on the first major movement: the notochordal wave and its interaction with mesodermal movement. The essential steps described include brain growth and flexions, as well as kidney, bladder, and uterine movements, highlighting their importance in embryological development. Practical demonstrations illustrate how these movements influence osteopathic manipulation, particularly through specific movements involving the peritoneum and liver, allowing students and therapists to deepen their understanding and enrich their care.

EMBRYOLOGICAL TAI CHI: RECAP

The first major **Tai Chi** movement focuses on the **notochordal wave**. This movement encompasses the entire accompanying **mesodermal movement**.

So we have the notochordal wave, with all the mesodermal movement beneath it.

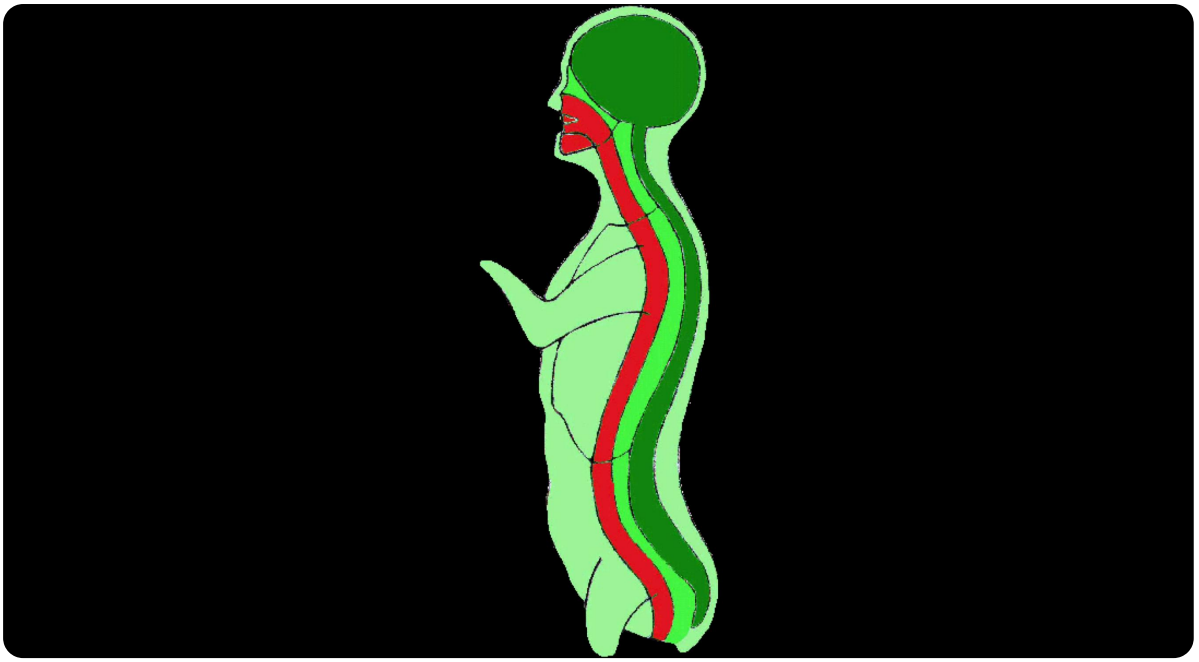


FIG. — Digestive and nervous system

Next, we focus on the **brain**, following these key steps:

1. **Brain growth**

2. **Cephalic flexure**

3. **Cervical flexure**

4. **Pontine flexure**

5. **Telencephalisation**

The movement of the **kidneys** follows, which is an **exorotation** movement.

We will now address the movements related to the **bladder** and **uterus**, before returning to the previous steps.

For the **peritoneum**, the right hand is placed in front. I take the loop at this point and bring it back towards my **hepatic** side.

The hand then descends and completes the rotation. Thus, I launch what is in front of me, and it grows.

I pull this hand and come to find the loop here to bring it back towards my **liver**.

Then, the hand at this point descends, ascends, and the other hand descends to form the **terminal face**.

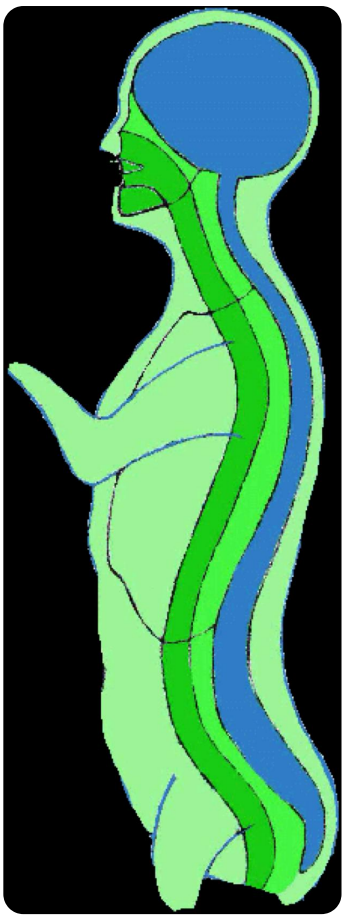


FIG. — Embryonic germ layers

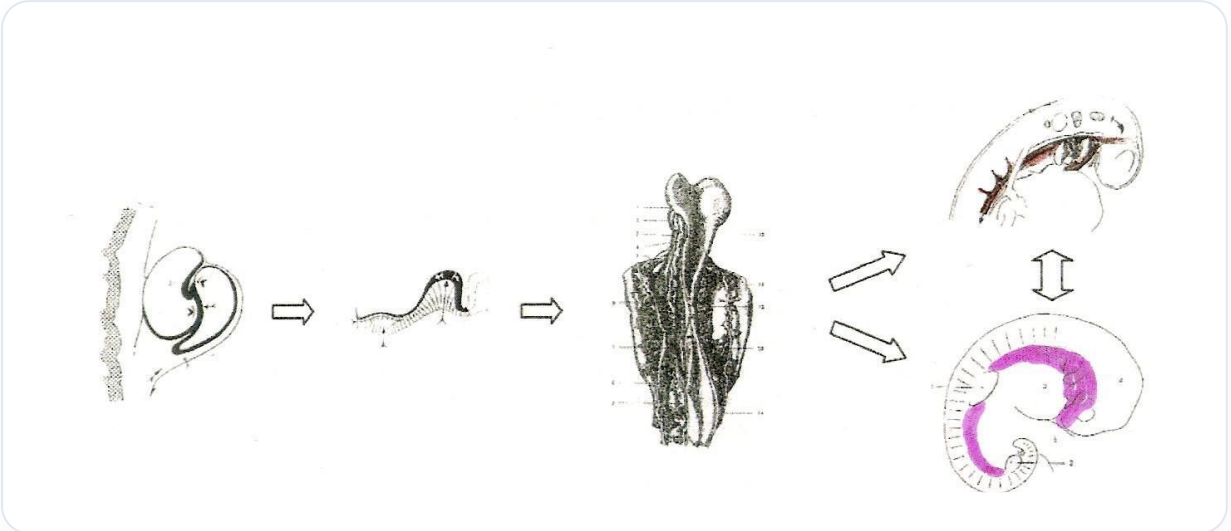


FIG. — Development of the nervous system

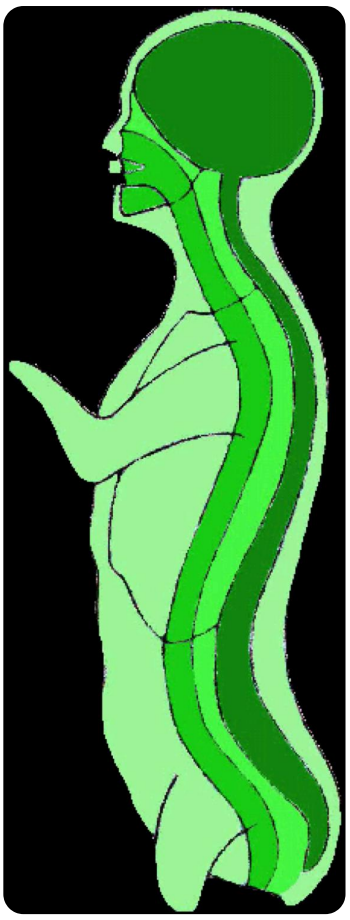


FIG. — Neurocranium and spinal column

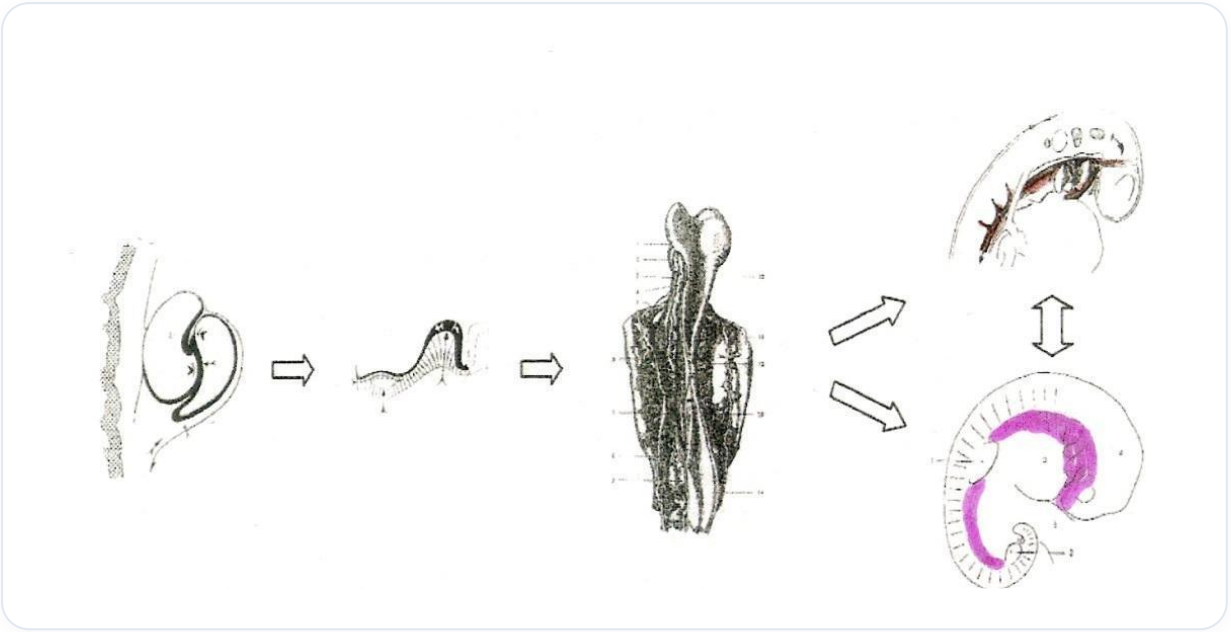


FIG. — Development of cervical muscles

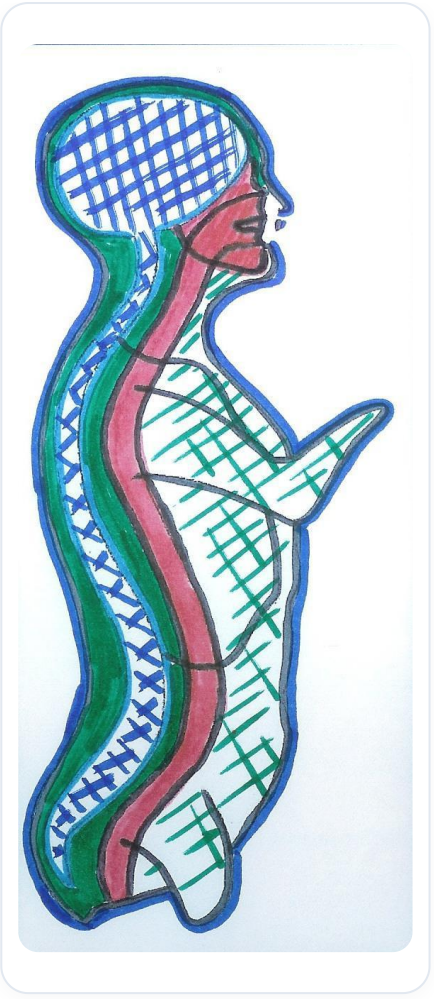


FIG. — Human body diagram

38. PRACTICE 2 : THE GENITAL SYSTEM

DURATION : 09:17 STUDY SHEET

COURSE SUMMARY

This video delves into practical techniques related to the reproductive system in biodynamics, teaching how to position oneself, feel, and respond to the patient's body needs. Key concepts include the importance of the neutral, conscious contact, and exploring delicate areas such as the peritoneal sac and the utero-prostatic zone. Students will learn how to engage the process of energetic reorganisation and recognise the body's signals, while creating a space conducive to harmony and healing, ideal after events such as difficult childbirths.

PRACTICE 2: THE GENITAL SYSTEM - BIODYNAMICS

INITIAL APPROACH: POSITIONING AND PERCEPTION

I position myself in a state of **neutrality**, focusing on my breathing and welcoming the **prana**.

I place my hands delicately on the body, in contact with the **tubercle of the third inductor**. My approach is not to impose, but to settle myself.

I identify the person's **birth axis**, ensuring it is well-centred. My attention is drawn to the surrounding space, as it is what guides me.

I perceive a slight **compression** on one side, without attempting to correct it immediately. The opposite side appears more relaxed. This observation is a preliminary step.

ENGAGING CONTACT: THE PELVIC BOWL

I change position for a new point of contact. My hand rests lightly on the **buttock**, my thenar eminence oriented towards the **trochanter**.

I form a contact like a **bowl**, encompassing the pelvis. My hands are aware of their position, but 90% of my attention is directed outwards, towards the space.

I wait for the person's neutral to gradually settle. The goal is the meeting of our neutrals, allowing the process to unfold naturally.

If this meeting occurs, I let the process happen. If it does not, a different intervention might be necessary. The life around is the most effective guide.

PERCEPTION OF MOVEMENT AND HARMONY

A subtle movement of space manifests. I move towards this point, where a process begins.

I ask the person about their sensations. They describe a **wave** from right to left, a wave followed by rebalancing.

I check my own sensations. It is different, but **harmony** is not yet established. **Symmetry** exists neither in the body nor in life, but harmony is sought.

I continue the exploration, inviting the person to change sides if needed.

DEEPENING CONTACT: PELVIC SUPPORT AND FULCRUM

I take the **pelvis** again, but this time lower, at the level of the **ischia**. The person lifts a buttock to facilitate the placement of my hand, which must be flat and relaxed.

The principle is the same: they are as if carried in space, while I remain very relaxed. This requires **patience**.

I then create a **fulcrum** at the top of the head, forming a triangle with my hands. This allows working with the space around the midline.

A slight movement, like a **tide**, indicates that the person's intrinsic process is active.

THE UTERO-PROSTATIC AREA AND THE PERITONEUM

I position myself just above the **pubis**, with a specific hand. Little by little, the person settles.

It is essential to feel the entire **peritoneal sac**, the lines of force, including the **Müllerian duct** and the **Wolffian duct**, which converge towards the midline.

As my hands settle, everything seems to gather and refocus at the prostatic or uterine level. It is a concentration of the peritoneum.

I am aware of the **terrestrial ligament**, the **falciform ligament**, and the entire peritoneum, as well as the posterior parietal peritoneum.

This contact is particularly interesting after a **difficult childbirth** or a **caesarean section**, when the uterus has not regained its energetic integrity.

ENERGETIC REORGANISATION AND HEAT

When the process engages, a **vibrational** change is perceived, often accompanied by a sensation of **heat**, a sign of the body's thermodynamics.

Initially, some patients may feel slight pain, requiring a very relaxed hand.

I always focus on the **micro-crystalline field** and observe the person's breathing. On exhalation, I adjust my position.

The system seems to gather transversally, like the **azygos system** or the **lumbar veins** that reorganise blood circulation. A small change in heat is perceptible.

I remain attentive to the space, increasing the quality of my listening and reducing my words, observing the gradual reorganisation.

THE NEUTRAL, PRIMARY RESPIRATION AND POTENCY

The heat concentrates, then dissipates, as if the body is seeking further what it needs. We both seek to return to the **neutral**.

To help, I synchronise with **thoracic breathing**. My attention is on the end-exhalation phase, allowing for **transmutation** or transformation towards a new form.

The process gathers, but a transverse force remains. I continue to accompany until the end.

It is important to have a representation of the **heart**, of this meeting towards the midline, of the **looping** and the **lateral tilting**, with the movements of **supination** and **pronation**.

EMBRYOLOGICAL UNDERSTANDING AND PRACTICE

We also discussed the developmental dynamics of the **liver**, **stomach**, the lesser sac of the **omenta**, and the **intestine**.

The posterior wall was explored in depth, with the establishment of the **renal lodge**, the **adrenals**, the **gonadal system**, the **uterus**, and the **prostate**, all converging towards the midline.

The birth axis and the way of carrying the pelvis differently were discussed, bringing a notion of **calm** and **rebalancing**.

This notion of the neutral, of the **"meeting point"**, the meeting point of two neutrals, is beginning to settle.

It allows the emergence of **primary respiration**, which gradually activates the therapeutic process (the tide). It is about being present and accompanying.

There is a **power** within you, the **"potency"**, which must be allowed to awaken.

IMPORTANCE OF MEDITATION AND IMAGINATION

It is essential to practice **meditation** to stabilise the mind. This brings more peace and is recommended for everyone, even the most distracted.

Imagination is a form of intelligence. One must know how to blend it with reality, without falling into one extreme or the other.

Reviewing **embryology** offers a better understanding of anatomy. Embryology is the establishment of living anatomy, connecting us to structure, function, and form.

39. THE LOCOMOTOR SYSTEM : DEVELOPMENT OF THE SOMITES

DURATION : 09:06 STUDY SHEET

COURSE SUMMARY

This video thoroughly explores the development of somites within the context of musculoskeletal system formation, emphasising the importance of the notochord, neural tube, and lateral plates. It addresses fundamental concepts such as the creation of the dermatome, myotome, and sclerotome, as well as their respective roles in the development of skin, muscles, and bone structure. The processes of molecular induction and epigenetics are also explained, highlighting how genetic and environmental factors interact to regulate this complex development.

DEVELOPMENT OF SOMITES IN THE MUSCULOSKELETAL SYSTEM

Initially, we studied the establishment of the **notochord**, then the **ectoderm**, with its associated **mesoderm**. The importance of this supporting tissue, which here is the intermediate tissue, was also discussed.

Next, we examined the **vitelline system**, which corresponds to our entire digestive system. We also began to study the **lateral plates**, including the **peritoneum**, **somatopleure**, and **splanchnopleure**.

The lateral plate, which ultimately forms the **renal**, **adrenal**, and **genital systems**, is a somatic plate. It is induced by the **epiblastic**, **chordal**, and **neural tube** levels.

This induction will cause a reorganisation of the somatic plate in several forms:

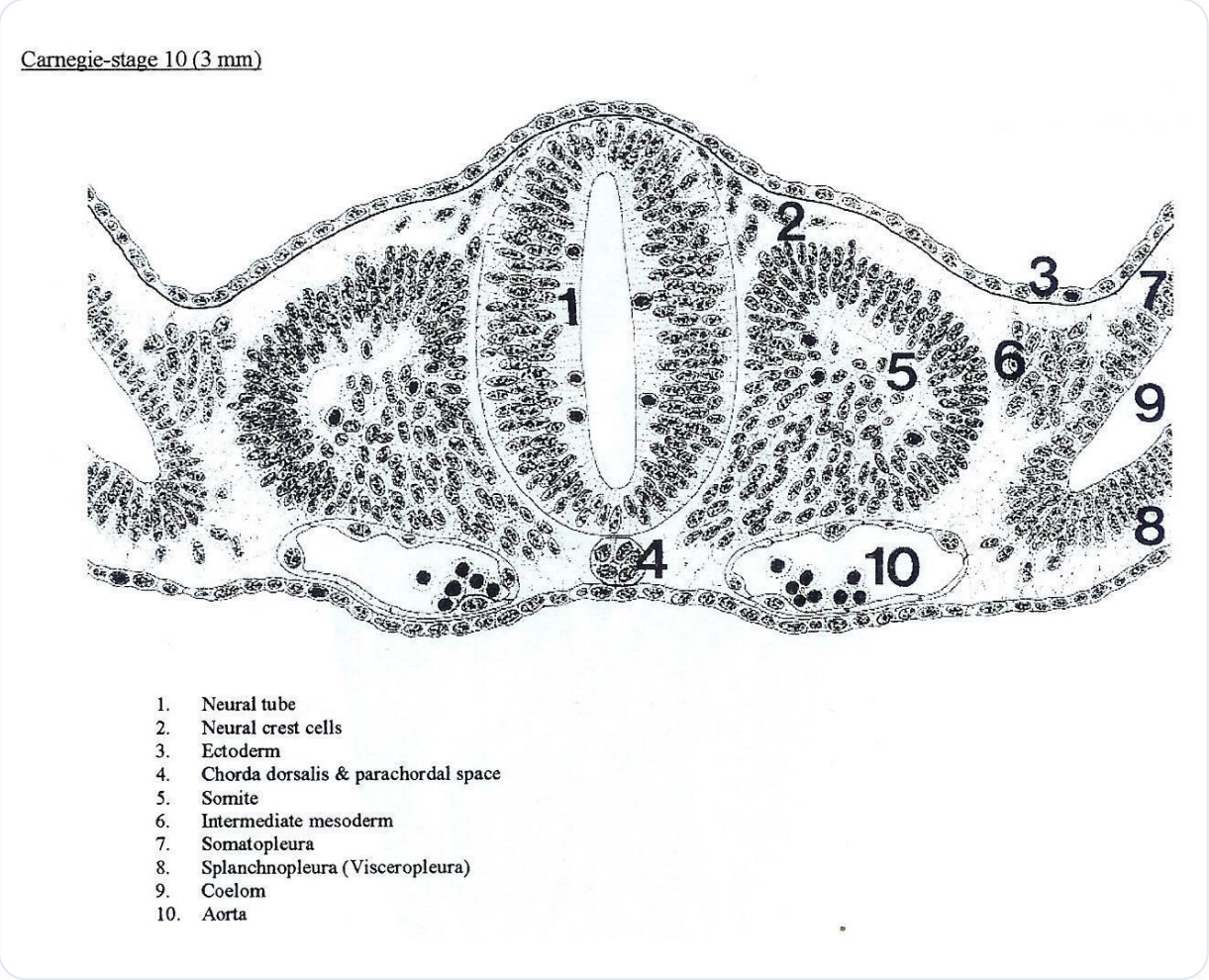


FIG. — Carnegie stage 10 embryo

- **Dermatome**: everything concerning the skin.
- **Myotome**: all muscles.
- **Sclerotome**: the vertebral axis.

The **fasciae** are found in the intermediate system, i.e., the peripheral tissue. They will be integrated by the myotome, which is muscular.

It is important to understand that there is **connective tissue** and **metabolic fields**. These fields are areas of dilation and retention, and the interaction between these systems will reorganise the whole.

It is a balance between the establishment of **bone** and **muscle**. Between the two, another field appears: a retention field, of a fascial nature.

The notochord influences the establishment of the neural tube, and this influence of the neural tube organises and reorganises the entire mesodermal structure, i.e., the somitic structure on either side of the neural tube.

We can therefore say that the notochord creates this **differential growth rate** within the peripheral and central parts of the neural plate, which will form a **groove**.

In this grooving process, a specific reorganisation of cells will occur. They will organise microscopically, with **induction factors**, to form the different parts of the somites.

The movement of the **aortas** influences the exorotation movement of the kidneys. At the same time, this movement will also play a role in the establishment of this somitic system.

The neural tube is made up of small cells. Do not forget that a tissue is made of cells, and cells are made of molecules.

So we have this whole combination. When we talk about **genetic factors**, we are at the molecular level. Cells will release new types of **proteins**, in the form of genetic **molecular enzymes**, such as **FGF8** or **sonic hedgehog (HH)**.

These proteins will provide the inductions, create the **ligands** and the necessary information to reorganise the system. The force comes from outside and will cause the reaction inside.

Genes are therefore not the engine, but the support for development. It is an **epigenetic** force.

As the aortas approach each other, through this differential growth rate and this movement of approximation, a kind of opening is observed, a movement where the cells, previously encapsulated, reorganise.

One part will form at the periphery through induction phenomena. Do not forget that there is an epiblast here. Molecular phenomena will occur, causing this part of the somite to become the **dermatome**.

The part located near the notochord will form the **sclerotome**. The sclerotome will form, among other things, the **vertebral disc**, and then the entire vertebra will take shape up to the spinous process, organised by this sclerotome coming from the somite.

Another intermediate part will follow another path: this will be the **myotome**, from which all muscles will derive.

So we have the **dermatome**, the **myotome**, and the **sclerotome**.

In this process of somite establishment, there is also a phenomenon of **segmentation**. Here, somites begin to appear. It is not just a tube; it becomes like small balls on either side.

This occurs when the embryo begins to lengthen, i.e., the neural tube develops. It is the neural tube that will induce.

But do not forget that at this moment, a phase sets in with the integration of the entire **vascular system**. Among other things, there was this system of **polarisation** where the ectodermal system, for the neural tube, was already in an assimilation field, beginning to create spaces.

It is therefore the vascular system that is at the origin of segmentation and not the neuronal system, as is often described. First, the neuronal system is needed, which will organise the vascular system, and from this neuronal system, the venous system will emerge.

Primary segmentation appears, as **Blechschmidt** very well described, through a region that develops as space and fragmentation of the system to form the somites. The embryo will have areas of **dermalgia**, and this is a segmentation: the somites segment.

It is "metamerised", if you will, it comes from the segmentation organised by the vascular system on a primitive plane, by the flexion and reabsorption of the aorta.

It is arterial, but in fact, at that level, these are **venous sinuses**. We will mainly say it is vascular, because initially, it is not yet truly defined as arterial or venous. It is a field of assimilation and dissimilation.

Do not forget that this **longitudinal growth** is allowed by aspiration, so it is polarity that performs it. Then, by enlarging, it will form what is called the **ectodermal ring**.

The embryo is like this. We are only just starting to have two or three first segmentations. And it is only when the flexion movement accelerates that blood will begin to enter the spaces that are being created.

Each level will begin to create what are called **somites**. There will be 33 pairs that will develop here.

This is an absorption by the vascular system, linked to a polarity between the ectoderm and the mesoderm. It is therefore polarity that ensures that each tissue, each space, must be nourished separately.

I cannot bring a vessel that is like this. No, it starts to happen like that. It is the vessels that create the space between the myotome and the sclerotome. It is the bone that has grown and pulls on the myotome. After that, it is the muscle.

So I have an inversion. Initially, the bone is fluid and grows. When it pulls and lengthens, it creates a metabolic field around it, which is a field of dilation. It is the bone that will become active first. And then, it will only be the muscle that will be active in a second phase.

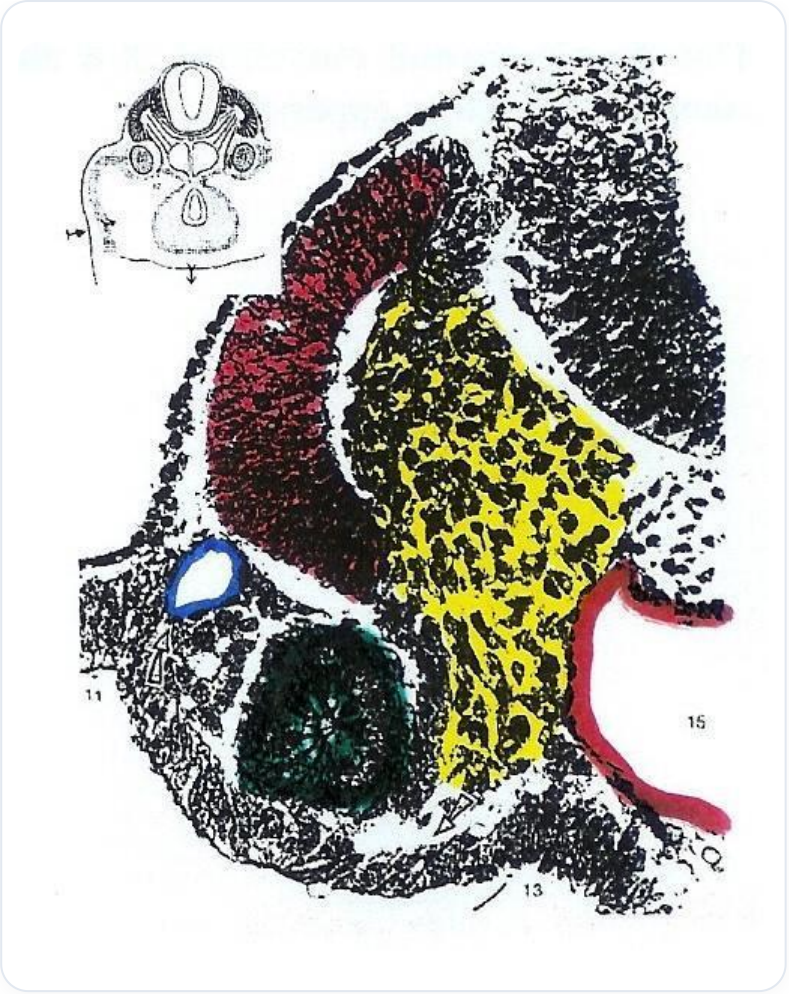


FIG. — Embryo: cross-section

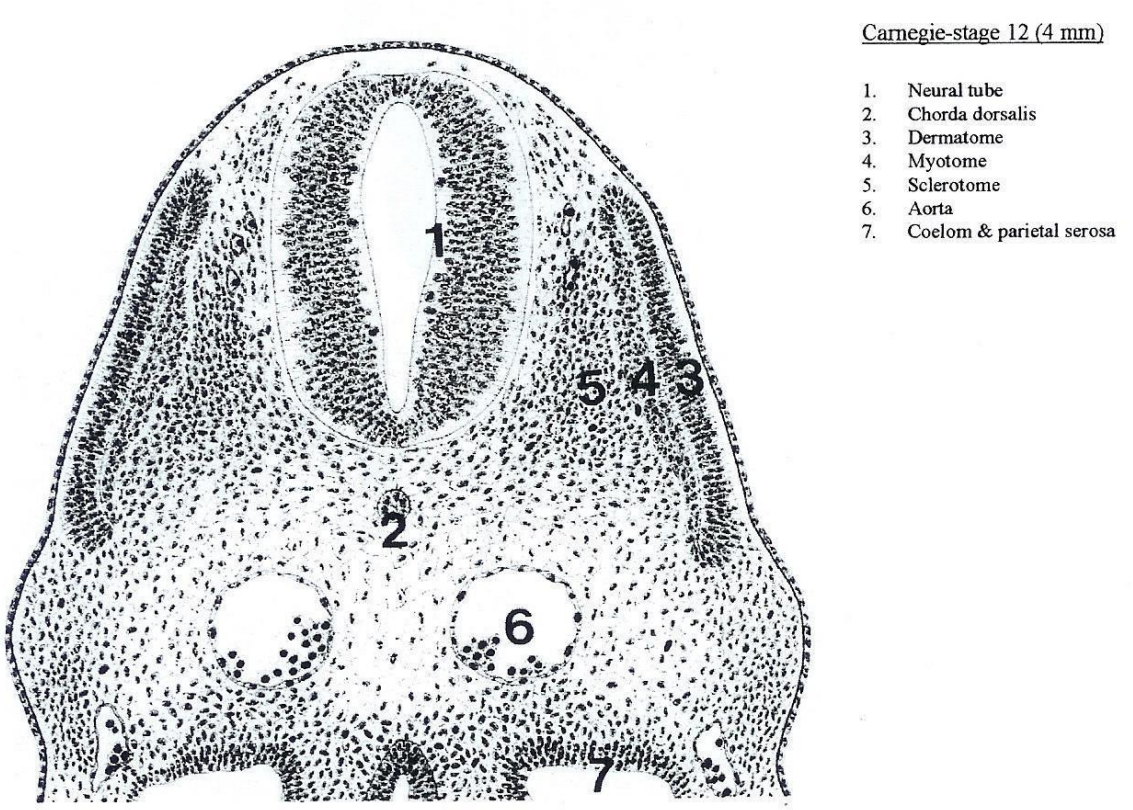


FIG. — Human embryo, cross-section, Carnegie stage 12

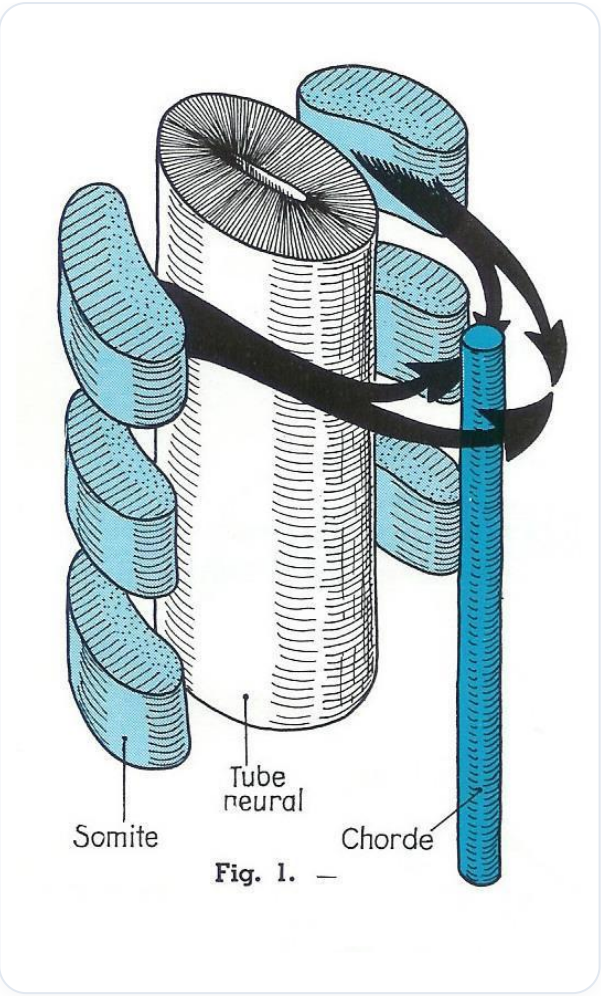


Fig. 1. —

FIG. — Neurulation and somites

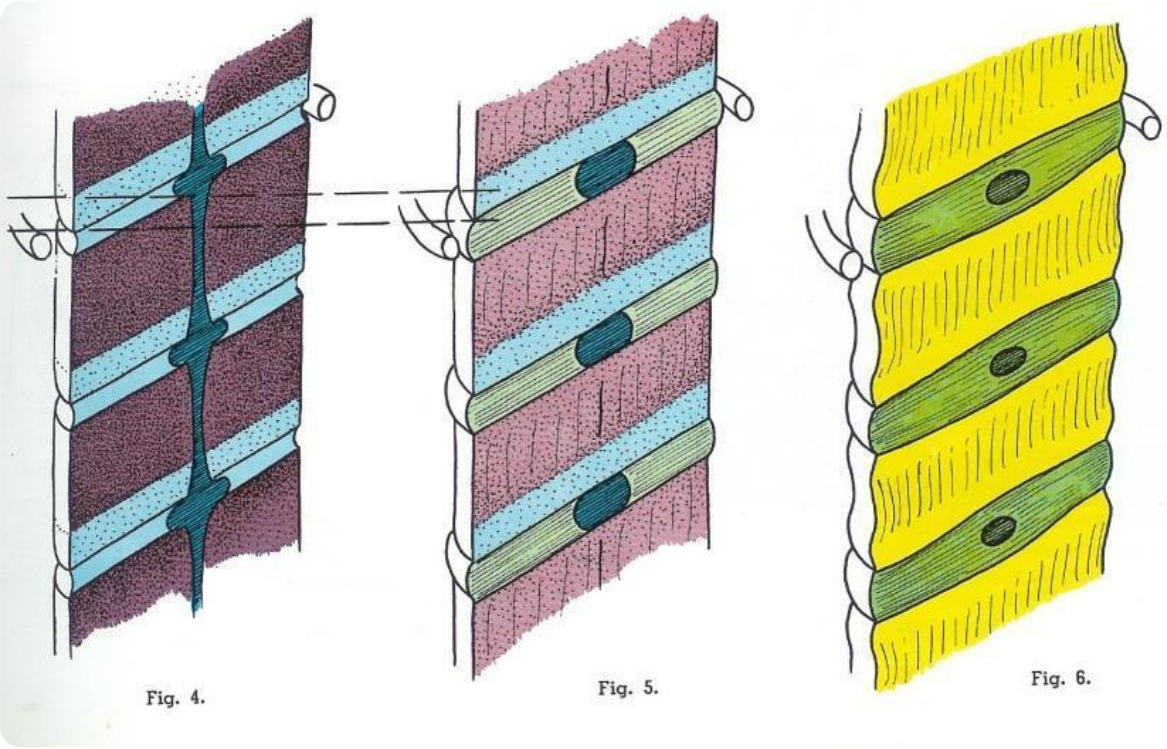


FIG. — Somite myotome dermatome

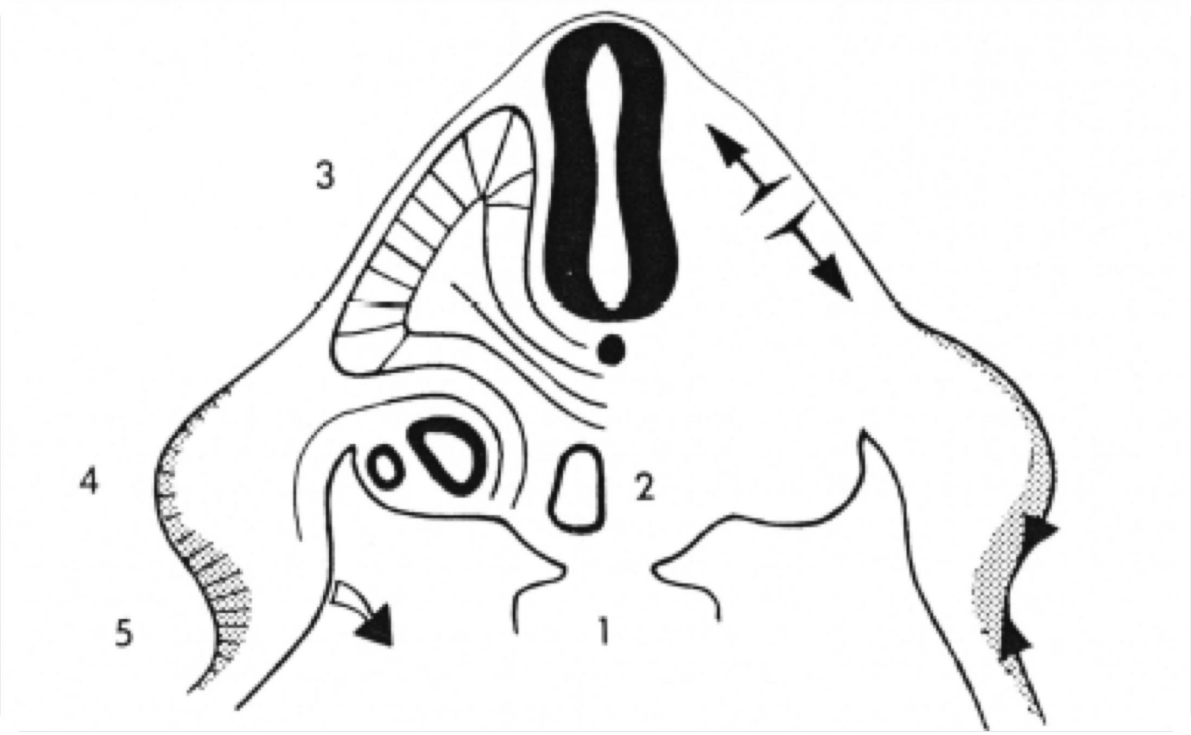


FIG. — Embryonic limb development

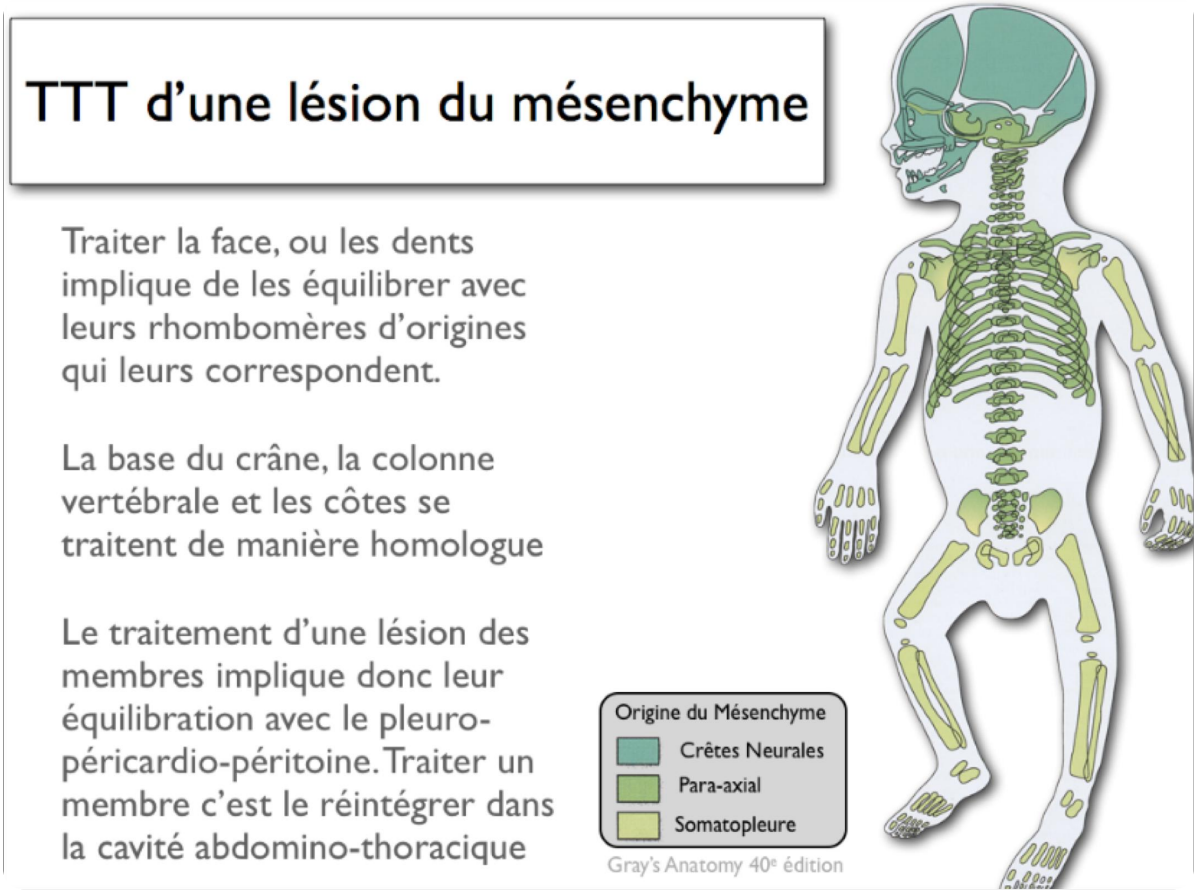


FIG. — Skeletal mesenchyme origin

40. DEVELOPMENT OF THE VERTEBRAE

DURATION : 06:41 STUDY SHEET

COURSE SUMMARY

This video delves deeply into the development of the vertebrae and the musculoskeletal system, detailing the segmentation that influences the formation of the axial skeleton, ribs, and limbs. It explores the establishment of the notochord and neural tube, crucial elements for embryonic structuring, as well as the interactions between the ectodermal ring and these structures. Key concepts include differential growth, the formation of intervertebral discs, vertebral bodies, and the importance of information from the neural tube and notochord in this process. Practically, therapists will learn how to palpate and work on these different structures to influence vertebral development and identify potential clinical problems associated with dermalgia.

VERTEBRAL DEVELOPMENT

The development of the **locomotor system**, and more specifically of the **vertebrae**, relies on precise segmentation. This segmentation influences the trajectories of the **axial skeleton**, **ribs**, and **extremities**.

The establishment of the **notochord** induces the formation of the **neural tube**, which progressively organises along the longitudinal axis and structures the vascular system. This process is accompanied by **differential growth** that becomes anterior, leading to a flexion movement. The anterior part creates what is called the **ectodermal ring**, an epiblast tissue that flexes, thus forming this ring.

This flexion is essential because it leads to significant changes for the organism, particularly the phase of **embryo delimitation**. The presence of the neural tube and **neural crests** is crucial, as these elements interact with the ectodermal ring, providing important information to the notochord.

The movement of the ectodermal ring generates compression that influences development around the notochord. This process is linked to the formation of the **intervertebral discs** and **vertebral bodies**. The upper part transforms into a disc, while the lower and upper parts begin to form the vertebral body, under the influence of the notochord and neural crest.

The notochord, a remnant of the **nucleus pulposus**, plays a key role in the formation of vertebral bodies from the **sclerotome**, derived from the somites. Thus, the notochord induces the formation of the vertebral body and the vertebral disc, while surrounding structures, such as the arch and spinous process, are also influenced by the notochord and neural tube.

There are two types of information in the formation of a vertebra. For the spinous process, the information comes directly from the neural tube, while for the disc, it is linked to the notochord. The notochord also induces the formation of the **vertebral chlorines** originating from the sclerotome, while the neural tube contributes to the formation of the surrounding mesoblast.

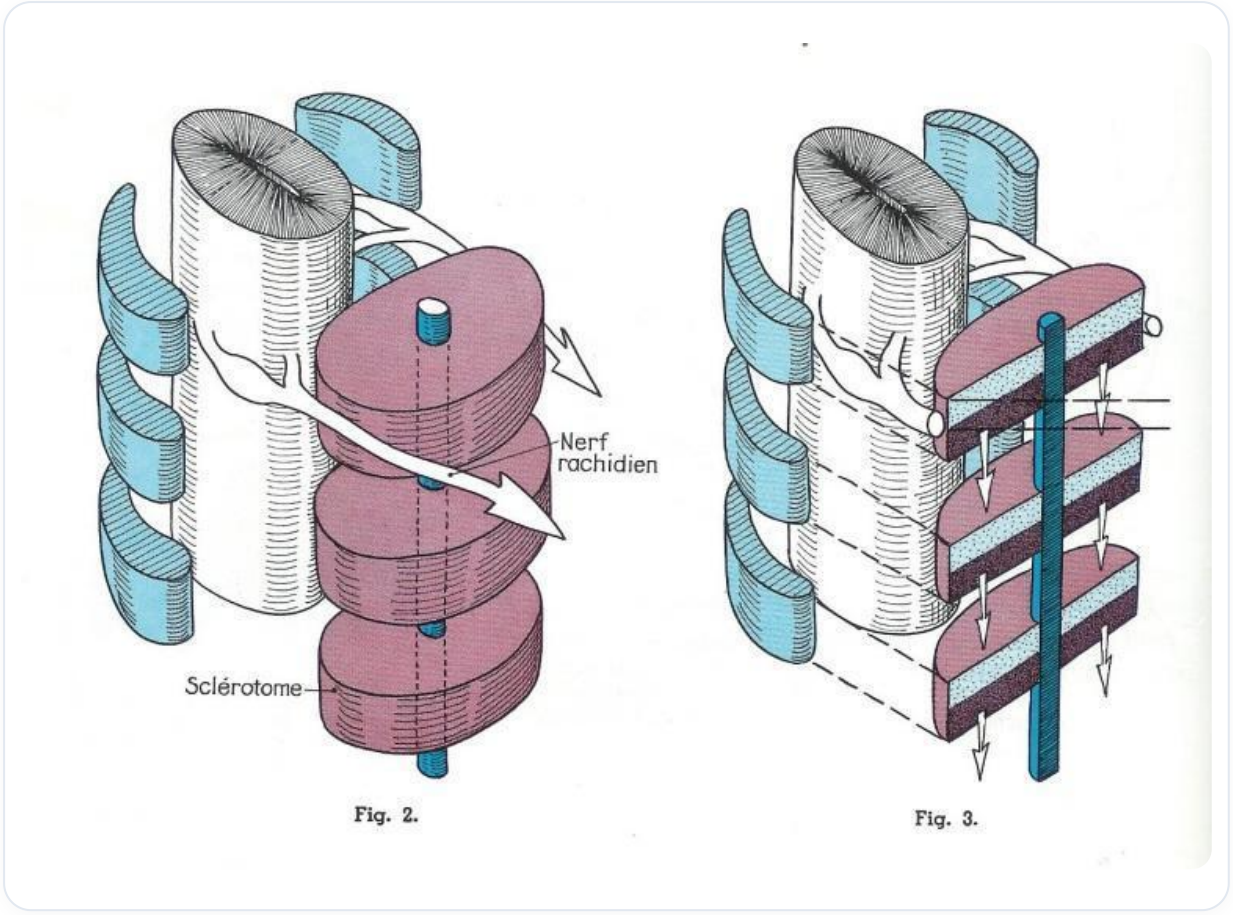


FIG. — Sclerotome, vertebrae, nerves

In complete formation, we observe the establishment of somites to form the sclerotome around the neural tube and notochord. The sclerotome is primarily responsible for the formation of the vertebral body, while a form of mesoblastic sclerotome contributes to the vertebral arch and spinous process.

The formation of the vertebral body and the vertebral arch occurs according to two planes: longitudinal for the vertebral body and transverse for the arch. This movement is essential for vertebral development.

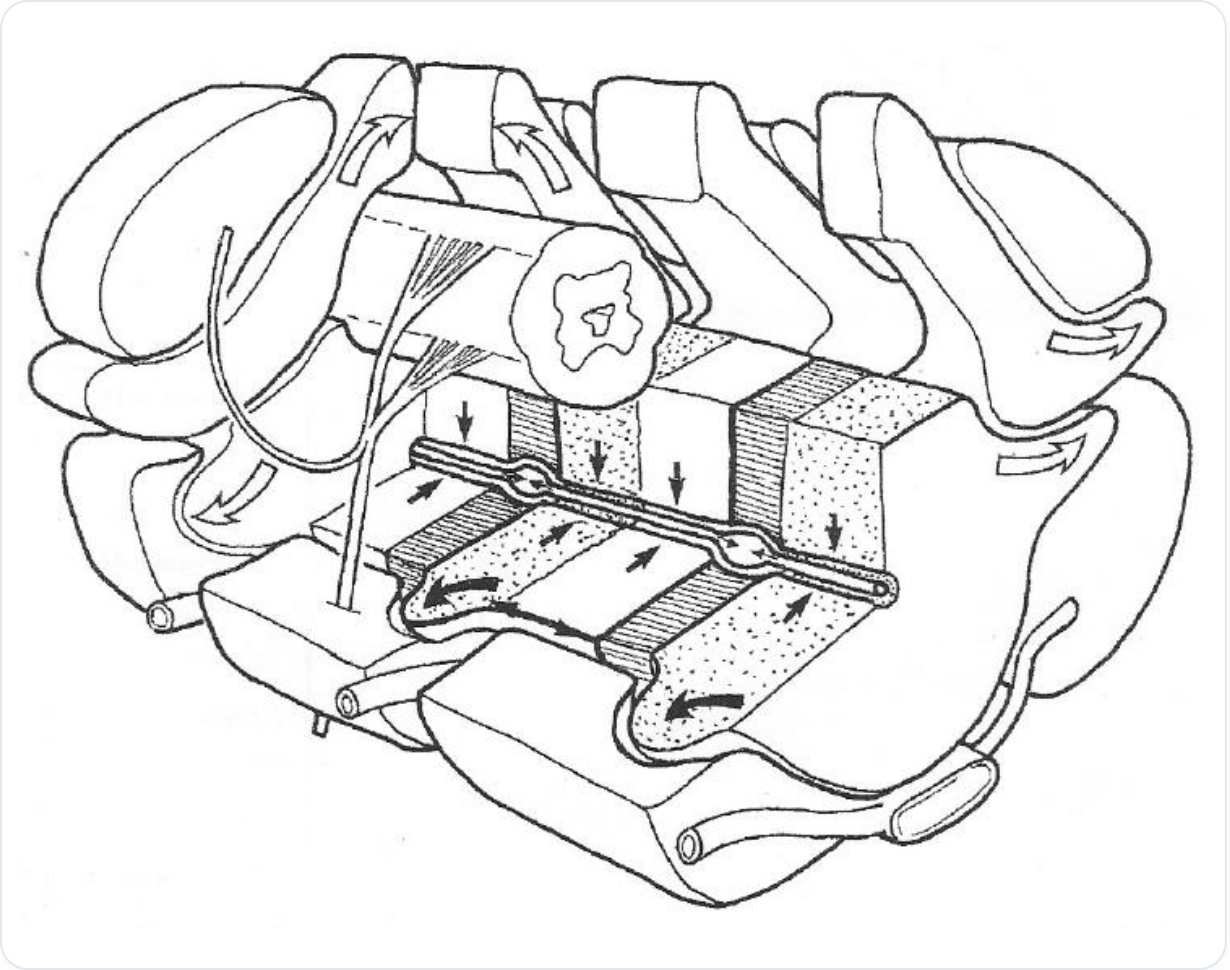


FIG. — Spinal CSF circulation

In practice, touching the body can be done at different levels: either at the level of the neural tube to work on the posterior arch, or on the longitudinal axis for the vertebral body, or in another plane for neural crest migration.

Each element, such as the skin, neural tube, and notochord, sends specific molecules to guide vertebral development. The notochord focuses on vertebra formation, while the neural tube deals with the arch and spinous process.

When development intensifies, it triggers specific inductions for the neural tube, leading to the formation of **dermatomes**, which are segmented throughout the body. **Dermalgia** on a dermatome can indicate a link with a viscus, while multiple dermalgias on the same dermatome may suggest a problem at the corresponding vertebra.

For example, dermalgia at level 9 can be associated with structures such as the **gallbladder**, **pancreas**, or **pyloric system**. In these cases, it is relevant to explore the associated vertebra to better understand the clinical implications.

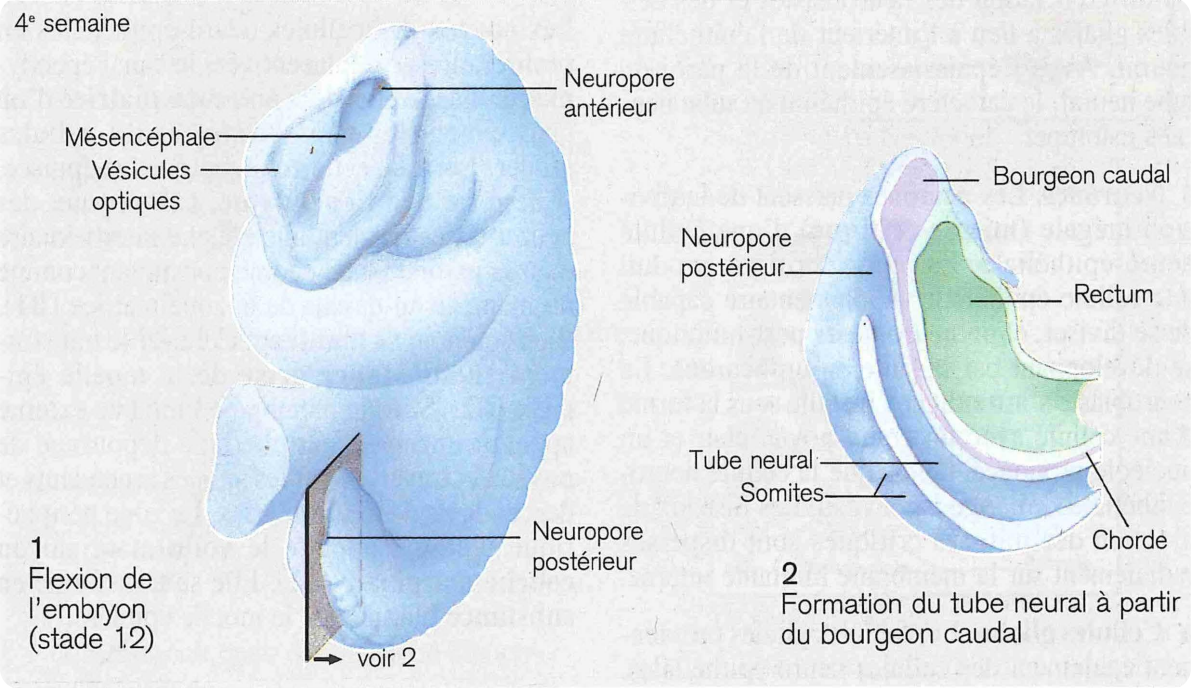


FIG. — Embryo 4th week

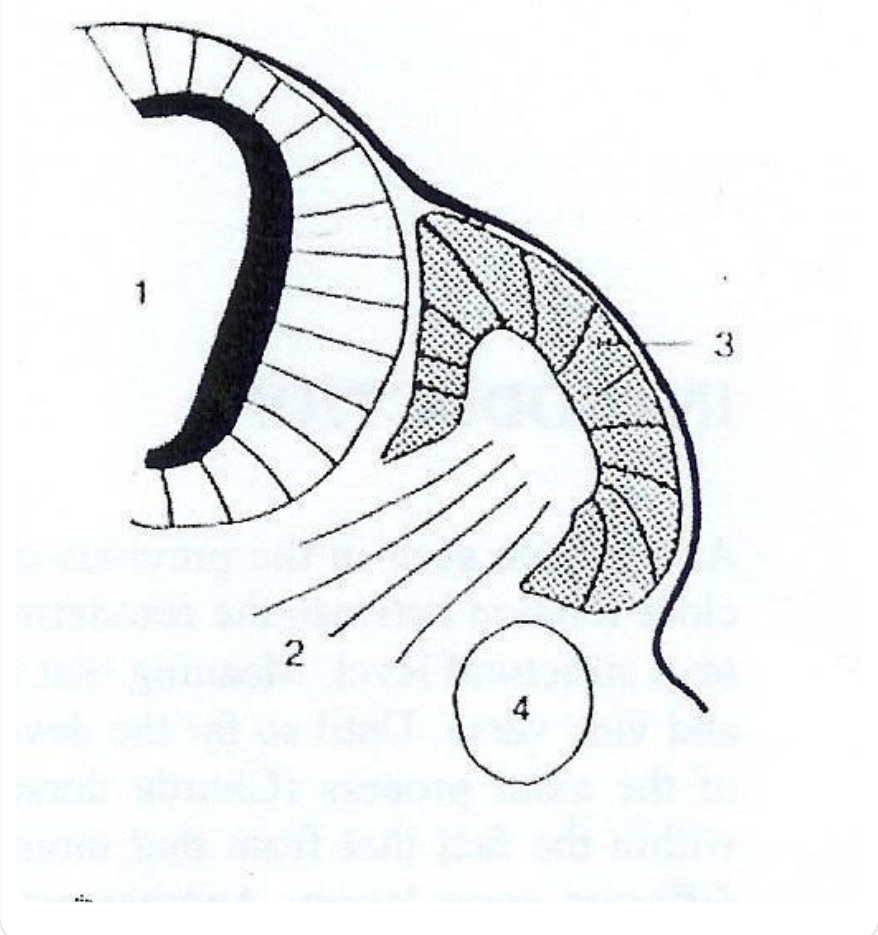


FIG. — Somites and neural tube

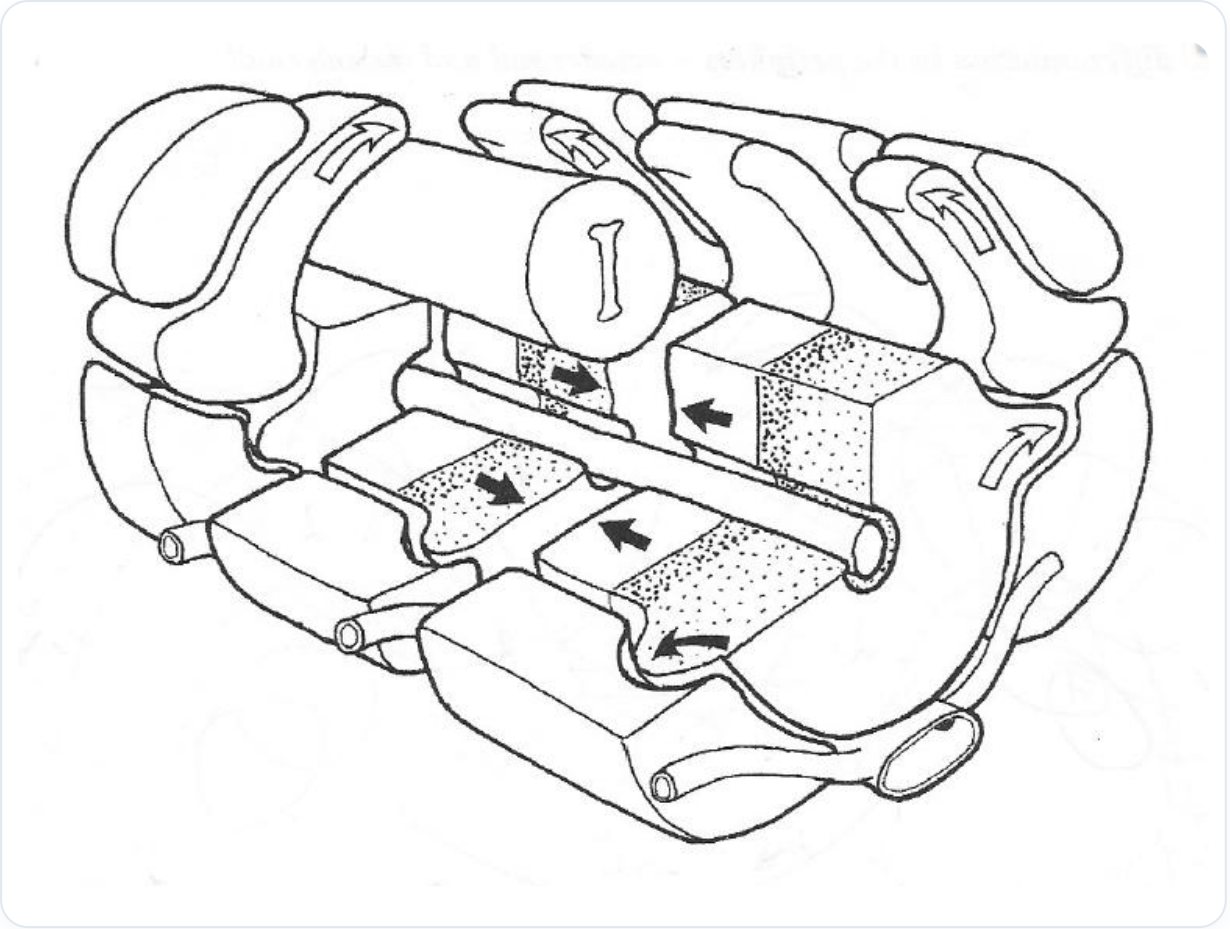


FIG. — Somites and neural tube

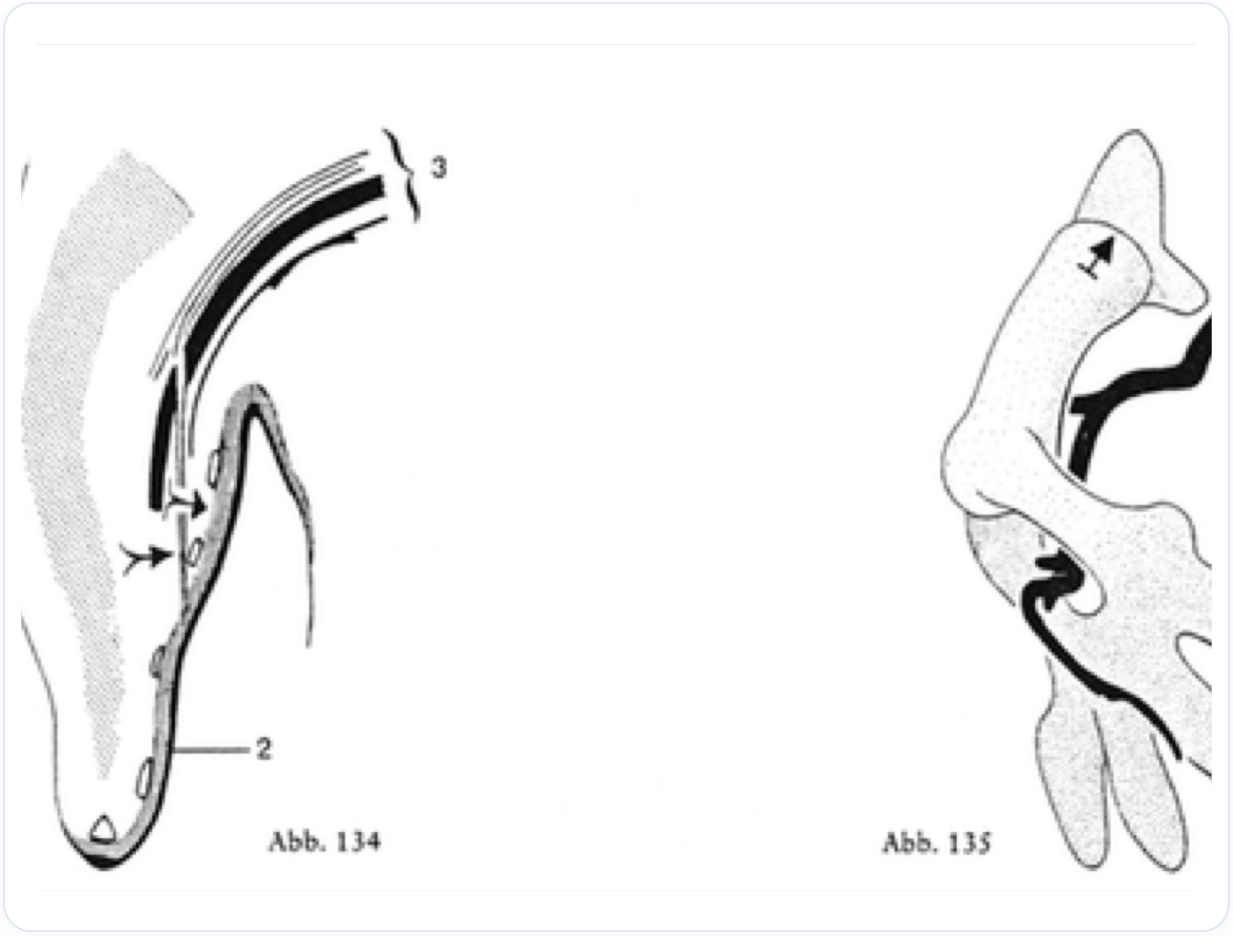


FIG. — Development of the digestive tract

41. DEVELOPMENT OF THE RIBS

DURATION : 07:22 STUDY SHEET

COURSE SUMMARY

This video delves into the embryological development of the ribs, linking elements such as the ectoderm and mesoderm to the formation of thoracic structures. It expounds on the concepts of complexity and emergence, and demonstrates how these processes can influence clinical conditions. In parallel, the importance of rib motility is highlighted, offering both direct and indirect techniques for working on each rib. The emotional aspect, particularly in relation to the fourth rib on the left, is explored, underscoring the correlation between respiratory movements and emotional balance.

RIB DEVELOPMENT

The **ribs** are linked to a complex developmental process. This development involves the **ectoderm**, **mesoderm**, **vascular anatomy**, **cerebralisation**, **segmentation**, as well as the development of the **gut**, **cavities**, **heart**, and **liver**. All these elements contribute to the overall emergence of the costal system.

This phenomenon is often described by **emergence theories**, which suggest that a simple process can give rise to more complex structures. At a certain stage, complexity increases, but there comes a time when the system can no longer absorb this complexity. It can then return to a state of **chaos** to rebuild itself. This process of deconstruction and reconstruction can be difficult to experience, just as **depression** can occur after a period of high pressure.

The ribs form as **densifications** between the segmented thoracic vessels, beginning around day 35 of development. The vertebral body and the growing rib arch lengthen, following the developmental movement of the embryo in its **amniotic cavity**. This process of **cephalisation**, **cardialisation**, **hepatisation**, and **delimitation** is essential for rib development.

From day 45, a **fracture** occurs, giving rise to a **chondrocostal** joint that progressively centres towards the midline, at the level of the **clavicle**. The sternal bars, which are mesenchymal condensations, also join at the angle of the ear, corresponding to the **mediastinum**.

After birth, the **caecum** completes its descent, and the embryo continues to develop in a similar dynamic. Structures around the vertebra, such as the **myotomes**, **dermatomes**, and **sclerotome**, gradually form, at the same time as the process of embryonic delimitation.

The ribs are also linked to clinical phenomena. For example, in boxers, a blow can lead to **intercostal evisceration**. If the rib is not broken, this may indicate an underlying problem. In parallel, rib development is accompanied by the **parietal pleura** and the internal fold of the **visceral pleura**, forming the **endothoracic fascia**.

Working on rib motility is essential. By deepening this approach, a connection can be established between the parietal pleura and the visceral pleura, particularly at the level of the **pulmonary hilum**. Freeing the ribs allows the lung to be freed, thus promoting essential exchanges.

It is possible to work on each rib individually or to adopt a more global approach using direct or indirect techniques. A free costal cage is crucial to allow a fluid circulation of exchanges.

An important point to note is the emotional impact of the **fourth rib on the left**, which is often associated with phenomena of anxiety and **precordialgia**. The ribs work in synchronisation with **inhalation** and **exhalation**. If a rib does not descend correctly, this can lead to emotional blockages.

Working on this area involves bringing the rib back into its inhalation and exhalation movement, correcting its position and balancing the pulmonary hilum. This process is essential to restore the balance and functionality of the costal system.

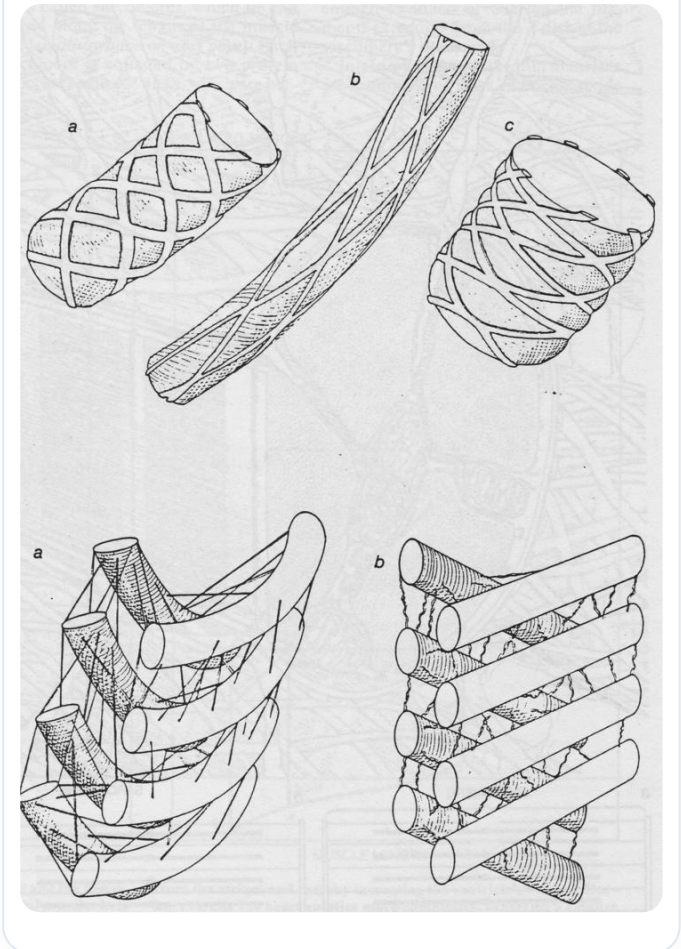


FIG. — Tensegrity

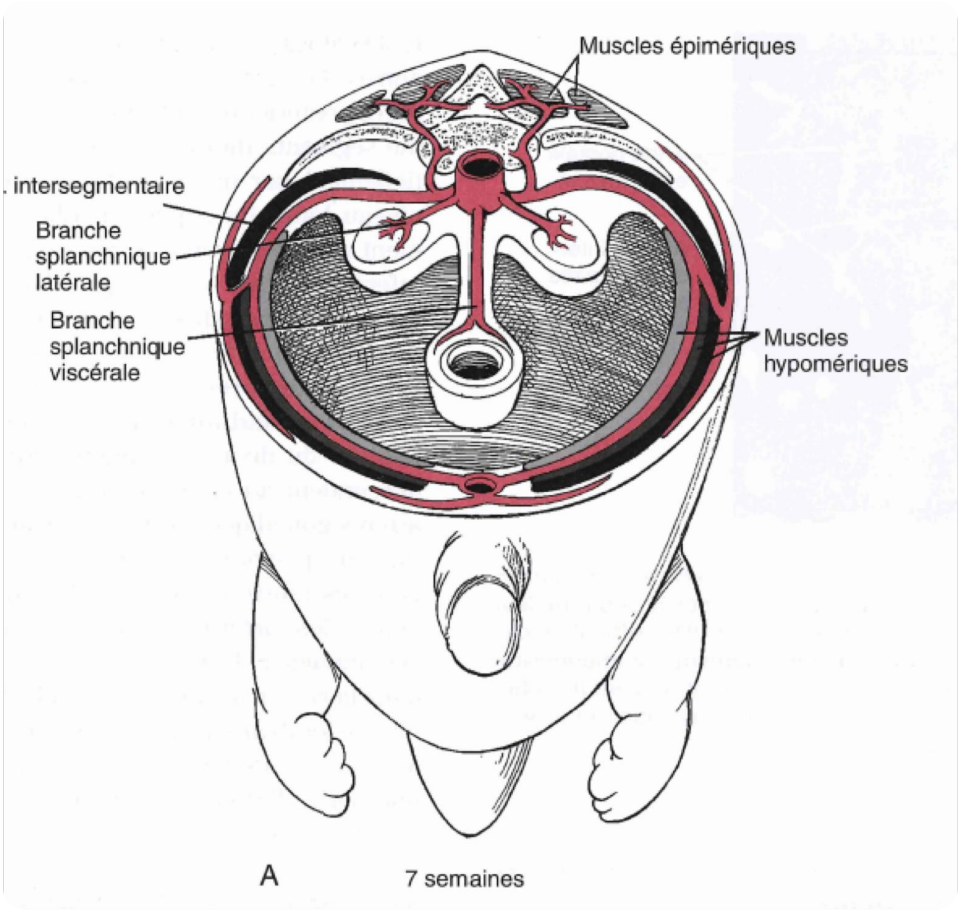


FIG. — Embryo 7 weeks, vessels

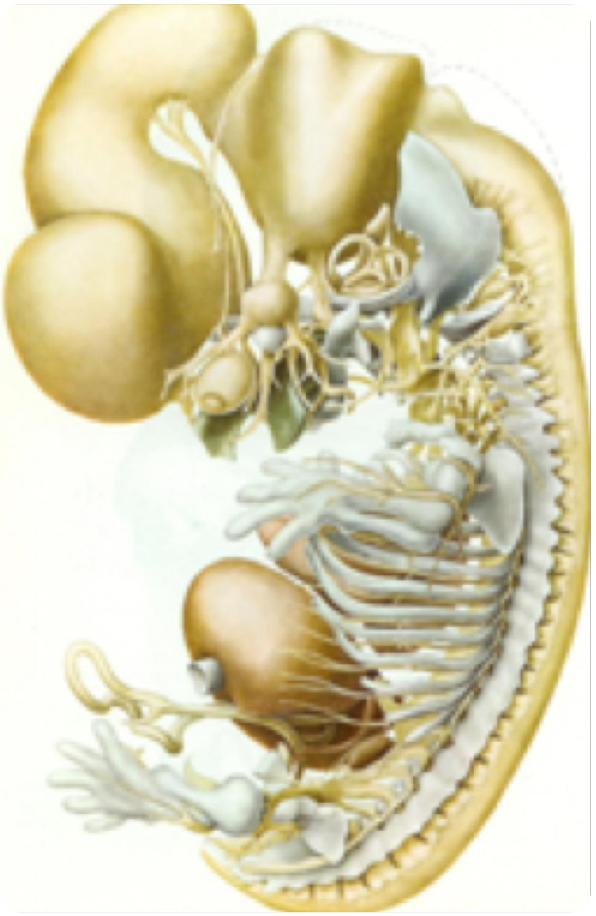


FIG. — Embryo nervous system

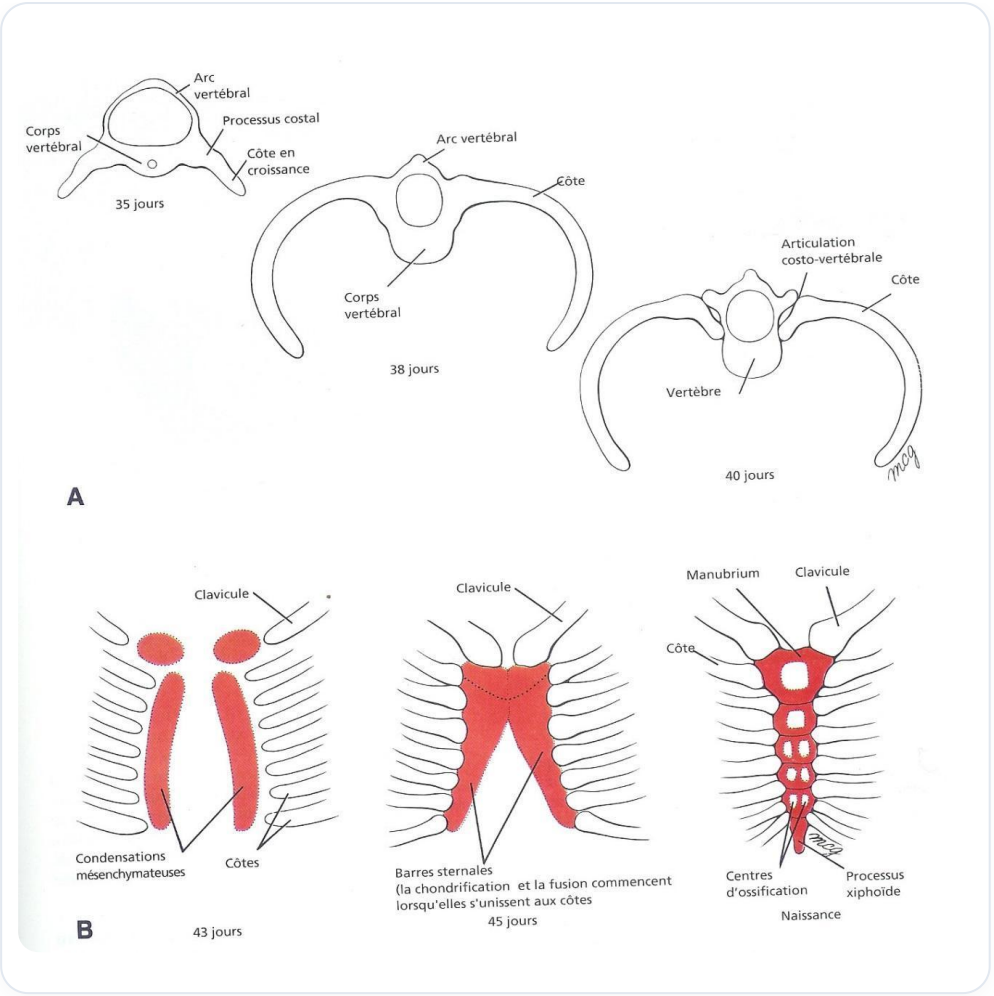


FIG. — Development of ribs and sternum

42. DEVELOPMENT OF THE LIMBS

DURATION : 10:30 STUDY SHEET

COURSE SUMMARY

This video highlights the fascinating link between limb development and internal structures, particularly the brain and digestive tract. You will learn how childhood injuries can influence current pain and how approaches in Chinese medicine and osteopathy can be integrated to treat tendinitis and other joint pain. Key concepts discussed include peritoneal dynamics, the importance of blood vessels in limb formation, and the interaction between bones and muscles during development. By understanding these complex relationships, you will be better equipped to effectively treat limb injuries and improve the overall health of your patients.

LIMB DEVELOPMENT

Limb development occurs in parallel with internal structures, particularly the **brain** and the **digestive tract**. After rotation, the lower and upper limbs form simultaneously.

Tendonitis can reflect a deficit in the rotation of the **peritoneum**. For example, a person experiencing wrist pain, often diagnosed as **tennis elbow**, may not have played the sport. It is crucial to explore **pulmonary** or **visceral** aspects to understand the origin of the problem. By analysing the organisation of the system, one can treat the visceral to free the limbs.

In Chinese medicine, specific peripheral points are stimulated to treat internal problems. The imprint of the peritoneum can reactivate to reuse previous functions. Major life injuries often occur before the age of 5, and some can even date back to the embryonic stage. What happens in the body can be reflected in the limbs.

The **cellulome** and the **neural tube** play an essential role in development. A specific angle creates a suction field, influencing growth. A metabolic field with areas of restriction and overgrowth leads to differential growth movements, influenced by the vascular system. Active metabolism generates a powerful flow in the **mesenchymal** tissue, promoting the growth of **ectodermal** tissue.

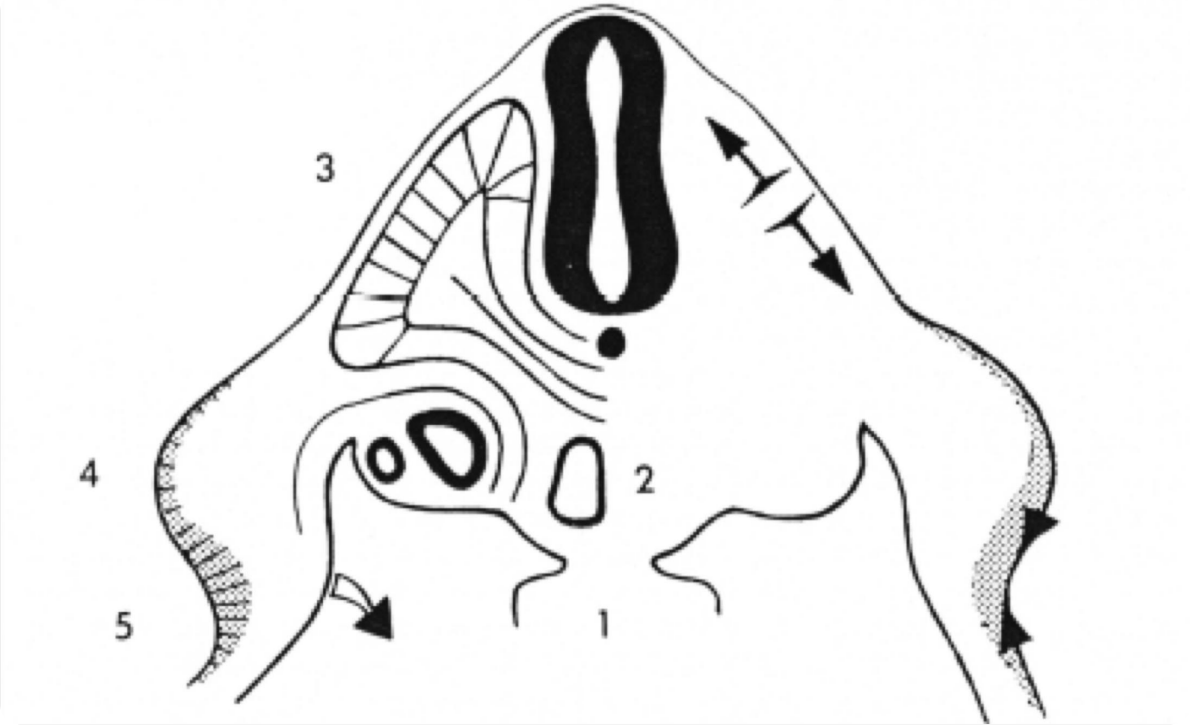


FIG. — Concept of aspiration and restriction fields, logically placed after the explanation of these mechanisms in cells and the neural tube

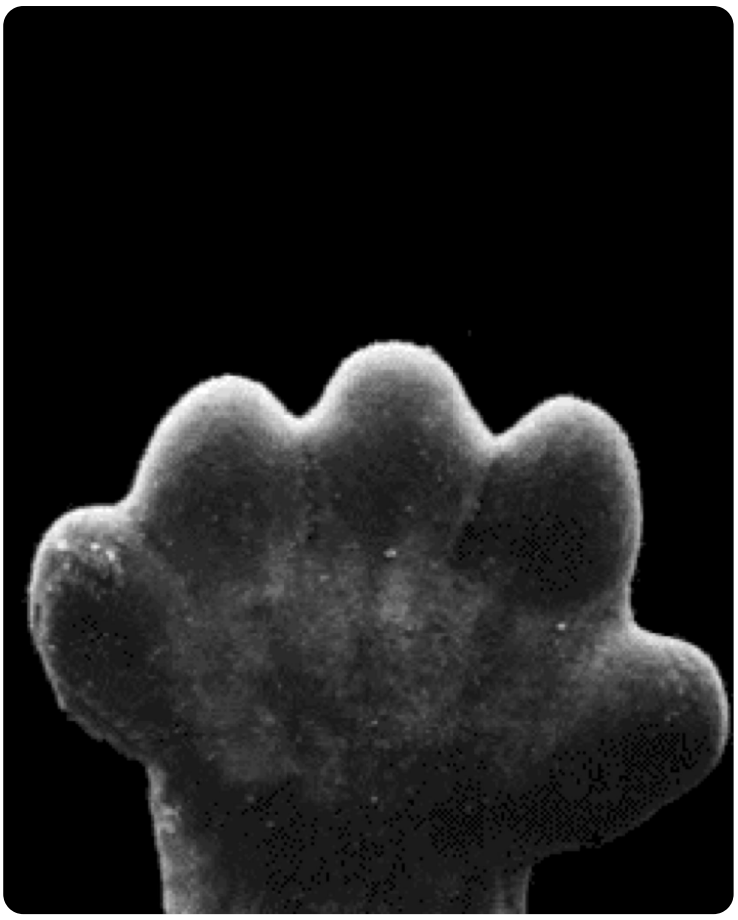


FIG. — Limb bud

The hand appears first in development, followed by the lengthening of the system. The flexion of the embryo is linked to the growth rate relative to the aorta. Joints follow a similar protocol, where the presence of blood vessels influences the density and shape of the structures. The external rotation movement is followed by an internal counter-rotation, orchestrated by the vascular field.

The **mesoblast** is responsible for the induction of the limbs, which are an expansion of the **cardio-pleuro-peritoneal sac**. Treating limb lesions requires their re-equilibration with the **pleuro-pericardio-peritoneal**. By freeing the limbs, better emergence of the upper and lower structures is allowed.

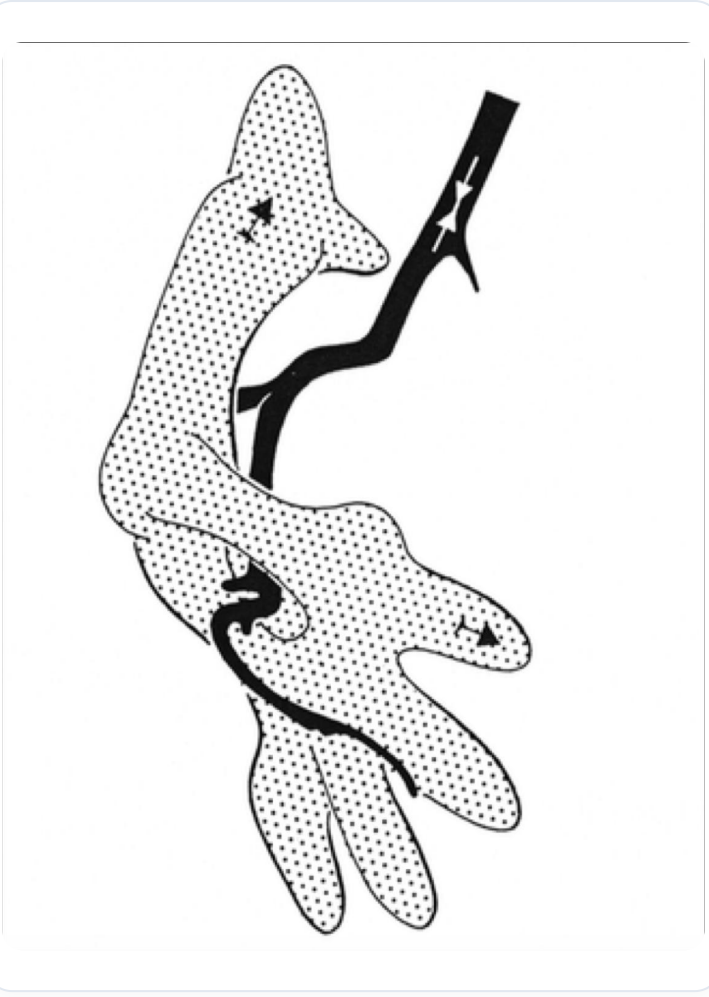


FIG. — Development of limbs

The vascular axis is fundamental for understanding metamerisation and limb arrangement. Each metamere is organised by the vessels and the neural crest, and each vertebra corresponds to an organ. Vertebral dermalgias are linked to specific organs, such as the liver or gallbladder.

The induction of the cellulome is oriented by the vascular field, influencing limb growth. The dynamic between bone and muscle is essential: initially, the bone is active, while the muscle is passive. Over time, the muscle takes over, defining the shape of the limbs.

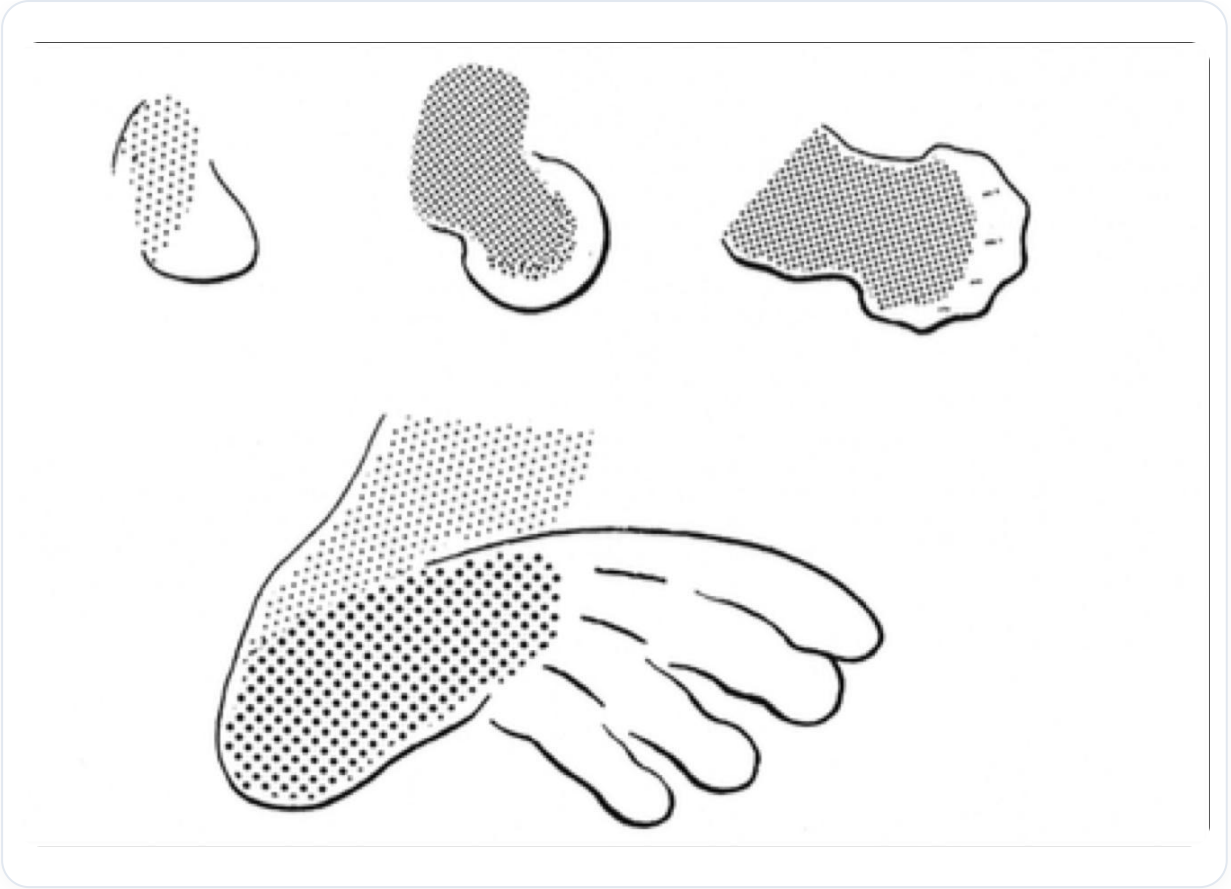


FIG. — Embryonic limb development

The development of the limbs, both upper and lower, is linked to **counter-rotation** and the organisation of the peritoneum. The base of the skull and the vertebral column derive from the **paraxial mesenchyme**, while the limb skeleton comes from the **somatopleure**.

Treating limb lesions involves reintegrating them into their peritoneal cavity, which is key to understanding the embryonic fulcrum.

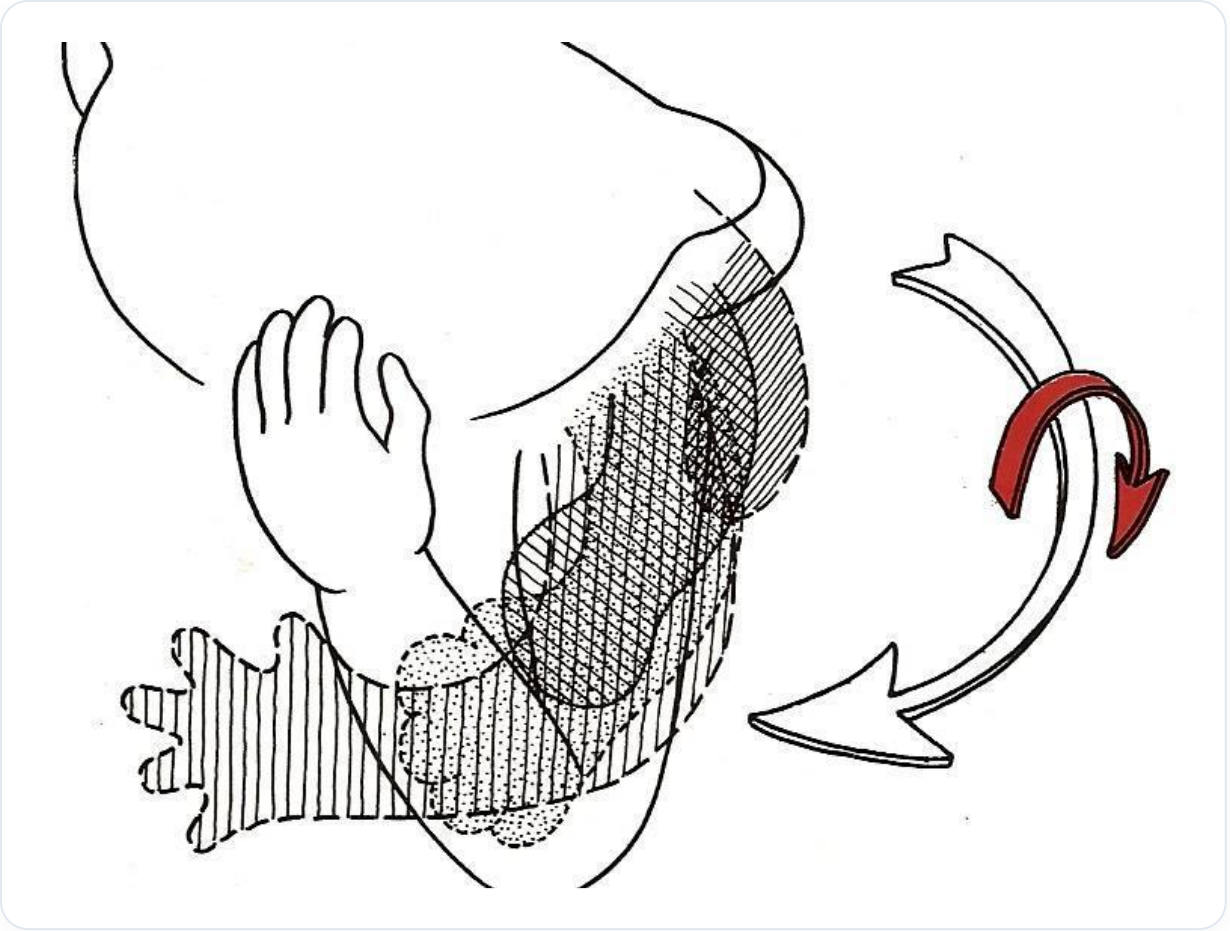


FIG. — Upper limb rotation

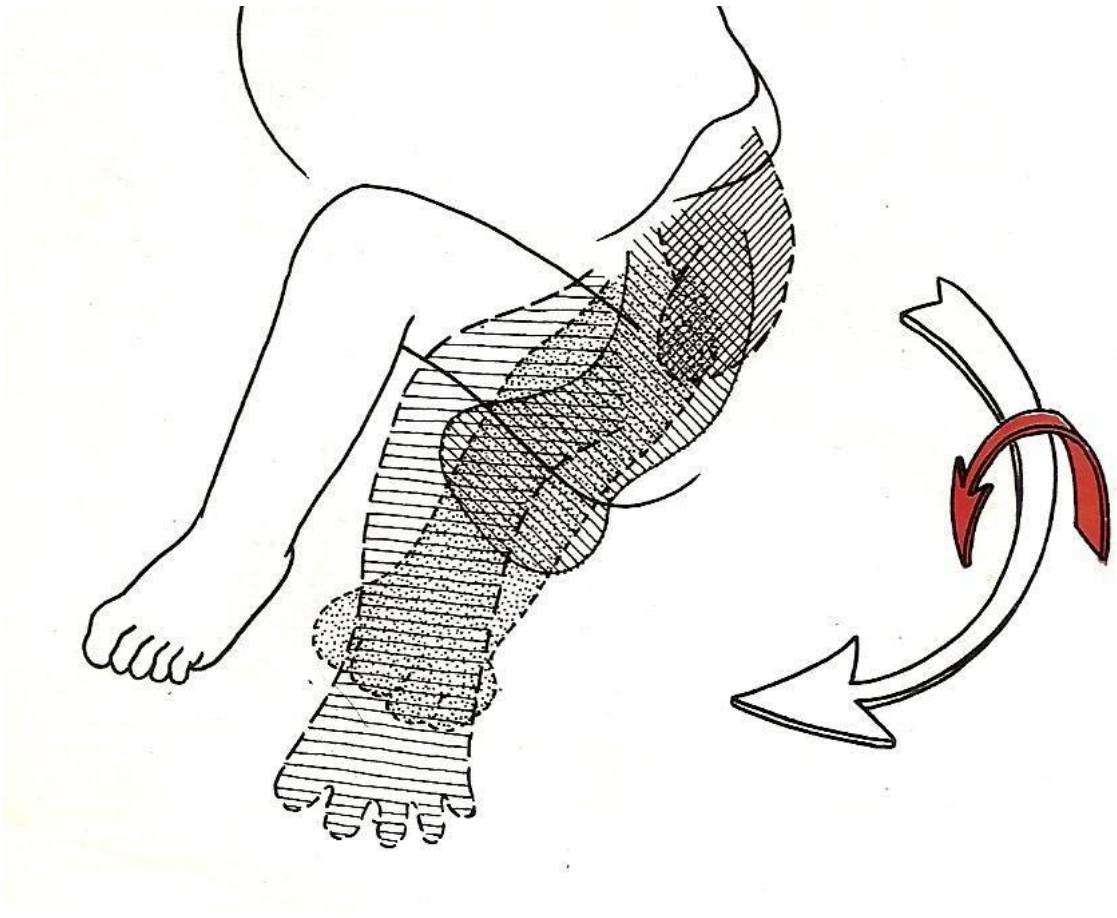


FIG. — Lower limb rotation

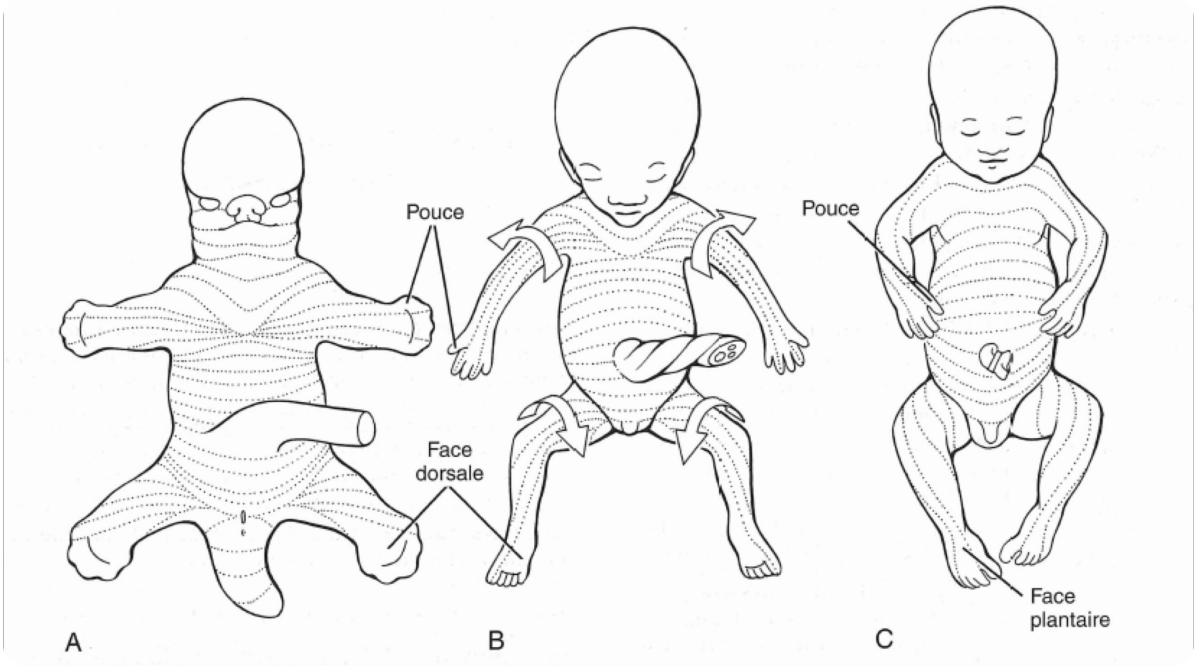


FIG. — Embryonic Langer's lines

43. PRACTICE : ACCESSING THE FLUID BONE : THE FIBULA

DURATION : 02:01 STUDY SHEET

COURSE SUMMARY

In this session, the focus is on exploring the fibula through a biodynamic approach. Practitioners will learn to feel the density of the bone, observe its structure, and develop sensitivity to the micro-crystalline field. By adopting a cellular then molecular awareness, participants are guided to align with the primary respiration, allowing bodily fluids to integrate harmoniously. The practice addresses key concepts such as the neutral, maturation, and the embryonic nature of this membranous bone. Finally, participants are invited to explore deeper realities ranging from the atomic to the luminous plane, highlighting the importance of water in this dynamic.

PRACTICE: ACCESSING THE FLUID BONE - THE FIBULA

In this practice, we will focus on the **fibula** to seek a sensation of **fluid**.

Begin by positioning yourself correctly. Hold the fibula as if you were holding a large feather. Observe its position and the structure of the bone.

It is essential to first focus on **density**. Adjust yourself to allow the bone to find its perfect spatial position. Concentrate on the **micro-crystalline field**. You will notice a change in warmth, a **thermodynamic** variation.

Next, move to **cellular awareness**. Imagine you are holding a packet of cells in your hands. Then, deepen your perception by adopting **molecular awareness**. Allow your primary respiration to guide you.

Remember to remain in your **neutral** and allow the person's neutral to mature. It is important that there is no maturation of the neutral on both sides. Sometimes, I invite a partner to position themselves at the base of the skull.

The bone is fluid and gradually integrates into the fluid body. Discovering this bone is interesting because it is a **membranous bone**, originating from a membrane that has condensed over time. It is an **embryonic** bone.

To go further, explore the **atomic plane**, then the **vibrational plane**, and finally the **luminous plane**. We have mainly worked on the molecular plane, where the **H2O** combination is found.

44. BIRTH

DURATION : 04:46 STUDY SHEET

COURSE SUMMARY

This video explores the profound impacts of birth on our lives and bodies, while offering practical methods to release tensions stored in bodily tissues. Key concepts include the importance of compression and decompression at the cranial, pelvic, and vascular levels, and how this affects the autonomous respiration of newborns. The speaker also highlights the central role of the spleen and oxygen transfer during the first breath, as well as the importance of fluid movement between the parietal and temporal bones to facilitate this crucial transition. Students will learn to better understand the link between bodily movements, spiritual dynamics, and the osteopathic approach, while integrating these principles into their therapeutic practice.

BIRTH: IMPACT AND MECHANISMS

The way you were **born** determines many aspects of your life. Every time you have to **be reborn** into something, like a job change, you can reconnect with that memory. It is sometimes necessary to release tensions in the bodily **tissue**, either by recompressing or decompressing it, in order to re-establish **respiratory disengagement**.

It is not enough to bring these memories to consciousness; they must be reborn in the tissue. In this approach, I will compress from the **parietals**. For a baby, if something seems necessary, I will work at the level of the **pelvis**. The child may need to be compressed or decompressed, and it is the feeling of the moment that guides this decision.

This process leads to a **spiritual** movement and dynamics in the body. At the **osseous-membranous** level, it is also crucial to consider compression at the **vascular** level. Imagine a sponge being compressed; this illustrates well what happens in the body.

The **spleen** plays an essential role during birth. If necessary, it will re-establish a flow, because a **rebalancing** is necessary throughout the vascular system, between the heart, the lungs, and the **umbilical cord**. This passage is fundamental, as it represents a major vascular change: you go from breathing and oxygenation via the cord to respiratory autonomy via your lungs.

The first **cry**, the first **breath**, is crucial. At that moment, you are in an umbilical axis, and blood enters through the **foramen ovale**. This is a major fluidic change. To facilitate this change, the **hepatic** and **splenic axis** intervenes, providing the necessary information. If this transition is not

well performed, some children may be born with **splenic** characteristics, often melancholic, because they did not perform this first breath correctly.

This passage of oxygen, of the global and fluidic compression of the system, as well as membranous compression, is essential. The importance of the **parietals** and the movement of the body should be emphasised.

The fluid movement between **parietal** and **temporal** is a first disengagement. This movement leads to a rotation of the **neck**, essential for releasing the body. The key moment to give the first breath is marked by a slight movement of the neck, followed by a spiral that forms.

This occipital movement, associated with a torsion in the axis, is crucial. Hyperextension of the neck and disengagement of the limbs are also elements to consider. Thus, there is compression in the membranes and compression at the fluidic level that occur simultaneously.

45. BONUS - LE MOUVEMENT RÉGÉNÉRATEUR

DURATION : 10:31 STUDY SHEET

COURSE SUMMARY

The Regenerative Movement (created by Marc Damoiseaux) is a powerful somatic practice where the body physically retraces its own genesis. Consciously, the practitioner relives the original polarization of the ovum, the cleavage of the first blastomeres, then traverses the gigantic embryonic folding (notochordal wave, heart looping, pulmonary shift, and rotation of the digestive fascia). This moving architecture ultimately brings back the perception of the Central Axis (Vertex - SSB - Heart - Sacrum), resetting the deep tissue balance of the median chain.

LE MOUVEMENT RÉGÉNÉRATEUR

Le mouvement régénérateur (création de Marc Damoiseaux) inspiré du mouvement développemental de l'embryon.

POSITION INITIALE

- Pousse de la main droite dans la main gauche, pieds parallèles. Largeur des épaules.
- Le dos est droit et détendu. Légère flexion des genoux.
- Et observation des différentes lignes médianes : La ligne médiane notochordale, postérieure et antérieure.

On observe bien la sphère neuro-sensorielle, rythmique, métabolique, urogénitale. Essayez de les ressentir, de vous équilibrer.

LES ÉTAPES DU MOUVEMENT

1. LA SPHÈRE

Dans un premier temps, le mouvement, l'ovule est sphérique. Vous dessinez dans l'espace une sphère devant vous qui représente l'ovule.

La première fonction de l'ovule en création est d'accepter. À l'intérieur, c'est l'acceptation de la fécondation pour créer cet œuf, le zygote.

2. LE MOUVEMENT TOROÏDAL

Une asymétrie s'installe. Dans un mouvement toroïdal qui se retourne tout le temps sur lui-même. Vous laissez ce mouvement parcourir votre corps pour faire le développement du corps.

3. LA NIDATION ET LE CORDON OMBILICAL

Le stade de la nidation. Au bout du 7ème jour et jusqu'au 21ème jour vient se mettre en place tous nos annexes : le cordon ombilical, le placenta, la cavité amniotique. Le mouvement est grand, protecteur.

4. LA LIGNE MÉDIANE

A ce stade-là avec de l'enroulement, de la formation du cœur, il y a un appel de la ligne médiane. Sentez ce retour vers votre ligne médiane, et ça se pose dans le cœur.

5. ROTATION ET COLONNE VERTÉBRALE

Cette asymétrie qui se met en place crée des phénomènes de rotation que ce soit avec la langue, que ce soit les bras, un membre va pousser vers le haut l'autre vers le bas. À ce stade-là notre tube neural s'est refermé.

Notre colonne vertébrale, notre moelle épinière est en place. Connectez la souplesse de votre corps, de votre dos en vous relâchant.

6. LE CŒUR ET LE PLISSEMENT

Le cœur avec toutes ces asymétries est obligé d'évoluer, il plonge, de quatre tubes il deviendra cœur. Avec ses pompes, reliez-vous avec souplesse dans ce mouvement du cœur où tout le corps se plisse pour se connecter et laisser l'énergie en place.

Il bat ce cœur...

7. RESPIRATION ET RETOUR

Le point de recueillement, ramener sur soi, puis laisser l'expansion. Sèche, la respiration. Relâchez-vous vers la fin, et laissez couler la liberté de la vie sur votre naissance.

L'important c'est de garder ce mouvement asymétrique et continuel, rien ne s'arrête jamais. Pour revenir sur l'essentiel, à la maison : l'humain. C'est vous, par la respiration.

Puis laisser reposer, refermer.